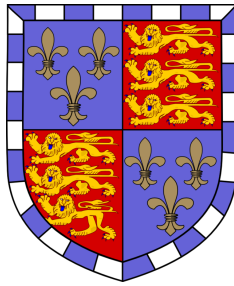




Scalable Autonomous Network Control using Minimally Informed Agents



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This thesis is submitted for the degree of
Doctor of Philosophy

Declaration

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or, is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text. It does not exceed the prescribed word limit for the relevant Degree Committee.

Yaniv Proselkov

April 2024

To my parents, the mentors who have guided and taught me, and the friends and colleagues who have supported me.

Abstract

This thesis contributes to solving the combined problem of dynamic scalable failure risk management in complex network infrastructures while returning tractable results. It does so by proposing methods of decentralised and distributed emergent risk identification and proactive mitigation, where assets, controlled by individual agents, use only limited network information to each inform live autonomous control.

This is completed in networks of telecoms - studied since information packet delivery speed is observable in computational environments - and supply chain financing (SCF) - studied since value for units is controllable. Measuring risk via criticality and viability, this thesis works around the challenge of managing a system without centralised observers, improving information delivery speeds (telecoms), and privacy (supply chains).

Studies are conducted by defining simulation environments, functions to measure baseline criticality, and then investigative analyses into the relationship of baselines with decentralised or distributed approximations of the baselines. This comparison is completed for a validation function that captures distributed dynamics of the given network, with respect to network measurement or control.

In telecoms, two criticality studies and one study into network changepoint analysis were conducted. The first study was direct propagation of linearly increasing criticality over a generated network. When telecoms networks cannot route data packets to an asset because it is processing other packets, they are queued. Upon asset disruption, queues are lost. Central assets must be protected because queues affect more neighbours. Graph operations on the topology of nearby assets quantify this. Restricting this to a limited region per asset can reduce response times and network pressure. The accuracy of limited measures for estimating criticality in dynamic simulations is studied, modelling criticality like an infectious disease. Accuracy benchmarking is theoretically derived. Information bounded criticality measures predict criticality.

A changepoint detection study is conducted on empirical UK backbone infrastructural dynamics data, comparing two multivariate time series data aggregation methods, and of this, several changepoint detection methods for this dataset. The first aggregation method is mean aggregation. The second conducts changepoint detection on all time series, clustering changepoints using kernel density estimation, returning changepoints as maximal distances between low and high likelihood kernel density estimates. This produces a decentralised method of changepoint detection.

The second criticality study proposes a model of packet routing with intercommunication, where performance of information bounded criticality estimates are measured by combining centrality measure timeliness and accuracy, creating a value of information metric. True criticality is determined using an empirical rather than theoretical benchmark. Findings show limited measures compute criticality estimates accurately and quickly, with high value of information, particularly in periodic intercommunication for degree centrality.

A model of supply networks is also introduced, similar to these, using orders instead of packets. These have cash values, constraining dynamics such that policies controlling financing dynamics are implicit of criticality, replacing measurement bias with performance targets. Two studies are completed, first defining the model, then investigating the effect of information limited distributed control over a modification of it.

The first defines a model of deep-tier complex SCF. A market node stimulates demand, propagating orders upstream, satisfied by deliveries in exchange for cash. Power asymmetries and payment delays induce financial stress on central but weak companies, causing failures which stress neighbouring nodes – the ripple effect. To survive this, individual agents finance themselves, dynamically changing financing benchmarks. Survival outcomes, depending on supply network topology, are explored, characterising the model. System survival times are found as bimodal, forming the basis for further investigations.

Limited range centrality measures from the telecoms studies are onto an advancement of the SCF model, informing financing benchmarks. Companies only share benchmarks, thus retaining privacy and giving dependence on distant shocks with appropriate damping. Nodes thus communicate instantly while avoiding full supply chain visibility. This mitigates against ripple effects, creating competitive, indefinitely viable supply chains. The relationship between visibility and long-term survival is found, expressed as the output of polynomial regressions. Findings show that in scale free topologies, information bounded distributed control enables long-term viability. Limiting visibility information is found to outperform full system visibility for long-term survival without affecting expected survival time of short-lived systems. Thus, it is shown that, by capturing only relevant information, limited information distributed decentralised control is a scalable and effective solution to autonomously mitigate against risk in infrastructural networks.

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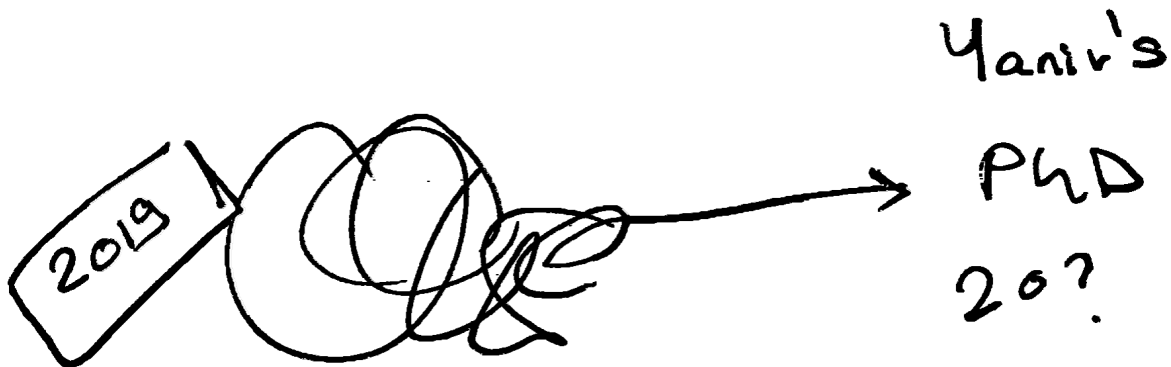


Fig. 1 PhD Progress, by Prof. Alexandra Melike Brintrup, 2022

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Nomenclature

Acronyms / Abbreviations

l_1	Least absolute deviation
l_2	Least squared deviation
BF	Bank Financing
CAS	Complex Adaptive System
CASN	Complex Adaptive Supply Network
CC	Constant Communication
CE	Criticality Estimate
CP	Communication paradigm
DAG	Directed Acyclic Graph
DTSCF	Deep Tier Supply Chain Financing
FRE	Financial Ripple Effect
GMM	Gaussian Mixture Model
IC	Instant Communication
IVM	Indefinite Viability Measure
KDE	Kernel Density Estimate
KDE	Kernel Density Estimate
KPI	Key Performance Indicator
LTVM	Long-Term Viability Measure
MSE	Mean Squared Error

OEM	Original Equipment Manufacturer
PC	Periodic Communication
PELT	Pruned exact linear time changepoint detection
RBF	Radial basis function
RE	Ripple Effect
RMS	Raw Materials Supplier
RSS	Residual Sum of Squares
SCF	Supply Chain Financing
SN	Supply Network

Centrality/Criticality

$A_{t,i}^{*(c)}$	Normalised aggregated network criticality for attribute $c \in C$, time t , simulation S_i
C_t^*	Normalised mean of an arbitrary CE at time time t over all nodes
ω_{vw}^t	Weighted sum of number of geodesics from u to v
$\overline{C_t}$	Mean of an arbitrary CE at time time t over all nodes
$\theta_{(uv)}$	Perturbation centrality of edge (uv)
$A_{t,i}^{(c)}$	Aggregated network criticality for attribute $c \in C$, time t , simulation S_i
$C(u)_t$	Arbitrary CE for node u at time time t
C_{BR}^E	Egocentric (Local) Bridging Centrality
C_{BR}^S	Sociocentric Bridging Centrality
c_b	Bridging Coefficient
C_B^E	Egocentric Betweenness
C_B^S	Sociocentric Betweenness
$C_B^{E(w)}$	Weighted egocentric (local) bridging centrality
$C_B^{WE(i)}(u)_t$	Weighted egocentric betweenness centrality for ego-net depth i of node u at time t
$C_B^{WS}(u)_t$	Weighted sociocentric betweenness centrality of node u at time t

$C_C(u)$	Closeness centrality of a given node
C_D	Degree Centrality
$C_D^W(u)_t$	Weighted degree centrality of node u at time t
C_E^E	Egocentric Eigencentrality
C_E^S	Sociocentric Eigencentrality
$C_E^{WE(i)}(u)_t$	Weighted egocentric eigencentrality for ego-net depth i of node u at time t
$C_E^{WS}(u)_t$	Weighted sociocentric eigencentrality of node u at time t
$C_K(u)$	Katz centrality of a node $u \in V$
C_L	Local Centrality
$C_L^{(w)}$	Weighted local centrality
C_W	Wehmuth Centrality
$C_W^{(w)}$	Weighted Wehmuth centrality
K_{uv}	Communicability between nodes $u, v \in V$
CM	Criticality Measures
IB-CM	Information Bounded Criticality Measures

Network Model Terms

β	Infection rate in an SIS mode
γ	Recovery rate in an SIS mode
$\mathbb{I}$	Set of Infected nodes
$\mathbb{R}$	Set of recovered nodes
$\mathbb{S}$	Set of Susceptible nodes
$B(n, m)$	Barabási-Albert preferential attachment network with n nodes and connectivity number m
$G(n, m)$	Erdős-Rényi random network model with n nodes m edges.
$G(n, p)$	Erdős-Rényi random network model with n nodes who each have a probability p of an edge between them.

BA	Barabási-Albert preferential attachment network
ER	Erdős-Rényi random graph of any sort
I	Infected node
R	Recovered node
S	Susceptible node
SI	Susceptible-Infectious disease model
SIR	Susceptible-Infectious-Recovered disease model
SIS	Susceptible-Infected-Susceptible model

Network Variables

(uv)	An edge from $u \in V$ to $v \in V$
$(uv)^{(i)}$	i^{th} edge between $u \in V$ and $v \in V$ in a multigraph
$\Delta\lambda_j(uv)$	Perturbation of the j -th eigenvalue, λ_j
δL	Laplacian of the perturbation matrix
$\Gamma_i(u)$	Set of nodes of geodesic i edges away from node $u \in V$
λ_2	Algebraic connectivity in connected networks
λ_i	i^{th} eigenvalue of L_N
$\mathcal{T}_u$	Set of nodes in the same tier as u
∇	Diameter of the graph G
$\overline{\mathcal{T}_u}$	Tier of the node u in a DAG
ρ_{uv}	Number of geodesics from u to v
$\mathbf{d}$	Vector of the degrees of nodes, $(d_1, \dots, d_n)$
$\mathbf{u}$	Vector of the nodes $(u_1, \dots, u_n)$
$\tilde{L}$	Weighted graph Laplacian
A	Adjacency Matrix
A^d	Adjacency matrix of G^d

$a_{i,j}^d$	Entry (i, j) of A^d
A_G^t	Temporal adjacency matrix at time t of the graph G
$a_{i,j}$	Number of edges between nodes u_i and u_j
$B_{t,i,u}^{(c)}$	Dynamic node attribute c for node u , time t , simulation S_i
C	Set of all dynamic node attribute types
D	Degree matrix
$d(u)$	Degree of $u \in V$
D^d	Degree matrix of G^d
d_l	Degree of $u_l \in V$
d_u	Degree of $u \in V$
E	Set of edges in G
E^d	Edges of G^d
G	A network graph
G^d	Directed multigraph incorporating node queues as edges
$H_i(u)$	Set of nodes of geodesic less than or equal to i edges away from node $u \in V$
k	Average degree
L	Laplacian Matrix
L_N	Normalised Laplacian Matrix
l_u	Mean distance of a node to all others
M_n	Square matrix of dimensions $n \times n$
P_{uv}^g	Set of geodesics between nodes $u, v \in V$
p_{uv}^g	Geodesic between nodes $u, v \in V$
P_{uv}	Set of paths between nodes $u, v \in V$
p_{uv}	Network path from u to v
V	Set of vertices in G

v	A node in V
V^d	Nodes of G^d

SCF Model Terms

β_{ij}^t	Available bank financing at time t
δ_{uv}	Payment delay between buyer u of power $\mathcal{P}_u$ and supplier v of power $\mathcal{P}_v$
γ	Discount factor for supply chain financing
$\hat{l}$	Proportion of l^* firms in the network
$\hat{m}$	Proportion of m^* firms in the network
$\hat{s}$	Proportion of s^* firms in the network
l_{ij}^t	Inventory at timestep i of firm u_{ij} at timestep t
κ_u^t	Current liquid cash of firm u at time t
κ_{ij}^t	Cash reserve of firm $u_{i,j}$ at timestep t
λ	Market demand distribution mean
$\mathcal{P}_{ij}$ or $\mathcal{P}_u$	Power and access to financing of firm u_{ij} or u
ρ_u^t	Bank financing debt limit of firm u at time t
ρ_{ij}^t	Given firm current unpaid debt
σ^2	Market demand distribution variance
τ	Supply chain financing product term
L	Set of all l^* firms
M	Set of all m^* firms
S	Set of all s^* firms
ξ_{ij}^t	Available supply chain financing at time t
b_u^t	Quantity of bank financing received by firm u at time t
b_{ij}^t	Given firm banking mandate limit at time t
$c(uv)$	Cost of one unit bought by firm u from v

c_u^t	Cash added to node u at time t
d_m^t	Demand generated by the market node m at time t .
d_{ij}^t	Demand at timestep i of firm u_{ij}
e	Delivery expenses
$f(\mathcal{P}_{ij})$ or $f(\mathcal{P}_u)$	Firm u_{ij} or u market share
f_u^t	Target safety cash stock of node u at time t
i_b	Annual bank interest rate measured in decimal values
i_s	Annual supply chain financing interest rate measured in decimal values
n	Number of tiers in a supply network
o	Operative expenses
o_c	Operating fee incurred by every firm in the network every timestep
P_a	Aggregate firm failure distribution
p_m^t	Market purchase order at time t
P_p	Piecewise firm failure distribution
P_{ij}^t	Vector of length 120 that records accounts payable for u_{ij} at time t
R	Total number of firm failures
R_i	Total firm failures in simulation run i
R_{ij}^t	Vector of length 120 that records accounts receivable for u_{ij} at time t
s^*, m^*, l^*	Firms with Small, Medium, and Large power, respectively
s_i, m_i, l_i	Number of failures on the i^{th} simulation of s^*, m^*, l^* firms, respectively
s_u^t	Supply chain financing for firm u at time t
S_{ij}^t	Supplier chosen at time t to supply firm u_{ij}
s_{uv}	Material unit value along edge (uv)
U_i	Set of firms at tier i
$u_{ij} \in U_i$	The j^{th} firm at tier i

Telecom Model Terms

$(uv)_q^t$	Data packet queue weight of of edge (uv) at time t
$\mathcal{Q}$	Set of all queue lengths
μ	Monitor interval
ϕ	Data packet backlog capacity for each node in a telco simulation
Φ^t	Total number of queued data packets in the whole network
ϕ_u^t	Size of the queue held by node $u \in V$ at time $t \in \Upsilon$
$\rho_{u,v}^{(w)}$	Number of geodesics from u to v , weighted by queue
Υ	Time series of all node data packet queues in a single simulation
$q(u)$	Queue length of u
q_{uv}^t	Queue at time t that node u perceives that v has
q_u^t	Queue at time t that node u perceives itself to have

Misc. Variables

ψ_{CP}^{*CE}	Normalised accuracy of a CE under a CP
SAD_i^*	Normalised SAD_i
THD_t^*	Normalised time series of differences in $MA_{\Phi,h}$ across h
ϑ_{CP}^{*CE}	Normalised timeliness of a CE under a CP
$\bar{M}^c$	Mean MSE over all simulations of attribute c versus Q
$\bar{x}$	Prominence of a peak x
η	Number of highest valued changepoints
Λ_D	The number of prominent peaks of D
$\langle X \rangle$	Mean of the set of numbers, X
$\langle v \rangle$	Mean value of information provided over all CPs for all CEs
$\langle v_{CE} \rangle$	Mean value of information for a CE over all CPs
$\langle v_{CP} \rangle$	Mean value of information from a CP over all CEs

$\langle \text{Comp}_{\text{CP}}^{\text{CE}} \rangle$	Mean computation time for a CE under a CP over all nodes and timesteps
$\mathbb{Z}$	Set of integers
$\mathbb{Z}^+$	Set of non-negative integers
$\mathcal{A}$	Semblance of R_c and R_p
$\mathcal{T}$	Set of all points circumscribed by the ideal time horizon
$v_{\text{CP}}^{\text{CE}}$	Spatiotemporal value of CP information for a CE
$\psi_{\text{CP}}^{\text{CE}}$	Accuracy of a CE under a CP
σ_D	Standard deviation of D
Err_t	Error from the benchmark at time t
IQR_D	Interquartile Range of the distribution D
IVM^X	Indefinite Viability Measure of a space of simulations X
LTMV^X	Long-Term Viability Measure of a space of simulations X
MAD_i^t	Absolute difference at time t of $\text{MA}_{\Phi;h_i}^t$ from Φ^t
$\text{MA}_{\Phi;h_i}$	Time series of moving average of total queues in network over window h_i
$\text{MA}_{\Phi;h_i}^t$	Moving average at time t of total queues in network over window h_i
RMSE	Root mean squared error
SAD_i	Sum of absolute differences of moving averages to total queues
THD_t	Time series of differences in $\text{MA}_{\Phi;h}$ across h
$\tilde{h}$	Bandwidth tuning parameter
ς	The inverse of the median of all pairwise distances in a given space
$\vartheta_{\text{CP}}^{\text{CE}}$	Timeliness of a CE under a CP
$\widehat{X}_D^Y(Z_D)$	Regression curve of variable X for topology Y with respect to variable Z over dataset D
Ξ	Prominence threshold
B_D	Breakpoint separating short-lived and long-lived simulation runs over the dataset D
B_D^{post}	Long-lived simulations over the dataset D

B_D^{pre}	Short-lived simulations over the dataset D
c_{rbf}	RBF cost function
c_{l1}	$l1$ cost function
c_{l2}	$l2$ cost function
D_{KL}	Kullback-Leibler divergence
g_i	Relative gain in error over windows i and $i + 1$
h	Ideal time horizon
h_{test}	Number of time windows tested
h_D	Modified Silverman Bandwidth
h_i	i^{th} time horizon window tested for a simulation
h_{val}	Validation threshold for determining h
M_i^c	MSE for simulation S_i of attribute c versus Q
P	Empirical distribution of failures per firm type
Q	Theoretical distribution of generated firms
R_c	Given region
R_p	Information covering region
r_s	Spatial/temporal Relevancy function
S	Set of all simulations in a test
S_i	i^{th} simulation in a test
T	Set of all runtimes in a test
$T^{(f)}$	Set of reduced runtimes in a test
t_{max}	Runtime of any test in the telco model
t_f	Time window of criticality in the augmented SIS
T_i	Runtime of the i^{th} simulation in a test
T'_i	Termination time of the i^{th} simulation in a test

$T_i^{(f)}$	Reduced runtime of the i^{th} simulation in a test
$U(x, y)$	Uniform random variable between x and y , such that $x \leq y$
x'	Placeholder for a variable x when conducting operations on it within a timestep
Comp	Computation time for a CE under a CP at time t on node u

Chapter 1

Introduction

“... In that Empire, the Art of Cartography attained such Perfection that the map of a single Province occupied the entirety of a City, and the map of the Empire, the entirety of a Province. In time, those Unconscionable Maps no longer satisfied, and the Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast Map was Useless, and not without some Pitiableness was it, that they delivered it up to the Inclemencies of Sun and Winters. In the Deserts of the West, still today, there are Tattered Ruins of that Map, inhabited by Animals and Beggars; in all the Land there is no other Relic of the Disciplines of Geography.

— Suarez Miranda, *Viajes de varones prudentes*, Libro IV, Cap. XLV, Lerida, 1658”

This short story, “On Exactitude in Science”, by Jorge Luis Borges, was written in 1946. Suarez Miranda never existed, “*Viajes de varones prudentes*” was never written, and the accreditation is completely false. The piece works as a metacommentary on the nature of simulation and modelling — a false story about a false empire creating a false version of itself by a false person in a false book. While the story illustrates the fallacy of seeking to represent a complex lived system at the same scale and precision as the system itself, the accreditation creates an entire plausible reality with only a single line, a snippet of information.

This works because we, as readers, already exist in the reality in which Borges constructs his false author, but the story exists in the simulation conjured by the false author. For it to have its power, a true approximation of reality had to be created. This is not a paradox — the map created by the false empire tried to represent that empire in unnecessary detail, while Borges uses just enough to get us to think about the nature of modelling and simulation. In creating models, one must recognise the complexity and richness one needs to know to capture particular effects.

A key effect this thesis investigates is the catastrophic breakdown of complex infrastructures. We are growing more reliant on complex, dynamic network systems to sustain our lives. These

grow larger and more codependent every day to efficiently service our needs. As their pervasiveness increases, the impact of their disruption grows as well.

Global systemic fragility is evident from the shock of the COVID-19 pandemic, the disruptions to international trade by the blocking of the Suez Canal by a single ship, financial shocks by financial services firms, including Greensill, Deutsche Bank, or SVB, and ever-increasing burdens on telecommunications by hybrid work culture and consumer AI tools such as ChatGPT or decentralised ledger technology. This, coupled with escalating international geopolitical instability, suggests infrastructural design and control should move their design focus away from efficiency and output, to risk mitigation.

These are built systems, and managing them in a way that accounts for their complex emergent dynamics and protects against failure could have greater modelling complexity than the systems themselves. As illustrated in, “On Exactitude in Science”, however, this is needlessly difficult, and utterly useless.

1.1 Problem Statement

Suppose a telecom service increases the number of nodes operating in its network to meet demand, or a supply chain adds more specialised companies as product complexity increases. Such activities are done to enable more effective service, however, they introduce greater complexity, which further introduces more management challenges to enable smooth operation. A potential solution to this would be to make network assets autonomous, such that the system becomes self-governing. In the context of telecoms networks and supply chains, this would involve enabling either nodes or supply chain companies to operate as independent decision-making agents with their own information gathering functions, such that they can make decisions that serve the system’s functions without some central controller. However, within infrastructural networks of collaborating agents, such as telecoms networks or supply chains, the classic assumption is that following the policy of maximising information allows one to minimise risk and variance. Under this assumption, to derisk the loss of control inherent within autonomy, each autonomous agent would need to increase the amount of information they take in as the system grows. This limits the size of the network. To remove such limitations, agents must have a scalable information collection policy, not growing as the network does. In this thesis, it is hypothesised that there are circumstances where information maximisation is disadvantageous, such that systems can improve performance by reducing information supplied to agents. This would allow for scalable autonomous control of infrastructural networks, enabling their unfettered growth.

This thesis therefore investigates methods of scalable autonomous control for infrastructural systems, broken down into control and analytics for the purpose of controlling against system failure. In particular, I investigate the relationship between information provision and network risk, studying how relevant certain types and quantities of information are to agents controlling that system. This

Table 1.1 Table of problems (on the axes) and associated solution techniques (in cells)

	Failure Risk Management	Complex Network Infrastructures	Scalability	Researchability
Dynamicity	Autonomous Control	Dynamic Networks	Relevancy	Ad Hoc and Post Hoc Analytics
Failure Risk Management		Network Criticality and Centrality	Decentralisation and Distribution	Simulation
Complex Network Infrastructures			Information Limitation	Multi-Agent Systems
Scalability				Validation

gives the research question: *How can information requirements for scalable autonomous control of infrastructural networks be reduced?*

Infrastructural network breakdowns are catastrophic events for human welfare, such that intentionally inducing one is infeasible and is thus avoided, making data on their dynamics sparse. To enable their study, I create several dynamic multi-agent complex multilayer network models, simulating infrastructural network breakdown, and employ these as test environments.

Complex multi-agent models are used because, as in reality, these models must be nonlinear, both in system and information dynamics, for agents to operate meaningfully with reduced information. In linear systems, control optimality grows with information provided. In nonlinear ones, this is not always the case. By creating hard to navigate environments — maps the size of empires — I create a way to learn the optimal model environment complexity that agents perceive within these larger environments.

There also exists the problem of measuring the dynamics of a system to condition against risk. Traditional measurement focuses on performance, while methods are needed that instead make a target function of the system dynamics. Existing metrics may be adapted for measuring risk management, including robustness (resistance to shock), resilience (recovery from shock), and viability (adaptation to improved function from shock), all extant in their application upon simulated and empirical systems. To manage these risks as they appear, dynamicity is necessary, demanding autonomous control in complex systems, in turn demanding risk measures to be extended into the dynamic space.

I argue that through use of network assets as cooperative controllers or sensors (where scopes of cooperation need not cover the network), autonomous networks infrastructural management can be possible, such that the requirements and the restrictions of achieving are reinterpreted as opportunities, and propose a collection of ways to use them, contained within this thesis. Each study needs dynamics which are **researchable**, giving solutions for **failure risk management in complex network infrastructures** which are both **dynamic** and **scalable**. Techniques and properties of such studies that satisfy this are shown in Table 1.1 and Figure 1.1. The following problems, and the solutions proposed for them, all fall within this nexus.

System agents represent network features that can act or be acted upon. To condition against risk, system agents should be endowed with new technologies and organisational dynamics to perform their

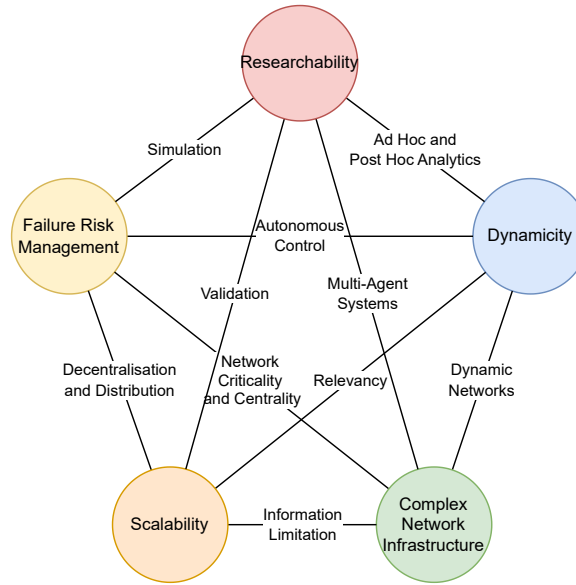


Fig. 1.1 Graph of problems (on the nodes) and associated solution techniques (on the edges)

functions quickly and effectively. Thus, the agents require delivery of key performance indicators (KPIs).

I further hypothesise that, by correctly restricting information to whatever is spatiotemporally relevant in a complex dynamic system, substantial accuracy and speed can be achieved in computing and delivering KPIs, to the point that methods of dynamic autonomous control become scalable and effective. I achieve this through decentralisation and distribution.

1.2 Research Gaps

System agents in abstract networks can be described as network agents. Here, agents are nodes, and their interactions are connective edges. In line with obtaining relevant spatiotemporal information to manage risk, core KPIs affecting risk for each agent are time to failure, likelihood of failure, and impact of failure — all are expectations. Time and likelihood of failure are largely functions and measures of asset condition, while impact of failure — criticality — is a function and measure of network conditions. Means to interpret and measure criticality in an information limited, distributed manner that satisfy the requirements posed in Figure 1.1 are therefore needed, with this research gap filled using the work in Chapter 4. This is compared against decentralised measures using network topological metrics of criticality and centrality, studied within a telecoms infrastructure network.

Often, agent-based simulation studies assume regular updates of network conditions, including criticality. However, dynamics in infrastructural systems are not necessarily constant, varying throughout the day. The unsolved problem of how to capture this irregularity in criticality dynamics in a manner satisfying the requirements posed in Figure 1.1 forms a research gap, solved in Chapter

5. I created a changepoint decomposition technique that separates epochs in a time series, taking advantage of network ensemble information, and produced findings on the intraday epochs of a national telecoms infrastructure.

Infrastructures have many agents within them with local scopes of interaction with respect to operation supported by the infrastructure, defined by how relevant that information is to operation. Finding ways to quantify relevancy in a manner that satisfies the requirements of Figure 1.1 is an unsolved problem that forms a research gap, which, if solved, can help define this scope. Telecoms infrastructures govern information dynamics, therefore information processing itself utilises network assets, so models can directly measure accuracy and timeliness with respect to information load. By imposing artificial limitations on communication rates, one can simulate communications lag without introducing extra factors. A method for doing so is defined in Chapter 6, based on information accuracy and timeliness of a given topological measure.

Under distributed control, rather than responding to a network wide instruction, network agents operate independently but cooperatively and can function with only short range knowledge of the network. When functioning distributedly, each node has fewer and closer sources of information and makes more decisions independent of higher level controllers. Agents can also work dynamically, meaning as network properties change, say, the distribution of items stored at the nodes, agent policies adjust. Doing so improves the ability to serve network functions at scale. Shorter information collection ranges reduce communication overheads, devoting fewer resources to information delivery. This requires cooperative systems to be possible.

One means of cooperation is a market, where trade dynamics are driven by demand and enabled by currency, such that value generation means sustained operation. This quantifies agent operational value at a given time, as a KPI. Balancing financial KPIs, say, liquidity, enables ad-hoc, distributed, live control. Thus comes the problem of modelling such a system with appropriate nonlinearity in a manner that satisfies the requirements of Figure 1.1, which forms a research gap. I solve this problem in Chapter 7 by defining a model of supply-chain financing (SCF) in an adaptive network, filling the research gap. In it, control policies are proactively and dynamically set as desired liquidity at a given timestep, as a function of historic financial position.

This does not answer the problem of information relevancy for control in a system that satisfies the requirements of Figure 1.1, which by being unsolved defines a research gap. I solve it in Chapter 8 by introducing a local decentralised control policy on the supply-chain model over complex networks, filling the research gap and reconciling all aforementioned KPIs through managing liquidity according to community contribution.

1.3 Research Aims and Objectives

I aim to advance knowledge in the field of dynamic, scalable failure risk management for complex infrastructural networks, considering the difficulty of obtaining data on catastrophic breakdowns of

infrastructural networks. Each factor of this — researchability, failure risk management, scalability, dynamicity, and complex network infrastructure — are particular problems to solve. Techniques for doing so exist along the intersection of problem pairs, and the union of all techniques describes each study in this thesis (Figure 1.1). I thus define research environments, methods, dynamics, and outcomes of systems employing each solution simultaneously.

This gives a collection of dynamic network simulation models with agents that employ decentralised and distributed methods for centrality and criticality calculation to monitor and mitigate approaches to catastrophic failure. Within this I study the effect of information limitation. This in turn permits autonomous control of network dynamics, contributing to our understanding of information relevancy, and together forms my research objective.

I investigate this with simulation models, which enable detailed analysis and full control over dynamics, improving the clarity of findings. Results are validated via a collection of post-hoc and ad-hoc analytics, representative of ways to conduct such control and analytics in real-life network infrastructures.

1.4 Contributions

This section describes contributions towards given research objectives within corresponding chapters.

1.4.1 RO1: Exploring existing solutions towards scalable autonomous control

C3: I generate a literature review of the state of the art of infrastructural systems modelling for risk management and associated methods to determine gaps in existing literature as well as collect together cross-disciplinary efforts in the task. I also review existing centrality and criticality measures, comparing and contrasting their computational complexity, showing the value and need for distributed measures for scalability.

1.4.2 RO2: Representing network failure dynamics while preserving meaningful nonlinearities

Nonlinear system dynamics are why an agent can restrict its information scope, but models require simplification to derive meaning from them. The following contributions deliver methods to represent network failure dynamics while preserving meaningful nonlinearities.

Models in Chapters 4 and 6 represent flow networks with routing and queuing in telecoms infrastructure. Chapter 5 gives methods to learn irregular inter-day changepoints. Chapters 7 and 8 build on these, introducing agency and control for trade networks in supply-chain financing infrastructure.

C4: A modification of the susceptible-infected-susceptible (SIS) model of network dynamics, investigated over a scale-free network. Nodes have fixed queue capacity, iteratively filled upon

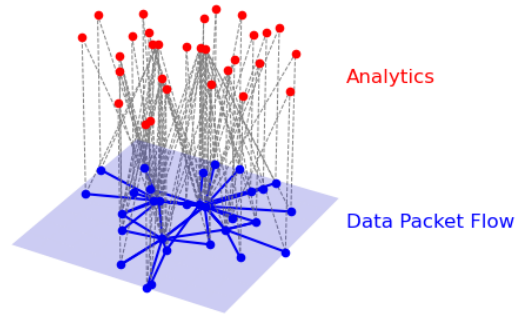


Fig. 1.2 Two-layer telecoms data packet flow and analytics model. Analytics do not affect the data packet flow layer, and do not interact

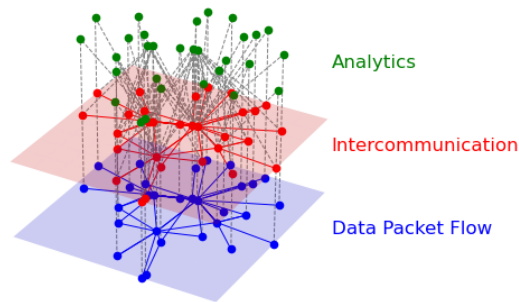


Fig. 1.3 Three-layer telecoms, data packet flow, intercommunication, and analytics model. Intercommunication does not affect the data packet flow layer, but does interact.

infection. Once full, nodes become infectious, and at any timestep, nodes can shrink their queues, representing backlogging and processing of queued data-packets. This defines an environment to investigate network queue system risk propagation. With analytics, this forms a two-layer multiplex dynamic network (Figure 1.2).

C5 A novel method of intraday epoch changepoint detection in telecom infrastructures. Using BT backbone datapacket traffic data, results are broken into 6 sections, useful for informing update dynamics in network analytical studies, as a basis for asynchronous or irregular agent updates.

C6: Three-layer model using an existing telecoms data packet generation and routing model, inducing a simulated agent-based communication model over this. Layer one routes data packets, two is internode communication, and three is analytics (Figure 1.3). Nodes simulate receiving lagged information about the queue states of other nodes. Information reaches nodes depending on their geodesic path length between one another and given communication paradigm. This defines an environment to investigate dynamic network analytics by network assets under information transmission rate constrained dynamics, such as for autonomously controlled telecoms networks.

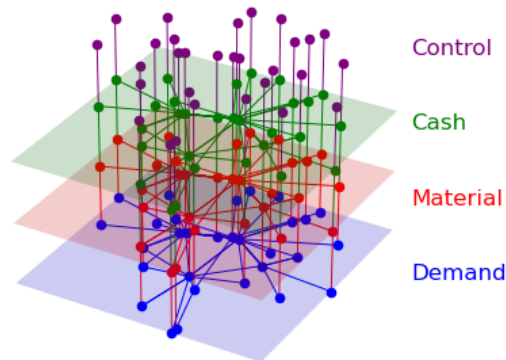


Fig. 1.4 Four-layer demand, material, cash, and control model. The control layer affects cash dynamics, but does not interact.

C7: A four-layer flow network with trade dynamics of deep tier supply-chain financing networks. Layer one propagates demand, two delivers materials in response to demand, three governs cash flow, and four distributedly injects liquidity ad-hoc with disconnected control nodes (Figure 1.4). Replaces path routing with heuristic, transaction routing. Orders stimulate material dynamics, enabled by cash exchange. This extends upon C4 and C6 by making layers influence each other — communication layer dynamics determine material layer dynamics, (analogous to data-packet routing), and the cash layer restricts material dynamics. Nodes also have agency via cash injection for liquidity, servicing the material layer. This model defines an environment to investigate dynamic network autonomy, where information transmission, affectation, and response rates are constrained by topology.

Investigations were also made into the effects of topology and power on dynamics and survivability, including power asymmetry in firms (represented by node agents), and tier width asymmetry. This gives theoretical findings into how catastrophic failure occurs within supply networks, including company failure distributions, and survivability distributions of the supply network.

C8: A five-layer model, building on C7. A collaboration layer gives nodes local information from ego-networks surrounding them (Figure 1.5). This moderates distributed control with scalable decentralised analytics, informing autonomous cash injection to enable long-term supply infrastructure viability. This defines an environment to investigate distributed, decentralised dynamic network control and autonomy, where information transmission is moderated by visibility, such as for a supply-chain with limited partner cooperation.

1.4.3 RO3: Network estimation methods leveraging distributedness in infrastructural networks to mitigate risk

C4: A collection of dynamic information bounded criticality estimates, estimating criticality through particular computation of ego-network structure. These can be applied to empirical network models to investigate criticality propagation in ways graph-learning derived measures cannot.

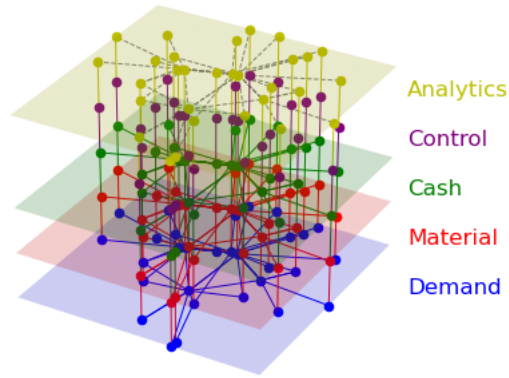


Fig. 1.5 Five-layer demand, material, cash, control, and distributed decentralised analytics model. The control layer affects both cash and analytics layer. The analytics layer interacts with itself and the control layer to then adjust the control layer.

C5: Two methods to extract epoch change information from dynamical network data in a decentralised manner. The first aggregates collections of time series and then treats them as singular. The second creates time series ensembles from time series of all network agents via kernel density estimates (KDEs) to detect shifts in network state, particularly variations in criticality, based on network agent similarity.

C7: Each node holds a dynamic benchmark criticality, informing distributed proactive financial control for each node. It is derived from internal moving averages of node states to create a proactive financing measure. Agents control for network survival, defining a distributed, dynamic form of network control to maintain viability.

C8: The benchmark criticality function is determined using the cash-states of a given agent and a local ego-network. This moderates distributed criticality control with scalable decentralisation where each actor uses a limited range of assets to inform control of itself, making the network viable.

1.4.4 RO4: Measuring risk for analytics and control for infrastructural networks

Throughout this thesis, I use the various models developed as proxies for real dynamics to investigate how information provision relates to criticality. This has been in various ways, and each has required methods to analyse the models and measures for their accuracy to each other and to characterise the models. With respect to the distributed, decentralised, and dynamic structure I am investigating, I developed various novel analytical techniques.

Forms of benchmark criticality were defined upon each risk propagation model. C4 and C6 define theoretical and empirical boundaries for temporal criticality impact measurement, while C7 and C8 control for criticality in time.

C4: Risk propagation dynamics and temporal criticality are defined according to simulation parameters. Through stimulating overload in each node, and returning these values, I create a temporal measure of the network's criticality to endogenous shock. Critical impacts occur over a time horizon, defined using a percolation argument as an infection rate function. This gives a dynamic network criticality measure defined by theoretical network dynamics. Local nodal decentralised criticalities are aggregated and compared against the systemic criticality benchmark in a time series, aggregated over the time horizon. Error over time from the associated benchmark is measured. This is a means to measure the accuracy of dynamic criticality estimates over time.

C6: Risk propagation is a nonlinear network function, so theoretical dynamics are unpredictable. Therefore, I develop an empirical dynamic criticality benchmark. The impact time horizon is a function of differences between time-series of network state moving averages. Dynamic benchmark criticality is a function of network states over the time horizon, computed in a decentralised manner. Differences between this and local decentralised criticality estimates over time gives the error, from which accuracy is computed.

The computational timeliness and accuracy of each agent is measured. Since computation time is work done to ensure accuracy for given criticality estimates, and information lag affects accuracy by reducing information relevancy, the value of information is a function of accuracy and timeliness. I generate it, developing a measure of the worth of a distributed dynamic analytics function for high speed control, given a communication paradigm.

The code for the simulation analysis uses agent based modelling techniques to permit measurement of computation for singular agents, and can be used in further analytics for different topologies, while the methodology can be used in investigations for installing decentralised analytics systems. It was shown that with less historic information, information value is better captured by simple, bounded measures than complex ones.

C7: I propose ways to study the relative influence of power distributions on network breakdown over multiple simulations. This can be used to characterise whether criticality is derived from supply-chain structure or from conditions the supply-chain is operating under.

C8: I define and explore two measures of long-term viability, relying on the methods from dynamic network analysis I construct that distinguish short-lived from long-lived systems using changepoint analysis and kernel density estimation. From this I build methods of survival space analysis, useful for characterising the nature of viability in supply-chains. This moves towards a theory of the relationship between present dynamics of infrastructures and their future performance. As long-term performance is enabled by distributed ego-network control, this investigation characterises their approach to an operationalisable benchmark of criticality.

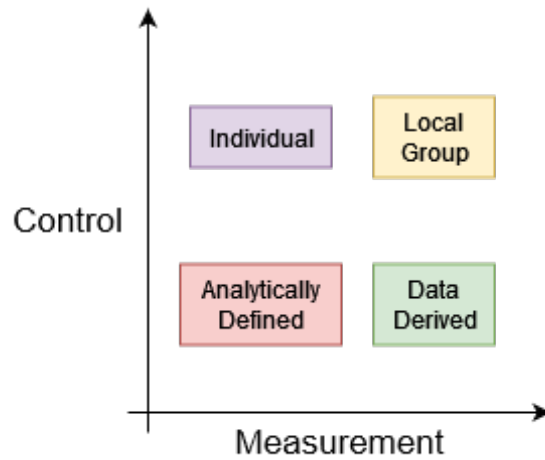


Fig. 1.6 Control-measurement complexity plane, illustrating the benchmarking techniques developed in this thesis as a function of how abstracted or restricted the measurement and control functions were in a given model.

These can each be used as methods to define dynamic criticality, depending on contexts applied. Abstracting away all dynamics other than linearly risk propagating ones, C4 can be applied. If incorporating communication limits or other nonlinearities that affect measurement capabilities, doing so for sensor design rather than control, C6 can be used. If simulating selfish controllable dynamics, C7 can be used. Finally, if investigating cooperative control, C8 model can be used. This can be expressed in the control-measurement benchmark complexity plane illustrated in Figure 1.6.

These measures each illustrate the capacity for limited information to give useful estimates of network criticality, thus providing scalable methods of network control and observation that do not overburden network operators with computational and data processing load.

The contributions come together to form a research outline, with a research question and four research objectives (Figure 1.7).

1.5 Thesis Structure

Following this, Chapter 2 contains important background material, necessary to understand many topics discussed in this thesis. I cover network scientific concepts in existing literature, introduce network model environments and generators, and data scientific methods.

Next, in Chapter 3, I review existing model environments and pertinent criticality and centrality measures, detailing their computational complexity, justifying the need for research into scalable methods. Following this is a brief review of factors important to information reach, then a review of infrastructural network studies, particularly in telecoms and supply-chain systems. Then is a comparative review of criticality measures and their complexity.

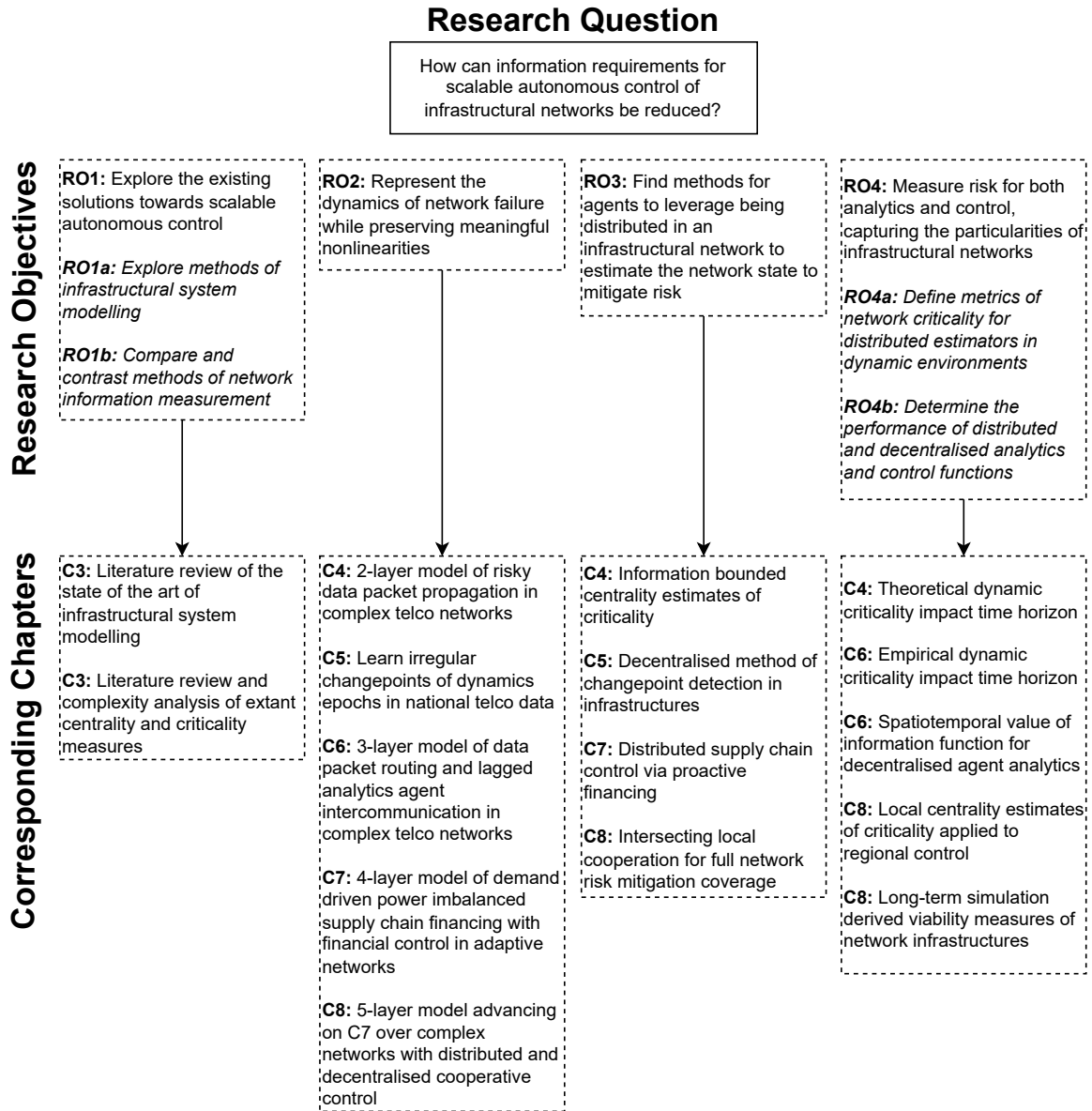


Fig. 1.7 Research Outline

Chapter 4 introduces three novel distributed measures of criticality which can be used to predict the behaviour of dynamic processes occurring in a network. Their performance is compared and tested on a simulated diffusive data transfer network.

In Chapter 5, I introduce a novel method of changepoint or epoch detection in the intraday case of a telecom infrastructure. This is tested on a case study dataset of infrastructural telecom dynamics to learn intraday behaviour of an entire network infrastructure.

In Chapter 6, I propose an agent-based model to assess the value of the information for calculating decentralised criticality metrics, assigning a data collection agent to each network element, computing relevant indicators of the impact of failure in a decentralised way. This is evaluated by simulating discrete information exchange with concurrent data analysis, comparing measure accuracy to a benchmark, and with measure computation time as a proxy for computation complexity. Results show losses in accuracy are offset by faster computations with fewer network dependencies.

In Chapter 7, I detail a model of financial dynamics with associated autonomous control functions. This is an agent based supply network simulation model capturing the effects of financial stress from late payments due to firm-to-firm power imbalances. I investigate structural conditions that make supply networks vulnerable to financial stress propagation and the resultant financial ripple effects using survivability analysis.

In Chapter 8, I propose a financial liquidity policy sharing method for firm-to-firm supply networks, introducing a scalable autonomous control function for viable complex adaptive supply networks. Collaborative set overlap makes all firms interdependent, reconciling cooperative and competitive aspects of the supply-chain and distributedly governing risk. I explore the relationship of visibility to viability in this using dynamic complex adaptive systems modelling. I found viability needs cooperation; visibility and viability grow together in scale-free supply networks; and distributed control where firms are supplied only limited partner information outperforms centralised control.

This thesis is then finished with general conclusions, summarising my contributions in detail, explaining limitations in my research, and from this proposing what broader impact my thesis means for wider literature and policy, and future work that can extend out from it.

1.6 Publication List

The following publications and preprints were produced over the course of the PhD:

- Chapter 8: **Y Proselkov**, L Xu, A Brintrup. Limited collaborative intelligence permits viable supply networks. Manuscript prepared for submission to *The International Journal of Production Research*.
- [319] L Xu, S Mak, **Y Proselkov**, A Brintrup. Towards Autonomous Supply Chains: Definition, Characteristics, Conceptual Framework, and Autonomy Levels. *Journal of Industrial Information Integration* (2024), 42.

- Chapter 7: [258] **Y Proselkov**, J Zhang, L Xu, E Hofmann, TY Choi, D Rogers, A Brintrup. Financial ripple effect in complex adaptive supply networks: an agent-based model. *International Journal of Production Research*, (2024), 823-845, 62(3).
- [320] L Xu, **Y Proselkov**, S Schoepf, D Minarsch, M Minaricova, A Brintrup. Implementation of Autonomous Supply Chains for Digital Twinning: a Multi-Agent Approach. *22nd IFAC World Congress*, (2023), 56, 11076-11081, Yokohama, Japan. Elsevier B.V..
- Chapter 6: [255] **Y Proselkov**, M Herrera, MP Hernandez, AK Parlikad, A Brintrup. The value of information for dynamic decentralised criticality computation. *14th IFAC Workshop on Intelligent Manufacturing Systems IMS 2022*, (2022), 55(2), 408-413, Tel-Aviv, Israel.
- [149] M Herrera, **Y Proselkov**, M Pérez-Hernández, AK Parlikad. Mining graph-fourier transform time series for anomaly detection of internet traffic at core and metro networks. *IEEE Access*, (2021), 9, 8997-9011.
- Chapter 4: [256] **Y Proselkov**, M Herrera, AK Parlikad, A Brintrup. Distributed Dynamic Measures of Criticality for Telecommunication Networks. *Service Oriented, Holonic and Multi-Agent Manufacturing Systems for Industry of the Future: Proceedings of SOHOMA* (2020), Paris, France.

Chapter 2

Preliminaries

This chapter details preliminary information that underpins language and concepts used in this thesis, to allow interdisciplinary readers to access all content within it. Section 2.1 introduces basic principles of network science relevant to this thesis, including mathematical concepts and notation. Following this, Section 2.2 discusses network scientific model environments which are used in this thesis, or material it is built upon or inspired by. This includes network generators and some infectious disease network dynamics. Finally, Section 2.3 introduces relevant data scientific methods that are applied for time series analysis or processing, including divergence analysis, smoothing, and changepoint detection.

2.1 Network Science

The Bridges of Königsberg The network approach to modelling was first introduced by Leonhard Euler to model the bridges of Königsberg (Figure 2.1), and learn how to walk them without repetition [117]. Landmasses were represented by nodes, and bridges by edges, creating the first graph model.

The infrastructures I investigate are often represented using network graph objects, with agents as nodes and their connections as edges [300]. The topology of a network can affect both the speed at which a data packet is routed — shorter distances result in quicker transfer — and the response of the network to disruption.

A given network graph is denoted as $G = (V, E)$, where V is the set of nodes or vertices and E is the set of links between nodes or edges between vertices. An edge between vertices $u, v \in V$ is denoted as $(vu) \in E$. Enumerated nodes u_i for $i \in \{1, \dots, n\}$ are denoted by the vector $\mathbf{u} = (u_1, \dots, u_n)$.

Euler found that landmasses need an even number of bridges to walk them all only once. In Königsberg, since there were seven bridges, this was impossible. The number of bridges incident on a landmass is represented by the degree of the corresponding node in the graph. This discovery is the first in graph theory, the precursor to network science. Since then, massive developments in network science allowed the study of larger scale and more complex objects. In this section, I describe some background material, necessary to understand the methodology introduced in this thesis.

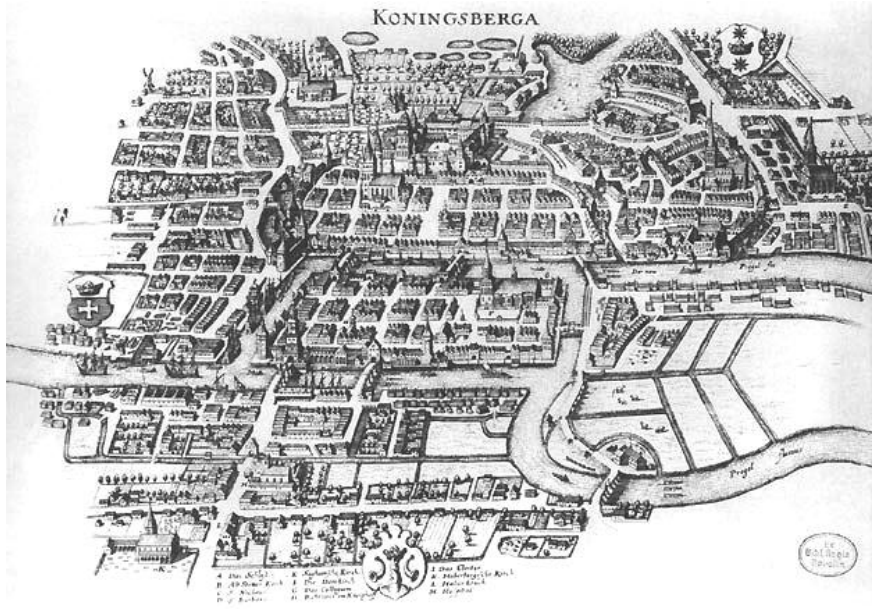


Fig. 2.1 A map of Königsberg, by Matthäus Merian, 1652

Network Connectivity The degree of a node u is denoted as $d(u) = d_u$. For enumerated u_i , $d(u_i) = d_i$, and with this, $\mathbf{d} = (d_1, \dots, d_n)$. This may be displayed on the leading diagonal of a matrix $D \in M_n(\mathbb{Z}^+)$, known as the *degree matrix*, where

$$D = (d_{i,j}) = \begin{cases} d_i \in \mathbf{d}, & i = j \\ 0, & \text{else.} \end{cases}$$

Complex network topologies have statistical properties. One of these is their degree distribution, which is, for some $k \in \mathbb{Z}$, the likelihood that any given node $u \in V$ has $d(u) = k$. They also have excess degree distribution, which is the likelihood that, having reached a node v from another, u , that node has $d(v) = k$.

Networks have been used to model animal territories [134], epidemics [284], opinion formation [308], and more. In Borůvka [51], the amount of transmission line cable used in the Moravian power grid was minimised by creating a minimum spanning tree.

The degrees of nodes in a network are the number of incident edges they hold. These are connections between nodes, so I denote the number of connections between the node u_i and u_j as $a_{i,j}$. For a simple network, nodes can only connect to other nodes, and may not have more than one connection between any node pair. These can be displayed within a matrix, $A \in M_n(\{0, 1\})$, known as the *adjacency matrix*, where

$$A = (a_{i,j}) = \begin{cases} 1, & (i, j) \in E \\ 0, & i = j. \end{cases}$$

The information contained within the degree matrix and the adjacency matrix of a network may be used to generate intrinsic measures of network structure, which may be compared between different structured networks. To achieve this, one must first produce the Laplacian [87] matrix, $L \in M_n(\mathbb{Z})$, where

$$L = D - A.$$

The Laplacian captures all the topological information of the network, normalising it such that

$$L_N = D^{-1/2} L D^{-1/2},$$

which has special properties valuable for comparison measures [15, 288, 223]. *Spectral analysis* can be performed on L_N , which involves looking at its eigenvector decomposition. This reduces it into its component eigenvectors, each associated with a member of the spectrum of eigenvalues.

For L_N , for $i \in \{1, \dots, n\}$, all eigenvalues $\lambda_i \in [0, 2]$ are such that $0 = \lambda_1 < \dots < \lambda_n \leq 2$. The number of connected components in a network is equal to the number of 0-eigenvalues in the spectrum, that is, $|\{\lambda_i : \lambda_i = 0\}|$.

In simple graphs, edges (uv) are undirected such that $(uv) = (vu)$, and nodes can only have one edge between. They are directed graphs, where $(uv) \neq (vu)$; multigraphs, where nodes can have multiple edges such that $(uv)^{(1)} \neq (uv)^{(2)}$; and directed multigraphs, with both features.

Network Directionality The directions in any graph can represent a space to traverse. If it is such that travelling over the directions of edges never allows for return to previously visited nodes, then it is a directed acyclic graph (DAG). These can be created by any total ordering over the edges in G , where each edge holds a value and edges are sequenced such that these values are monotone increasing. Taking this ordering, one can construct the DAG by, for any pair of nodes u, v , directing an edge according to whether the value for u or v is greater. By repeating this for all edges, a DAG is formed [289].

Network random walkers are probabilistic objects that traverse edges according to their direction, weight, and any other rules imposed upon them [217]. Walkers, at any position and time, can randomly select a new, adjacent position to move themselves to in the next time step. In a graph, these positions are nodes, and they may travel along edges to neighbouring nodes. A random walk is the sequence of nodes any walker takes over some time. If the random walk ends at a node without a means for the walker to leave the node, then it is an absorbing random walks.

This multiplicity and weights in networks can change in time, representing a temporal network, which can be used to describe a flow network. These are networks with a quantity of substance that travels across the network between nodes, generating flow across the edges and nodes. From these can be generated residual graphs, which are flow networks with capacity equal to the original capacity network minus the original current flow network. When the difference between the flow and the residual network is zero, no edge is generated in the residual network.

A network path, $p \subset V$, is any ordered sequence of unique vertices where consecutive members share an edge, such that for all $n_i, n_{i+1} \in p$, $(n_i, n_{i+1}) \in E$. For any two nodes $u, v \in V$, the shortest path between them is called the *geodesic*, denoted p_{uv}^g . It is, in the unweighted case, the path with the fewest elements that starts with the source node and ends with the destination node. There might not be only one p^g . If multiple unique paths share the same length and no other paths have fewer elements than them, these are all p^g . For nodes $u, v \in V$, the set of all paths between u and v are denoted as P_{uv} and geodesics as $P_{uv}^g \subset P_{uv}$, such that the quantity of unique p_{uv}^g is $|P_{uv}^g| = \rho_{uv} : V \rightarrow \mathbb{Z}^+$. For $w \in p_{uv}^g$, the quantity of unique geodesics p_{uv}^g that pass through w is $|\{p^g \in P_{uv}^g : w \in p^g\}|$, denoted $\rho_{uv|w} : V \rightarrow \mathbb{Z}^+$.

Considering travel from city to city, by representing roads as edges, and their intersections as nodes, one can use Dijkstra's algorithm [95] to efficiently find a shortest path to a destination. The diameter ∇ of the network is measured as, over all pairs of nodes $u, v \in V$, the longest geodesic the graph G , such that $\nabla = \max_{u,v \in V} p_{uv}^g$.

There also exists the notion of the graph cut of nodes u and v . These are sets of edges that, when removed, mean there are no paths between nodes u and v . The minimum cut of u and v is the smallest possible cut between them. In a flow network, the minimum cut is the cut that minimises total current flow for the edges in that cut, which can be found using the Ford-Fulkerson method [123]. This begins by fixing the flow in the residual network to zero. For a path connecting u and v without any zero weight edges, the edge with the smallest residual capacity is found, and then this flow is added to all edges on the path in a copy of the current flow graph. The residual graph is updated with this copy, and the process is repeated until a residual graph where u and v are disjoint is obtained. The missing edges are the minimal cut.

Another measure of relationships between two nodes is the communicability between node pairs [114], where for a given node pair, v, w , is given by

$$K_{vw} = \sum_{j=0}^{\infty} \frac{(A^j)_{vw}}{j!}.$$

Here, A^j is A raised to the power of j , and $(A^j)_{vw}$ is the value in A^j in the same position of the entry associated with v and w in A . Estrada and Hatano [114] denoted the numbers of paths between v and w , each divided by the total possible distinct sequences of the given path length. This naturally weights longer paths less, and converges to

$$K_{vw} = (e^A)_{vw},$$

which is the position for v to w in A of the matrix exponential of A .

Another form of network characterisation is k -core [276] or k -shell decomposition [105], which generates a series of subgraphs, such that the k -core is the maximal subgraph such that the degree of every vertex is at least k , while its k -shell are the subset of the k -core with exactly degree k . The

k^{th} core can be found by stripping away nodes with degree less than k , and then removing those nodes that have had their degree reduced to less than k , until there are no more of these [33]. K-core decomposition may be applicable as a method to design communication architecture for distributed computations, since it generates a clear network hierarchy quickly and simply.

2.2 Model Environments

Network Topology Generators Euler, Borůvka, and Dijkstra worked on topologically simple networks, constructed by placing nodes and assigning edge values. The next major breakthrough in network science came with the introduction of the Erdős-Rényi (ER) random graph models [113, 133], denoted as $G(n, m)$ and $G(n, p)$. These have n nodes, assigning edges randomly. The $G(n, m)$ model assigns m many edges randomly, while the $G(n, p)$ assigns an edge to a node with probability p , with independent likelihood. For an ER graph, for all $i, j \in \mathbb{N}$,

$$\langle |\Gamma_i(u)| \rangle = \langle |\Gamma_j(u)| \rangle.$$

The emergence of major links between the physics of complex networks, graph theory, biology, and sociology led to the birth of network science in the 1990s. This can be traced to Watts and Strogatz [309], on small world networks. Networks can model our social ties and the small world phenomenon [221] of how pairs of total strangers can be linked by a short chain of acquaintances.

Network science was then popularised by Barabási and Albert [27], on scale-free networks. These are networks that have a small number of interconnected “hubs”. In other systems, these can model the structure of the internet. For example, how large sets of unconnected websites are all linked to a search website, and how search websites link to each other. It was found in Albert et al. [6] that certain network configurations are more tolerant to targeted attacks, and others to random attacks. With attacks interpreted as failures, different topologies were more resilient to different methods of failure. When failure occurs randomly, scale-free networks show remarkable resilience to shocks, unlike random networks. The Barabási-Albert (BA) preferential attachment network configuration, $B(n, m)$ [5], is a model for producing scale-free networks. It defines a network of size $n > m$, and is generated by starting with m nodes, and then adding new nodes with m edges wired to existing nodes in the network until reaching n nodes. Edges from a new node u are randomly linked to an existing node v in a probability that scales to the degree of v .

These milestones are the subject of major reviews and textbooks [5, 234, 104, 47, 233, 106, 235, 253, 26]. Current work aims to broaden the class of systems that can be described in the networks framework. The major fields are synchronisation phenomena [18], adaptive networks [141], multilayer networks [147, 184], and temporal networks [158]. My thesis unifies these four topics, investigating the endogenously triggered breakdown of adaptive networks with multilayer temporal dynamics.

Infectious Disease Models An example of temporal network models are models of infectious disease, which in this thesis are used to represent risk propagation. Classic ones include the Susceptible-Infectious (SI) model, Susceptible-Infectious-Susceptible (SIS) model, and Susceptible-Infectious-Recovered (SIR) model [180, 54].

The SI model is a temporal model which can be used to model the spread of infectious diseases that, once an agent is infected, it is always and infectious cannot recover. Nodes can take one of two states, susceptible, S , and infected, I , respectively becoming members of the sets $\mathbb{S}, \mathbb{I} \subset V$. An I node, in time state, $u \in \mathbb{I}$ may, in random time, according to a Poisson point process of rate β , synchronise with a uniform randomly chosen neighbouring S node, say, $v \in \Gamma_i(u) \cap \mathbb{S}$, turning it into an I node, such that $v \in \mathbb{I}$.

The SIR model is a temporal model which can be used to model the spread of infectious diseases that confer immunity after recovery. Nodes can take a third state — recovered, R — becoming members of the set $\mathbb{R} \subset V$. It follows SI dynamics, but here, an I node, say again $u \in \mathbb{I}$, may also, according to an independent Poisson process of rate γ , recover and convert into R node, such that $u \in \mathbb{R}$. Together, these represent the dynamics of a disease that does not confer immunity.

The SIS model is a temporal model which can be used to model the spread of infectious diseases that do not confer immunity, but can be recovered from, such as flu. In this case, an I node, once again $u \in \mathbb{I}$, may also, according to an independent Poisson process of rate γ , recover and convert into S node, such that $u \in \mathbb{S}$. Together, these represent the dynamics of a disease that does not confer immunity.

2.3 Data Scientific Methods

One element investigated in this thesis is the change in dynamics invoked by given conditions in the simulation models. This is determined using various data scientific methods to determine thresholds and magnitudes of difference.

Kullback-Leibler divergence Kullback-Leibler divergence [188], denoted D_{KL} , measures the statistical distance of the difference between the probability distribution P and the reference probability distribution Q . For a state space X , it is defined as

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right).$$

Savitsky-Golay filter smoothing The Savitsky-Golay filter is a method to smooth a dataset [271]. It is achieved by convolution, fitting successive subsets of adjacent data points using a low-degree polynomial with linear least squares. Upon equally spacing the datapoints, one finds an analytical solution to the least-squares equations found, using a singular set of convolution coefficients, applicable

to all data subsets, giving estimates of the smoothed signal and its derivatives, or orders, at the central point of each subset.

Changepoint detection In a time series, the times of abrupt change in behaviour are known as *changepoints*, detected through changepoint detection algorithms. I investigate how to extract these changepoints systematically, quickly, with minimal computation cost, and in a way that makes use of the data concerning many nodes within a network.

Changepoint detection algorithms consist of a cost function that is minimised according to some search method for changepoints within the signal, subject to a set of constraints [296]. For cost functions, the *least absolute deviation*, written as $l1$, is a base case. This cost function measures for median-shifts in signal. For any given signal $y = y_1, \dots, y_T$ on the interval I , this is calculated as

$$c_{l1} = \sum_{t \in I} ||y_t - \bar{y}_{\text{median}}||_1.$$

As an advance on this, another cost function, *least squared deviation*, written as $l2$, is also tested in this thesis. This cost function measures for mean-shifts in signal. This is calculated as

$$c_{l2} = \sum_{t \in I} ||y_t - \bar{y}_{\text{mean}}||_2.$$

The last cost function tested in this thesis is built on a *radial basis function* (RBF) [128]. For any pair of numbers, a, b , this is

$$\text{rbf}(a, b) = \exp(-\zeta ||a - b||^2),$$

where ζ is the inverse of the median of all pairwise distances in a given space. Let a window from y_a to y_b be denoted $y_{a:b}$. Then

$$c_{\text{rbf}}(y_{a:b}) = b - a - \frac{1}{b - a} \sum_{s, t=a+1}^b \text{rbf}(a, b).$$

This cost function is able to detect changes in the distribution of an independent identically distributed (iid) sequence of random variables. As a non-parametric function, it can be applied to a wide range of tasks [19].

There are two kinds of changepoint methods: offline for distinguishing behaviour regime changes, online for anomaly detection. Since this thesis investigates internal dynamics, I study endogenous shocks, which do not occur due to anomalies but due to regime changes [299, 287]. Thus, offline changepoint detection methods are studied in this thesis.

Searching for changepoints involves minimising some function of the segments generated, derived from the cost function used. One may either assume a known number of changepoints or not. In the context of this thesis, there is no clear knowledge of the number of changepoints, requiring only methods that can handle this relaxed constraint. The classic search method with this property is the

pruned exact linear time (PELT) method [182]. However, this method may have high computational overhead since it is an optimal method. Alternatives include approximate methods, such as binary segmentation [24] and window-based detection [13]. In this thesis, these three methods are used since they are classic, canonical methods on which methods that use them may be studied, while still allowing for a breadth of results. In addition, they already exist within the Python ruptures package, enabling convenient application of them.

Changepoint methods used include binary segmentation [24], window based detection [13], and the pruned exact linear time (PELT) method [182].

Suppose there exists a time series, $\mathbf{y} = y_1, \dots, y_T$. *Binary segmentation* works by iteratively applying a given cost function upon splits of the data. The algorithm first assigns a changepoint onto y , say, y_t . Then it assigns changepoints onto the regions before and after the previous, y_t , and repeats this until it is found that, for a cost function, $c : y \rightarrow \mathbb{R}$, and penalty value, β ,

$$c(y_{1:t}) + c(y_{(t+1):n}) + \beta \geq c(y_{1:n}).$$

This method is fast and simple, with computational complexity of $O(n \log(n))$.

Window based detection works by sliding two windows across a signal. The signals within each window have their statistics compared against a discrepancy measure. For a given cost function c , one defines a cost gain, or discrepancy, measure $d : y \rightarrow \mathbb{R}$ as

$$d(y_{a:b}, y_{b:c}) = c(y_{a:c}) - c(y_{a:b}) - c(y_{b:c})$$

for $a < b < c \in \mathbb{N}$. If the sliding windows $y_{a:b}$ and $y_{b:c}$ have large overlap or are very similar, their statistical properties are similar and the discrepancy between the first window and the second window is low. If the sliding windows fall into two dissimilar segments, the discrepancy is significantly higher, suggesting that the boundary between windows is a change point. For n samples and a sliding window of length w , the discrepancy is defined for all indices $t \in [\frac{w}{2}, n - \frac{w}{2}]$, with complexity of $O(nw)$, where maxima indicate changepoints.

PELT is based off of the optimal partitioning method [327, 172]. For some cost functions, c , and penalty value, β , this method looks to minimise

$$\sum_{i=1}^{m+1} c(y_{u_{i-1}+1:u_i}) + \beta, \quad (2.1)$$

where m is the number of changepoints. Define $\mathbf{v}_s = \{\mathbf{u} : 0 = u_0 < u_1 < \dots < u_m < u_{m+1} = s\}$ as the changepoint vector set. Define the minimisation function of Equation 2.1 as $F : y \rightarrow \mathbb{R}$, where

$F(0) = -\beta$, and for n many datapoints, it is such that

$$\begin{aligned} F(s) &= \min_{\mathbf{u} \in \mathbf{V}_s} \left\{ \sum_{i=1}^{m+1} c(y_{u_{i-1}+1:u_i}) + \beta \right\} \\ &= \min_t \left\{ \min_{\mathbf{u} \in \mathbf{V}_s} \left\{ \sum_{i=1}^m c(y_{u_{i-1}+1:u_i}) + \beta \right\} + c(y_{t+1:n}) + \beta \right\} \\ &= \min_t \{F(t) + c(y_{t+1:n}) + \beta\}, \end{aligned}$$

showing a recurrence relation in F . This can be solved for $s \in \{1, \dots, n\}$, giving the optimal partitioning method.

The PELT method adds pruning to the optimal partitioning method to reduce the expected computational complexity from $O(n^2)$. If it holds that for some constant, K , and for $s \in (t, T)$

$$c(y_{t+1:s}) + c(y_{s+1:T}) + K \leq c(y_{t+1:T}),$$

such that adding a changepoint reduces the total cost of all segments, then if

$$F(t) + c(y_{t+1:s}) + K \geq F(s),$$

then t is not the optimal changepoint. Since the algorithm checks pruning for all changepoint pairs, it is globally optimal within the solution space, and so does not require multiple runs. The algorithm finds all t such that this holds, thus reducing the space of possible changepoints, yielding an optimal yet more efficient algorithm.

These three methods have the advantage of existing in stable, reliable builds within the changepoint detection package that contains them, `ruptures`, allowing for reliable results. Despite their variability in computational complexity, choice of changepoint detection method depends on expected shape and data structure. Within this thesis, all three are investigated and compared.

Chapter 3

Literature Review

In the following, I deliver a comparative literature review of the joint principles underpinning research within this thesis and how they have been studied. These are divided across: empirical contributions, which are reviewed here broadly under principles towards modelling infrastructural networks and controlling them; theoretical contributions, which are reviewed here under methods of modelling criticality, a core network risk metric; and the intersection of the two which is present in each broader section, depending on focus.

I justify the case studies conducted in my investigations, that of telecommunications and supply chain financing, as complementary but sufficiently distinct, and appropriate for the study of scalable methods for dynamic failure risk management in complex network infrastructures. I demonstrate the research gap, and thus novelty of this within the contexts investigated in this thesis.

3.1 Modelling Infrastructural Networks

To research scalable and dynamic failure risk management in complex network infrastructures, one must use methods that are appropriate to the dynamics governing a given infrastructure. Generalisable principles exist, however agents and subsystems within larger infrastructural systems interact non-linearly, demonstrating emergent properties that are unique to their particular dynamics as a consequence of these interactions. Hence, while methods applied for one system can yield insight into another, doing so constitutes highly nontrivial findings that are research contributions unto themselves. I therefore review literature regarding various infrastructural network control models, case applications, and principles for methods of scalable control in those models.

3.1.1 Information Relevancy

The relationship of information relevancy to agents is meaningful to control of the agent itself and control when agents are cooperating. For this topic, agent-based models of collaboration have been investigated [324] but not considering hosts of collaborative agents, and complex network models

have been used to model collaboration [108]. Temporal information relevancy has been investigated in the context of internet information decay for online social networks [22]. Books on methods of cooperative control in dynamic multi-agent network topologies have been written [331, 314], but did not cover complex model dynamics over these systems.

These investigations into information relevancy and control architectures are all defined by their embedding into the given system they are involved in. The dynamics governing systems of mobile communication sensors are distinct from those for energy-sharing clusters in a power grid, and require different control systems and architectures, such that the nature of information in space and time is distinct, which changes the fundamental notions of information relevancy. There are core ontological threads with degrees of generalisability amongst all these systems, however, and these commonalities justify a treatment of the breadth of possible investigations and their particularities. This serves to explain the cases investigated in this thesis, and clarifies their similarities.

For the pursuit of scalability, the questions are “what” and “how much” information each agent must learn to effectively operate in a distributed manner without sacrificing network performance.

- **“What”** is answered by a correct choice of measure
- **“How much”** is answered by the ideal information reach

Learning this will allow engineers and policymakers to create scalable adaptive systems which manage risk effectively and autonomously. The research case in this paper, within telecommunications, is networks with quick response times to mitigate emergent risks. Extra data may improve results, but it increases computational overhead, slowing calculations [256, 255]. Care must be taken to ensure overhead doesn’t grow so much that the network is too slow, while still effectively computing or estimating criticality.

Within supply chains, as they are systems of still competitive agents, cooperating on a service goal, the information reach range must ensure an admissible balance of competitiveness and cooperation with appropriate supply chain players for it to be feasible to adopt. For computational agents, to avoid overworking the computational load of an agent and the resultant lag or inefficient use of company resources, a balance must be therefore struck. This trade-off is a core focus of this thesis. The following examines sources yielding insight into this.

Organisational Distance There exists a broad overview of the nature of information for minimally cognitive agents through the perspectives of information theory and dynamical systems theory [38]. One can construct a hierarchical network that incorporates small world structure between branch members in the hierarchy to maximise robustness against congestion under information exchange dynamics [309, 98]. In a branching hierarchy network, two nodes, $u_i, u_j \in V$ share a common supervisor, r_{ij} , and respectively have a distance, or depth, from their common supervisor of $|p_{ir_{ij}}^g|, |p_{jr_{ij}}^g| \in \mathbb{N}$, while the supervisor has $|p_{r_{ij}o}^g|$ many steps from the highest authority, or origin node, o . Assuming that nodes at the same depth from their supervisor are connected by default,

organisational distance is defined as $x_{ij} = \sqrt{|p_{ir_{ij}}^g|^2 + |p_{jr_{ij}}^g|^2} - 2$, for all i, j such that $|p_{ir_{ij}}^g| + |p_{jr_{ij}}^g| \geq 2$, similar to the Euclidean metric. Then, due to homophily of near nodes, one can generate the probability of establishing a link, $\mathbb{P}((u_i u_j) \in E)$ that monotonically decreases with increasing x_{ij} . $\mathbb{P}((u_i u_j) \in E)$ also decreases monotonically with lower $|p_{r_{ij}o}^g|$. The following stochastic rule governing formation of new interactions is defined as

$$\mathbb{P}((u_i u_j) \in E) \propto e^{-|p_{r_{ij}o}^g|/a} e^{-x_{ij}/b},$$

where a and b are tunable parameters that represent characteristic lengths in x_{ij} and $|p_{r_{ij}o}^g|$, respectively, where for all greater values, connections become unlikely. This formulation however is not applicable to complex nonlinear interactions within a dynamic environment. Nonetheless, the principle of homophily that it shows suggests the existence of *relevancy*.

Relevancy Relevancy, as a function of usefulness, is described in Bisdikian et al. [45] as a function of information quality, which, by defining operations in an infrastructural network as services and users as customers, can be measured as a function of the quality of service, that is, the positive influence information has on agent or system capacity to serve its functions. The interplay of function, information, and action are detailed in a theoretical ontology of when and where constituting why for application, engaging on who and how to constitute what for information. Bisdikian et al. [45] propose that spatial and temporal relevancy for region R_c and information covering region R_p conform to the relevancy function,

$$r_s = \frac{f(\mathcal{A}[R_p, R_c])}{f(\mathcal{A}[R_c, R_c])},$$

for some positive, non-decreasing function $f(\cdot)$, while $\mathcal{A}$ is the semblance of R_c and R_p , such as the area of overlap. This is valuable for post hoc measurement, but in a system of distributed agents, $\mathcal{A}$ demands scope beyond a cover region, which is not delivered in a distributed system. Region types are spatial or temporal, depending on if this is spatial or temporal relevancy.

- In my work, I instead determine relevancy according to performance metrics against some ground or assumed truth, either an evaluation of information itself, or according to systemic performance.
- This reorders the ontology such that the assumed truth forms a function of the measurement.
- This means I measure relevancy according to embedded impact on maximising live performance on all elements of the ontology, as embedded in an interactive system of control and cognitive agents. In my methods, relevance is thus agent-wise and systemic, rather than dyadic as r_s

Another candidate solution for ideal information reach is made by, for every node, for a given depth, extracting the induced minimal spanning tree of the subset [98]. Then one finds the likelihood of the given configuration of all other branches within the subgraph. One then estimate the ideal depth

as the one that maximises the likelihood over all nodes within the network, since this creates the set of most robust subgraphs. One may fix a and b to conform to the subgraph topology, where max depth from first branch is a and max distance to a supervisor is b .

- I used this principle in constructing the DAG for the final supply chain model in this thesis through nonlinear approximations and extended to non-tree topologies.

This field has been grounds for further relevancy research, providing a definition of information relevancy that compares node importance as a function of the nodal importance and global importance, internode selection, and community structure [298, 322], however such methods are centralised, and further work is necessary to apply them to dynamic model cases.

Alternative formulations may consider a cost/benefit tradeoff for a given measure, maximising the ratio between robustness and information gain, finding the least robust subgraphs possible that still preserve suitable network awareness, since less robust networks have fewer links. This has been done using robustness, a particular sort being connectivity robustness, defined as $R = S/(N - N_r)$, where S is the size of the largest component, and N_r is the number of removed nodes [6, 59, 84]. For a measure, these are the same as the number of routes that are needed to extract information, justifying this metric. Alternatively, one could maximise the ratio between information reach and some other metric, either a centrality estimate of criticality, or a criticality measure.

- This justifies the creation of weighted distributed dynamic measures of criticality for viable networks, as in this thesis.

3.1.2 Infrastructural Network Control

There are diverse applications for network control, and their particular conditions each mean a distinct control manifestation appears per application. Despite this, infrastructural network control is a generalisable principle. The boundaries and aspects of this are illustrated in the following with a collection of infrastructural network control examples. This shows that different solutions exist for different cases, and I discuss their relationships as well as how one solution space can be applied to another context. In showing the diversity of related case studies, I frame my contribution in this thesis as to the general space of scalable infrastructural network control, via the two illustrative examples of telecoms and then supply chain financing.

Navel Auxiliary Systems Scheidt [273], framed this topic in the wider context of defining computational architecture for intelligent autonomous agent-based control. In this, control for agents is a three step process of estimation, planning, and execution, which is an agent-centric advance on the system comprehension model [110]. That is, first agents compute some belief function, then plan what to do with it, and then actuate commands to realise this plan. Agents can have some awareness of their surroundings and other agents, which have peer or hierarchical relationships. Information relevancy in the spatiotemporal sense within networks is thus determined by the relationship between

an agent's capacity to fulfil its three functions with respect to other control agents and the network itself. This paper defined a distributed control architecture for naval auxiliary systems, not broader infrastructural networks of any sort, but argues strongly in favour of distributed computation for such complex networks as I investigate. Furthermore, it proposes that such control behaviours could be emergent, and enable long-term operational capacity, in agreement with my results. In this sense, the boundary for relevancy is the limit of any local information, subject to any abstractable information axis, thus describing an epistemology of how agents learn what they know [11].

- This thesis explores various models of complex network dynamics with autonomous agents operating at each of the three steps of intelligent autonomous agent-based control, and unlike Scheidt [273], proposes architectures for large scale complex infrastructural network cases. Distributed control paradigms have further been defined in [89, 17].

Ad-hoc military communications networks Other work [131] applies the older system comprehension model to ad-hoc communication network control in military contexts, with a focus on group self-assortativity via adjusting position in a physical space. This is a focus on control for dynamics of the network via spatial embedding rather than on it as a KPI for operational interaction, as this thesis investigates. It uses machine learning methods as the focus is operational effectiveness rather than discovering information relationship properties.

Military command networks Further military applications advance on prior concepts of robustness, which is resistance to shocks, and propose to engineer interacting systems of systems for resilience, defined as the return to operability after shocks [25, 295]. Similar work has been completed in the case of unmanned aerial vehicle swarms, a spatially embedded case [23].

Industrial production networks Further work has been applied into the field of workstation specific production methods in industrial production networks [144]. Theory for this is highly case specific — information relevancy is hard to generalise for the complicated and varied tasks for workstation operators — so artificial neural networks were applied for prediction for task control.

- This presents a degenerate case of information relevancy research, as such methods return no conceptual insight, showing that to understand the principle more strongly, maximal agent simplicity which conserves nonlinear dynamics is necessary, thus justifying complex network systems I study with agents of uniform function.

Transport networks Researching into cascade disruptions of transport networks, Dekker et al. [92, 93] creates a means of deriving cascade effects from complex rail network delay datasets. These are methods for interpreting existing data, not projecting forward.

Geological flow networks Investigations into scalable algorithms, as applied to geological flow networks [90] report a set of algorithms designed for scalable, autonomous control in dynamic network environments, but specifically lattices, and no notion of risk is present.

Software defined networking Many of the models defined above are implemented using software defined networking [41] which is an approach to network management that uses dynamic and efficient network configuration to improve network performance and monitoring by dynamically adjusting network resource flow. This is a centralised paradigm, however, which limits scalability and other factors. Work has been completed to enable distributed use of this [303], shortening the pipeline of research to application of work developed in this thesis, and others.

This has been the subject of review [14] from the perspective of how resilience and dependability is managed in distributed environments for edge and fog computing, which are distributed evolutions of cloud computing. Different cases are reviewed here, including control for urban infrastructure [282, 326], drone networks [60, 209], vehicle networks [159], medical monitoring [206], and telecoms [302], with promising results for hybrid environment using blockchain to underpin distributed communication and control [163, 189, 219, 88].

Blockchain Blockchain, as a communication architecture, imparts value to communication through the work required for cryptographic hashing [270].

I argue that telecoms and finance each present a simplified case to blockchain with similar structure. Telecoms with data flow, and finance with cashflow, but divorced of restriction of telecommunication dynamics. From this is derived the following:

- Telecoms case where information dynamics is studied in this thesis as a feature of the model to determine information value
- Financing case where information value is studied in this thesis as a feature of the model to generate control schema.
- It thus presents a unification of telecommunications and financial infrastructure. These are the two cases I research in this thesis, producing simulation models of their dynamics and investigating techniques for autonomous scalable control and their relationship with restricted information in their contexts.

Telecoms Telecoms is studied in particular because of its restriction as an information network, such that communication and control is enacted in the network at a timescale comparable to its operation, which is unique among infrastructural network systems. Thus it presents an edge case where spatiotemporal information relevancy can be directly studied within simulation models in ways that are representative of their physical dynamics. Due to advances in software defined networking, these can be readily transposed into application spaces.

Finance Finance, particularly supply chain finance, is studied because objects flowing within such systems have value and the system itself generates value quantifiably. Information relevancy can therefore be abstracted into a literal *value of information*, which can be used to inform control architecture.

In the following is a review of contributions in the fields of telecoms and supply chain finance, introducing concepts used throughout this thesis and the state-of-the-art in modelling each, with respect to methods toward autonomous scalable control against network risk, and what risk actually means in these contexts. The concepts pertinent to this thesis are described first in Section 3.1.3 for telecoms and then in Section 3.1.4 for supply chains and supply chain financing. The mechanisms of the function and dynamics of these systems, the problems encountered, the state-of-the-art in solving these problems, core principles in their study and development of these solutions, and resultant gaps are introduced and discussed. From that, what topics, concepts, and theory this thesis uses and investigates are explained, thus preparing the reader to further engage with the content in this thesis.

3.1.3 Telecoms

Manufacturing processes have become more data-driven and dependent on interconnection of multiple facilities for efficient decision-making, thus telecom infrastructure is as pervasive a component of manufacturing industries as the powergrid and other critical infrastructures.

Telecom infrastructures are physical networks that support internet, telephony, and other digital services by facilitating data transfers between users [107]. For example, when watching a film on some video streaming service, a user sends a request to the service's servers as a signal which is routed through the infrastructure network. In response, the streaming service sends the data of the film back from the server to the user's device.

Protocols Data communication can be described in the overarching abstraction of the TCP/IP model [53, 52], with application, transport, internet, and network access layers. In a simplified explanation, the application layer is how data is applied and used, the transport layer is what matches hosts at the ends of the network, the internet layer is in charge of the routing, and the network access layer dictates how data is physically transferred within the network. There exists ongoing work towards decentralising the internet infrastructure that TCP/IP governs, however it is non-trivial where and how to apply such methods in tandem with existing centralised solutions [48]. The layers of communication, control, and flow in TCP/IP inspire the layers of dynamics expressed in each model in this thesis.

There are many kinds of data that can be sent through the infrastructure network. In addition to single streaming services, group communication applications can use IP multicast [91] to send data packet information to multiple sources. The methods, reviewed in [260], comparing three varieties of group key management control approaches: centralised, where a single entity controls the whole key

distribution; decentralised, where there is a hierarchy of subgroup managers; and distributed, where all members distribute and generate keys to each other. Methods toward making distributed control scalable have been generated in recent years [8], providing an architecture for distributed control.

Advances in network protocols for handling complex data packet structures form major foundations for the variety of network services available today, for example, architecture for supervising shared network calls [297]. These advances are required to support advanced networking standards, such as 5G [116, 328], which are being built to satisfy increasing communication demands [76].

Countering Risk with Distribution An advantage of distributed measures rather than centralised measures, specifically in networks with communicative agents with computational capabilities, is that they can minimise data transmission load [74]. In centralised computation, each node requires information to be transmitted from all nodes on the network, whereas, in a distributed approach, nodes only require information from a small region. If choosing which routers to navigate data packets through under stress, for example, a DDOS attack, this information would be valuable when looking to counter risk through proactive decision-making and would improve reaction speed [75].

Network Representation Telecoms infrastructures are often represented by graphs, with network elements such as routers or switches as nodes, and connections as edges [300]. Network topology affects routing and resilience to disruption since shorter distances give quicker transfers. Optimal network design has received much attention [5, 234, 47], especially for resilience [201], particularly in complex communications networks [316, 321, 218].

Dynamic Network Models for Resilience Network systems with items travelling within them are flow networks. In flow networks, such as telecommunications infrastructures, data packets arriving at rates greater than nodes can emit them, means they queue up to be emitted later. If queue growth rate is positive, congestion occurs, causing node failure, which, cumulatively, can make networks inoperative. Robustness to congestion can be maximised with certain classes of multiscale networks [98], but adaptivity disallows fixed network structure. Modern alternatives solve this with time-varying networks [313]. It was shown that in a theoretical communication propagation model, consensus to reject malicious operations can be theoretically made robust when agents only hold information on neighbours.

- However, this is restricted to a game theoretical case, and does not measure against operational targets. I therefore investigate complex simulation models of routing, such that the complexity of their dynamics warrants nonlinear measures and analytics.

Criticality, defined as the impact of a node's inactivity on the operation of a network, evaluated in telecoms by network connectivity, is a key factor in understanding network resilience [208], and can be estimated using the current network state [256]. Criticality can inform prioritisation in network prognostics for proactive maintenance.

- This motivates the metrics applied in this thesis, where risk is measured according to current criticality.

Dynamic Network Risk Failure of a structurally critical node in an otherwise underused region may affect overall data traffic less than a structurally non-critical node in an overused region, since in the latter case, present data traffic must be serviced immediately. Criticality thus depends on dynamics and thus is not a static value. A goal of observing criticality in telecoms is to temporally find nodes under stress so that failed nodes may be repaired.

It is thus necessary to generate dynamic measures of real criticality. To achieve this, one must understand failure dynamics. The dynamic of node failure from congestion is studied in [307], showing it can be expressed as a three stage process, from generation, to diffusion, and then dissipation. Accurate and quick control of network behaviour is important in networks that are functioning at or near capacity, as is expected for backbone networks of the near future [225]. This is also important for networks operating under irregular stress, the taxonomy of which is defined in [66]. Within this framework, the COVID-19 outbreak may be defined as a major socio-political challenge and demonstrates a clear and present need for advances in methods that maximise resilience to deal with such problems as they appear. Cascade failure may also occur within otherwise regularly functional systems, due to effectively random errors, as in January 1990, where 114 switching nodes of the AT&T network went down in succession due to the wrong resetting signal [249].

- Thus, since networks are under constant risk at any given timestep, I investigate the endogenous risk of network dynamics at those timesteps.

How to monitor the impact of disruptions such as congestion on the network — criticality — is important for fast and accurate network behaviour control [225]. More generally, telecoms infrastructure is used for routing data packets from source node to destination node, similar to supply chains and the material units transferred over them. These systems all fail when paths between nodes become unusable, and cascade failures are a key problem in this regard.

Routing Model Criticality Measures Nodes that are more critical reduce system resilience and viability, since their failure will more greatly disrupt the operation of these paths and the network as a whole. Methods to identify node criticality are therefore required to accurately inform operations policy. This motivates the creation of criticality measures (CMs) that track disruptions to pathways within a network. In centralised network monitoring and criticality computation, the central computational resources need information on the whole network [268, 118].

There exists work to efficiently compute measures of criticality of certain types, but done so with centralised measures [148]. Centralised CMs require live and dynamic node topology and attribute data to respond to behavioural shifts. The increased data cause longer computational times, so conclusions arrived at a given point in time lose relevance, and promote critical events if too much

data is transferred. This is not scalable with network size, complexity, and performance requirements, but must remain accurate and, in the particular case of telecoms, with minimal data transmission.

A distributed measure of congestion was defined in [307], however this does not use local information, rather computing a parameter based on node state and certain predefined parameters. This is insufficient, since it does not capture the effect the congestion will have on the rest of the network.

- Therefore this thesis uses centrality measures over local graphs instead.

Telecoms Asset Communication Nodes within telecoms networks provide information of their state either by transmitting to a supervisory node, which facilitates centralised centrality calculation, or with each other, which facilitates distributed centrality calculation. They can achieve decentralised communication through broadcasting to all neighbours their node ID, the value, and topological information including the travel history of the data packet, and previously broadcasted packets that are known to remain in motion, [198]. This takes at least the minimum distance between two nodes to be completed in reality.

The use of decentralised computation to find the node most critical to network robustness is a field in network criticality literature, core to this thesis [310, 68, 230, 181, 96].

- In this thesis, following from the above, forms of analytics are developed to account for the issues of congestion on both network criticality and operability. This makes the amount of data used for a CM scalable while preserving meaning.
- This is achieved by computing information in a decentralised manner, permitting one to impose a limited information boundary around any given data collecting node. This makes data paths across the network finite and stops computational complexity from ballooning. These are *information bounded* CMs (IB-CMs), their performance studied within this thesis.
- Metrics used are accuracy and timeliness, but others exist, such as energy measurement [215].

3.1.4 Supply Chains

Supply Chain Representation Supply chains are networks that may begin with firms extracting raw materials, which are sold on and processed multiple times through different firms. Supply chains have been modelled using AI in many ways, which has been subject to recent review [173], given their complexity and expense that necessitates this. For control and data collection on stress-testing and viability, human-machine hybrid digital twin models were proposed [167], followed by multi-agent systems combined with foundation models which integrate human and computer systems for fluid control and can build the basis for autonomous supply chains [317].

Modern supply networks (SNs) are increasing in complexity as more needs are met with more solutions [145] meaning risks from parties outside a given firm's control may propagate through the

network unpredictably [155, 121]. Case studies of infrastructural failures have been reviewed with their associated risk mitigation strategy [227]. A notable example of a risk generating event is the Suez Canal's clogging by an Evergreen container ship in 2021, stalling 12% of world trade [197] due to the canal's strategic position. Supply chain risk management can benefit from modelling and analysis of supply chain structures [77, 194, 34], and exploring how complex network structures within supply networks affect material flow [101, 323]. Advanced artificial intelligence techniques for this have been the subject of recent review [186], and the value of this is shown by its implementation into supply chain management theory [21].

- This describes major elements of the studies conducted in this thesis.
- Risk propagation effects are generated from simulations
- Adaptive complex networks topologies are investigated
- Material flow dynamics are simulated
- Multi-agent network intelligence is used to control network dynamics

Financial Squeeze For each transaction, one expects firms with more bargaining power to create payment terms that are most favourable to them [212, 78]. For example, paying for goods and services as late as possible, maximising the buying firm's liquidity and current assets [130]. Although not necessarily decreasing the price of goods and services, it reduces the liquidity they impart. This functions as a *financial squeeze*. Squeeze is when, due to market dominance, a buyer pressures their supplier through more difficult to fulfil orders, resulting in more efficient buyer operations, but less efficient supplier operations [305, 275, 85, 291].

Attempts at modelling the effects of financial squeeze through direct simulation have been created, though currently only price squeeze has been modelled [202].

- Therefore, I present in this thesis a simulation model to exhibit an alternative form of financial squeeze, *liquidity squeeze*.

Financial squeeze, while advantageous in the short term for buyers, may disadvantage suppliers because they must survive until payments that can be expected to be up to 120 days late, and may take longer for cost-cutting focused firms, under financial stress, and yet still holding lots of bargaining power [156]. These liquidity issues may be propagated up the supply chain, resulting in late payment practices reaching suppliers deep in the chain [222].

Firm Power Balance Risk propagation due to network structure, in supply chain management literature, is nexus risk. This is the risk to the wider supply network by a firm as a function of its positioning and embedding [323], that is, its topological importance, as measured using centrality measures [37, 162, 264].

- Four classic centrality measures used for supply chains in this thesis each cover distinct aspects of nexusness [277].
- Degree centrality is a node's influence over its partners [50]
- Betweenness is its position as an information and material broker [183]
- Eigencentrality is its provision of indirect linkages to critical players in supply networks [323]
- Closeness is its access to information [183].

A supplier with topological importance is a nexus supplier.

Firms in a supply chain have different degrees of bargaining power, affected by their market share of supplier clientele, market sector control, and other factors [67]. Firms use their power over partners or the wider supply chain to transfer risk onto the supply chain and its actors, and selfishly mitigate against risk [212]. They impose risk by paying for materials late, where the greater the firm's bargaining power with respect to its supplier, the later it may pay as an expression of their market control [179, 139].

For supply chains, confidentiality helps maintain competitive advantages and independence, so exposing internal information, such as cash or material stock held, is undesirable, though methods for effective transparent supply chain management and control have been proposed [9] relying on decentralisation for financing. The importance of independence and distribution can be seen in the difference between the old, "Fordist" style of vertical integration, which incorporates the full supply chain, and the modern, Apple style of vertical reintegration, whose suppliers are held to Apple standards but are independent entities. This enables superior flexibility and efficiency [224].

- Avoiding late payments motivates information exchange, however privacy concerns argue against this. I reconcile this with limited range information exchange, where overlapping collaborative regions ensure privacy preserving network-wide information exchange.

Supply Chain Financing

As a firm's source of liquidity is reduced, they finance themselves, for example, through bank loans, which are difficult to secure because of advanced due diligence, banking mandates which restrict available customers, and increasing financial regulation [29, 32, 31]. These have created the modern regulatory environment of increased competition, tightened margins, and slower transactions in exchange for superior security [146].

To alleviate these difficulties, firms may instead use supply chain financing (SCF). This mitigates against risk propagation imposed by financial stress [265], where more powerful firms use bank loans, which are traditional, reliable, and low expense, while weaker firms use higher interest SCF solutions, such as reverse or invoice factoring.

SCF has become more available due to increased speed and automation of financial services (and risk assessment operations), particularly for customers such as small to medium enterprises, because of improved financial technology [315, 220]. This has been driven by modern regulatory conditions, coming from the 2008 financial crisis, which revealed systemic financial risks within the market, spurring greater regulation and risk assessment requirements, making traditional lending often unprofitable below a critical threshold [306]; more recently the COVID-19 pandemic and lockdowns, which stimulated movements toward global financial digitisation [16]; and general ongoing drives towards globalisation, where international cooperation in supply networks has promoted more consistent accounting and business practices [64].

SCF provides financing focused on a given business deal between two firms. Invoice factoring is a buyer focused example, measuring risk through accounts receivable. An invoice is sold to another party, typically a financial institution, in return for a discounted cost of the original invoice, where discounting is calculated as a function of counterparty risk, days to payment, and other factors [332, 247]. Invoice factoring can be accessed using an online platform, such as a peer-to-peer marketplace that matches supplier firms to financiers [12].

An alternative method is reverse factoring (often referred to as supply chain financing), which involves resale of an invoice but uses the credit risk level of the buyer instead of plain invoice factoring, where one bases rates on the credit risk of the supplier firm. These methods are limited by the existing presence of transactions, and cannot be sought out without them. The presence of a supply chain financing infrastructure has been noted to be increasingly seen as a strategic asset unto itself, where the management of finances by the organiser in charge of reverse factoring allows for insights into the dynamic flow within the supply network [200].

- The comparative performance of supply chain financing and bank financing, as mitigation functions for supply chain ripple effects form a core topic of study in this thesis.

Recent research has looked into advances on existing SCF, which is dyadic in the sense that products are based off invoices, to Blockchain backed SCF (BCF), where products are functions of the history of trades [283].

Deep Tier Supply Chain Financing Modern supply chains have multiple supply lines [279], an example of which is the Daimler-Chrysler supply chain [80]. It is an automotive supply chain, beginning with a raw material processing firm and finishing with an original equipment manufacturer and supplier. It is an upstream manufacturing supply chain for a resilient ecosystem that produces products that use non-perishable materials which are worked and recombined as parts up to market. There exists work unifying supply chain and financing dynamics in simulation models, however not modelling supply chain depth as well [325].

SCF solutions are called *deep tier SCF* (DTSCF) when used to support firms with many tiers between themselves and focal firms [238, 81]. The diamond supply topology can be an example of a deep tier topology. This is first defined in Choi and Hong [80]. This has a single member of the first

tier, many members of the second tier, and a single member of the third. This was extended following Shao et al. [277], on the Honda supply network. They observed that as one moves from the centre of the chain to its peripheries, the number of suppliers per firm decreased on average.

- To capture the interfirm dynamics relationships, the supply chain models studied within this thesis are all deep tier.

Power Asymmetry SCF, and particularly DTSCF, facilitates a method where all firms in each supply chain tier have similar access to financial services as those with superior credit scores at the ends of the supply chain, thus improving supply chain competitiveness [251]. Deeply embedded firms have higher dependency on their partners, so tend to be more affected by financial squeeze [202], for example, when the focal firm does not pay its supplier in a timely manner, which is a type of liquidity squeeze. A buyer risks supplier default when it pressures its supplier, suggesting the supplier is not critical to its immediate buyer, and that the supplier has many competitors. It does not have enough bargaining power to control payment practices [211], else it would be protected or dominate relations.

This further suggests strong embedding and high connectivity within supply chains. They are thus highly susceptible to changing market conditions, and hence correlated to them. This selects for unidentified nexus suppliers [55, 194, 20], so companies with important positions are also least likely to be identified, which further risks bankruptcy.

- Power asymmetry, expressed as information asymmetry is thus both a strong indicator and driver of shock propagation. To encapsulate this, all models in this thesis incorporate asymmetric power in companies, affecting their negotiation protocols [122].

Ripple Effect Due to the bankruptcy changing the network, the original embeddedness of the bankrupt company induces similar conditions for nearby companies, initiating a failure cascade known as the ripple effect (RE) [171, 286, 101, 100, 207, 203, 166, 257]. Since the shock reduces supplier availability and increases operational constraints through reducing markets size and hence cashflow, it is a bidirectional ripple [204, 99]. To avoid this, remain liquid, and preserve the supply chain, companies seek out financing with credit from bank loans, SCF [129, 86], and other financial objects.

In financing themselves, firms forecast according to surges in their cash receivables relative to their cash position, financing too much, reducing their long-term cashflows and access to financing, and passing liquidity issues onto suppliers. This causes a financial squeeze, resulting in a financial bullwhip [72], analogous to the bullwhip effect [69].

The bullwhip effect occurs when a sudden demand shift for a product catches out a supplier, who then forecasts further future demand, and to prevent missing the demand surge, orders a larger quantity [195]. This occurs for each supplier, so amplifies upstream, leaving suppliers with too much stock. Such events affect supply and demand in supply chains. Companies estimate business

ecosystem responses with forward projections, their inaccuracies magnified each tier up the supply chain, generating the shock from the bullwhip effect [196].

Bullwhips and financial bullwhips travel up the supply chain, eventually hitting nexus suppliers, thus propagating throughout the network, generating an RE. These disruptions cascade through supply network tiers, impacting the performance of the entire supply network [101] and reducing liquidity, counteracted by SCF [130], where buyers pay suppliers later after receiving products of services. This creates a liquidity gap [40], inducing a financial squeeze on small-to-medium sized enterprises (SMEs), both due to their fewer resources and weaker bargaining positions [46], as well as their positions between large firms [226, 120] in deep tier supply networks [251].

Inappropriate SCF policies can amplify ripples. For example, material bullwhips can propagate upstream, contributing to liquidity gaps, demanding financing [71]. Financing policy is generated from predictions of financial conditions as a function of cashflow prediction, subject to the same principles as bullwhips. This results in a financial bullwhip propagating downstream [10, 63, 240, 72]. This means that when the ripple begins, it propagates over a more risky system, thus is amplified.

This creates an RE for financing — a financial RE (FRE) [101], which, coupled with more financial stress for upstream suppliers, may render the nexus supplier inoperative, perhaps bankrupt. This weak link may result in network-wide breakdown, and there may be far more of these broken weak links than expected if the entire system is stressed and squeezed [151]. This has been explored across modern empirical markets, with findings showing that the bullwhip effect for cashflow was experienced by 37% of retailing, 43% of wholesaling, and 81% of manufacturing firms in a sample of 763 U.S. public companies from 2010 to 2019 [246].

- The financial ripple effect is the primary risk function explored in this thesis. The proposed control measures therein are designed to mitigate against it.

Supply Chain Financing Benefits Recent work has gone towards showing how SCF introduces a link from supply networks to financial markets and business ecosystems, where it affects asset pricing and can be used to support progression to greater financing [135, 138]. It can also support the wider economy by enabling a means to reconfigure the supply chain in a *creative destruction process*, where inappropriate contractual links with suppliers are broken with late buyers without cutting liquidity from suppliers, thus improving the supply chain [254, 73].

The benefits of SCF suggest it is important to find methods to accommodate its risk [156], motivating my supply chain investigations. Early warning systems help ensure expenses are directed and used in a timely manner, and are important to agents involved in such networks, achieved through network analytics and integrated to network control [43, 176]. Reverse factoring is also a solution, involving the financial cooperation of the focal company and its suppliers. This involves information sharing, which undermines competitiveness. In addition, management of these processes is cost intensive and amplifies with supply chain size.

Opportunities in Decentralised Control This is problematic because many actors servicing infrastructural networks have limited scope, so at certain scales of network complexity or operative speed, a single, centralised controller is inefficient and unscalable. An alternative — decentralised control — is computationally scalable [274], using agents as sensors of network dynamics to mutually contribute to an analysis of the system.

Aligning firm to community goals via real time policy sharing will overcome these problems as an expression of the shared value principle [254], where firms focus on the joint value they can generate rather than individual. Real time policy sharing does not reveal private information, but allows for effective policy coordination, so all companies could base their financing policies off of dynamics affecting the whole of the supply chain.

This solution demands information sharing between firms to infer excess partner structure via extended supply chain visibility. This is challenging due to many reasons, including budget constraints, data standards, privacy, conflicts of interest, skill requirements, and supply chain complexity [175]. Possible solutions for effective policy coordination are forms of collaborative intelligence [94].

- This describes the distributed, decentralised control architecture proposed within this thesis, which has not yet been characterised in deep tier, multi-agent complex network models of the sort used here.

Thus, the smaller, squeezed companies that would most benefit from this have the least means to enact it. Companies have sought ways to forecast supply chain dynamics without any shared information [7] and there is active work on privacy preserving methods of learning on supply chains using federated learning [334, 185].

The sensory aspect has been studied in telecommunications as tied to its model, whereby limiting information to the relevant regions of operation, and using centrality weighted by data packets held at nodes in these regions to approximate criticality, it was shown that decentralisation well approximates systemic criticality dynamics [256, 255].

- I contribute to literature a model of infrastructural control applied to the supply chain simulation model designed within this thesis that uses advantages provided by decentralised analytics, also characterised within this thesis, and integrates it into distributed control, thus capturing the benefits of supply chain financing against the financial ripple effect.

This in some cases has been examined with the Blockchain, with BCF has been proposed as a solution for the information sharing problems of deep tier systems, in the form of Deep Tier BCF [103].

Complex Adaptive Systems

Supply network infrastructures are systems of interdependent assets that produce or relay units amongst themselves to achieve some goal, supplying material goods, products, and services [137].

Key to a supply chain's function are its firms, who make decisions regarding assets based on internal policies, motives, and emergent conditions. In computational models of interfirm supply network infrastructures, they are represented by agents, quantifying these factors. Network agents are interdependent, relying on and responding to each other's operations. Due to the complexity of interactions and agents' capacity to adapt to change, supply networks and their models are defined as complex adaptive systems (CASs) [79].

Adaptivity, unpredictable behaviours in an interactive system, and the nexus supplier concept are all features of a CAS, which is a system with dynamic behaviour in response to changing conditions that modifies the behaviour of embedded actors [80]. Such a perspective has value to researchers for integrating existing supply chain management research into a structured body of knowledge with a framework for studying new theories relevant to real-world supply networks [244]. Several authors framed supply chains using CASs, primarily modelling the dynamic flow of materials [292, 132, 333]. The evolutionary dynamics of complex adaptive SNs (CASNs) are described and categorised in Pathak et al. [245], who advocated the use of multi-agent simulation for modelling and analysing such systems, used in works including Ho Seung et al. [150], Ledwoch et al. [194], and Li et al. [203].

There exist various classes of supply chain simulation models based on selfish multi-agent system dynamics [109], with risk generation and mitigation functions [34, 194, 43, 77]. These financial solutions make firms affect each other as they transfer material downstream. Firms thus simultaneously exist in a supply network and a financial network, acting as manufacturers and financial institutions. This complexity and the dual nature of SCF mean that a viability measure [165] is best for learning how it affects long-term supply chain dynamics. Viability is best measured using survivability [164].

- Works within this thesis use complex network multi-agent CASs to investigate and measure viability, as suggested by these works, which prior to this thesis had not been done for systems that are financial and material.

Many existing RE studies have been on material flows, and in my work I advance on each of the following papers, filling a research gap through combining their focuses to investigate new emergent phenomena.

- The structural robustness of a randomly connected SN was investigated in [143], assessing the probability of RE formation, however they did not study networks inspired by real supply network structure, and no risk mitigation strategies were employed.
- Since supply chain resilience methods measure returns to normality, they are more complex, sorting into adaptivity or stability methods, either reactive or proactive [168].
- Network "reachability" was proposed in [286] as a mediator for formation of ripples, also introducing a graph theoretical way to assess metrics determining dynamic performance of a supply chain under REs.

- Current viability literature [169, 170] suggests a need to focus on a singular combined metric to capture more information under conditions of less certainty.
- Inventory injection was studied as a function of inventory position and network topology to prevent disruption cascades [193]. It would be of interest to study cash-injection in such contexts, however it has not been done prior to work within this thesis.
- A cash-constrained multi-agent model was also studied in [280], but they do not assess behaviour over a complex network.

Network rewiring and a proactive strategy where buyers sever connections with critical suppliers after observing a potential cascade formation was proposed [333].

- This thesis focuses on more complex environments, which require further research to produce such a network level proactive strategy and instead cover reactive cutting.

Findings suggest that steering strategies, which do not fully control at the network level, are more appropriate [1]. Network structure influences RE formation more significantly in the short-term than longer-term [204]. Thus, it is more appropriate to study short term financial objects such as SCF. REs of blackouts have been captured and then related to viability [166]. This more complex measure requires more outcome focused measurement, and is best studied in complex adaptive systems.

- In this thesis the supply chain models developed use networks inspired by real supply network structure.
- Resilience and robustness are captured using mitigation strategies with both reactive and proactive elements.
- These are not explicitly measured — instead, viability is measured, focusing on a singular combined metric to capture more information under less certainty.
- Cash flows are incorporated as a network dynamic, studying cash injection as the mitigation strategy.
- Business relationship cuts are reactive.
- Cash injection is in the form of a short term financial object, SCF.

Stability

A supply network enters a failure state where goods cannot reach market without changing the network when it becomes disjoint [82]. Using CASN terminology, network dynamics cause supply network failure [157]. There, however, exist resilient and non-resilient network topologies [263], where if resilient, networks may return to functionality quickly, suggesting their adaptations after a shock

minimise propagation of said shock. From a systems dynamics perspective, amplifying factors are smaller than moderating ones.

Resilience [250], robustness [59], and efficiency all are indicators of performance, but efficiency is a function of current operation, robustness is resistance to shock, and resilience is capacity to return to prior operations after shock. Viability is positive adaptation after shock, and as a complex network risk measure, is most appropriate to measure in a CASN [161, 259]. Since economies grow, viability most strongly conditions for long-term effectiveness, and due to high degrees of nonlinearity, can be measured using simple metrics, particularly survivability [165].

- Since a CASN may be modelled using both systems dynamics and agent-based dynamics, and complex agent-based modelling creates a complex network, dependent on topology characteristics [292], I study the simulation models using system dynamics arguments [228].

There should thus exist thresholds from unstable states to stable ones upon moderating network operating parameters for a given network topology. Simulation methods allow the empirical investigation of such dynamics while completely capturing network operational configurations, letting one observe the relationship between topology and network configuration [257].

The critical transition point from stable to unstable network configuration is at a phase transition, which in a CAS is at a bifurcation point within the network's state space [140]. Bifurcations correspond to unstable equilibria, such that the likelihood of the network being at those points is always a local minimum for a given state space [141]. Taking ideas from local search [239], assuming one begins in either a stable or unstable environment, it is difficult to see via moderating configuration parameters which states define the complementary stability state. Thus, the system dynamical perspective runs into challenges explicitly defining a CASN as a dynamical system to determine stability.

- Therefore, in this thesis, the outcome space of the CASN simulations is directly analysed without reducing it into a dynamical systems model, but system dynamical arguments are still used for extracting properties out of the outcome space.

Trade credit is abstracted in [34] to combinations of accounts payable and accounts receivable.

- Extended in this thesis is a model which provides buffer periods and mitigation strategies, such that the time a payment must be made and the time a firm is removed are not just a point process.

This thesis also contains a model testing the effectiveness of a collaborative intelligence control method in a dynamic complex adaptive systems model [79] of SCF [257].

- Testing over multiple network topologies, parameterisations, and different nexus measures, I investigate the relationship between visibility, as a function of the network, and viability, as measured by incidence of long-lived supply chains.

- It is evident that the topic of studying the scope of cooperative control regions in complex network environments for complex nonlinear dynamic models is a gap in literature, which I address in this thesis.

The next section of this literature review investigates network scientific methods of measuring criticality, which is the key measure of risk investigated in this thesis.

3.2 Criticality and Centrality Measures

In infrastructural networks, assets — for example a router in a telecoms system, or a firm in a supply chain — can be represented and controlled by agents situated at nodes in a network. Their loss can have dramatic knock on effects in the network, inducing it to break down. This can be represented as both the topological approach to disconnection or as the stalling of network dynamics. To find ways to mitigate against such events, analytical methods are needed to create appropriate measures to operate upon. In this thesis, I advance upon how to monitor the impact upon node failure on the network, using criticality.

With a notion of the varied impacts on complexity of varying information, the question of information relevancy can be defined as the ideal depth for a region surrounding a node for measuring criticality at some time given some dynamic. This can be framed as one of the value of information, or what one would claim to be the ideal information reach of a distributed or decentralised algorithm. The following section discusses previous work in this field and possible applications for this question.

Criticality, which is the measure of how risky a node is to the operation of the network, is similar to centrality, which is the measure of how deeply embedded a node is within the network. By using *centrality measures* as *criticality estimate* (CE), one can approximate a criticality measure (CM) because both define importance within a multicomponent system [44]. This is why many criticality measures are extensions or variations of centrality measures.

Centrality measures are used to capture many different forms of importance for network elements, and may estimate criticality. This is always an estimate based on the network state. The criticality is the impact of the removal of an element, which can only be found explicitly through empirical data or repeated simulation. It is also computationally expensive. In addition, the concept of “impact” is still context dependant. A compromise can be reached, where criticality is measured using a proxy for simulation, a criticality measure. Here, the network element is physically removed from the network, and the impact is estimated through some schema.

Removing a network element is a form of network dynamic, and can either be a dynamic *of* the network or *on* the network. Dynamics *of* networks are topological changes, while dynamics *on* networks are state changes [236]. Of interest within this thesis is criticality from dynamics from both on and of the network, which is the result of the behaviour of a given model that changes properties assigned to network elements, and then the topology of the network. In the case of flow networks with capacity, removing an element can be achieved through congesting it in the telecoms case, or

bankrupting it in the supply chain financing case, rendering it inoperable in either, removing the node. Measures that address this dynamicity and change with it are needed, since these can be used to detect nodes under more stress at a given time, or to inform whatever control function to mitigate against that stress. Within this thesis, the simulation models and metrics applied in their analytics and for their control can be understood as a formalism to create such criticality measures.

3.2.1 Criticality

The **bag-of-paths** criticality measure [192] is based on the bag-of-paths framework [124], where all paths within the network are assigned a probability of being selected according to some distribution. Each node pair has a set, or bag, of paths that have them as their endpoints. A probability distribution is defined for each of these sets, where the likelihood of selecting a path corresponds to the solution to minimising the total expected path cost for all node pairs, while relative graph entropy remains fixed. Path costs are varied. A path for a given node pair is hence less likely to be chosen if its constituent edges are typically on a longer path. Path choice follows a Gibbs-Boltzmann distribution of energy [278].

The bag-of-paths criticality is then found by initialising on the graph state found by the bag-of-paths framework, deleting a node, recalculating, and then calculating the Kullback-Leibler divergence (a measure of difference between two probability measures) [188] for the relative entropy in the previous probability distribution and the new probability distribution. Therefore, to evaluate the criticality of a particular node in a fixed graph, one must find the difference between the global criticality after deleting the node and the initial global network criticality. This is a powerful method, but is a global measure, so not useful in a distributed computation setting. In addition, since it relies on graph entropy, it is highly sensitive to graph topology, so a distributed version of this measure cannot be used.

The **dynamic perturbation criticality** [148] is designed to determine changes in intraday link criticality within a temporal information exchange network, such as a backbone communications network. It does so by first generating the weighted graph Laplacian, defining a change at a single edge per time step on the network as a perturbation on the Laplacian, and updating the graph Laplacian as the sum of the Laplacian of the perturbation matrix, $\delta L \in M_n(\mathbb{Z}^+)$, and the previous Laplacian, such that

$$\tilde{L} = L + \delta L.$$

Perturbation is edge-centric, so for an edge, $(uv) \in E$, the eigenvalues of $\tilde{L}$ may be computed as the sum of the eigenvalues of L and its perturbation $\delta L(uv)$ [65], such that $\tilde{\lambda}_j = \lambda_j + \Delta\lambda_j(uv)$, where $\Delta\lambda_j(uv)$ is the perturbation of the j -th eigenvalue, λ_j . In addition $\Delta\lambda_j(uv)$ can be represented as an eigenvector decomposition, where eigenvectors of $\tilde{L}$ can in turn be represented as a perturbation on the eigenvectors of L where $\Delta q_j(uv)$ is the perturbation of the j -th eigenvector, q_j , corresponding to

the removal of the edge, (uv) . These relationships are used to quickly and simply compute the spectra of the Laplacian over time, needing only to compute an initial Laplacian at the macrograin.

This allows the use of **perturbation centrality** [65]. The measure is based on the sum of the K smallest eigenvalues of the Laplacian perturbation matrix. Letting this perturbation be the removal of edge (uv) , this becomes an edge criticality measure, such that

$$\theta_{(uv)} = \sum_{j=2}^K |\Delta \lambda_j(r)|,$$

where $\theta_{(uv)}$ is the perturbation centrality of edge (uv) . Since the smallest eigenvalues of the Laplacian inform graph connectivity, this describes the variation in graph connectivity after the removal of an edge, (uv) . This measure is therefore sensitive to edges that bridge neighbourhoods. Case studies of intraday data traffic within the UK backbone network have been conducted, where they applied the perturbation centrality to learn new insights into how data demand is distributed within a day [148].

Perturbation centrality has numerous advantages — its ease of computation, sensitivity to naturally critical edges, and structural interpretation. Due to it acting on a perturbed (and in this case removed) edge, it is a true criticality measures. It also has limitations — it is not distributed, nor node-centric. One could theoretically redefine it to satisfy these properties. The principles of perturbation of edges that underpin this are used within the simulation models and dynamics of information bounded centrality measures (IB-CMs) defined within it.

3.2.2 Centrality Measures

These methods, while effective, are too complex to be employed at pace and over limited regions, which motivates the approximation of criticality dynamics with centrality measures. Centrality measures may be roughly separated into three categories [126, 49], either degree-like, closeness-like, or betweenness-like. This distinction is important when selecting which measure to use for a given system, however they may also be defined by the information they take. Centrality measures for a node can be computed sociocentrically, over the whole network; or egocentrically, over a limited region surrounding the node. I document here the classic and some more advanced measures and compare them against eachother with a focus on complexity, providing the basis of the family of measures defined within this thesis, justified by gaps in existing literature.

If one is limiting information computed for each node via egocentric centrality, distributed or decentralised computation is necessary for each node to operate using all network information [174, 213]. There can be methods to compute global measures in a decentralised manner, or the measure itself can be decentralised [198]. In the first case, the measure is computed from total network information, however this may be bounded above through the use of incremental algorithms, or those that seed from a known collection of nodes. In the second, each node defines a local subregion, and computes centrality within this region, so that no node necessarily has total network awareness. Nodes

then communicate their outputs, and use a separate algorithm to learn their relative centrality. The simplest is a linear comparison, either of intervals or ranks [290].

A distributed centrality measure is such that nodes in the network do not need full network information to determine their centrality [330]. This is valuable in distributed computation architectures, for several reasons. Distribution may allow for the same compute power as a centralised system, but more efficient, since less investment in costly computational devices or labour is necessary [39] to process more and more distant information. In a distributed system of information bounded agents, all agents are situated a restricted distance from the dynamics they measure or control, such that it is the most relevant information for them. In addition, when computing centrality from a nodewise perspective, distribution reduces overlaps and overhead on the network, since fewer nodes take data from the same source, further increasing efficiency [329].

This section covers the various centrality measures which are adopted or modified to estimate criticality in this thesis. The measures are compared according to the key characteristic required for creating distributed, dynamic, estimates of criticality for networks where elements can have weight. They are also compared according to approximately how long one would expect it would take to compute the measure in the worst case, that is, time complexity. This has been previously surveyed [272] for a collection of classic measures, though orders of complexity were not given. Any given algorithm has a set of parameters that are computed in some way. For each dimension of the parameters, as the number of data points for that dimension grow to very large sizes, those that have more contribution will dominate in complexity. For example, let an algorithm take $f(n) = n^2 + n$ much time to compute. As n grows large, then $f(n) \approx n^2$, since $n^2 \gg n$. Then one says that the algorithm has $O(n^2)$ time complexity. This is called “big-oh” notation. Less time complexity is preferable, and if an algorithm does not scale to network size, then autonomous control of the network is scalable.

To have a clear notion of information bounding, I introduce appropriate notation. The information boundary around a node is defined by the geodesic distance. For a node, $u \in V$, the set of nodes such that $\text{len}(p_{u.}^g) = i$ is denoted as $\Gamma_i(u) \subset V$, where $\Gamma_1(u)$ is the neighbourhood of u . The set of nodes such that $\text{len}(p_{u.}^g) \leq i$ is denoted as $H_i(u) = \bigcup_{j=1}^i \Gamma_j(u)$. If u has an information boundary at distance h , it takes information from $H_h(u)$. Now follows an introduction of the various centrality measures.

Centrality is a concept used when one wishes to identify the most influential node, often in cases such as social network analysis, telecommunications network analysis or supply network analysis. It is applicable in any field where one may use a network representation of a given system. Early use cases of centrality explain differences in performance of communications networks and constituent members [35, 36, 191].

Other papers give examples of limited range criticality and centrality for static measures and for dynamic distributed criticality measures [310, 68, 230, 181, 256]. All show accuracy despite limited boundaries. However, works such as these have conducted no large scale analysis on the relative efficiency via computation time and accuracy for dynamic criticality measures, which I detail.

Degree Centrality for a node u counts the number of neighbours. It is defined as $C_D(u) = |\Gamma_1(u)|$. It is the simplest centrality measure and least computationally expensive, where for each node and connected edge it counts for one more timestep. It has time complexity $O(m)$ because $k = \frac{2m}{n}$ and it scales with mean degree and node count.

Local centrality [68] extends the degree centrality, and was originally designed to determine the level of embeddedness of a node within a network. To compute the local centrality of $u \in V$, denoted $C_L : V \rightarrow \mathbb{Z}^+$, first compute its neighbourhood, $\Gamma_1(u)$. Second, compute the neighbourhoods of each node in $\Gamma_1(u)$. Then sum up for each node, w , in each $\Gamma_1(v)$, the number of nodes in the set of all nodes at most two edges away from w , such that

$$C_L(u) = \sum_{v \in \Gamma_1(u)} \sum_{w \in \Gamma_1(v)} |H_2(w)|. \quad (3.1)$$

This counts nodes for a local region, which is linear. For a given node, the algorithm needs the neighbours of neighbours, which takes $O(k^2)$ information. The measure may cover nodes at most h many edges away, requiring, $O(k^h)$ information, and does so for all nodes, n . Since $k = \frac{2m}{n}$, this simplifies to complexity of $O(mk^{h+1})$.

Sociocentric Eigencentrality [190] captures the connectivity of the network, valuing nodes with more connections to well-connected nodes. Denoting the adjacency matrix of G as A_G , the eigencentralities of the nodes in the network are found for the largest eigenvalue, λ_n^G , where for $\{x_1^G, \dots, x_n^G\} = \mathbf{x}_G$,

$$A_G \mathbf{x}_G = \lambda_n^G \mathbf{x}_G. \quad (3.2)$$

The sociocentric eigencentrality for $u_i \in V$ is thus the solution to Eqn. (3.2) such that $C_E^S(u_i) = x_i^G$. The time complexity of computing the sociocentric eigenvalues and eigenvectors of a matrix is $O(n^3)$ if numerically solved according to von Mises, or “power” iteration [304], since this is approximately n vector operations on the n^2 sized A_G .

Egocentric Eigencentrality adjusts Eqn. (3.2) to be over $H_h(u)$ rather than over G , creating

$$A_{H_h(u_i)} \mathbf{x}_{H_h(u_i)} = \lambda_n^{H_h(u_i)} \mathbf{x}_{H_h(u_i)}. \quad (3.3)$$

For node $u_i \in V$, the egocentric eigencentrality is $C_E^E(u_i) = x_i^{H_h(u_i)}$. Assuming the excess degree distribution equals the degree distribution, the time complexity of computing the egocentric eigenvalues and eigenvectors of a matrix is $O(m^{3h})$.

Katz centrality [177] is an extension of eigencentrality that takes advantage of a unique property of adjacency matrices, which is that the $(i, j)^{th}$ position in the k^{th} power of A is an indicator of if nodes u_i and u_j are a distance of k from each other. Then the Katz centrality for a node sums for all adjacencies for powers of adjacencies of k over all positive integer k , and for each k multiplies by

some constant, α , bounded above by the largest eigenvalue of A , giving

$$\begin{aligned} C_K(u_i) &= \sum_{k \in \mathbb{Z}^+} \alpha^k \sum_{j=1}^n (A^k)_{ij} \\ &= \left((I - \alpha A^T)^{-1} \right)_i, \end{aligned}$$

which is the i^{th} element of the leading diagonal of the matrix, $(I - \alpha A^T)^{-1}$. The upper bound for α means that the further away from a given node another is, the more its contribution to centrality is penalised. Matrix inversion takes $O(n^3)$ time to complete, while all operations are linear. Since this reads elements off of the computed matrix, this is the complexity of this centrality measure.

PageRank centrality [241] builds on the recursive network of influence that defines eigencentality and Katz centrality, but is tailored for the World Wide Web. It represents hyperlinks towards a webpage as a directed edge towards a node. The centrality of a page is the sum of the neighbour's centralities, each divided by the number of pages they connect to (degree). This method accounts for meaningful connections by reducing the impact of populating a webpage with hyperlinks to many unimportant pages, as in a directory. There is an issue, however, with this first formulation, in that links to closed cycles make the solution to it impossible. In this thesis, the two case studies investigated are telecoms infrastructure and supply chains. In the former case, edges are undirected, so PageRank is inappropriate. Both are interested in criticality, but in both cases, moderation for hubs does not help represent criticality. This is more complex than eigencentality due to the weighting of meaningful pages, but only linearly, and so still has the same time complexity $O(n^3)$.

PageRank centrality is not good for criticality computation. It is designed to promote pages that have the most unique connections to themselves and others who are similar. In PageRank, having many connections to nodes with few connections will not greatly improve the score. However, this is for purposes of exploring the network, not considering network cohesion and criticality. Therefore, PageRank is not fit for use in controlling for risk.

Wehmuth centrality [310] gives a measure that can be computed in a distributed manner, without each node requiring full network knowledge, and use information on the local structure to determine criticality. To learn how well connected a component is within itself, one uses the smallest non-zero eigenvalue, called the *algebraic connectivity*. For connected networks, this is the second-smallest eigenvector and has eigenvalue λ_2 . This defines the Wehmuth centrality, a distributed criticality ranking measure.

For some node, u , the Wehmuth centrality, denoted $C_W(u)$, finds the algebraic connectivity of the induced subgraph of nodes at most h edges away from u , denoted λ_2^u , and divides this by the logarithm of the degree of u , unless the node is a leaf, in which case it is non-critical. If $h = 2$, this finds how well connected a node is with respect to its immediate area and how integral it is to the connectivity of the surrounding area. It divides by the binary logarithm of the degree of u because this

stops non-critical hubs from being ranked too high. The formula for the Wehmuth centrality is thus

$$C_W(u) = \begin{cases} \frac{\lambda_2^u}{\log_2(d_u)}, & d_u > 1 \\ \infty, & d_u = 1. \end{cases} \quad (3.4)$$

This algorithm works over a limited region of the size $1 + k^h$, n many times. Diagonalisation per region takes $O(k^{3h})$ time, while all other operations are linear. Therefore, the Wehmuth centrality has $O(nk^{3h})$ time complexity.

Sociocentric betweenness [125], denoted $B_s : V \rightarrow \mathbb{R}$ is one approach to track pathway disruption potential, calculating the fraction of geodesics between all node pairs in a network that pass through a given node. Higher betweenness suggests higher centrality, and is calculated as

$$C_B^S(u) = \sum_{\substack{v,w \in V \\ u \neq v \neq w}} \frac{\rho_{vw|u}}{\rho_{vw}}. \quad (3.5)$$

This requires finding all shortest paths in the network, and then completing a linear operation, the mean. As a base case, let the shortest paths be found using Dijkstra's Algorithm [95], which has complexity $O(m + n \log(n))$. Repeat this for all pairs, showing that sociocentric betweenness has $O(n^2(m + n \log(n)))$ complexity.

This measure can be modified into **egocentric betweenness**, which measures the betweenness of a region surrounding a node, and then compares the valuations between nodes in the network. It correlates strongly with sociocentric betweenness [214], and is computable in a decentralised manner. For each node, say $u \in V$, it measures the betweenness of the induced subgraph of $H_h(u)$, such that

$$C_B^E(u) = \sum_{v,w \in H_h(u)} \frac{\rho_{vw|u}}{\rho_{vw}}. \quad (3.6)$$

In centralised computation, egocentric shortest path betweenness would be the same type of computation as sociocentric shortest path betweenness, restricted to paths of length less than or equal to $2h$. However, egocentric betweenness is applied in a distributed computing sense, which requires redundancies in computation. Per node, this measure finds the subgraph of nodes with shortest paths of at most h from the node. Pathfinding per subgraph takes $O(k + \log(k^h))$ which can be written as $O(\frac{m}{n} + \log h(\frac{m}{n}))$, since the factors of 2 become linear in logarithms. This is n many times, for $(\frac{2m}{n})^h$ many pairs, thus having $O(mk^h + \log h(\frac{m}{n}))$ complexity.

Higher betweenness suggests greater criticality since being a passing point for a larger proportion of paths suggests, if a node were to fail, it would cut a larger proportion of all paths in the network. Betweenness is powerful, but is designed to detect centrality, not criticality, and assumes shortest path traversal. It can be elaborated upon.

Traditional betweenness assumes all network travellers take the shortest path, but this is not always the case. A workaround for this is **flow betweenness centrality** [127]. It differs from shortest

path betweenness in two ways: it works for weighted networks, and it is scaled to all the independent paths between all node pairs, rather than the shortest paths. It works specifically for flow networks, where all edges have substance travelling on them, and a maximum capacity for the substance. Such networks can be said to be defined by two layers, the capacity network, where edges are weighted according to their capacity, and the current flow network, where edges are weighted according to their flow. No edge in the current flow network can exceed its counterpart in the capacity network. This measure uses graph cuts, found with the Ford-Fulkerson method. By ordering it, it can be solved in $O(n^2m)$ time [97]. Repeating this $n(n-1)$ times finds the minimum cuts for all pairs, however there exist only $n-1$ many distinct minimum cuts [4]. There exists another reduction result that can solve the minimum cut problem for a pair in $O(n^2)$ time, however this requires generating a tree of minimum cuts of the network, which is infeasible in dynamic systems. Therefore, flow betweenness has $O(n^3m)$ complexity.

This measure can be used on weighted networks, however, the information obtained through this formulation will decrease very quickly with smaller graphs. If this was converted into a distributed context, then the consideration of indirect paths is lost, thus devaluing the measure. Both betweenness and flow centrality consider graph structure. However, weighting alternative paths can give counter-intuitive results [237]. The following measure is designed to fix this issue.

Random walk betweenness centrality [237] builds on flow betweenness by directly computing the passage of information or substance through the network. In it, a node's centrality is calculated by counting how often a node is traversed by a random walk between two other nodes. For some node, i , it sums the probabilities that an absorbing random walk that starts at j and ends at $k \neq j$ crosses $i \neq j \neq k$ for all j, k pairs, returning the random walk betweenness. This measure does not require simulations to be calculated, and can instead be found through matrix operations on the graph Laplacian. Applied in a flow network, the random walk betweenness for a node u is its average current flow to all other nodes. This is found through inverting the Laplacian after some operations to make it non-singular, which takes $O(n^3)$ time, and itself takes $O(mn^2)$ time to be computed since it runs over all edges, and all node pairs. This means random walk betweenness has $O(n^2(n+m))$ complexity.

This method lies at the extreme of the family of betweenness measures, calculating for all possible paths a node can take. The following measure sits between this and measures that use the shortest path.

Building on from the idea using walks while reducing weighting of longer paths from flow centrality, **communicability betweenness centrality** [115] strikes a balance that can more confidently measure centrality. The communicability centrality of a given node u , as in Equation (3.5), measures the sum over all node pairs not including u of the commonalities between node pairs, conditioned on passing through u , denoted as G_{vuw} , divided by the communicability of v and w . The complexity of using a matrix as an exponent is $O(n^3)$, and this must be done for all nodes, giving a complexity of $O(n^4)$.

This measure may be useful as a criticality measure, since to calculate G_{vuw} , one computes communicability of v and w on the graph where incident edges of U are removed. The absence of matrix inversion is also appealing, however this requires the matrix exponential, which has similar complexity. Next is a measure that extends one of the betweenness measures into a viable estimate of criticality.

If one considers local network structure that suggests positioning at a critical point, one can obtain a robust criticality measure. The structural metric used is the *bridging coefficient* [160, 231], which describes embedding of a node within a connected component. It is defined as the reciprocal of the node's degree, divided by the sum of degrees of its neighbourhood. For a node, u , the formula of the bridging coefficient, $c_b : V \rightarrow (0, 1]$, is

$$c_b(u) = \begin{cases} \frac{1/d_u}{\sum_{i \in \Gamma(u)} d_i}, & d_u > 0 \\ 1, & d_u = 0. \end{cases} \quad (3.7)$$

By multiplying the sociocentric betweenness centrality and bridging coefficient, we obtain the **sociocentric bridging centrality** [160], $C_B : V \rightarrow \mathbb{R}$, such that

$$C_{BR}^S(u) = C_B^S(u) \cdot c_b(u). \quad (3.8)$$

As two independent calculations, the complexity of this is the sum of the complexity of computing the bridging coefficient, and the betweenness centrality, which is already established. The bridging coefficient is just degree information, and hence is $O(m)$ complexity. This means it has the same complexity as sociocentric shortest path betweenness centrality, which is $O(n^2(m + n \log(n)))$.

This can be changed into the **local bridging centrality** [230] (or egocentric bridging centrality) by replacing the sociocentric betweenness in Equation (3.8) with the egocentric betweenness, and rewriting it into

$$C_{BR}^E(u) = C_B^E(u) \cdot c_b(u). \quad (3.9)$$

The complexity of this algorithm is the sum of computing the egocentric betweenness and distributedly computing the bridging coefficient. In this case, the complexity of the bridging coefficient is $O\left(\frac{m^2}{n}\right)$ since it computes graphs of mean size k^2 , n many times.

The **closeness centrality** [36] is the mean length of the shortest path between the given node u and all other nodes in the network, which can be defined as the reciprocal of the sum of shortest paths from u to all others. Denoting closeness centrality as $C_C : V \rightarrow \mathbb{R}$, it is such that

$$C_C(u) = \frac{1}{\sum_{v \in V} \rho_{u,v}}.$$

This is a measure that treats a node as the source and sink of information within a network. It requires the same information as sociocentric betweenness centrality, that is, all shortest paths in the network. The method computes the output differently, but this is a linear operation, and so is the same complexity, with $O(n^2(m + n \log(n)))$.

Experimental evidence suggests increased computational efficiency and satisfactory performance of information bounded network measures [112]. This details the relationship of the depth of the information bound and size of the value distribution of the associated bounded betweenness measures. The value distribution increases exponentially with the depth of the bound up to mean geodesic length before decreasing, suggesting meaningful sensitivity at the mean geodesic length. For scale-free and random graphs, which only have one cluster, it is expected that the ideal depth may be the mean cluster geodesic length.

The following measure uses a very different centrality formulation, and may not be useful for centrality calculation, but could be applicable for node communication architecture. This is the **k-core centrality** [62], or the k-core index in the case of internet systems, where a node's index is the k of its highest valued k-core. For connected networks, this has a time complexity of $O(m)$, counting edges in the same way as degree centrality. The advantage of this method is that one may extract subgraphs from the network based on their common centrality, useful for identifying categories of importance. This is an extremely simple method, but is at a very rough grain and is centralised, relying on the specific topology of the network, not key indicators, such that distance reduction leads to accuracy reduction. It, however, is an interesting criticality indicator since it reveals core network structure.

Table 3.1 compares all the centrality measures discussed in this literature review.

Table 3.1 Comparative Table of Centrality Measures. A checkmark in “Criticality” denotes that the centrality measure can approximate criticality, a mark in “Distributed” denotes if it distributedly computed, and in “Weighted” and “Dynamic” denotes if it returns meaningful information for such networks. “Order of Complexity” denotes the big-oh time complexity of the respective algorithm. For $G = (V, E)$, $n = |V|$, $m = |E|$, and for a distributed measure, i is the depth. The complexity calculation has been done with minimal assumptions, which would match maximal entropy, which corresponds with an Erdős-Rényi random graph. The weighted local, weighted local Wehmuth, and weighted local bridging centralities are novel measures developed for this thesis, published in Proselkov et al. [256].

Centrality Measure	Info.	Bounded	Weighted	Dynamic	Order of Complexity
Degree	✓				m
Local [68]	✓				mk^{h+1}
Closeness [36]					$n^2(m+n \log n)$
Sociocentric Betweenness [125]					$n^2(m+n \log n)$
Egocentric Betweenness [214]	✓				$nk^{2h}(\frac{m}{n} + h \log \frac{m}{n})$
Flow Betweenness [127]			✓		$n^3 m$
Random Walk Betweenness [237]			✓		$n^2(n+m)$
Communicability Betweenness [115]			✓		n^4
Bridging [160]					$n^2(m+n \log(n))$
Local Bridging [230]	✓				$\frac{m^2}{n^3} + nk^{2h}(\frac{m}{n} + h \log \frac{m}{n})$
Sociocentric Eigencentality [190]					n^3
Egocentric Eigencentality [190]	✓				m^{3h}
Katz [177]					n^3
PageRank [241]					n^3
k-core [105]					m
Wehmuth [310]	✓				nk^{3h}
Weighted Local [256]	✓		✓	✓	mk^{h+1}
Weighted Local Wehmuth [256]	✓		✓	✓	$n8^h k^{3h}$
Weighted Local Bridging [256]	✓		✓	✓	$\frac{m^2}{n} + nk^{2h}(\frac{m}{n} + h \log \frac{m}{n})$

3.3 Summary

In this literature review is discussed how infrastructural network modelling is presented in current literature with mind towards scalable autonomous control, including a discussion on information relevancy, which is the key factor towards scalability focused on in this thesis, and a broader review of infrastructural network control techniques and cases, considering their contribution to the combined research gap this thesis concerns. It is argued that, given modern solutions combine valuation mechanisms with high-speed communication, telecoms and supply chain financing research are the most appropriate cases to investigate, with methods to measure temporal information dynamics and spatial information dynamics with valuation, respectively.

This is followed by an in-depth review of these two cases. First is modelling telecoms infrastructures, discussing the state of the art, network risk and criticality, and methods and research concerning asset communication structures in telecoms. Following this is modelling supply chain infrastructures, discussing the state of the art, principles of firm power balancing, supply chain financing, supply chains as complex adaptive systems, and supply chain stability. The methods in this thesis for studying analytics for telecoms and control of supply chain systems are explained and justified.

Then is a review of algorithms for criticality measurement in network systems, their estimation via centrality measures, with a comparative review of the computational complexity and applicability of these measures.

Together, these provide a basis and justification for the research described in this thesis. I research methods for scalable dynamics analytics using centrality measures over simulation models of complex telecoms network infrastructure systems, owing to it being the most information restricted case, and then methods for scalable autonomous control using principles derived from this research over simulations of complex supply chain financing network systems, owing to these having value pegged dynamics of material goods.

Chapter 4

Decentralised Dynamic Criticality Measurement over a Failure Dynamics Model

4.1 Introduction

This chapter proposes methods of dynamic decentralised measures of nodal criticality in a telecoms risk propagation model. In this, three distributed measures of nodal criticality are created, modified off of originally structurally defined measures, and augmented with dynamically changing node weights, representing stress incurred directly on the node.

I define a manner of determining system criticality from this dynamic criticality propagation model, and a means to use dynamic information to determine the accuracy of a given criticality estimate.

Section 4.2 describes methods to define an augmented SIS model [180, 54], where nodes have internal counters of the degree to which they are infected, before becoming infectious themselves. This represents the propagation of load within a multi-agent system where members have fixed storage capacity, where once this capacity is reached, data packets intended to be sent for processing in at-capacity nodes are instead held in queues in the neighbour they were being transmitted from, described in Subsection 4.2.1. This thus models an incremental infection. Then the mechanism to investigate criticality for this effect is described in Subsection 4.2.1. I then define a form of dynamic criticality derived from network dynamics in Subsection 4.2.1 and propose three approximations of this for nodes in Section 4.2.2, using bounded ego-networks to compute the approximation. The boundedness means methods are scalable with network size in application. These are benchmarked against each other using methods outlined in Subsection 4.2.3, showing strong similarity, presented in Section 4.3. Conclusions to derive from this are mentioned in Section 4.4, with a summary of

contributions in Section 4.5. All measures and methods produced in this chapter are derived from Proselkov et al. [256] except when indicated.

4.2 Methods

In telecoms infrastructure, where nodes represent devices such as routers or switches, if data packets arrive at a rate greater than the node can emit them, they may be queued up, to be emitted later. Weighted networks are used here to model this. This has the advantage of extra weighting being given to local regions with a high number of queued data packets.

4.2.1 Simulation Model

Augmented SIS

The simulation model used is based off of the SIS model of random infection over a network. This represents network systems where nodes can propagate infection, but randomly return to healthy non-infectious states, which can equally have infection propagated unto them, chosen because congestion can. This is chosen, in comparison to the SI or SIR models as congestion is a state than can be left, but having been congested in the past does not stop congestion in the future.

To estimate the accumulation of queued data packets, suppose each node contains a counter denoting queue length, $\mathcal{Q} : V \rightarrow \mathbb{Z}^+$, where if for a given node, the queue reaches the infection threshold, I , it is declared infectious. That is, for a given node, u , if $\mathcal{Q}(u) \geq I$ then $u \in \mathbb{I}$, else if $\mathcal{Q}(u) < I$ then $u \in \mathbb{S}$. In the augmented SIS, infection and recovery steps take the form of counter additions and reductions. An infected node, u may increase the counter of a neighbouring susceptible node, v according to the same infection dynamics as the SIS at rate β . All nodes may exhibit recovery behaviour, where they reduce their counter by one according to a Poisson process of rate γ .

The infection step represents a data packet failing to travel from the susceptible v to the infected u since u has a full queue and cannot process it. The packet is returned to v , increasing its queue. The recovery step represents the resolution of one packet, and does not represent an addition to any other node's queue. This model estimates packet transfer, so data packets are not agents and do not physically affect it.

Simulation Procedure

To isolate each node's impact on the network, n many simulations are run. Each simulation $S_i \in S$, where S is the set of all simulations, is initialised by bringing the queue length of node $u_i \in V$ up to its threshold. Thus one may determine that any failures within the network occurring during simulation S_i are the consequence of attacking node u_i . For the runtime of simulation S_i , denoted $T_i \in T$, where T is the set of all runtimes, $T_i = \{0, \dots, T'_i\}$, where T'_i is either a fixed cutoff point or when there are no more queued data packets.

True Criticality

Suppose the true criticality in the network can be captured by the sum of the node queues for a short future time horizon. This is justified by recognising that each measure proposed within this thesis is designed to compute the risk at a given moment in time that a node poses to the operation of the network. Specifically, this is the impact of the failure or reduced service of that node, which is represented in this model by the node being able to infect neighbouring nodes and so increasing their data packet queue length. Since this increases the likelihood of the failure of neighbouring nodes, which would occur after some random time, the impact of a node's failure would occur with a delay. In a well maintained infrastructure network, it would then be repaired after some time. This results in a peak of total queued data packets, which I claim to occur approximately at the half life of an infectious node, assuming independent node lifespans. Using a mean field approximation, the future time window is defined as

$$t_f = \left\lceil \left(\frac{1}{\gamma} \right)^{1/2} \right\rceil.$$

The aggregation for the real criticality is a sum across all nodes at a given timestep, and all nodes at most t_f steps into the future. Since no computed measure is defined outside of T_i , this analysis is only performed for the shorter timescale $T_i^{(f)} \in T^{(f)}$, where $T^{(f)}$ is the set of reduced runtimes, such that $T_i^{(f)} = \{0, \dots, T_i' - t_f\}$.

4.2.2 Criticality Measures

I now define three CEs to learn which approach most accurately estimates criticality. These measures are chosen because each uses a different fundamental network feature while being flexible enough to logically incorporate dynamic node states. The first metric counts the degrees of local nodes to determine criticality, the *local centrality*, derived from [68]. The second metric uses local structural measures to compute criticality, the *Wehmuth centrality*, derived from [311]. The final metric considers possible paths through a given region to compute criticality, the *local bridging centrality*, derived from [230]. The original versions are defined in Chapter 3.

They may not suggest criticality in an absolute sense, but as a comparison measure they show which node at a given time poses more risk to the network than others. Measures are normalised from 0 to 1, where being closer to 1 suggests greater criticality. This has the advantage of both preserving ranking as well as scaling.

Weighted Local Centrality

Consider local centrality in the case of incremental weighting. Let us cover the queues of nodes v which are i steps away from u , and cover the queues of nodes w which are each i steps away from at

least one v . Then Eqn. (3.1) can be rewritten as

$$C_L^{(w)}(u) = \sum_{v \in \Gamma_1(u)} \sum_{w \in \Gamma_1(v)} c H_i(w)^t.$$

Weighted Wehmuth centrality

Wehmuth centrality is extended to include queue lengths, which will vary with time, doing so through the dynamic processes acting on the network, such as the movement of data packets within a telecoms network. Wehmuth centrality uses network structure and node degrees, both of which are independent of queue lengths within nodes. To account for the queue lengths, the simple undirected network is redefined as a directed multigraph $G^d = (V^d, E^d)$. This allows the original Wehmuth centrality to be redone on this new graph, instead dividing by the log of the out-degree of a node.

For a simple graph with weighted nodes, node information is converted into topological information by transforming it into a directed multigraph. Each undirected edge is replaced with a pair of directed edges. Let, for each node, the number of out-edges to be multiplied by the node's queue length. The properties of the Laplacian still hold for such graphs. The formulation for this approach includes redefining the degree matrix for G^d as $D^d = cD$. To create the adjacency matrix of G^d , I define

$$A^d = (a_{i,j}^d) \in M_n(\mathbb{Z}^+), \quad \text{where} \quad a_{i,j}^d = a_{i,j} \cdot c_j.$$

Since the laws of the network Laplacian are consistent for directed graphs, I apply the whole of the Wehmuth centrality procedure onto the network defined by D^d and A^d , obtaining the weighted Wehmuth centrality.

$$C_W^{(w)}(u) = \begin{cases} \frac{\lambda_2^u}{\log_2(d_u)}, & d_u > 1 \\ \infty, & d_u = 1. \end{cases} \quad u \in V^d \quad (4.1)$$

Weighted Local Bridging Centrality

Since telecoms route data packets, they must traverse a path. To determine criticality, I quantify the significance a node has on a path. Sociocentric betweenness centrality does this, but needs full network information. Egocentric betweenness centrality shrinks the range but loses information. Bridging centrality is designed to estimate if a node exists at the boundary of two clusters with few other connections, and does so with local information, so may be safely shrunk and still find major chokepoints. I wish to make this dynamic.

To create a measure that changes over time, I incorporate the dynamically changing queue lengths contained within nodes, denoted as $q \in \mathcal{Q}$ such that $q : V \rightarrow \mathbb{Z}^+$, where queue length of u is $q(u)$. This so that a node with more data packets is recognised as more critical to the network. The local bridging centrality is thus augmented with node weight associated with criticality. The measure has two factors, which are the egocentric betweenness and the bridging coefficient. The bridging

coefficient is designed to estimate the likelihood that a node is on a bridge between clusters. It is unclear how node weight correlates with this, so incorporating node states into the bridging coefficient is left for further research, however the egocentric betweenness may be naturally extended. Since node queues are made up of data packets, which follow paths, and the egocentric betweenness is a path measure, each path may be weighted by the sum of its constituent nodes' queue lengths. This is achieved by redefining $\rho_{u,v}$ and $\rho_{u,v|w}$ to create

$$\rho_{u,v}^{(w)} = \sum_{P \in P_{uv}^g} \sum_{x \in P} q(x) \quad (4.2)$$

and

$$\rho_{u,v|w}^{(w)} = \sum_{\substack{P \in P_{uv}^g \\ w \in P}} \sum_{x \in P} q(x). \quad (4.3)$$

By inserting Eqn. (4.2) and Eqn. (4.3) into Eqn. (3.6), and this into Eqn. (3.9), one obtains the weighted local bridging centrality, $C_B^{E(w)}$.

4.2.3 Determining Criticality Accuracy

Criticality Estimation Function

Each measure is node and time dependant, so varying the node and the time the node is measured changes the value computed by any given measure. This time dependency is the result of the dynamic behaviour of the network model used in testing. To be able to compare the different measures, one must aggregate across at least one of the dimensions. Since these are dynamics measures, of chief interest is the performance of the measure for all nodes and their performance in time. Therefore per dynamic measure, for each given simulation, I sum over all nodes at some timestep and return an aggregated measure. I denote the set of all dynamic node attributes as $C = \{\mathcal{Q}, C_L, C_W, C_B\}$, and the instance of a dynamic node attribute $c \in C$ for a given node $u \in V$ at time $t \in T_i$ for simulation S_i as

$$B_{t,i,u}^{(c)} : \{T_i^{(f)}, S, V, C\} \rightarrow \mathbb{R}$$

From here, the *aggregated criticality measure* is defined as

$$A_{t,i}^{(c)} = \begin{cases} \sum_{u \in V} B_{t,i,u}^{(c)}, & c \in \{C_L, C_W, C_B\} \\ \sum_{u \in V} \sum_{\tau=t}^{t+t_f} B_{\tau,i,u}^{(c)}, & c = \mathcal{Q}. \end{cases}$$

Note that $A_{t,i}^{(c)}$ is a decentralised network criticality measure. To allow for comparative analysis, the values of $A_{t,i}^{(c)}$ are normalised with respect to t for each simulation and attribute using min-max feature scaling, generating $\hat{A}_{t,i}^{(c)} : \{T_i, S_i, V, C\} \rightarrow [0, 1]$. This creates a decentralised network estimate of criticality derived from dynamic nodewise measures.

Benchmarking Procedure

Herein is outlined the benchmarking procedure used to measure the relative accuracy of the dynamic distributed measures of criticality.

I compute the mean squared error (MSE) across time for each normalised, aggregated dynamic measure, $\bar{A}_{t,i}^{*(C_L)}$, $\bar{A}_{t,i}^{*(C_W)}$, $\bar{A}_{t,i}^{*(C_B)}$, versus the normalised aggregated queue length, $\bar{A}_{t,i}^{*(Q)}$, and denote it as

$$M_i^c = \text{MSE} \left(\bar{A}_i^{*(c)} \right), \quad c \in \{C_L, C_W, C_B\}.$$

Then the mean value across all simulations for each measure is found, therefore finding the overall accuracy to which each measure estimates the criticality of each node within the network, such that the final result is

$$\bar{M}^c = \frac{\sum_{i=0}^n M_i^c}{n}, \quad c \in \{C_L, C_W, C_B\}.$$

4.3 Results and Discussion

All analyses were completed on Google Colab Pro, a Jupyter notebook service that provides a Python 3 Google Compute Engine backend of an adaptable memory of up to 32GB RAM with 2 virtual CPUs, Intel(R) Xeon(R) @ 2.20GHz. The packages used were `networkx`, `random`, `numpy`, `matplotlib`, `pandas`, `math`, `json`, `scipy`, and `sklearn`.

The augmented SIS is applied to a scale-free $B(n, m)$ graph because this is the most representative graph topology of telecoms infrastructure networks [149].

The outputs are analysed according to the proposed validation procedure on a $B(n, m)$ network with $n = 15$ and $m = 3$. For both, infection rate $\beta = 0.9$, recovery rate $\gamma = 0.5$, and infection threshold $I = 3$. This is to simulate and compare two simple networks at and beyond capacity. Network G is visualised in Figure 4.1. Node size corresponds to magnitude of Q per node, which, for demonstrative purposes, has been assigned randomly.

Let us explore the output of simulation S_1 , initialised from the node highlighted in green in Figure 4.1. The values of $\bar{A}_{t,1}^{(c)}$ are presented in Table 4.1.

Time	Queue Length	Static			Dynamic		
		Local	Wehmuth	Bridging	Local	Wehmuth	Bridging
0	26	380	13.509	0.068	253	29.011	131.882
1	25	380	13.509	0.068	200	4.166	58.548
2	24	380	13.509	0.068	266	34.048	130.119
3	17	380	13.509	0.068	215	28.409	113.635
4	11	380	13.509	0.068	158	23.517	108.101
5	6	380	13.509	0.068	81	0	32.147

Table 4.1 Example aggregated attributes for a simulation seeded from node 3.

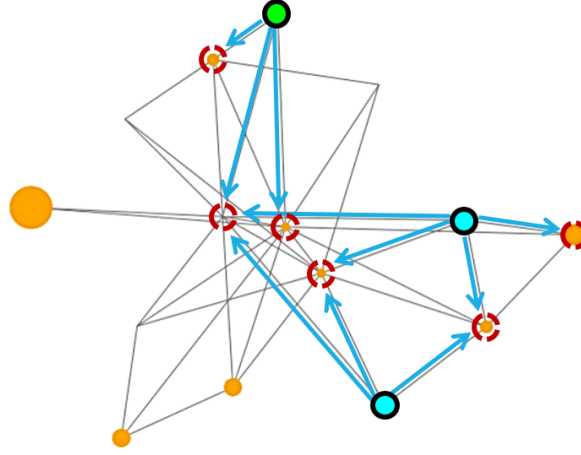


Fig. 4.1 A Barabasi-Albert preferential attachment network G with node count $n = 15$ and edge attachment number $m = 3$, with randomly weighted nodes, representing data packet queue lengths. The green node u_1 seeds for simulation S_1 , and blue u_2, u_3 seed for S_2, S_3 . Arrows and red rings are possible congestion candidates per timestep. All nodes have a corresponding simulation.

Measure	$\bar{M}^c$
Dynamic Wehmuth Centrality (C_W)	0.197
Dynamic Local Centrality (C_L)	0.102
Dynamic Bridging Centrality (C_B)	0.162

Table 4.2 Average mean squared error for normalised aggregated dynamic attributes, across all simulation instances.

Table 4.1 shows that by weighting measures with queue lengths, they more closely track the progression of data packets within the network, and since the aggregated queue length is forward looking, the correlation suggests that the dynamic criticality measures can detect risky nodes. The similarity between the queue length and the dynamic measure is seen in Figure 4.2, which uses normalised values, allowing for comparative inspection.

However, this is only information for a single simulation instance. It is still not obvious which measure more closely estimates future progression. I combine the information of mean squared error of measures from queue lengths for each simulation instance, obtaining the averaged mean squared error, $\bar{M}^c$ (Table 4.2).

Dynamic local centrality performs best, with $\bar{M}^{C_L} = 0.102$, followed by dynamic bridging centrality, with $\bar{M}^{C_B} = 0.162$, and dynamic Wehmuth centrality is the worst, with $\bar{M}^{C_W} = 0.197$. Speculatively, this may be a result of the model used to test their accuracy, since momentary rate at which a node obtains data packets is strongly determined by the number of packets held by the given node's neighbours, and dynamic local centrality directly counts this.

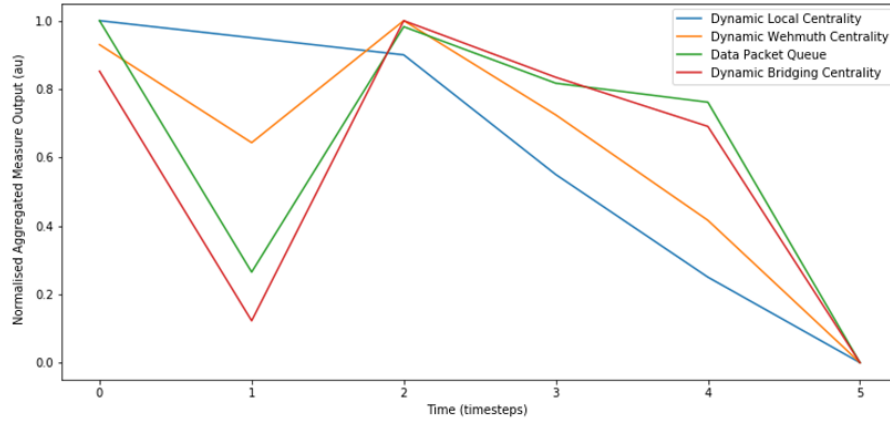


Fig. 4.2 An example plot showing the relationship between normalised aggregated measures and queue length from a single seeded simulation from some node.

Since all values are bounded within $[0, 1]$, one can see that each measure computes criticality within a network with between 80% and 90% accuracy within the contexts of this test, which is quite accurate.

4.4 Conclusions

Dynamic and distributed approaches had not previously been combined in such a manner as in this chapter. Each measure was tested within an augmented SIS scenario, and it was found that within this test, they performed with high accuracy. Dynamic local centrality most accurately predicts criticality within this test, though the reason for this is yet unclear.

No such measures which approach the problem of dynamically and distributedly predicting criticality of nodes in this manner have been previously developed before the work described in this chapter, and it is exciting that they have such high predictive accuracy, suggesting there is value in researching more dynamically obtained measures for predictive purposes. Such new measures are necessary to deal with increasing data traffic demands, where greater network requirements are suddenly imposed on an already at capacity system, as during the COVID-19 outbreak.

Applications of such measures include: integration within telecommunications systems to inform predictive maintenance; integration into autonomous vehicle networks to inform predictive routing to minimise traffic jams; supply networks, where actors are often only able to obtain primary or secondary connection information; and any system of dynamically communicating agents. With such measures in place, agents will be able to quickly and reliably establish their short term criticality, allowing for swift, inexpensive predictive action to ensure ongoing network function.

Robust statistical analysis will have to be performed to establish the true accuracy of the three measures, including: test repetition; alternative regression metrics; comparison of quality versus static measures and classic network measures; and multidimensional analysis.

4.5 Summary of Contributions

In this chapter, an environment to simulate the propagation of criticality through a telecom routing infrastructure was developed, based off an augmented SIS model. Upon this, a means to isolate criticality from a single node using dynamic network analysis was defined. I then defined a notion of true network criticality as a function of impacts over a given timeframe, formed a method to derive this timeframe, and then from the dynamic information imparted by this, determined the network's true criticality.

Three dynamic and information bounded measures were defined that estimate criticality of given nodes within a network. Their dynamic outputs were aggregated to create a dynamically defined measure of system criticality, which is computed in a decentralised manner. A means of measuring the accuracy of such a decentralised measure against true criticality was introduced.

This forms a two-layer multiplex network model, which can be used to define an analytics layer in telecoms infrastructural systems for determining criticality directly. Its structure means that minimal work is needed to apply this model for general systems where criticality can be approximated to propagate as an infectious disease.

Chapter 5

Decentralised Changepoint Detection for Intraday Telecom Data Traffic

5.1 Introduction

This chapter uses a dataset of intraday data packet demand for telecommunications traffic [148]. Behaviour within a day changes according to the time or task placed upon the telecommunications backbone network. The times of abrupt change in behaviour are **changepoints**, detected through changepoint detection algorithms. I show how to extract these changepoints systematically, quickly, with minimal computation cost, and in a decentralised way that makes use of the data concerning many nodes within a network. These methods can be incorporated into autonomous telecoms network control.

This chapter is structured as follows. In Section 5.2, the method for changepoint detection proposed in this chapter are outlined, describing first a mean curve based method in Subsection 5.2.1, then a more advanced, KDE based method in Subsection 5.2.2, the first shown as a base case to which the KDE method is compared. Both results and a discussion thereof are presented in Section 5.3 for each changepoint detection algorithm covered in the methods section. Conclusions to be drawn from this are presented in Section 5.4, and the chapter is finished with a summary of its contributions in Section 5.5.

5.2 Methods

To conduct changepoint analysis, changepoint detection methods are used that do not use predetermined numbers of partitions. These are binary segmentation [24], window-based detection [13], and the pruned exact linear time (PELT) methods [182].

I apply these methods onto a dataset of data demand over time as a function of packet generation within a telecommunications backbone network for each node in the network (Figure 5.1). This is an

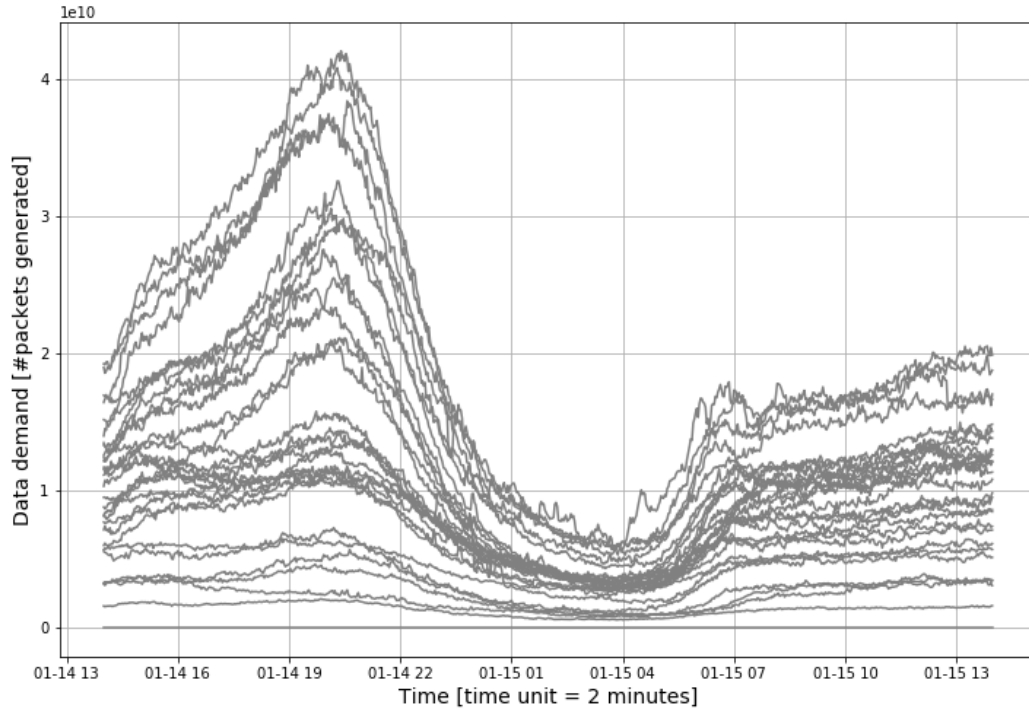


Fig. 5.1 Graph of all data demand signals for each node in a backbone telecommunications network over the course of a day.

ensemble of well-correlated time series, where behaviour of each node depends on the others, but is not deterministic. Given their similar behaviour, data from each node can be used as a sample of the population, so that the statistical properties of the set can be used to make predictions on typical node behaviour. The x-axes for all other graphs denote the x number of 2-minute increments past the datetime of 1-14-14-0.

The purpose of this is to find regime changes to inform any simulators how they should behave when trying to represent a given time of day. Only the statistical properties of the segment between two changepoints should be used for the same timeframe. The samples are aggregated in two ways to generate data to compare their relative performance. Given that this data is derived from a system of network agents, this is a decentralised method of changepoint detection.

5.2.1 Mean Curve Inspection

The first method takes the moving average of all signals (Figure 5.2). Possible changepoint detection methods and cost functions are then cycled over, the results of which are shown in Figure 5.3. Visual inspection suggests that PELT with the rbf cost function has the best performance.

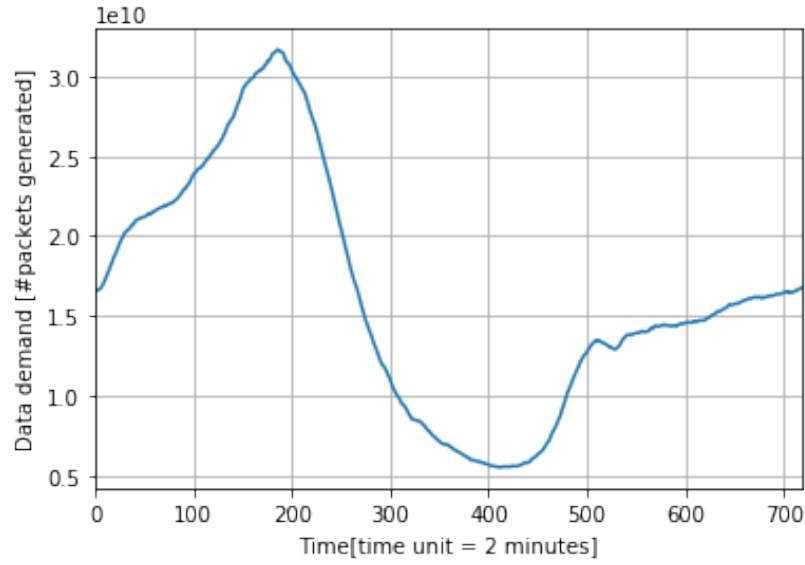


Fig. 5.2 Mean data demand curve for all signals for each node in a backbone telecommunications network over the course of a day.

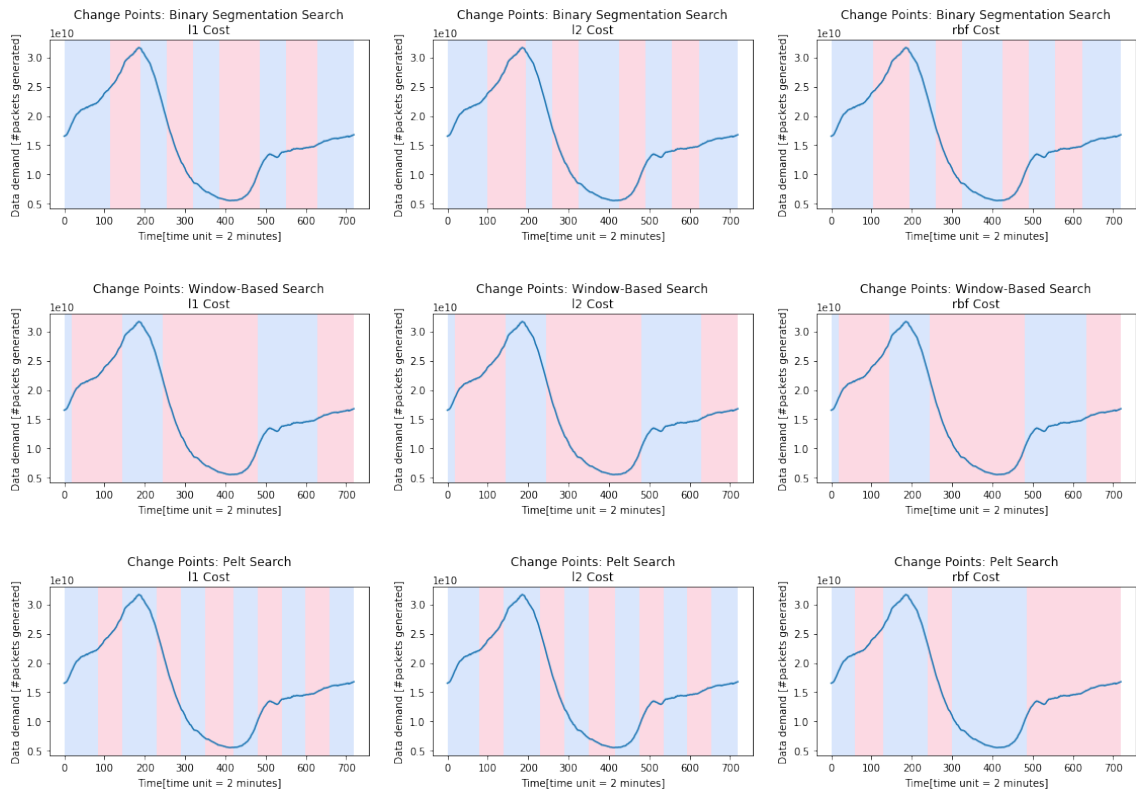


Fig. 5.3 Collection of mean curve changepoint detection outputs depending on method and cost function.

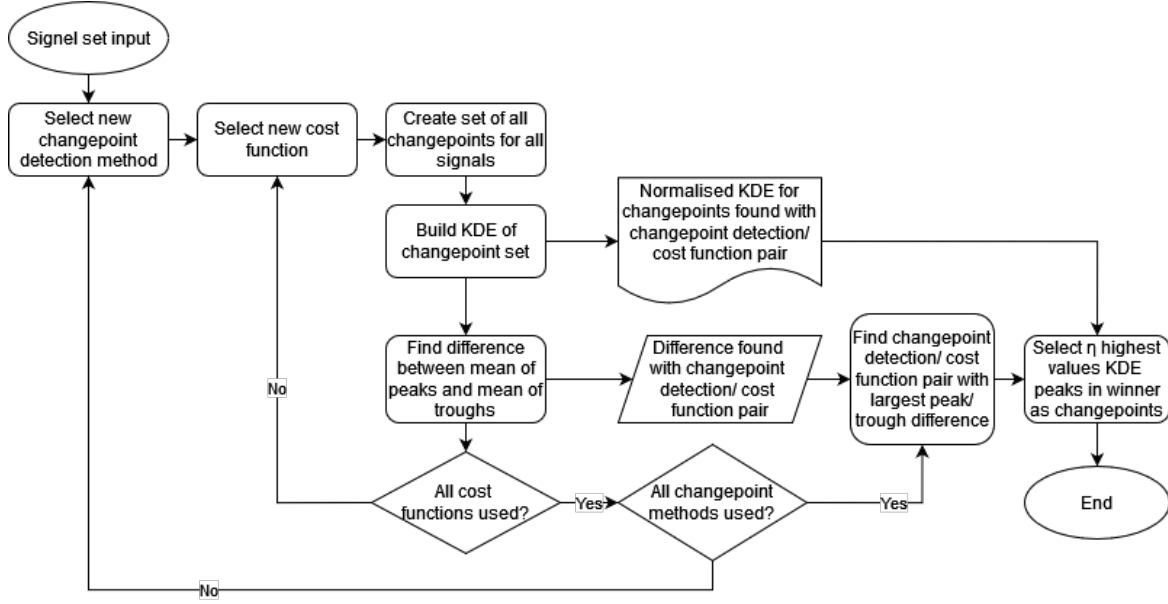


Fig. 5.4 Flowchart explaining the second changepoint detection procedure.

5.2.2 KDE Signal Decomposition

The second method is to apply all changepoint detection methods with all cost functions onto each signal within the sample (Figure 5.4). Then all breakpoints for all signals per method, per cost function, are bagged together for the purpose of performing kernel density estimation (KDE) on each set, assuming that each signal, associated with a given node, is an instance of the same distribution.

The purpose of this is to find where changepoints occur most often for a changepoint detection algorithm for a given cost function. The reasoning is that each signal, associated with a given node, is assumed to be an instance of the same distribution. This claim is supported by Figure 5.1, where it can be seen that all signals resemble each other under local deformation. The boundaries of these deformations would be the most desirable changepoint. The method/cost function combination that finds changepoints with the most confidence is the most accurate. This confidence is determined as function of the strength of the clustering of predicted changepoints for the sample.

This clustering is found via KDE, which is a non-parametric way to estimate the probability density function of a random variable. KDE smooths data by making inferences about the population, based on a finite data sample, in this case, the set of changepoints. I assume that these are univariate independently and identically distributed, and are drawn from some distribution, Y , using the probability density function (pdf), $f : Y \rightarrow [0, 1]$. The KDE of f , $\hat{f}_h : Y \rightarrow \mathbb{R}$ is defined as

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - y_i}{h}\right),$$

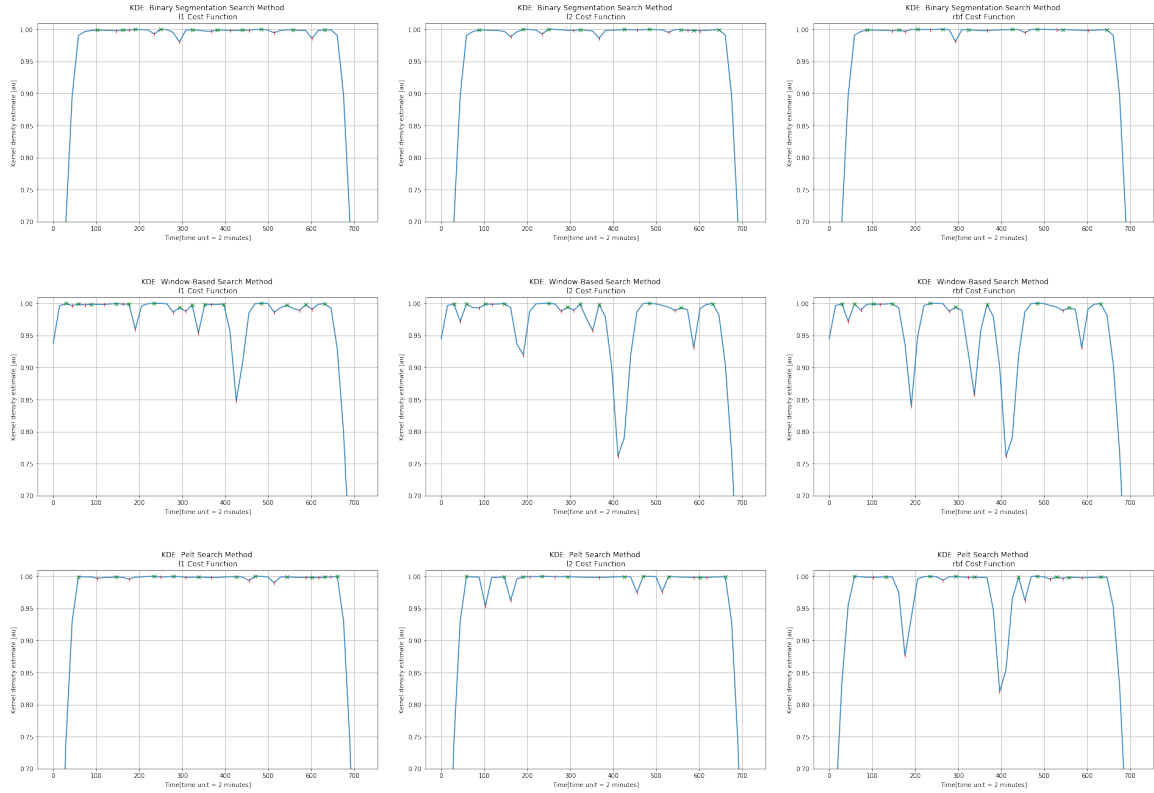


Fig. 5.5 Collection of changepoint distribution kernel density estimations.

where K is the Gaussian kernel function, and h is the bandwidth, in this case found for each method, cost function pair through searching the hyperparameter space through exhaustive grid search using the `GridSearchCV` function from the `sklearn` Python package on the `KernelDensity` function from the same package, with 20-fold cross validation.

Each KDE is generated for each method/cost function pair, and then normalised to $[0, 1]$, with peaks representing clusters of changepoints, and troughs representing cluster boundaries (Figure 5.5). The curve with the greatest confidence is found by taking the difference of the mean of the peaks and the mean of the troughs, and selecting the largest value. This is because higher peaks suggest stronger clustering, while lower troughs suggest wider boundaries. Together this gives a picture of a changepoint detection method that has both high precision, through the peaks, and low bias, through the troughs, and so the most accurate. The mean is used to ensure there is no bias towards larger numbers of changepoints. From that, the η highest valued changepoints are selected, where η is chosen according to visual inspection of the time series. The output of this is shown in (Figure 5.6).

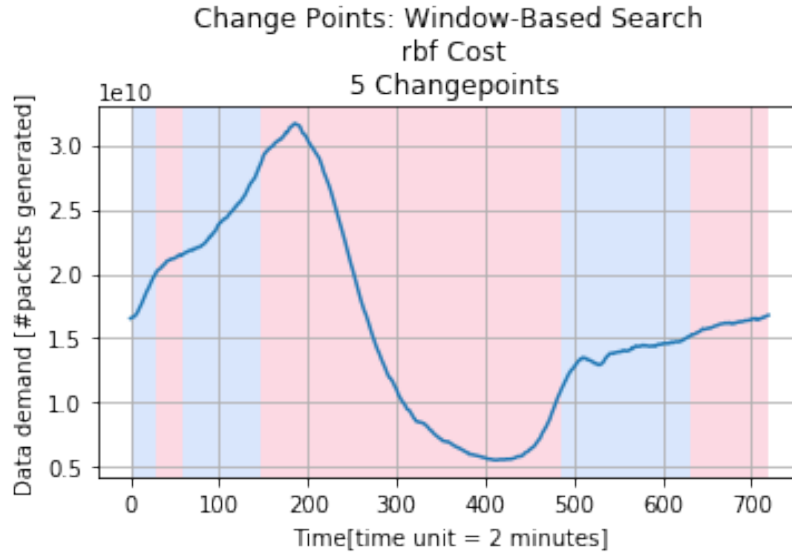


Fig. 5.6 The most accurate changepoints found with KDE for each changepoint detection algorithm and cost function, according to my problem construction. The most accurate method cost function combination was window-based search with kernelised mean change. $\eta = 5$ changepoints were selected. The mean curve has been overlaid for visual comparison.

5.3 Results and Discussion

Simulations were completed on an Armari Magnetar workstation, with a 3.5GHz Intel Xeon E5-1620 CPU, 64GB RAM, in Windows 10 Pro. Modelling and results were completed in Python 3 with the pandas, matplotlib, ruptures, numpy, datetime, scipy, and sklearn libraries.

5.3.1 Mean Curve Inspection

Figure 5.3 shows the outputs of the mean curve changepoint algorithms. The PELT method with rbf cost function partitions the mean curve approximately over a point where the rate of change stops decreasing and begins increasing, then where it begins to decrease again surrounding a peak, followed by approximately covering a region of constant rate of change, partitioning as rate of change starts to increase, then partitions at a sharp bend for a rough but constant rate of change region. Most important is the treatment of the final rough constant region as its own partition, which is not observed in other curves. This suggests this is the best method/cost function pair, as the constant slope suggests no regime change.

The PELT method with RBF cost function partitions the mean curve near inflection points of rate of change and yields the most accurate detector of changes in regime by encapsulating a common dynamic region. This is because it most closely matches points where the magnitude of rate of change is maximised, while letting similar continuous curves remain whole segments. The changepoints are 1-14-14-40, 1-14-18-50, 1-14-22-10, 1-15-6-0, 1-15-11-10, 1-15-14-0.

	Least absolute deviation	Least squared deviation	Kernelised mean change
Binary Segmentation	0.025	0.030	0.029
Window-Based	0.173	0.272	0.372
PELT	0.026	0.042	0.082

Table 5.1 Table of differences for all method cost function pairs, in au, rounded to 3 DP. The cell with the highest difference value, Kernelised mean change for window-based changepoint detection, is highlighted in green.

5.3.2 KDE Signal Decomposition

Figure 5.5 shows the outputs of the changepoint kernel density estimation operation. The KDE with the greatest confidence was found by taking the sum of peaks minus the sum of troughs and choosing the largest value. This was found to be window-based changepoint detection using kernelised mean change (Table 5.1).

After taking the $\eta = 5$ highest valued peaks in the KDE using window-based search and kernelised mean change, the set of changepoints shown in Figure 5.6 are obtained, overlaid on the mean curve. These are 1-14-14-58, 1-14-15-57, 1-14-18-53, 1-15-6-9, 1-15-11-3. Limitations of this method are seen as the first three changepoints are very close, and the soft corner just above 1-14-20-40, 200 points in, is not noted as a changepoint. Since the final changepoint cuts through a region of constant rate of change, it may be incorrect, however, given this is a data driven network method, results suggest determining regime change with rate of change may be an inappropriate method.

5.4 Conclusions

This chapter proposes methods of decentralised intraday changepoint detection in telecoms infrastructures, comparing various changepoint detection methodologies and proposing a kernel density estimate based method of data aggregation. It applies this to a UK backbone telecom dataset supplied by British Telecom (BT).

With these changepoint detection methodologies, any set of signal samples can be fed into the method to be used to inform a data packet movement simulator. As more data is obtained, this can also be used in the future to create a general model of intraday and interday data packet behaviour within a network, thus improving the quality of tools for mid-term network prediction.

The value of the second method is that it is non-parametric, uses the whole set of data in a way that incorporates the structure of the network signals rather than taking the mean curve, and determines the best changepoints detection method and cost function pair systematically. The principles governing this schema — kernel density analysis, peak analysis, changepoint detection — underpin work in later chapters, and can be used in practice to inform timings in update schedules of agent perceptions of the network.

This study was completed prior to the COVID-19 pandemic, and findings on the given dataset may be outdated, given modern hybrid work culture and widespread usage of distributed digital technologies such as blockchain or AI, though this is to be seen. An updated study would be of great value to give data on how things have changed since.

5.5 Summary of Contributions

In this chapter, I defined a pair of decentralised methods of changepoint detection and applied them to a telecom backbone infrastructural dataset. Findings can be used to propose a characteristic day of dynamics, which can inform both simulation model and control update timescales. These methods can be applied to general decentralised infrastructural analytics for scalable autonomous control. Findings for this given dataset closely track the work and rest periods of human activity during a characteristic day for telecoms usage in the UK.

The methods and findings in this chapter contribute to solving the problem of irregular changepoints for epochs in telecoms, and more generally infrastructural network systems. It provides tools to inform the timescale over which certain analytics should be conducted, and demonstrates the effectiveness of kernel density estimate changepoint detection methods as support tools to remove human bias. This also forms an exploratory analysis into methods of consolidating multiple network derived time series datasets for decentralised analytics, as well as the nature of epoch change in these systems.

Chapter 6

Measuring Value of Information for Dynamic Decentralised Criticality Computation in a Telecom Dynamics Model

6.1 Introduction

The use of centralised network monitoring and criticality computation means central computational resources need information on the whole network, requiring live and dynamic node topology and attribute data to respond to behavioural shifts. This increases computational times, slowing them down and reducing relevance of the results, promoting critical events if too much data is transferred.

The more accurate and more efficiently computed an IB-CM is, the more valuable the information used in its calculation is, where the *value of information* is a multivariate measure composed of the accuracy and efficiency of a given IB-CM under some information provision.

To investigate in depth if decentralised network monitoring can improve the value of information in a realistic context, this chapter builds upon the decentralised monitoring defined in Chapter 4 and uses the knowledge that network dynamics have variable paradigms, established in Chapter 5. It does so by investigating the effect of differentiating analytical update times on accuracy for analytical agents, through defining a network intercommunication layer. The methods to achieve this are presented in Section 6.2. I first define the specific simulation model used in this study in Subsection 6.2.1. I then outline a method to assess the accuracy and computational efficiency of IB-CMs that change with time, with respect to a novel benchmark estimate of dynamic criticality under different CPs, in Subsection 6.2.2. This generates the value of information. These IB-CMs are designed for homogeneous flow networks. A prototype is presented that uses classic centrality measures as a stand-in for CMs and IB-CMs, with these defined in Subsection 6.2.3. The results

of applying this to a real network topology on the use case of a telecom simulation model are then presented in Section 6.3, and conclusions derived from this are in Section 6.4. All material in this chapter has been published in Proselkov et al. [255] except where indicated.

6.2 Methods

6.2.1 Telecom Simulation Model

A discrete information packet exchange simulation is run using the Python package, `anx` [205], with short range dependence, meaning random nodes generate data packets independently according to a Poisson distribution with random destinations [301]. As this is an information flow network, a timestep is how long it takes for information to traverse one edge. Packets traverse the network, stored and routed along nodes on the way to their destination, where they are removed from the system. Nodes and edges each have a fixed capacity which gets filled up over time since it takes time to process packets at nodes and transmit them between nodes, with processing and transmission time as fixed model input parameters. This process is terminated after either a fixed number of timesteps or until the network is too congested to function. The nodes each have a backlog capacity of ϕ . The simulation produces a time series over Υ of each node's queued up data packet backlog, where the size of the queue held by node $u \in V$ at time $t \in \Upsilon$ is ϕ_u^t .

I then examine how nodes would behave if they were receiving and processing network data in real time in an agent-based simulation using the Python package, `mesa` [178], with an independent agent situated at each node. Depending on the monitoring data CP, which determines how up-to-date information is (*currentness*), different nodes get different information regarding others depending on their position in the network. Three CPs are investigated, named *instant*, *constant*, and *periodic*.

6.2.2 True Criticality

Over this model, I first define a measure of true criticality, obtainable post-hoc, using data that is neither limited by the imperceptibility of the future nor communication constraints. Linear functions of the total queue sizes of the whole network are taken using $\Phi^t = \sum_{u \in V} \phi_u^t$. A time range is found for which there is sufficient confidence that all network states are sufficiently dependent on each other.

Ideal Time Horizon

This is a moving time window, bisected by the present timestep, where the window's start sufficiently influenced all timesteps up to the present, and the present will sufficiently influence all timesteps up to the window's end. With it can be found, for all timesteps t , how far into the future sufficiently captures both the current network state and its influence. I iterate over a fixed number, h_{test} , of time horizon windows, h_i , less than half the simulation length, t_{max} , where $h_i = it_{\text{max}} / (2h_{\text{test}})$, and take

moving averages over Φ^t for each width h_i , so

$$\text{MA}_{\Phi;h_i}^t = \begin{cases} \sum_{t-i}^t \Phi^t / i, & t \geq i; \\ \emptyset, & t < i, \end{cases}$$

and $\text{MA}_{\Phi;h_i}$ is the time series made up by $\text{MA}_{\Phi;h_i}^t$. Then for all t such that $\text{MA}_{\Phi;h_i}^t$ exists, I take the absolute difference between $\text{MA}_{\Phi;h_i}^t$ and Φ^t , such that

$$\text{MAD}_i^t = \begin{cases} |\text{MA}_{\Phi;h_i}^t - \Phi^t|, & t \geq i; \\ \emptyset, & t < i, \end{cases}$$

and get the sum of absolute differences, $\text{SAD}_i = \sum_t \text{MAD}_i^t$. Normalised, this is

$$\text{SAD}_i^* = \frac{\text{SAD}_i}{\max_{i=1}^{h_{\text{test}}} \text{SAD}_i}.$$

Iterating through SAD_i^* in ascending i returns $g_i = h_{\text{test}}(\text{SAD}_{i+1}^* - \text{SAD}_i^*)$. The ideal time horizon is where the relative gain in error by a wider window is large enough to suggest that all smaller window sizes cover regions with significant influence over each other. Beyond that, since error gain slows down, one cannot confidently claim events are the direct consequence of the current time. This confidence, the *validation threshold*, is an independent parameter, h_{val} , with which is defined the ideal time horizon, h for the first i where one of the following conditions is fulfilled, where if the last case is reached, more windows must be tested, or the confidence threshold must be increased:

$$h = \begin{cases} \lfloor i/2 \rfloor, & g_i \leq c; \\ \lfloor (i-1)/2 \rfloor, & g_i < 0; \\ \emptyset, & i = h_{\text{test}}. \end{cases}$$

Benchmark Criticality

To test the accuracy of information limited measures, a ground truth is needed. Within the context of the simulation, the ground truth is readily accessible as the state of the whole network for the length of the simulation run. The agents approximating this with limited information do so over a subset of the network and over a window of the simulation timesteps. I define criticality as the intensity of change in network operation induced by any network state changes. Following that h is the timescale over which these changes happen, the true, or benchmark criticality is defined. This uses the entire network state over the full timescale as input, such that there is no information limitation, enabling us to investigate information limitation. As it uses the entire timescale, it would be impossible to implement live.

Dependencies are sufficiently large for all timesteps at most h timesteps far from each other, so impacts occur over a meaningful timescale of h . Impact at time t is the change over h timesteps across t scaled by the built-up queues at time t , since a heavily used system has more to lose than an underused one. A moving average with window width h , $\text{MA}_{\Phi;h}$, is obtained, producing a time series of scaled differences across a time horizon,

$$\text{THD}_t = \text{MA}_{\Phi;h}^t \left(\text{MA}_{\Phi;h}^{t+h} - \text{MA}_{\Phi;h}^{t-h+1} \right).$$

This is normalised to $[0, 1]$, creating the *criticality benchmark*,

$$\text{THD}_t^* = (\text{THD}_t - \min_t \text{THD}_t) / (\max_t \text{THD}_t - \min_t \text{THD}_t).$$

Communication Paradigms

If there exists a path p_{uv} , u gets information about ϕ_v^t . The queue at time t that the agent at u believes v has is the *perceived queue*, $q_{uv}^t \in Q$. If $u = v$ this is q_u^t . Centrality calculation takes time, so the time from transmission to output is always greater than transmission to receipt between nodes, thus nodes must compute centralities at a lower frequency than the CP dictates to avoid losing currentness. I call this period between calculations the *monitor interval*, μ .

Instant communication (IC) is a simplification, assuming monitoring data is transferred instantly, where for all $u, v \in V$, $q_{uv}^t = q_v^t$. This is a base case, only achievable if monitoring data transfer became so fast as to be insignificant.

Constant communication (CC) has nodes declare their queues every timestep. This data traverses the network normally, because this declaration is a low-bandwidth operation. For a shortest path (geodesic) of length n between u and v , information from v takes n timesteps to reach u , so u perceives v late by n timesteps, so $q_{uv}^t = q_v^{t-n}$. Distant nodes thus give less accurate, valuable or relevant information.

Periodic communication (PC) has nodes declare their queues and perceived queues at the same frequency as they calculate their centrality. This corresponds to some aggregation of the functions within higher order control functions that use centralities as inputs. For monitor interval μ and u and v with length n geodesic, $q_{uv}^t = q_v^{t-\mu n}$, as each queue pass is μ timesteps.

With each CP, nodes receive perceived queues of others in the network. These values are used to inform dynamic, queue-dependent CEs, which are calculated with the adjacency matrix. They are therefore calculated with respect to edge weight rather than node weight. The first step of calculating this is redefining the graph as directed. Then, for a node u , for all $v \in \Gamma_1(u)$, the queue weight of edge (uv) at time t is

$$(uv)_q^t = \frac{q_u^t}{|\Gamma_1(u)|}, \quad (6.1)$$

because larger neighbourhoods give nodes more chances to emit data packets and distribute load among them, and it accounts for the respective queues of node pairs since for all (uv) there exists a (vu) produced under the same rules.

The following subsection describes the data analysis carried out with the data provided according to each CP.

6.2.3 Centrality Measure CEs

According to each of the above CPs, the data delivered to each agent situated at a node is used to compute centrality measures over time as proxies and estimates of criticality. In this initial study, standard centrality measures are used, with weighted and bounded extensions. The unweighted measures only take topological data, whereas weighted measures adjust their outputs according to the perceived queues for each node. Unbounded, or *sociocentric*, measures take information from the whole network and stand in for CMs, while bounded, or *egocentric* measures take information from a limited region around a given node and stand in for IB-CMs.

Weighted Degree Centrality counts each node as many times as their perceived queue lengths. It is dynamic and defined as $C_D^W(u)_t = \sum_{v \in \Gamma_1(u)} q_{uv}^t$.

Weighted Sociocentric Betweenness Centrality uses weighted shortest path variables. Using Eqn. (6.1) these are created as

$$\begin{aligned}\omega_{vw}^t &= \sum_{p_{vw} \in P_{vw}^g} \sum_{\substack{(st) \in p_{vw} \\ (st) \in E}} (st)_q^t; \\ \omega_{vw|u}^t &= \sum_{u \in p_{vw} \in P_{vw}^g} \sum_{\substack{(st) \in p_{vw} \\ (st) \in E}} (st)_q^t,\end{aligned}\tag{6.2}$$

giving weighted sociocentric betweenness centrality as

$$C_B^{WS}(u)_t = \sum_{\substack{v, w \in V \\ u \neq v \neq w}} \frac{\omega_{vw|u}^t}{\omega_{vw}^t}.\tag{6.3}$$

Weighted Egocentric Betweenness Centrality takes Eqn. (6.3) but over $H_i(u)$, such that

$$C_B^{WE(i)}(u)_t = \sum_{\substack{v, w \in H_i(u) \\ u \neq v \neq w}} \frac{\omega_{vw|u}^t}{\omega_{vw}^t}.\tag{6.4}$$

Weighted Sociocentric Eigencentrality uses the directed network with edge weights as defined by Eqn. (6.1). The adjacency matrix becomes dynamic and temporally dependent, such that for $A_G^t \in M_n(\mathbb{Z}^+)$ for the graph, G ,

$$A_G^t = (a_{i,j}) = \begin{cases} (u_i u_j)_q^t, & (i, j) \in E \\ 0, & i = j. \end{cases}\tag{6.5}$$

A_G in Eqn. (3.2) is then replaced by A_G^t in Eqn. (6.5) to create $C_E^{WS}(u)_t$.

Weighted Egocentric Eigencentrality uses A_G^t from Eqn. (6.5). For node u_j it is over $H_t(u_j)$, not G , creating

$$C_E^{WE(i)}(u_j)_t = \left(A_{H_t(u)}^t \mathbf{x} \right)_j = \left(\lambda_{H_t(u)}^t \mathbf{x} \right)_j.$$

These CEs are used as proxies for criticality measures. To compute the value of information as processed through each measure, I now outline a validation method.

Comparison Accuracy Function

The measures above must be validated as correctly approximating dynamic criticality within the network. To do this, the similarity of the CEs to the criticality benchmark must be quantified. This is not trivial, as the benchmark aggregates information over the whole network and all time, while the CEs represent the output of a limited spatio-temporal information subset, thus they are not measured according to the same units. A validation function to transform CE outputs must be defined. This validation function must determine at any timestep the similarity of the CE to a benchmark and its period of relevance. Criticality measures the impact of failure, which must be defined, and for how long the effects of some action can be said to have been caused by a previous one.

Information from CEs must be aggregated. To make it agnostic to the relative importance of each node and isolate such effects to the CEs, mean averaging is used. For a CE, $C(u)_t$, the network mean is calculated as $\bar{C}_t = \sum_u C(u)_t$, normalised to get $\bar{C}_t^*$, which can be compared to THD_t^* . Let

$$\mathcal{T} = \{ \tau \in \Upsilon : \tau = k\mu, k \in \mathbb{Z}^+ \}.$$

The error from the benchmark is thus $\text{Err}_t = \bar{C}_t^* - \text{THD}_t^*$, and the root mean squared error is

$$\text{RMSE} = \sqrt{\frac{\sum_{t \in \mathcal{T}} \text{Err}_t^2}{|\mathcal{T}|}}.$$

RMSE is used as it provides a measure of error that is in the same unit as the target variable, simplifying interpretation and comparison. Additionally, RMSE penalises larger errors, making it sensitive to outliers. The lowest RMSE gives the most accurate measure since it most closely follows the benchmark criticality.

Value of Information

For a given telecom model simulation run, for a given CP, data is generated by simulating what information nodes would be provided in a live environment. For a given CE and CP, at each timestep per node and associated agent, the computation time $\text{Comp}(u, t, \text{CP}, \text{CE})$ is also recorded, averaging

over all timesteps and all nodes to get

$$\langle \text{Comp}_{\text{CP}}^{\text{CE}} \rangle = \frac{\sum_{u \in V, t \in \Upsilon} \text{Comp}(u, t, \text{CP}, \text{CE})}{|V||\Upsilon|}. \quad (6.6)$$

From (6.6), I derive the accuracy of calculation and the timeliness, respectively defined as

$$\psi_{\text{CP}}^{\text{CE}} = \frac{1}{\text{RMSE}_{\text{CP}}^{\text{CE}}}; \quad \vartheta_{\text{CP}}^{\text{CE}} = \frac{1}{\langle \text{Comp}_{\text{CP}}^{\text{CE}} \rangle}. \quad (6.7)$$

Together, (6.7) gives a spacial and temporal measures of how valuable information is for a given CE. Normalise each gives

$$\psi_{\text{CP}}^{\text{CE}*} = \frac{\psi_{\text{CP}}^{\text{CE}} - \min \psi_{\text{CP}}^{\text{CE}}}{\max (\psi_{\text{CP}}^{\text{CE}} - \min \psi_{\text{CP}}^{\text{CE}})}; \quad \vartheta_{\text{CP}}^{\text{CE}*} = \frac{\vartheta_{\text{CP}}^{\text{CE}} - \min \vartheta_{\text{CP}}^{\text{CE}}}{\max (\vartheta_{\text{CP}}^{\text{CE}} - \min \vartheta_{\text{CP}}^{\text{CE}})}. \quad (6.8)$$

Multiplying these together gives a variable that grows with reduced error and increased time efficiency, giving a spatiotemporal measure of *value of information* from a CP for a CE. Respectively normalising returns

$$v_{\text{CP}}^{\text{CE}} = \psi_{\text{CP}}^{\text{CE}*} \vartheta_{\text{CP}}^{\text{CE}*}; \quad v_{\text{CP}}^{\text{CE}} = \frac{v_{\text{CP}}^{\text{CE}}}{\max v_{\text{CP}}^{\text{CE}}} \quad (6.9)$$

since $\min v_{\text{CP}}^{\text{CE}} = 0$. With this, I can assess how useful information provision of a certain variety is under a given method of interpretation, and even create

$$\langle v_{\text{CP}} \rangle = \frac{\sum_{c \in \text{CE}} v_{\text{CP}}^{\text{CE}*}}{|\{c \in \text{CE}\}|}; \quad \langle v_{\text{CE}} \rangle = \frac{\sum_{c \in \text{CP}} v_{\text{CP}}^{\text{CE}*}}{|\{c \in \text{CP}\}|}; \quad \langle v \rangle = \frac{\sum_{c_1 \in \text{CE}, c_2 \in \text{CP}} v_{\text{CP}}^{\text{CE}*}}{|\{(c_1, c_2) \in \text{CE} \times \text{CP}\}|}$$

to measure the value of information for each CP, each CE, and across all tested formulations over the telecom simulation model.

6.3 Results and Discussion

I compared simulation results of instant, constant, and periodic CPs for the accuracy of decentralised, dynamic, and information bounded centrality measures for estimating criticality. Three simulations, one for each CP, using the real topology of the UK outer backbone infrastructure network for a UK telecoms service provider (Fig. 6.1). Simulation inputs are in Table 6.1. All analyses were completed on Google Colab Pro, a Jupyter notebook service that provides a Python 3 Google Compute Engine backend of an adaptable memory of up to 32GB RAM with 2 virtual CPUs, Intel(R) Xeon(R) @ 2.20GHz. 5000 timesteps are skipped to avoid degeneracy since simulations are initialised on empty networks.

Fig. 6.2 shows plots of all centralities and the criticality benchmark, NTHD_t , for each CP, filtered using a first order Savitzky-Golay filter. This graph shows substantial difference between outputs for

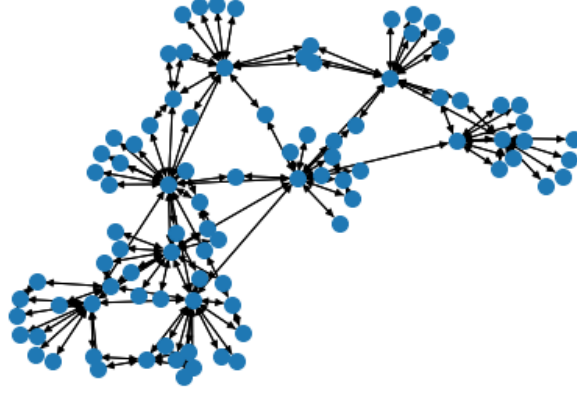


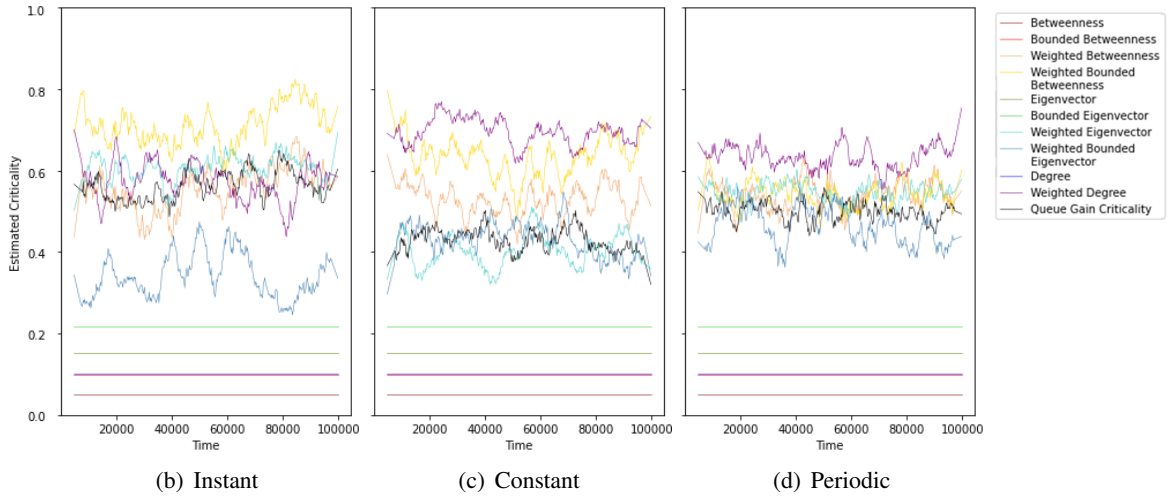
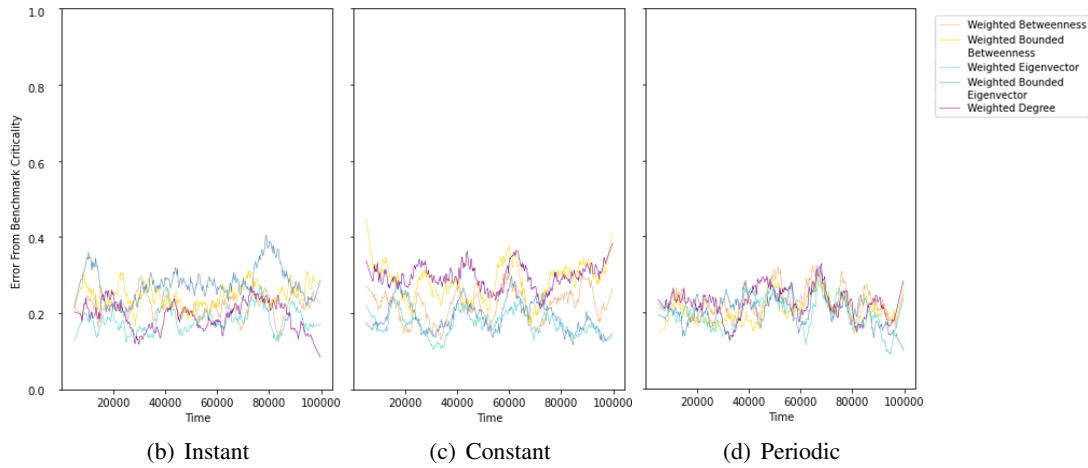
Fig. 6.1 Outer backbone UK infrastructure network for a large UK service provider

Table 6.1 Simulation Inputs

Runtime $t_{\max}$ (hundredths of a second)	100000 timesteps
Monitor Interval μ	250 timesteps
Packet generation rate	19 packets/timestep
Processing delay	13 timesteps
Queue check time	1 timestep
Transmission time	30 timesteps
Node capacity ϕ	128 packets
Link capacity	1024 packets
Information visibility boundary	2 hops
Validation threshold c	0.1 relative difference
Window widths tested h_{test}	20 windows
Ideal time horizon h	10000 timesteps

weighted and unweighted measures across all CPs, but further analysis will show similar accuracy. The absolute error, $|\text{Err}_t|$, from NTHD_t was calculated, the results plotted in Fig. 6.3. These plots are only for weighted measures, since error from static values is a trivial transformation of the criticality benchmark. This shows the relative accuracy of each curve, showing similar accuracy between bounded and unbounded measures. Periodic CP readouts are more closely clustered in terms of accuracy. Error plots are filtered using a first order Savitzky-Golay filter.

Table 6.2 shows normalised accuracy, $\psi_{\text{CP}}^{\text{CE}}$, towards the criticality benchmark for each CE and each CP. Weighted, dynamic CEs largely performed much better than their static counterparts. Boundedness minimally impacted accuracy. Typically, CEs were most accurate for constant CP, followed by periodic then instant CPs. Of the weighted bounded CEs, in periodic CP betweenness was best with large variation between CPs; in constant CPs eigencentrality was best; and in instant CP, degree, and much better than periodic and constant, which show wide difference. This suggests different CEs can be used for different CPs. The best average weighted bounded CE is $\langle \psi_E^{\text{CE}} \rangle = 0.6396$, and worst $\langle \psi_B^{\text{CE}} \rangle = 0.5392$. Despite this, the best weighted bounded performance for PC was betweenness, with

Fig. 6.2 CEs and NTHD_ts for simulations of each CP.Fig. 6.3 Error for Weighted CEs and NTHD_ts for simulations of each CP.

0.7265, for CC was eigencentality, with 0.8009, and for IC, degree, with 0.8131. Degree also has the widest range at 0.4479 and worst performance of the weighted bounded CEs, with 0.3652 for CP.

Boundedness and the time horizon are spacial and temporal efforts to increase relevancy of a given calculation. A sufficiently small information boundary also reduces computational complexity, allowing calculations to take place within the relevant period. In application, the monitor interval should be bounded above by the relevant period, and is typically bounded below by the computation time. This motivates analysing computation time of each measure under each CP.

Fig. 6.4 shows computation time plots. Limiting information most affects weighted betweenness, which unbounded can take over 0.07 seconds, but bounded may be less than 0.01, close to weighted bounded eigencentality. Instant and constant computation times are similar for all measures, though instant CP shows variability and intermittent spikes during network congestion, where queues grow due

ψ_{CP}^{*CE}	Instant	Constant	Periodic
Betweenness	0.0000	0.1969	0.0733
Bnd'd. Betweenness	0.0600	0.2987	0.1482
Wtd. Betweenness	0.6792	0.6446	0.5982
Wtd. Bnd'd. Betweenness	0.5108	0.3802	0.7265
Eigencentrality	0.1428	0.4383	0.2496
Bnd'd. Eigencentrality	0.2662	0.6543	0.4021
Wtd. Eigencentrality	0.8839	1.0000	0.7830
Wtd. Bnd'd. Eigencentrality	0.4182	0.8009	0.6998
Degree	0.0658	0.3065	0.1556
Wtd. Degree	0.8131	0.3652	0.6304

Table 6.2 Normalised Accuracy, for each CP and CE. Red is the least accuracy, white is the 50% percentile, blue is most accuracy.

to build-up exceeding processing speed in certain regions. Periodic CP was uniformly faster, which, since it places a lighter memory load through lower frequency, may be an artefact of computational stress on the computer during simulation. Dynamicity increases computation time for complex measures, but minimally impacts degree centrality, computed nearly instantly. Means of timeliness, the inverse of calculation time, for each CE and CP are shown in Table 6.3.

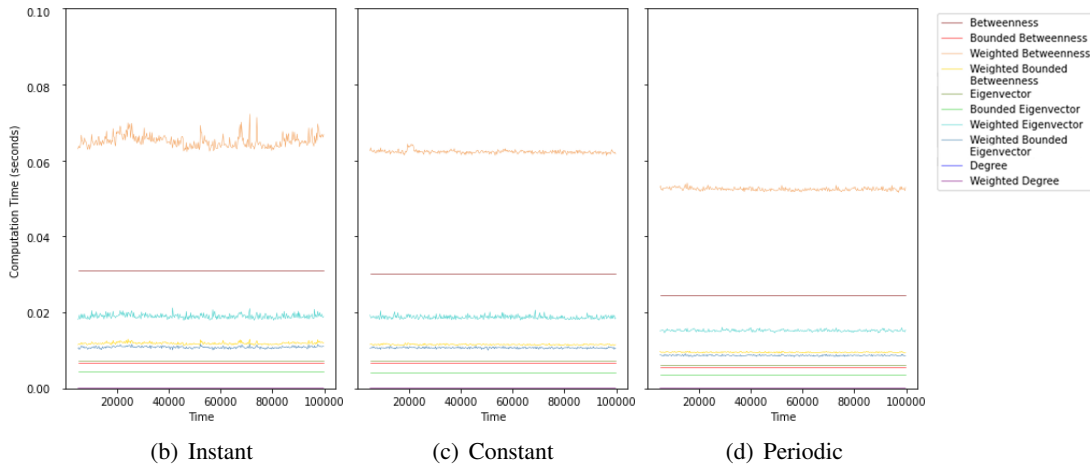


Fig. 6.4 Computation time for all CEs for simulations of each CP, measured in seconds.

Joining timeliness and accuracy together gives the value of information measure, v_{CP}^{*CE} , shown in Table 6.4. Information has little value for unbounded betweenness measures, as the product of their error and computation time is high, relative to others. This suggests routing from distant nodes has little impact for short term criticality, as defined by my ideal time horizon. Degree and weighted degree value information highly, the latter with greater value. This suggests criticality, in this context, spreads like a contact process and is best measured through the states of nearby nodes.

ϑ_{CP}^{*CE}	Instant	Constant	Periodic
Betweenness	0.0001	0.0001	0.0001
Bnd'd. Betweenness	0.0007	0.0007	0.0009
Wtd. Betweenness	0.0000	0.0000	0.0000
Wtd. Bnd'd. Betweenness	0.0004	0.0004	0.0005
Eigencentrality	0.0006	0.0006	0.0008
Bnd'd. Eigencentrality	0.0011	0.0011	0.0014
Wtd. Eigencentrality	0.0002	0.0002	0.0003
Wtd. Bnd'd. Eigencentrality	0.0004	0.0004	0.0005
Degree	0.3992	0.4761	1.0000
Wtd. Degree	0.2835	0.3760	0.7543

Table 6.3 Normalised timeliness, for all CEs for simulations of each CP. Computation time, from which this is derived, is measured in seconds. Red is worst, white is the 50% percentile, blue is most timely.

Looking at $\langle v_{CE} \rangle$ shows that generally, betweenness uses information poorly, however bounding it to a limited range, the value of information increases dramatically, and weighting it does so further. Eigencentrality performance is middling, improved by bounding but less clearly by weighting. Degree centrality uses information most effectively across all CPs, and weighting doubles its effectiveness. For $\langle v_{CP} \rangle$, information is of most value under a periodic CP, most reflective of reality, suggesting diminishing returns in reducing information latency versus improving handling of limited information. Finally, I observe that the value of information across all CEs and CPs is very low.

v_{CP}^{*CE}	Instant	Constant	Periodic	$\langle v_{CE} \rangle$
Betweenness	0.0000	0.0000	0.0000	0.0000
Bnd'd. Betweenness	0.0001	0.0004	0.0003	0.0003
Wtd. Betweenness	0.0000	0.0000	0.0000	0.0000
Wtd. Bnd'd. Betweenness	0.0004	0.0003	0.0007	0.0005
Eigencentrality	0.0002	0.0006	0.0004	0.0004
Bnd'd. Eigencentrality	0.0006	0.0016	0.0012	0.0011
Wtd. Eigencentrality	0.0004	0.0004	0.0004	0.0004
Wtd. Bnd'd. Eigencentrality	0.0004	0.0007	0.0008	0.0006
Degree	0.0553	0.3069	0.3273	0.2298
Wtd. Degree	0.4849	0.2888	1.0000	0.5912
$\langle v_{CP} \rangle$	0.0542	0.0600	0.1331	$\langle v \rangle$ 0.0824

Table 6.4 Normalised value of information, for all CEs for simulations of each CP. This is the product of timeliness and accuracy. Red is worst, white is the 50% percentile, blue is most valuable. Mean values of information over CEs and CPs are printed in the far right and bottom columns, each with their own colour scaling.

6.4 Conclusions

This chapter investigated network criticality, communication paradigms, and existing IB-CMs. A model of simulated decentralised online network measurement under three CPs was introduced. The IB-CMs used in my analysis (actually CEs) were then defined. The validation method that determines the error and from there, value of the information was then outlined. Simulations for the various CEs under different CPs and their results were then detailed, showing the viability of information bounded and dynamic criticality estimation. Together, this research provides a framework to develop more advanced CEs and gives insights into the relationship of measurement and provided information. Bounding information visibility is a viable method for scalable measures that preserves accuracy while speeding up calculation.

Each CE was shown to behave similarly. This holds true for error curves too (Fig. 6.3). In fact, error shows to be more constant for the periodic CP, suggesting an advantage in terms of control for that CP.

Timeliness was found to have similar ordering among CEs between CPs, decreasing in variance and magnitude from instant to constant to periodic communication. This may support the argument for decentralised computation, since less information frequency improves computation speed in singular agents. Alternatively, this may be an artefact of simulating decentralised agents with a central computer. Analysis of accuracy shows CE high performance of bounded dynamic (weighted) measures, sometimes outperforming static or unbounded measures.

Unifying them to give value of information clarifies things. Dynamic degree measures are best, balancing accuracy and timeliness. The best results were on periodic paradigms, showing that maximally simple though flexible metrics are most effective and are best in information delayed environments. Thus, extrapolating to more general systems, in cases where information is slow to arrive and requires processing, the most valuable approach is quick analysis with frequent updates.

6.5 Summary of Contributions

In this chapter, I defined a notion of empirically derived dynamic benchmark criticality, and a collection of dynamically changing approximations of criticality using limited range weighted centrality measures. I further explained several communication paradigms that describe proposed communication dynamics of live decentralised criticality approximating computational agents. For each criticality estimate over each communication paradigm, a decentralised dynamics measure of criticality was derived, and the value of the information used to obtain it was measured as a function of the time taken to compute it and its accuracy.

Through the application of an intercommunicative layer over a data packet flow layer, and conducting analytics over information in the intercommunications layer, in this chapter I define a three-layer simulation model of live, information bounded network analytics for a telecoms system, which can be repurposed for study of other network topologies. It was applied to the BT outer

backbone UK infrastructure network, and suggests that if a live, scalable, decentralised analytics architecture were to be incorporated into it, the most cost-effective analytics methods, regarding information cost, should minimise complexity.

Beyond the telecom case, this analytic framework will be applicable to other systems with dynamic flow and independent cognitive agents, such as business networks, mail networks, river networks, and more, each a critical support network for any manufacturing system.

Chapter 7

Financial Ripple Effects in Complex Adaptive Supply Networks: An Agent-Based Model with Distributed Control

7.1 Introduction

The previous chapters have been concerned with modelling minimal information for distributed autonomous measurement of complex infrastructural network dynamics, in the context of telecoms. However, the matter of control has not been solved in this context. This is where the limits of simulations of telecoms start to appear, since for scalable autonomous control of the infrastructure, agents need to make decisions. For an agent to be decisionmaking, it must be able to compare options and select the optimal one according to an internal logic. With regard to packet flow over the network, this would suggest some kind of value to choosing some action or another based on packet flow. The issue is that it is not clear how valuable a data packet is to a given agent with respect to criticality. The agent wants to reduce criticality while maximising network service, which can be conflicting goals.

Instead, I build on the prior case study by generating a model of unit flow over a network, but units have a cash value that helps permit agent operation. Now agents can make choices according to this value, enabling the study of scalable autonomous control with minimally informed agents. The case study investigated is that of a supply chain financing infrastructural network.

This chapter defines a multilayer model of complex agent-based supply chain dynamics that demonstrates the effects of reduction in liquidity of suppliers through late payments, called *liquidity squeeze*.

Since the purpose of a firm is value generation, and supply chains are systems of value adding relationships for these firms, finances form an intrinsic feature of supply chains. Their effects are

difficult to model in an agent-based manner over a complex supply network since only product creation is the outcome of the system, while profit is what enables participation, so its effects are secondary observations. Financial squeeze is an outcome of the nonlinear interaction of financial and material flow, so the assertion of this chapter is that it is best captured by nonlinear multi-agent models. The research question associated with this work is, “What effect does the financial ripple effect have on the viability of supply chain firms and how does supply chain financing affect it?” I therefore create a cash constrained multilayer complex agent-based model that captures the effects of financial squeeze as a consequence of endogenous nonlinear interactions.

In response to FREs or to maximise liquidity, firm bargaining power is exploited to delay repayment of accounts payables [119]. This is how the FRE is modelled here; through buyers trying to avoid bullwhip effects by maximising their liquidity in response to material demand changes via issuing invoices with length dependent on the buyer’s, now debtor’s, negotiating power over the supplier, now creditor. Since longer delays can only appear with more power, longer repayment times are assumed to mean more power. The effect of this on the survivability of a supply chain with a given topology, simulating the structural effects of a financial ripple due to the slowed distribution of liquidity.

Following this is Subsection 7.1.1, introducing the key concept of complex adaptive systems as networks. This underpins the work of this chapter. Next is Section 7.2, which covers the methods of the supply chain financing model defined here. It introduces how interfirm relationships are defined and what dynamics take place over the simulated supply network. This opens with Subsection 7.2.1, explaining the topologies studied, the dynamics of different network layers, asymmetric power relationships between firms, and the function of proactive financing. The analytical methodology is then presented in Subsection 7.2.2, detailing how simulation outputs are studied. Results are described in Section 7.3, with the parameters and test conditions used for testing listed. Of key concern are the initial conditions upon network generation and the distribution of outcomes, with a focus on firm failure distributions, system failure distribution, and the regression analysis. Results are discussed in Section 7.4, with distinct discussions on firm, or company level outputs, and system level outputs. This includes both graphical and data analysis, examining the cause for behaviours in the results, and their implications within the model. In Section 7.5, the impacts of the findings are listed, and their meanings are discussed, contextualised within actual application in business, but with respect to the limitations of the simulation study. Finally, the contributions within this chapter are detailed in Section 7.6. All material introduced in this chapter is derived from Proselkov et al. [258] except where stated.

7.1.1 Complex Adaptive Systems as Networks

Using the CASN concept for a deep tier supply chain, firms may adapt to changing market conditions through adjusting their stock buffers. Unlike supply chain models that restrict analysis to limited firms and their relationships, [142], the CASN approach allows analysis of supply chain topology and

evolution over time. Inaccuracies in adjusted behaviour during this evolution propagate through the supply chain as each actor responds inaccurately to the inaccurate responses of their partners. Such problems only appear in deep tier structures, where players in the supply chain have ignorance about other players, generating the RE. Similarly, in DTSCF, inaccuracies in financing due to the associated costs of material flow management cause the FRE.

Unlike systemic risk for the RE, one models systemic cascades in a supply network context and observe longer term impact. The RE is defined as the “disruption-based impact on SN performance and scope of changes on the SN structure and parameters” [171]. It can include mitigation strategies to avoid or respond to propagation of disruptions. Accordingly, I investigate how firms on a complex topology generate financial ripples through late payments, resulting in SN breakdown, and then incorporate response strategies, investigating the impact of supply chain and bank financing (BF) solutions to mitigate the RE.

Financial flow occurs via monetary transactions upon the transfer of materials between firms. Cash flows facilitate material flows, which directly affect cash flows in a supply network. Therefore, cash and material flow must be studied in tandem. I thus develop an agent-based supply network ripple effect model, where firms adaptively rewire when partners have insufficient cash flows to remain as players within the given supply chain.

7.2 Methods

The model introduced here belongs to the class of supply chain simulation models based on selfish multi-agent system dynamics [109], with information flow stimulated by some function generating risk, and a risk mitigation function. The stimulation function is demand, randomly generated at the market level to first tier Original Equipment Manufacturers (OEMs), who then pass this demand up the supply chain.

This model focuses on supply chain transactions through information flow, without explicitly modelling production. Holding and material purchases have associated costs, where holding removes cash from the supply chain system, and purchases transfer cash from buyer to seller. Materials are assumed to be homogeneous, where one material unit bought from a supplier u by firm v allows for one material unit to be sold by v to buyer w .

Firms have different degrees of bargaining power as a function of their market share of supplier clientele. Risk mitigation capabilities are based off of this, spreading risk to other supply chain members. They impose risk by paying for materials late, where the greater the firm’s bargaining power with respect to its supplier, the later it may pay as an expression of their market control. This captures the effects of power differentials [67, 212, 179, 139].

Risk during financial stress is mitigated by financing [265], including reliable and low interest bank loans accessible to more powerful firms, and higher interest SCF solutions, such as reverse factoring. Firms both engage in complex financial solutions that directly affect each other and transfer

material down the chain so they simultaneously exist in a supply network and a financial network, acting as manufacturers and financial institutions. This thus defines a multilayer network model with layers for material flow, demand flow, cash flow, and financing control.

On this model can be learned how supply chain financing affects the supply network in the long-term while capturing selfishness. A viability measure is used [165], based off of survivability, measured in time to removal from the supply network. Removal is generalised to having insufficient cash for expected continued operation in the supply chain, capturing long-term supply network recovery, or lack thereof, as in [164]. Examples of removals are bankruptcy or the firm choosing to exit a supply chain for another due to sustained unprofitability of serving a particular market.

7.2.1 Model

I investigate two topologies, with full connectivity among firms in adjacent tiers:

- Lattice: tiers are homogeneous, holding equal firm numbers.
- Diamond: a deep tier network. Each tier is one firm smaller per tier from the centre.

Financial squeeze is simulated in a multilayer complex supply network material flow model with demand and cash flow layers. Adaptive supply chain financing mitigates against risk, simulating the effects of dynamic cash injection on system health. Relative firm bargaining power depends on the number of firms within a given tier, determining payment cycle control, lengthening them and moderating access to financing.

Material Flow Layer

The material flow model follows the structure of [34] where trade credit is abstracted to combinations of accounts payable and accounts receivable. This extends with buffer periods and mitigation strategies, such that the time a payment must be made and the time a firm is removed become more than just a point process.

The lattice and diamond topologies are shown in Fig. 7.1. The lattice is a network with dimensions $m \times n$, where m is the number of tiers and n is the number of firms per tier, such that there are mn firm nodes. This represents an upstream manufacturing supply chain for a resilient ecosystem producing products with non-perishable materials, worked and recombined as parts up to market, such as the Daimler-Chrysler [80] supply chain, with multiple supply lines [279]. It begins with a raw material processing firm and finishes with an OEM. Tier i 's width is denoted $|i|$. The diamond has m tiers, such that

$$\left\lceil \frac{m}{2} \right\rceil = \left\lfloor \frac{m}{2} + 1 \right\rfloor = \max\{|i| : i \in \{1, \dots, m\}\},$$

$$|m| = |1| = \min\{|i| : i \in \{1, \dots, m\}\}$$

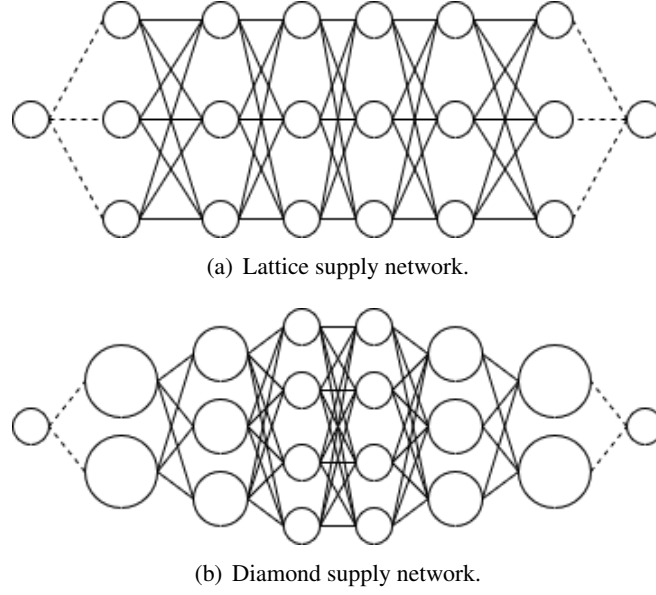


Fig. 7.1 The lattice and diamond topologies I analyse, where $m = 6$ and $n = 3$. Dotted lines are to dummy raw material and dummy market nodes.

and, denoting $\langle X \rangle$ as the mean of any set X ,

$$\left\langle \left\{ |i| : i \in \left\{ 1, \dots, \left\lceil \frac{m}{2} \right\rceil \right\} \right\} \right\rangle = \left\langle \left\{ |i| : i \in \left\{ \left\lfloor \frac{m}{2} \right\rfloor + 1, \dots, m \right\} \right\} \right\rangle = n.$$

Tier 1 firms represent OEMs, and firms at tier m are the Raw Materials Suppliers (RMSs). OEMs have an associated dummy consumer market node that generates demand for a number of material units and introduces it into the supply chain, such that all further demand experienced by the rest of the chain depends on this and the response by the OEM. RMSs have an associate dummy resource node, treated as having infinite resources, from which raw materials are worked to meet demand from customer firms downstream. Altogether the graph may be represented $G = (V, E)$, where $V := \{U_i : i \in \{0, \dots, m+1\}\}$, U_i are tiers and $|U_i| = n$.

For $i \in \{1, \dots, m\}$, the tier, and $j \in \{1, \dots, n\}$, the index in a single tier, firm $u_{ij} \in U_i$ stores demand at a given timestep t as d_{ij}^t and have ι_{ij}^t inventory. OEM nodes, u_{1j} , receive demand from the market nodes, u_{0j} , according to a normal distribution with mean λ and variance σ^2 . OEMs satisfy demand by reducing their inventory by $\min \{d_{1j}^t, \iota_{1j}^t\}$, selling as much as they can of what the market desires, such that

$$d_{1j}^{t+1} = d_{1j}^t - \min \{d_{1j}^t, \iota_{1j}^t\}.$$

If $d_{ij}^{t+1} > 0$, then each firm $u_{ij} \in \bigcup_{i=1}^m U_i$ randomly selects a connected supplier from $\{u_{i+1,k} \in U_{i+1} : (i+1, k) \in E\}$. All unfulfilled demand is backlogged, such that

$$d_{i+1,j}^{t+1} = d_{i+1,j}^t + d_{ij}^{t+1},$$

summing over all buyers that have chosen this firm. This occurs for all firms other than at tier m , the RMSs, who have no firms to pass demand onto. Instead they increase their inventory to meet new demand such that

$$I_{mj}^{t+1} = I_{mj}^t + d_{mj}^t.$$

For each firm other than the OEMs at tier 1, that is, $\forall u_{ij} \in \bigcup_{i=2}^m U_i$ each firm $u_{i-1,k}$ chooses a random connected buyer from $\{u_{i-1,k} \in U_{i-1} : (i-1, k) \in E\}$. It then ships as much material as it may send, sending out $\min\{d_{ij}^{t+1}, I_{ij}^{t+1}\}$, the limits of its resources and available market. This is repeated for all firms, creating a steady material flow of goods down the supply chain. This forms the first two layers of the model. The next section introduces a cash constraint extension for purchases and discusses the cashflow layer.

Cash Constraint Extension and Cashflow Layer

The above version only includes material flow, however since there is no material acquisition limit, demand will be satisfied exactly as it enters the system, if only with market lead times of $2m$. This is because it takes m steps to reach the RMSs from the OEMs, and there is a dyadic lead time of 1.

Purchasing practices must be restricted and match real conditions. A cash value is imposed onto purchases, stored in an edge $(uv) \in E$, denoted $c(uv)$, and introduce the cash reserves of a given firm, κ_{ij}^t . For buyer firm u and selected supplier firm v , purchasing which is transferring a material unit in exchange for cash from v to u costs $c(uv)$ many cash units for u , and grows those reserves by the same amount for v , such that

$$\begin{aligned} & (I_u^{t+1} = I_u^t + x) \\ \iff & (I_v^{t+1} = I_v^t - x) \\ \iff & (\kappa_u^{t+1} = \kappa_u^t - xc(uv)) \\ \iff & (\kappa_v^{t+1} = \kappa_v^t + xc(uv)). \end{aligned}$$

A global operative cost is also assigned to all firms, denoted o_c , which is subtracted from the cash reserves of each firm, each timestep. This has the effect of justifying the need for cash, rather than simply the desirability of profit, and represents the difficulty of survival within a given supply network. This forms the third layer of the model. The operative cost captures costs such as taxes, paying employees, rent, and is implicit of production costs, irrespective of firm size. The operative cost thus introduces restrictions which allow for complex system behaviours, but first the concept of power dynamics must be introduced, as in the following section.

Power Dynamic Extension

Firms in a supply network are not always equals when dealing with eachother, such that one actor in any buyer-supplier dyad may control the relationship more and have more influence than its

competitors. This may be due to being a key supplier, the market dominant brand, or having more market share. Key suppliers are non-existent in my model, since it has homogeneous material flow. Therefore, bargaining power is let to be an independent parameter, corresponding to market share and access to financing, assuming that greater bargaining power suggests greater ease in negotiating bank mandates. This assumption is used to investigate the effect of varying market share for a given power position in an isolated manner.

Firms may have small, medium, or large amounts of power, represented as $\mathcal{P}_{ij} \in \{s^*, m^*, l^*\}$ for all $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$, which corresponds to access to financing.

Homogeneous material flow and firms have no unique characteristics in the model, so bargaining power is assumed to be tied to market share. As such, a firm's power has a mapping that determines its market share, that is, how frequently it is selected among the suppliers of a buyer, such that any m^* or l^* supplier is selected respectively p_m and p_l times more than any s^* supplier. Using the mapping, $f : \mathcal{P}_{ij} \rightarrow \{1, p_m, p_l\}$, gives

$$\begin{aligned} f(s^*) &= 1; \\ f(m^*) &= p_m; \\ f(l^*) &= p_l. \end{aligned}$$

Defining the random variable S_{ij}^t as the choice of supplier for firm u_{ij} at time t , the probability mass function is

$$\mathbb{P}(S_{ij}^t = u_{i+1,k}) = \begin{cases} \frac{f(\mathcal{P}_{i+1,j})}{\sum_{(i+1,k) \in E} f(\mathcal{P}_{i+1,k})}, & (i+1,k) \in E; \\ 0, & (i+1,k) \notin E. \end{cases}$$

This variable dictates throughput over firms. Having a closed system with homogeneous product flow means, on average, each tier has the same total material flow. If a tier has fewer firms, then each firm has more material flow through it, resulting in greater market share in my model. To account for the linear relationship between power and market share, I define a function that, for thinner tiers, increases firm tendency to be defined as more powerful. Width of a tier is relative, such that, for a uniform random variable between 0 and 1, denoted $U(0, 1)$, and for

$$x = \frac{U(0, 1) \min(|i|)}{|i|}, \quad (7.1)$$

we have that

$$\mathcal{P}_{ij} = \begin{cases} l^*, & x \in [2/3, 1]; \\ m^*, & x \in [1/3, 2/3]; \\ s^*, & x \in [0, 1/3). \end{cases}$$

Thus, market share is the de-facto measure of power in this system, which is expressed in the dyads through late payments. Firms typically pay 60-120 days late [265] for purchases, and the more powerful a firm, the later they may pay to maximise their own liquidity. This is summarised in Table

7.1, with the delay in purchase along (uv) , denoted δ_{uv} , which depends on the power of the buyer, on the horizontal axis, and the supplier, on the vertical. This means that how much more powerful a buyer is than its suppliers determines how late it can pay, since they remain an important customer with a large market share.

	s^*	m^*	l^*
s^*	60	90	120
m^*	30	60	90
l^*	30	30	60

Table 7.1 Table of invoice terms (x,y) where agents with financing access x pay agents with financing access y after (x,y) timesteps, representative of days.

Using the mapping $f : \{s^*, m^*, l^*\} \rightarrow \mathbb{R}$ such that $f(s^*) = 1, f(m^*) = 2, f(l^*) = 3$, delay is defined

$$\delta_{uv} = \max(30(\mathcal{P}_u - \mathcal{P}_v) + 60, 30).$$

Delayed payments recording gives a meaningful ledger of accounts payable and accounts receivable such that, given the latest a firm is meant to pay for its goods is 120 days, there is an expected inflow and outflow of cash that does not use extrapolated data. These are vectors of length 120 for each firm that change in time, denoted P_{ij}^t and R_{ij}^t respectively.

Conditions for removing a firm from the system can be defined according to payment ledgers. Let a firm be always operational if $\kappa_{ij}^t > 0$ since it is currently liquid, meaning it may change its conditions by purchasing more goods to sell at profit. If $\kappa_{ij}^t \leq 0$, then it has no liquidity, but if it expects inflows that are net positive, that is, $R_{ij}^t + P_{ij}^t > 0$, then it is still a profit making firm and may continue until market conditions change; for example, competing suppliers may be removed and thus more demand will be received. If however $\kappa_{ij}^t \leq 0$ and $R_{ij}^t + P_{ij}^t \leq 0$ then the firm is considered unprofitable and inoperative, such that it cannot continue functioning, and is removed from the network. This is not necessarily a bankruptcy, as actors within the network, in real life, may be subsidiaries of larger profitable firms. It simply means that all contractual agreements between this firm and its partners in this network are severed — for example, the firm may be absorbed into another firm in another chain, or may alter its output to become profitable.

A state of network inactivity can now also be defined, and thus a definition of network lifespan. If there is no route from any RMS to any OEM, then that means there is no way to bring new material goods to market. A network with n tiers is declared inactive if there does not exist any path $(u, \dots, v)$ where $u = 1$ and $v = n$. In the $m \times n$ topologies, this means for any $k \in \{1, \dots, n\}$, $u_{kj} \notin V$, exemplified in Fig. 7.2.

This justifies reasons for a firm to find ways to remain liquid, even at the cost of profitability. This leads to the financing extension, which also forms the fourth layer of the model, focusing on control.

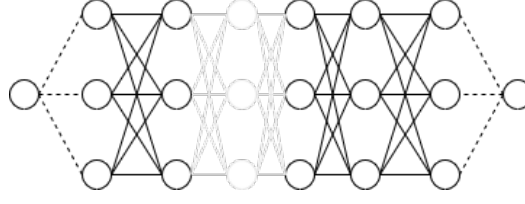


Fig. 7.2 Broken Lattice supply network.

Financing Extension and Control

A firm avoids removal from the network if it is either liquid, or expects profits. It is more valuable to have cash now than later [40] due to the real life variability of conditions, competition, ability to grow in power through purchasing further assets or paying more for a more skilled workforce, and other factors. I therefore introduce a financing function that tries to stop removal for illiquidity, hypothesising that a cost to overall profits is worth the gain in current profits.

First is *bank financing*. If a firm is illiquid, that is, $\kappa_{ij}^t \leq 0$ at the end of a timestep, then it will seek financing from a traditional bank loan. The amount of financing available, or banking mandate limit, denoted b_{ij}^t , is dependent on the credit worthiness of the firm, which its bargaining power allows it to negotiate. Credit worthiness is approximated as scaling to firm power and current cash, such that it is a product of a measure of power and liquidity. That is,

$$b_{ij}^t = \begin{cases} 2\kappa_{ij}^t, & \mathcal{P}_{ij} = s^*; \\ 3\kappa_{ij}^t, & \mathcal{P}_{ij} = m^*; \\ 4\kappa_{ij}^t, & \mathcal{P}_{ij} = l^*. \end{cases}$$

A separate variable, firm debt, denoted ρ_{ij}^t , refers to the size of the overall debts owed by a given firm, including interest, that has not yet been repaid. If $\rho_{ij}^t \geq b_{ij}^t$, then a firm cannot incur more debt. This is another expression of a power disparity — greater market control gives greater freedom to remain illiquid for longer. Denoting available BF as β_{ij}^t , the function is such that

$$\beta_{ij}^t = \max(\min(b_{ij}^t - \rho_{ij}^t, |\kappa_{ij}^t|), 0)$$

then, denoting my existing κ_{ij}^t as $\kappa_{ij}^{'t}$ then I can update κ_{ij}^t such that

$$\kappa_{ij}^t = \kappa_{ij}^{'t} + \beta_{ij}^t.$$

This manner of appending variables with a ‘prime’ is used through this paper to denote the same variable during an operation within a timestep. As an approximation of fixed loan repayments over the long-term, all interest is assumed paid by the longest date into the future my model takes — 120

days. The annual interest rate for bank loans as denoted i_b , such that

$$\begin{aligned} P_{ij}^{t+120} &= P_{ij}^t + \beta_{ij}^t \cdot i_b / 365 \cdot 120; \\ \rho_{ij}^t &= \rho_{ij}^{t'} + \beta_{ij}^t \cdot i_b / 365 \cdot 120. \end{aligned}$$

Thus, there are cases when the banking function does not provide sufficient funds to be liquid. To solve this issue of inaccessible bank financing, I introduce *supply chain financing* (SCF), modelling SCF from alternative sources that allow for focal firms with neither a monopoly nor a stable market.

In this case, invoices are sold at some discount in exchange for immediate liquidity, moving a discounted amount of accounts payable from a given invoice term from the future to the present. A global term, denoted τ , is a parameter defining the number of timesteps into the future accounts payable are reduced from. Instead of being restricted by b_{ij}^t , available supply chain financing, denoted ξ_{ij}^t , is restricted by $R_{ij}^{t+\tau}$, such that

$$\xi_{ij}^t = \max \left(\min \left(|\kappa_{ij}^t|, R_{ij}^{t+\tau} - |\kappa_{ij}^t| \right), 0 \right).$$

This gives

$$\kappa_{ij}^t = \kappa_{ij}^{t'} + \xi_{ij}^t.$$

Interest is assumed to be paid at the invoice due date, such that the size of the invoice is removed from the accounts receivable, and the size of the interest - an annual rate of i_s - is added to the accounts payable. This is defined as:

$$\begin{aligned} P_{ij}^{t+\tau} &= P_{ij}^{t+\tau} + \xi_{ij}^t \cdot i_s / 365 \cdot \tau; \\ R_{ij}^{t+\tau} &= R_{ij}^{t+\tau} - \xi_{ij}^t. \end{aligned}$$

These functions mitigate versus failure within the network, using financing. However, firms have ways to estimate their future earnings and demand based on previous behaviour, and from there can estimate their financing requirement. This motivates the addition of a proactive, rather than reactive, extension of the financing functions.

Proactivity

Incoming costs can be estimated by computing a moving average of demand and multiplying this by the average purchase cost among suppliers. An iterative computation is used to estimate the moving average thus avoiding any information storage. I define cash requirements, r_{ij}^t , such that

$$r_{ij}^t = \max \left(0, d_{ij}^t \left\langle \{c(u_{ij}, v) : (u_{ij}, v) \in E\} \right\rangle \right),$$

then include a global window size parameter, w , that determines widths of moving average windows. The moving costs average $\overline{c}_{ij}^t$ is defined recursively, such that

$$\overline{c}_{ij}^t = \begin{cases} r_{ij}^t, & t = 1; \\ \left(\overline{c}_{ij}^{t-1}(t-1) + r_{ij}^t \right) / t & t \leq w; \\ \left(\overline{c}_{ij}^{t-1}(w-1) + r_{ij}^t \right) / w & t > w. \end{cases}$$

This gives a dynamic estimate of required cash to meet demands at any given moment, which can be used as a proactive threshold, giving two cases for each financing method, that of *reactive BF*, *proactive BF*, *reactive SCF* (analogous to invoice discounting from firms seeking alternative financing), and *proactive SCF* (analogous to reverse factoring from contractual agreements with supply chain financing firms).

Proactivity expedites financing, reducing profits due to the interest incurred, i_b and i_s . As a reward for proactivity and seeking financing while the firm still has resources, say, the annualised interest rates are 1% less than they would be if reactive. This means if $\kappa_{ij}^t \leq \overline{c}_{ij}^t$, then for proactive banking,

$$\begin{aligned} \beta_{ij}^t &= \max \left(\min \left(b_{ij}^t - \rho_{ij}^t, \overline{c}_{ij}^t - \kappa_{ij}^t \right), 0 \right); \\ P_{ij}^{t+120} &= P_{ij}^t + \beta_{ij}^t \cdot (i_b - 0.01) / 365 \cdot 120; \\ \rho_{ij}^t &= \rho_{ij}^t + \beta_{ij}^t \cdot (i_b - 0.01) / 365 \cdot 120 \end{aligned}$$

and for reverse factoring,

$$P_{ij}^{t+\tau} = P_{ij}^{t+\tau} + \xi_{ij}^t \cdot (i_s - 0.01) / 365 \cdot \tau.$$

This concludes the simulation methodology. A visualisation representing each layer and their interactions can be seen in Fig. 1.4. My analytical methodology is outlined in the following subsection.

7.2.2 Analytical Methodology

I run repeated simulations across a spectrum of model parameters, exploring a wide selection in the experimental space, analysing the lattice and diamond topologies. Demand is assumed to be independently identically normally distributed for each market node.

I first analyse the relative significance of different parameters upon the survivability of the supply network and constituent firm types by determining the theoretical distribution of firm types per topology, then measuring the deviation from this distribution.

The null hypothesis is that the theoretical distribution of generated firms, Q , equals the empirical distribution of failures per firm type, P . My alternative hypothesis is that topology affects the failure distribution, such that $Q \neq P$. This is determined using the Kullback-Leibler divergence, D_{KL} , where the state space X is the proportion of firm failures by bargaining power.

The number of failures on the i^{th} simulation for firms with small (s^*), medium (m^*), and large (l^*) amounts of power is denoted $s_i, m_i, l_i \in \mathbb{N}$. These sum to the total firm failures in simulation run i , $R_i = s_i + m_i + l_i$, which sums to the total number of failures, $R = \sum_i R_i$.

Simulation parameters are varied deterministically, but by treating simulations as a sample, relative risk can be calculated in two ways. First is in *aggregate*, finding the proportion of failures over all simulations, denoting aggregate P as P_a , such that, for $x \in (s^*, m^*, l^*)$, and x_i as the failure count of a given firm type at simulation i ,

$$\mathbb{P}(P_a = x) = \frac{\sum_i x_i}{R}.$$

Second is *piecewise*, finding the proportion of failures per simulation, summing together proportions, then dividing by total likelihood. Piecewise P is denoted as P_p , giving

$$\mathbb{P}(P_p = x) = \frac{\sum_i \left(\frac{x_i}{R_i} \right)}{\sum_i \left(\frac{s_i}{R_i} + \frac{m_i}{R_i} + \frac{l_i}{R_i} \right)}.$$

D_{KL} is then calculated for P_a and P_p versus the theoretical generated firm types. D_{KL} is useful for analysing significance and difference, but gives no insight into the nature of a distribution. For that, the histograms of failure are directly studied and probability distribution curves are fit to them.

For systemic analysis, Kernel Density Estimates (KDEs) [243, 267] are used over the systemic survival times histograms to produce curve fittings for inspection. Since KDEs showed multiple inflection points, the unimodal Gamma and bimodal Gaussian mixture distributions [262] are fitted for the system survival times.

This model mitigates against ripple effects using financing. I explore the effect of varying parameters associated with financing, including interest rates, repayment term times for supply chain financing, and whether financing is proactive or reactive. With reactive financing, firms seek financing whenever they have zero liquid cash. With proactive financing, a firm, before it has zero liquid cash, predicts future cash flows with a moving average window to determine a dynamic threshold, below which it seeks financing. I also investigate the effect of adjusting window length.

System and firm survivability are examined. The model may heterogeneously distribute access to financing, and thus associated financing function, or treat all firms homogeneously. A moderated regression analysis is conducted, with $p = 0.05$ as a significance threshold of parameters governing the financing. Regression analysis is further broken down by moderating whether the system is financed and under which paradigm.

7.3 Results

Experiments were conducted over the space of model parameters in Table 7.2. Simulations were completed on an Armari Magnetar workstation, with a 3.5GHz Intel Xeon E5-1620 CPU, 64GB RAM, in Windows 10 Pro. Modelling and results were completed in Python 3 with the `math`, `random`,

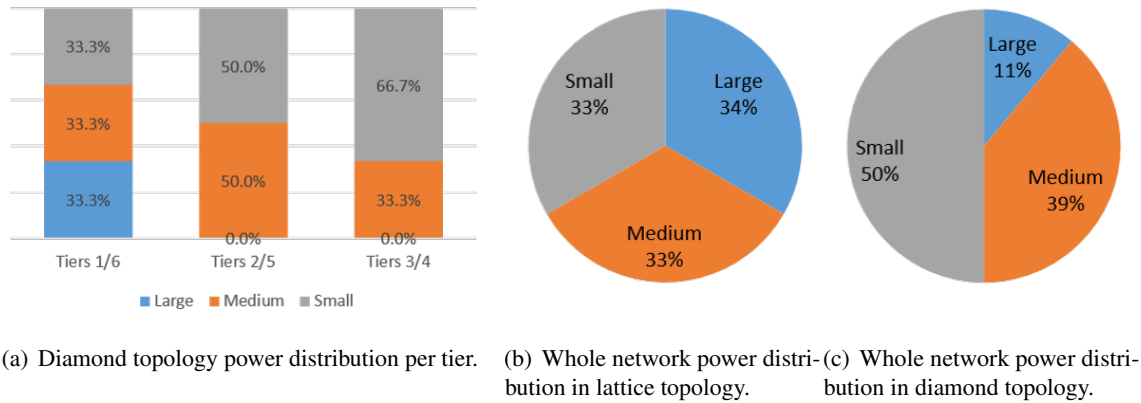


Fig. 7.3 Distributions of firm powers upon generation of a network topology of a given type. Since the diamond topology is heterogeneous, I also show the distribution per tier.

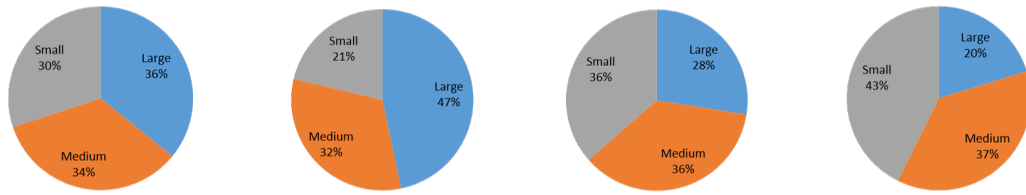
csv, itertools, os, numpy, statistics, matplotlib, pandas, seaborn, fitter, and scipy libraries.

Parameter	Values
Operation Fees	5, 10, 15
Average Demand	5, 20, 50, 100
Demand Standard Deviation	5, 10, 15
Firm Power Distribution	Homogeneous, Heterogeneous
Financed	True, False
Paradigm	Reactive, Proactive
Supply Chain Financing Annual Rates	1.01, 1.03, 1.05, 1.07, 1.09, 1.20, 1.50, 2
Bank Annual Rates	1.01, 1.03, 1.05, 1.07, 1.09, 1.20, 1.50, 2
Terms	5, 30, 60, 90, 120
Window Sizes	5, 30, 60, 90, 120
Medium Power Market Share	1, 2, 5
Large Power Market Share	1, 3, 10, 20
Topologies	Lattice, Diamond

Table 7.2 Table of the parameters tested. Note, large power market shares that would be smaller than medium power market shares are not tested

I investigate the topologies in Fig. 7.1(a) and Fig. 7.1(b). Upon generation, power is distributed in the lattice equally across tiers and through the network as in Fig. 7.3(b), and in the diamond, as a result of Eqn. (7.1), is distributed according to Fig. 7.3(a), resulting in Fig. 7.3(c).

To understand the relative risk incurred upon a firm with given bargaining power, the distributions of generated firms are compared to that of failed firms by looking at distributions in *aggregate* (Fig. 7.4(a), Fig. 7.4(c)), then *piecewise* by averaging proportions of failures for each simulation run (Fig. 7.4(b), Fig. 7.4(d)). The exact values are in Table 7.3. Next is calculated the D_{KL} between theoretical distributions of generated firms and empirical distributions of firm failures (Table 7.4). I then inspect the histograms in Fig. 7.5 and fit probability distributions to them (Table 7.5). Table 7.6 captures mean days survived of each firm type under each topology.



(a) Lattice topology aggregate failure distribution. (b) Lattice topology piecewise failure distribution. (c) Diamond topology aggregate failure distribution. (d) Diamond topology piecewise failure distribution.

Fig. 7.4 Distribution of power among firm failures in lattice and diamond topologies, visualised, as well as in total or piecewise.

		Small	Medium	Large
Lattice	Aggregate	30	34	36
	Piecewise	21	32	47
Diamond	Aggregate	36	36	38
	Piecewise	43	37	20

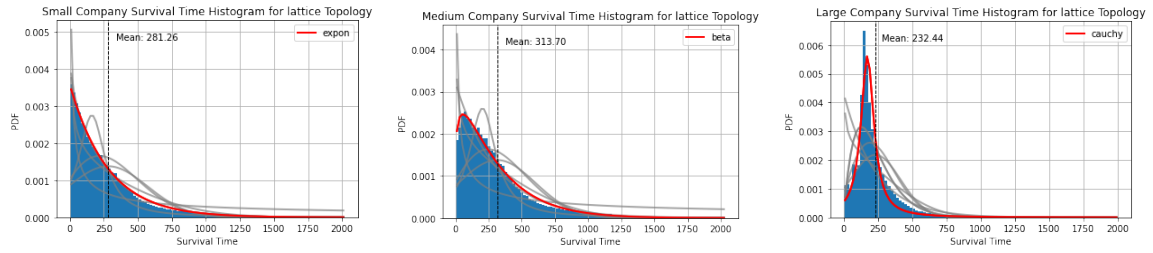
Table 7.3 Distribution of power among firm failures in lattice and diamond topologies, as well as in total or piecewise. More intense green mean a larger proportion of firm types failed for a topology and measurement type.

The unimodal Gamma distributions and bimodal Gaussian mixture model distributions are fitted over the histograms of systemic survival times of lattice and diamond networks (Fig. 7.6). KDEs are shown in Fig. 7.7. Relative accuracy of the unimodal and multimodal distributions was compared using residual sums of squares (RSS) (Table 7.7). Moderated regression analysis for each dependent term is summarised in Table 7.8.

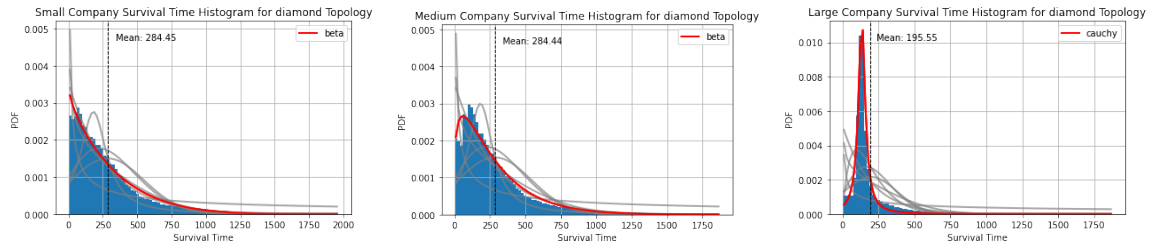
7.4 Discussion

In this work I created a supply network simulation model where selfish actors with different degrees of bargaining power try to maximise their liquidity through negotiating longer repayment terms and other financing functions. I study the resultant financial ripple effects using survivability analysis.

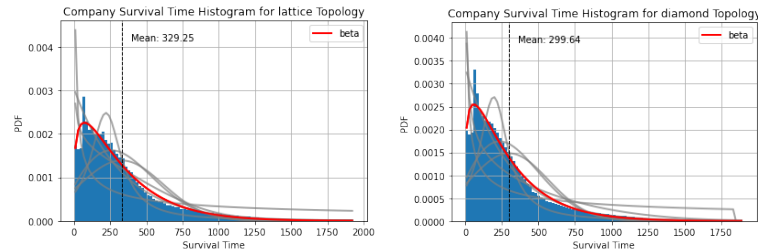
Companies are distinguished by power, endowing them with the capacity to set terms. If a firm needs financing, it must choose to set terms that maximise availability of financing at minimum rates. Similarly, since later payments mean higher liquidity, if firms are selfish, then it logically follows more power means longer repayment schedules and larger mandates. Therefore companies have small, medium, and large amounts of bargaining power, which is coupled with access to financing and repayment times. This follows existing power or size classifications such as the concept of small, medium, and large enterprises.



(a) Small firm survival times on a lattice. (b) Medium firm survival times on a lattice. (c) Large firm survival times on a lattice.



(d) Small firm survival times on a diamond. (e) Medium firm survival times on a diamond. (f) Large firm survival times on a diamond.



(g) Homog. firm survival times on a lattice. (h) Homog. firm survival times on a diamond.

Fig. 7.5 Histograms of firm survival times divided by topology and firm type, with fitted curves. Survival time is in days. Red curves are the most accurate, and their distribution type is shown in the legend. The mean of the histograms is also superimposed on the graph, with a dashed vertical line at its position.

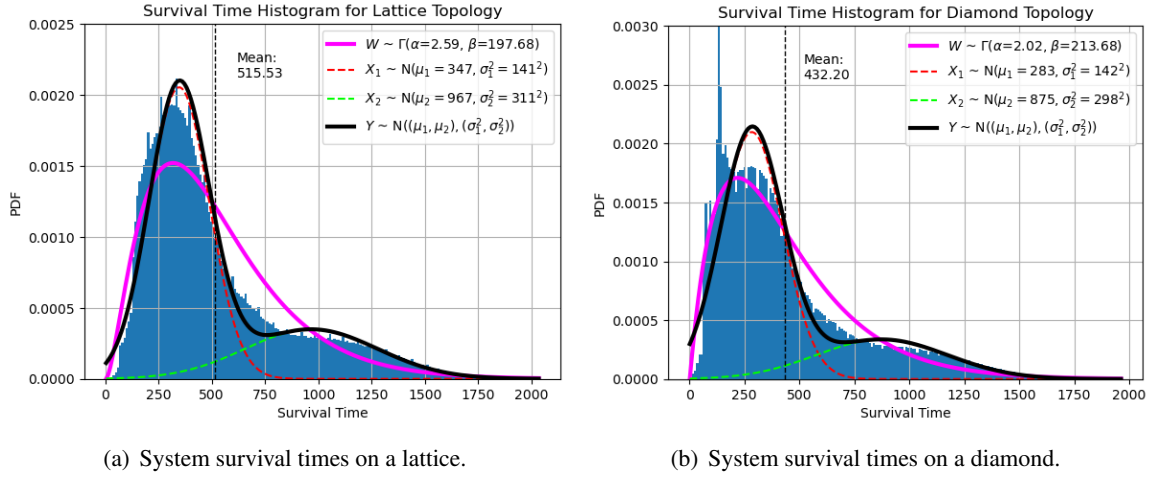


Fig. 7.6 Histograms of system survival times divided by topology, with fitted bimodal gaussian mixture curves. W is a unimodal Gamma distribution, which found to be most accurate of the tested unimodal distributions. X_1 and X_2 are the first and second gaussians, with associated means λ_1 and λ_2 , and variances σ_1^2 and σ_2^2 , respectively. Y is the mixture distribution using X_1 and X_2 .

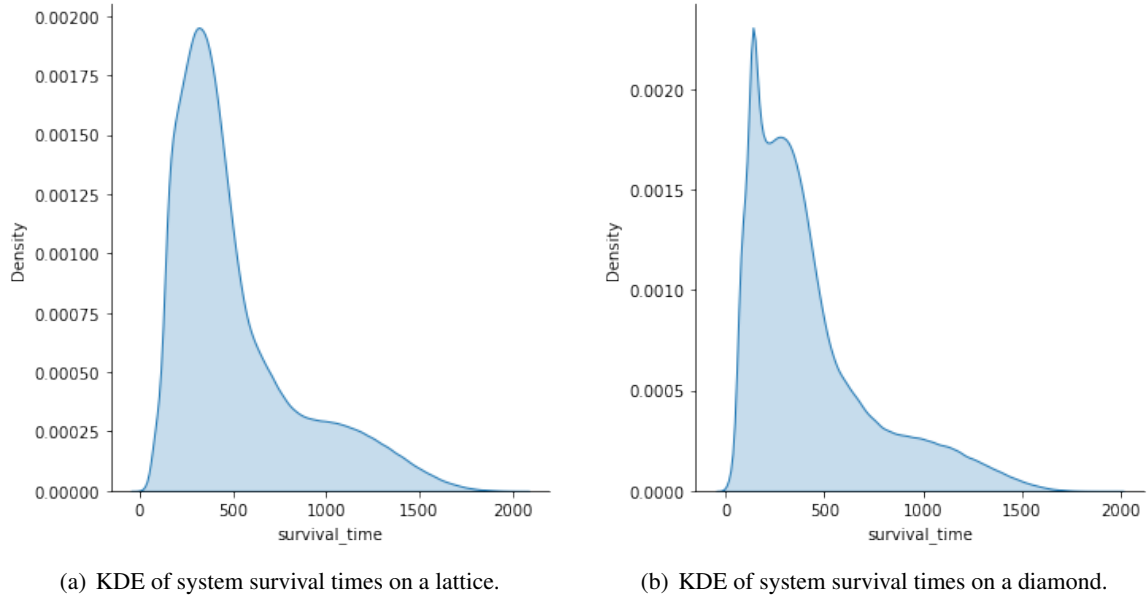


Fig. 7.7 Kernel density estimates of system survival times for lattice and diamond topologies in days.

	KL Divergence	Small Difference	Medium Difference	Large Difference	
Lattice	Aggregate	0.04897	-3%	1%	2%
	Piecewise	0.4078	-12%	-1%	13%
Diamond	Aggregate	0.1094	-14%	-3%	17%
	Piecewise	0.0361	-7%	-2%	9%

Table 7.4 Difference in failures to generated firms

Firm Type	Topology	
	Lattice	Diamond
Homog. Firm	Beta($\alpha = 1.27, \beta = 58.9$)	Beta($\alpha = 1.24, \beta = 1.29e5$)
Small Firm	Exp($\lambda = 280$)	Beta($\alpha = 0.97, \beta = 7.28$)
Medium Firm	Beta($\alpha = 1.21, \beta = 1.56e8$)	Beta($\alpha = 1.26, \beta = 2.05e11$)
Large Firm	Cauchy($x_0 = 174, \gamma = 56.6$)	Cauchy($x_0 = 134, \gamma = 27.9$)

Table 7.5 Fitted probability distributions of firm survival time histograms for each firm type. Time is in days.

Using the bargaining power argument, I investigate if topology significantly redistributes risk according to bargaining power. Since greater bargaining power yields greater access to BF, this gives a measure of the risk trade-off in return for superior financing.

7.4.1 Company Level

Results in Table 7.4 suggest that in exchange for greater power, large companies absorb risk from smaller companies, perhaps due to greater market exposure, where in the case of this thesis, industrial effects are stronger than firm strengths, as in [120]. Additionally, company creation may be a misleading systemic risk measure, since it overestimates risk to small companies, but underestimates risk to large ones, which have more impact. Measures of reduced operation, such as failure, are needed for accurate measurement [2].

Risk is predicted accurately by generation on the lattice in aggregate where $D_{KL} < 0.05$. Piecewise for the lattice, however, has a $D_{KL} > 0.4$, such that generation does not predict risk. In the diamond, risk is not predicted by generation in aggregate, where $D_{KL} > 0.1$, but is predicted piecewise, where $D_{KL} < 0.04$. Failures aggregation closely matches theoretical generation, suggesting configuration does not change the shape of the distributions from generation, but drastically changes the number of failures, captured by piecewise measurement. This may be due to the linearity of the lattice, manifesting in linear changes. Diamond topology aggregation analysis is significant, but piecewise is not, suggesting configuration affects the number of failures in a run, but not the distribution. This may be because of the overriding influence of the deep tier topology, where medium firms tend to connect

Topology	Firm Type			
	Homog. Firm	Small	Medium	Large
Lattice	329.25	281.26	313.70	232.44
Diamond	299.64	284.45	284.44	195.55

Table 7.6 Mean number of days survived for a given firm type in a given topology

Topology	RSS	
	Gamma	GMM
Lattice	7.60e-6	3.68e-6
Diamond	7.71e-6	1.20e-5

Table 7.7 Residual sums of squares (RSS) of Gamma and Gaussian Mixture Model (GMM) distribution fits for both lattice and diamond topologies.

to both small and large ones, and their stronger relationship with industry structure affects the rest of the system [120].

Homogeneous company failures (Fig. 7.5(g), Fig. 7.5(h)) are beta distributed for both the lattice and diamond. The diamond has less deviation with a smaller scale parameter of $\beta = 1.29\text{e}5$ versus the lattice's $\beta = 58.9$ (Table 7.5). Both have similar shape parameters ($\alpha = 1.27$ and $\alpha = 1.24$ for the lattice and diamond respectively) suggesting homogeneous company failure distribution is resistant to topology. This follows the right skewness of graphs of failures versus age in [187], though at a smaller timescale. Time is measured in days.

Small companies in the lattice (Fig. 7.5(a)) are exponentially distributed, modal around 0 days, with exponential decay. In the diamond they formally fit to a beta distribution, also modal around 0, following approximately the same shape as the exponential curve (Fig. 7.5(d)), suggesting small company failures are generally exponentially distributed.

Medium companies are beta distributed, which is not necessarily symmetric, capturing their skewed failure distributions (Fig. 7.5(b), Fig. 7.5(e)). Shape parameters in the lattice and diamond are $\alpha = 1.21$ and $\alpha = 1.26$, respectively, with smaller scale for the diamond. Medium shape parameters match those for homogeneous companies, suggesting medium companies most characterise company survival.

Large companies are Cauchy distributed. This suggests failures for large companies are symmetric but shifted down (Fig. 7.5(c), Fig. 7.5(f)). This could be an artefact of the long large company failure tails which match the Cauchy distribution.

Respective modes of company survival times suggest the first companies to fail are small, then medium, then large. Large companies have the worst mean survival time, while homogeneous companies are top performers. Small, medium, and homogeneous companies have fat tails, while large ones have thin tails, meaning the system has adapted such that large companies fail quickly. Since only medium firms empirically have been found to be more affected by the supply chain or

Network Object	Topology	Paradigm	Regression Coefficient			
			Supply Chain Financing Product Terms	Annualised Interest Rates		Proactivity Window
				Supply Chain	Bank	
System	Lattice	Reactive	*	*	*	x
System	Lattice	Proactive	-2.53	*	*	*
System	Diamond	Reactive	*	0.53	*	x
System	Diamond	Proactive	-3.2	0.54	*	-0.01
All Firms	Lattice	Reactive	*	-0.19	*	x
All Firms	Lattice	Proactive	-2.57	-0.09	-0.72	-0.01
All Firms	Diamond	Reactive	*	0.17	-1.09	x
All Firms	Diamond	Proactive	-3.21	0.18	*	-0.01
All Firms: Het.	Lattice	Reactive	*	-0.14	-0.83	x
All Firms: Het.	Lattice	Proactive	-2.7	-0.15	*	-0.02
All Firms: Het.	Diamond	Reactive	*	0.3	-1.02	x
All Firms: Het.	Diamond	Proactive	-3.24	0.18	*	-0.01
All Firms: Hom.	Lattice	Reactive	*	-0.19	*	x
All Firms: Hom.	Lattice	Proactive	-2.36	-0.13	*	*
All Firms: Hom.	Diamond	Reactive	*	0.07	-1.35	x
All Firms: Hom.	Diamond	Proactive	-3.11	0.15	*	-0.01
Large Firms	Lattice	Reactive	*	-0.51	*	x
Large Firms	Lattice	Proactive	-2.1	-0.54	*	-0.01
Large Firms	Diamond	Reactive	*	*	*	x
Large Firms	Diamond	Proactive	*	*	*	*
Med. Firms	Lattice	Reactive	*	0.2	*	x
Med. Firms	Lattice	Proactive	-3.47	0.28	*	-0.02
Med. Firms	Diamond	Reactive	*	*	*	x
Med. Firms	Diamond	Proactive	*	*	*	-0.01
Small Firms	Lattice	Reactive	*	0.36	*	x
Small Firms	Lattice	Proactive	-3.05	0.34	*	-0.04
Small Firms	Diamond	Reactive	*	0.49	*	x
Small Firms	Diamond	Proactive	-3.71	0.3	*	-0.02

Table 7.8 Table of moderated regression coefficients, assuming a system with financing. ‘-’ means the *p*-value of the variable in question for that configuration is below 0.05. ‘x’ means that the variable in question is degenerate for that configuration.

industry than their own firm attributes, this suggests exceptional integration between firms in the supply chain. Further analyses on more sparsely connected or purely empirical graphs should be explored.

This model also suggests greater financing for companies with greater bargaining power means more debt, which reduces overall cash. Greater risk exposure caused by dependence on more weak companies results in a misuse of funds, since partner risk is not considered in financing.

Company survival times are worse in the diamond than the lattice (Table 7.6). I propose the diamond has a higher risk exposure because stronger companies are positioned closer to the market and raw materials supplier in thinner tiers, thus relying on a larger number of weaker companies, who in turn suffer from less liquidity because these powerful companies are paying late too often. This highlights the combination of bullwhip effects and financial ripple effect in action [203]. As inefficient weaker companies get eliminated, prior systemic adaptations now mean such poor market conditions that previously strong companies fail in quick succession.

These results combined suggest that increased heterogeneity in both topology and characteristic powers, while improving the relative survivability of larger companies since they fail after small ones, overall reduces systemic survivability. Since large companies fail after small ones, this may explain the large company risk skew — their advantages undermine the system, whose survival dominates theirs.

Having analysed company survivability and observed how these affect systemic survivability, I now examine the survivability of the system itself.

7.4.2 System Level

KDEs of histograms in Fig. 7.7 have multimodality via multiple inflection points for both lattice and diamond topologies. Analysis is therefore conducted with the Gaussian Mixture Model (GMM) in mind.

The first mode of the lattice network matches the mean of lattice company survival times (Fig. 7.6(a)). This is true for diamond system survival times also. Bimodality suggests system survival times may have different distributions after the mean focal company failure times than those before.

The GMM is compared to the best performing unimodal distribution, found similarly to the distributions in Table 7.2, and proving multimodality. This is a gamma distribution, likely owing to the irregular skew of system survival times (Table 7.1). RSS were calculated to determine relative distribution accuracy (Table 7.7). The GMM and gamma distributions represent the lattice topology better than they do the diamond. The lattice GMM outperforms the gamma. The diamond GMM performs worse than the gamma. This is due to the high density of early systemic deaths. The GMM performed well overall, suggesting bimodality is correct, if not ideal for curve fitting.

Looking at the X_1 normal curves (Fig. 7.6(a), Fig. 7.6(b)), the mean survival time is σ_1 above λ_1 , while the X_1 and X_2 intercept is around $2\sigma_1$ above λ_1 , and σ_2 below λ_2 . This gives a basic topology

of the survivability space and what a given simulation's survival time suggests about the stability of a supply network.

The survival space has a wide range. To understand this further and explore the survivability space, I conduct a moderated regression analysis of the mitigating functions and their variables in this model.

Mitigation of Ripple Effects

This subsection examines regression coefficients of a moderated regression analysis of a spread of mitigation function parameters (Table 7.8). Operation fees, average market demand, and demand standard deviation always significantly change the survivability of any network object under all other parameter combinations, so they are the primary drivers of the simulation model. Interest rates for SCF and BF, and repayment terms for SCF depend on there being financing. The window to inform the proactive financing threshold depends on financing being proactive. Moderation is thus restricted to network object, topology, and financing paradigm.

Supply Chain Financing Product Term Term is the length of time after which SCF products are repaid. Term only has significant impact in the proactive financing case, despite not being degenerate for reactive financing. When it is significant, longer terms of repayment for invoices tend to have the largest impact on survivability. This suggests shorter terms promote survivability only for companies with predictive ability. Under proactive financing, term has significant impact on survival time for all lattice network objects. In the diamond, term under a proactive paradigm is significant for all expect medium and large companies, who survive independent of term.

Supply Chain Financing Interest Rates Higher interest rates positively correlate with diamond topology system survival time and all small company survival times but are insignificant for medium and large companies. Higher rates have positive impact for the 'all companies' category in a diamond topology, but negative impact in the lattice topology. Large companies are negatively impacted by higher rates in the lattice. Together, this suggests that system survival time, as a function of supply chain interest rates, is driven by large companies. When the negative effects of higher rates are insignificant for large companies, however, small company survival time dominates as the system survival driver, which is improved by higher rates. In the topologically homogeneous lattice, higher rates simply impose more restrictions on the whole system. In the topologically heterogeneous diamond, large companies are more dependent on their supply chain than any financing functions. In this case there is less survival time variation (Fig. 7.5, Fig. 7.6), and a tendency for earlier system failure (Fig. 7.6(b)), suggesting fewer viable configurations. I propose that higher rates eliminate companies in inefficient positions within the system, bringing more business to small nexuses, achieving more stable configurations sooner. This is a consequence of the adaptivity of the model.

Upon initialising a simulation, longer terms give a longer grace period to ill-configured firms. Increasing terms reduces total system cash but extends the period of inefficient configuration of the model. This leads to smaller cashflows to otherwise efficient companies, thus uniformly reducing survivability. Increased SCF rates give no such grace period.

Bank Financing Interest Rates BF interest rate has either a negative regression coefficient or is insignificant. Both system survival time and survival time of particular company types are unaffected by changing BF interest rates. I propose this is due to the structure of repayment within the model; companies need only pay back cash 120 days into the future rather than the shorter terms typical of SCF. As the repayment date is past the most common removal time of any company, nodes do not feel the effects of repaying BF.

Increasing BF interest rates, unlike increasing SCF interest rates, always negatively affects survival times when it is significant. It may be that financing is only repaid after the network has settled into a reliable topology. If it had effect before stabilisation of the supply chain, then it would have the option to help remove inefficient companies. However, since it always has negative effects for increased rates, I conclude it is only affecting efficient companies.

Proactivity Window The moving average, or proactivity window, has the smallest impact of all mitigation factors and is always negative. It has no impact on survival for lattice systems, homogeneous lattice companies, and large diamond companies. It significantly affects survivability for all other moderated tests.

The regression coefficient is small because changing the proactivity window does not change actual flows, it skews the financing threshold to older information. Information used in a shorter proactivity window is always contained within a longer one, so the relative amount of new information decreases as the window length used increases. This means, as long as the threshold changes linearly with the window, there is little scope for change. The ranges for which new information contributes meaningfully to the threshold are small and close to the present time, so deviate only a small amount from the threshold derived from smaller windows.

Moderated Regression Analysis Summary Not all mitigation function parameters are significant. The most impactful parameter is the length of supply chain financing (SCF) repayment term, with a consistently negative regression coefficient. Second is the interest rate for bank financing, meaningful for reactive systems. Third is the SCF interest rate, significant for most configurations. The least significant is proactivity window length, always negative. Key observations on each parameter are:

- minimising time to repayment of SCF products should improve survivability of the system and supply chain participants;
- higher SCF interest rates may remove inefficiently distributed companies, producing a positive effect on system survivability and network participants;

- higher bank financing (BF) interest rates do not affect most network objects under most conditions, and most likely only affect efficient and otherwise stable systems. Hence, they may work as a useful financing source in stress periods if available;
- the width of the proactivity window has little effect, and only negative, so a more advanced and nonlinear method of processing past information is necessary to study this relationship.

Finally, regression coefficients for mitigation functions are often unintuitive or insignificant. I propose that this is a consequence of the model being both closed and non-evolutionary. It is closed since the supply chains investigated do not affect external systems that influence the supply chain itself - they are assumed to function independent of the supply chain. It is non-evolutionary since there is no growth function of firms or the supply chain. The function of financing is to allow growth to the point of profitability through the stages of unprofitability. Since this system only exhibits adaptivity through rewiring upon failure, financing can only be used to carry a system through to stability after cutting of inefficiencies. This limitation has the benefit of ensuring failure after a certain time, allowing for survivability analysis of the sort conducted here, however further research should incorporate growth functions and external relationships to better emulate the use of financing.

7.5 Conclusions

To investigate the financial ripple effect (FRE) on supply networks, I created a simulation model of a complex adaptive supply network with financing, where companies vary repayment schedules, generating financial squeeze which in turn causes disruptions. This is an agent-based supply network simulation model with homogeneous material flow, cash-constrained transactions where companies are differentiated by their negotiating power, which affects their access to financing and how late they may pay for purchased goods. Companies seek financing to fill the liquidity gap caused by delayed payments, and financing can be sought out proactively ahead of expected financial ripples using estimates of future demand. Companies are removed from the system whenever they run out of cash, whereupon buyers switch to sourcing from alternative suppliers. In modelling this, I capture the decay effects of a re-configurable supply chain [102].

Tests were completed using two characteristic supply network topologies: a lattice, with homogeneous tiers; and an extended diamond, with heterogeneous tiers and deep tier structure. A moderated regression analysis was conducted to analyse parameters which govern financing, designed to mitigate against FREs. In studying this using a complex adaptive system as a network, I was able to observe the relationship between agent properties and system resilience.

Policymakers should also note topology has a significant impact on financial resilience suggesting that increased visibility of supply chain structure is important. Powerful companies tend to be overexposed to risk because less powerful companies fail first, turning the network topology unstable, whereupon powerful firms fail in quick succession before complete system failure. This suggests firms with more power should support weaker firms to prevent a failure cascade.

The diamond topology is worse for company and system survival time than the lattice. This is due to increased partner risk, which financing functions do not consider. Thinner tails suggest that greater heterogeneity limits configurations that can survive in the long-term and results suggest homogenising company bargaining powers to ensure this. Also, deep tier structure, when not correctly accounted for, is detrimental to survival. I also found that survival may be distributed bimodally, suggesting there may be a categorical difference in resilience between some configurations and others.

Using moderated regression analysis on system survivability, I found the following:

Longer repayment terms for SCF products in a closed system only negatively impact survivability for all network objects when a prediction function is in place (as it is likely to be in reality) else it has no impact.

Liquidity, within this closed system, appears to be less valuable than total cash, suggesting invoice terms should be minimised to improve member survivability in a supply chain.

Only small companies depend on term times with the largest coefficient magnitude of all tested. Large and medium companies are affected only by terms in the lattice.

Unexpectedly, greater interest rates for SCF actually improve survivability of the diamond topology and of firms embedded within a diamond topology, however reduce survivability in lattice topologies. This may be because high interest rates remove ill fitting firms in a diamond for the good of the whole. Therefore, supply chain financing firms have reasons to increase interest rates and shorten repayment terms for products in deep tier type systems. Else they should be more lenient to ensure survival of their customers.

Bank financing rates have little clear impact in a closed system. Maximising the amount of bank financing does not reduce survivability of the system. However, focal companies should take care to minimise rates since, when they do have an effect, this is a long-term negative impact on all companies.

The proactive financing predictive window should use limited information scaled to typical repayment terms, due to the dynamicity of supply chain health and cash distribution.

I propose that the impact of longer moving average windows is negative because a company is selecting its financing threshold based on more out of date information. This suggests that, in the moving average, the value of greater quantities of temporal information [255] is negative since they reduce the relevance of information to the network's current state, matching existing research into credit shock propagation within supply chains [3]. To induce a larger contribution that is not necessarily negative, a nonlinear function, possibly using machine learning, should be introduced.

The results have significance for multiple types of stakeholders of supply chains. Non-focal supply chain participants can use these insights and simulation design to help forecast their own survivability under different configurations for the purpose of strategic planning. Focal companies may use the model in this chapter to inform supply chain planning and design, as well as provide insight into what sort of supply chain financing to organise. Finally, regulators and policymakers can use this to inform

policy as to which supply chain configurations should be promoted and as a toolset to conduct further research.

7.6 Summary of Contributions

In this chapter I introduced a multilayer agent-based model of supply chain financing dynamics, specifically of cash-constrained demand-driven material flow, where cascade failure in the form of the ripple effect is mitigated against by autonomous distributed financing control. This defines a four-layer network simulation model, with demand flow, material flow, cash flow, and control on different layers. This model can be used to simulate dynamics in different topologies to investigate supply chain financing dynamics.

In the model, failure cascades are generated by financial bullwhips, triggered by late payments due to power imbalances among firms. The dynamics of such cascades are illustrated and studied for two deep tier topologies, a lattice (constant tier width) and diamond (thin tiers at the ends of the supply chain), which operate and break down as complex adaptive systems.

I studied the contribution to supply network breakdown from power imbalances, financial and market demand conditions, as well as the distribution of survival time under different conditions of different network objects, including firms and the supply chain itself. I found that survival time is strongly clustered to be early, and homogeneity of tiers and firm power conditions for longer survival time.

This model has the benefit of explicitly calculating routing of a material unit as a network function, that of the cash and demand layers. This allows for control functions in the form of cash injection to be incorporated, which forms the basis for the next chapter into scalable autonomous control for supply network viability.

Altogether, I have that:

- greater bargaining power for buying firms and its benefits are achieved by greater risk exposure;
- the extended diamond topology is generally worse for company survival than a lattice;
- skew of risk to powerful companies is greater in the diamond than the lattice;
- powerful firms should protect weaker ones in their supply chain to prevent their own failure;
- both deep tier SCF and BF without considering partner exposure may result in inefficiencies;
- homogeneous configurations are resistant to the negative effects of the diamond topology, but still perform better in the lattice as well.

Chapter 8

Enabling Viable Supply Networks with Cooperative Distributed Local Decentralised Control

8.1 Introduction

This chapter unifies the work throughout this thesis to define a scalable autonomous control architecture in complex supply-networks, combining both distributed and decentralised control layers to enable long-term viability. It answers the research question, “What effect does local cooperation for financing guidance have on supply chain viability as a function of the range of cooperation”?

To preserve independence and facilitate cooperation, one needs anonymised and distributed data sharing. Such methods take advantage of the nature of inter-firm supply chains, as collections of agents who seek both profit and longevity. Using the previous chapter’s model of cash-constrained dynamic trade and financing in a supply network [257], I investigate the effect of such distributed functionality on supply chain survivability.

This model also integrates work in previous chapters in decentralisation in telecoms and applies it to supply chain financing, expressed in the form of nexus supplier risk measures [323].

This chapter solves the problem of financial squeeze on disadvantaged companies via real-time policy sharing, shown through simulations, measuring effects on long-term supply chain survival and viability. I propose to share policies as a function of the policies of agents visible in a local region — or ego-network — around given agents. This is achieved via limited range broadcasting of financial policies which dynamically change according to dynamic ego-network data. This does not reveal private information.

The hypothesis is that limited visibility used to determine scope of collaboration for financing in a complex adaptive system can solve issues of long-term survivability. Given that this control function generates collaborative intelligence with limited visibility, it is designed to mitigate against critical

failure. A simulation approach is used to directly study survivability, which is most affected by this function, and thus viability.

Since survival time is the most appropriate measure of viability [167], I also introduce the methodology to construct the Long-Term Viability Measure (LTVM) and, subset on it, the Indefinite Viability Measure (IVM).

The chapter opens with Section 8.2, which outline the methods used in this study. First, the network simulation dynamics are introduced in Subsection 8.2.1, explaining how the network behaves over time, building upon the methods of the previous chapter. Then, in Subsection 8.2.2, the method of constructing the network topology used in this study is explained. Following this, in Subsection 8.2.3, the mechanism of financial control in the network is explained, elaborating upon the previous chapter with a collaborative control mechanism. Then, the experimental and analytical procedure is presented in Subsection 8.2.4 showing how time series information is decomposed and studied to understand long-term and viability dynamics in the study. Results are listed in Section 8.3, detailing initial conditions and testing parameters. Results are presented for both topology types studied, with regression analyses for each on relevant variables. What these results mean and how they come about is discussed in Section 8.4. Conclusions are delivered in Section 8.5, discussing implications of results both conceptually and in practical or business applications, with mention of limitations. Finally, contributions of this chapter are summarised in Section 8.6.

8.2 Methods

8.2.1 Network Simulation Dynamics

This investigation is conducted using a modification of the simulated environment from Chapter 7 [257], where a network, $G = (V, E)$ of suppliers interconnects acyclically. This initiates on a complex network and has sufficiently distinct dynamics so as to explain in depth here. Connected to the nodes without out-degree is a raw materials node, r , and to those without in-degree, a market node m . From m , every timestep t , a quantity of demand is generated by the market node according to a Poisson distribution, such that the demand at timestep t is defined as $d_m^t \sim \text{Poisson}(\lambda)$. A node u upstream of m , such that $(m, u) \in E$, is randomly selected according to its market share, the function $f(\mathcal{P}_u)$, and that demand is propagated upstream to u . An example of this is shown in Fig. 8.1. The market share of u is based on its power, $\mathcal{P}_u = \{s^*, m^*, l^*\}$, or “small”, “medium”, and “large”, respectively [248].

The node u has a quantity of available inventory, i_u^t . A purchase order is made by the market equal to $p_m^t = \min(d_m^t, i_u^t)$. A buyer u uses the relative power between u and seller v to determine how many timesteps δ_{uv} before u pays v its owed cash amount, stored for v as a receivable $R_v^{t+\delta_{uv}}$, and for u as a payable $P_u^{t+\delta_{uv}}$. The $\mathcal{P}$ and δ relationship is shown in Table 8.1. Between a company and m or r , there is no delay, so for any $u, v \in V$, $\delta_{mu} = \delta_{vr} = 0$.

Each edge, including those attached to m or r , has a material unit value parameter, s_{uv} , such that for a sale from u , it gets cash $c_u^t = P_v^t s_{uv}$, added to its receivables, updating with $R_v^{t+\delta_{uv}} = R_v^t + \delta_{uv} + c_u^t$,

		$\mathcal{P}_u$		
		s'	m'	l'
$\mathcal{P}_v$	s'	60	90	120
	m'	30	60	90
	l'	30	30	60

Table 8.1 Table of invoice terms where buyer u with power $\mathcal{P}_u$ pays supplier v with power $\mathcal{P}_v$ exactly δ_{uv} days late after v makes a delivery to u .

while losing stock equal to the purchase order. The buying firm loses the same amount of cash, added to its payables with $P_v^{t+\delta_{uv}} = P_v^t + \delta_{uv} + c_u^t$. The market node, m , has infinite cash, while the raw materials node, r , has infinite stock.

Nodes use this cash to make further purchases upstream of their suppliers. They also use it to pay for operative expenses, o , paid as a fixed fee per timestep, and for delivery expenses e , which are incurred as a fee for every unit sold, so firms with less activity incur lower costs, while those with higher activity incur higher costs.

On the next timestep, whatever is left in u is propagated upstream to a supplier, v , according to the same rules. Nodes will source materials from the raw materials node, by issuing orders to r . A company, u , is removed from the system when their current liquid cash, $\kappa_u^t = R_u^0 - P_u^0 \leq 0$ and their total receivables are less than their total payables, where $\sum_{t=0}^{120} R_u^t \leq \sum_{t=0}^{120} P_u^t$. Information is stored only up to 120 days to the future. When a company is removed, all its edges are also removed, and the relative market shares of competing companies are recalculated using market shares of remaining companies.

Every timestep, receivables and payables are updated such that $R_u^{t+1} = R_u^t$ and $P_u^{t+1} = P_u^t$, modelling the passing of a single day.

To avoid bankruptcy while satisfying payments to partner firms and operational expenses, companies may seek external financing to increase their liquidity at the cost of long-term cash flow.

One option is bank financing, which is constrained by debt ρ_u^t and the prior cash position multiplied by the banking mandate, which is a scale factor dependant on company power. When a company receives bank financing b_u^t , it adds it to both to κ_u^t and to ρ_u^t (where $\rho_u^0 = 0$), where b_u^t is limited with

$$\rho_u^t + b_u^t \leq \begin{cases} 2\kappa_u^{t-1}, & \mathcal{P}_u = s; \\ 3\kappa_u^{t-1}, & \mathcal{P}_u = m; \\ 4\kappa_u^{t-1}, & \mathcal{P}_u = l. \end{cases}$$

The bank financing increases current liquid cash at the loss of future earnings. This loss is expressed as companies needing to pay loans back in full, with interest, β , as a payable, where $P_u^{t+120} = P_u^t + (1 + \beta)b_u^t$. Because this reduces cash over the long-term, companies seek the minimum quantity of financing to bring their current cash, κ_u^t , to a given threshold, f_u^t , which works as a safety

cash stock. I apply the decentralised collaborative intelligence control method upon the choice of the financing threshold.

If bank financing is not enough due to the debt limit, where $\kappa_u^t + b_u^t < f_u^t$, then a node uses SCF. It reduces receivables a fixed time in the future τ to increase liquid cash by that amount, discounted by a factor of γ . For supply chain financing, s_u^t , where $\kappa_u^{t'} = \kappa_u^t + s_u^t$, where $s_u^t \leq R_u^{t+\tau}(1 - \gamma)$ and $R_u^{t'+\tau} = R_u^{t+\tau} - s_u^t(1 + \gamma)$.

The following section contains a description of how to construct the network topology for study in a complex deep tier acyclic network.

8.2.2 Complex Network Topology

The $G(N, p)$ ER random graph model is used, visualised in Fig. 8.1. This is treated as a base case, capturing a long chain with complex unstructured design. This however does not capture the heterogeneous degree distribution that real supply chains have.

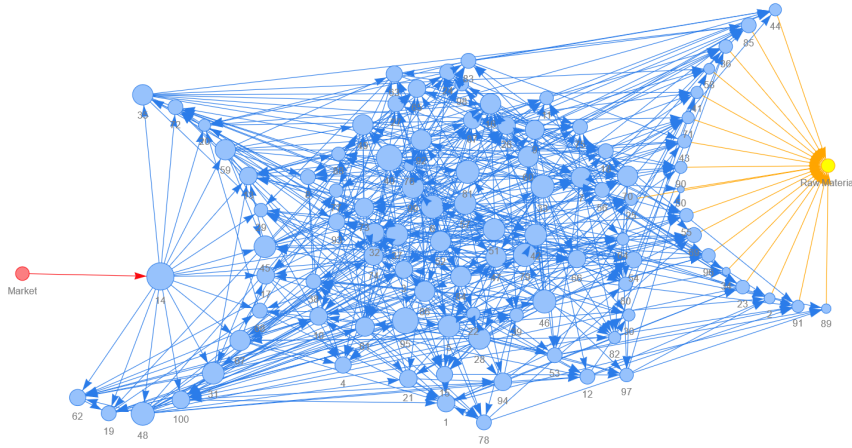


Fig. 8.1 Visualisation of a random, Erdős Rényi, $E(n, p)$ graph topology with structural modifications to represent a deep tier supply chain. $n = 100$, $p = 0.1$, and diameter, $\nabla = 7$.

I compare the ER, which is random complex, with the scale-free BA model, which is structured complex [27], visualised in Fig. 8.2. It is defined by $BA(N, m)$. Both the ER and BA models produce an undirected graph, which is converted into a directed acyclic graph (DAG) to retain a crisp notion of tiers to determine buyer-supplier relationships and the direction of cashflow.

To create the DAG, a unique total ordering is applied to the set of nodes, so each pair of nodes is uniquely comparable. The chosen ordering is the mean distance of each node to all others, l_u . When there exists an undirected edge $(uv) \in E$ and $l_v > l_u$, replace it with a directed edge (uv) , so edges point to nodes closer to all others in the network. If $l_v = l_u$ then follow a preassigned random index ordering and use the directed edge (uv) . This preserves the total ordering because the index ordering is also a total ordering, enveloped by the mean distance total ordering. This creates a DAG [289].

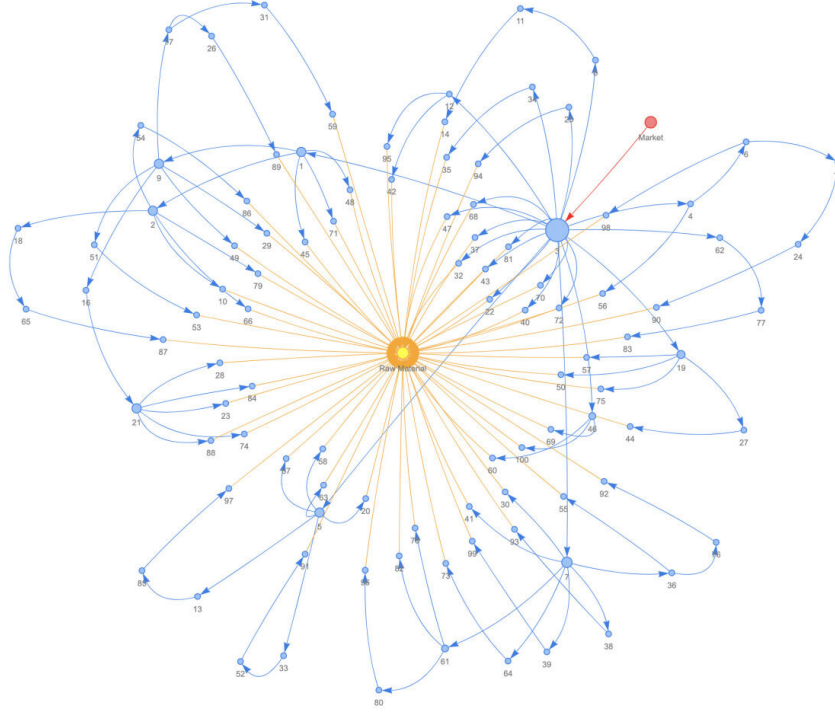


Fig. 8.2 Visualisation of a scale-free, Barabási-Albert, $BA(n, m)$ graph topology with structural modifications to represent a deep tier supply chain. $n = 100$, $m = 3$, $\nabla = 3$.

Then m is attached to all nodes with in-degree 0, and r to all nodes with out-degree 0, so nodes most closely connected to the network are focal nodes, while those at the fringes produce the wide collection of basic goods, such as in an automotive manufacturing network.

The power of nodes depends on the number of nodes in the same tier, $|\mathcal{T}_u|$. For the sets $\mathbf{S} \ni s^*$, $\mathbf{M} \ni m^*$, $\mathbf{L} \ni l^*$, for variables $\hat{s}, \hat{m}, \hat{l} \in [0, 1]$, where $\hat{s} + \hat{m} + \hat{l} = 1$, fix a proportion of the network to be small, medium or large, such that $|S| = \hat{s}|V|$, $|M| = \hat{m}|V|$, $|L| = \hat{l}|V|$. Tiers of nodes, denoted $\mathcal{T}_u$, are equal to the length of the shortest directed path from the market node m to the node u in the starting configuration. Then I fix the starting cash position of nodes $\kappa_u^0 = 50|\mathcal{T}_u|$. Then, create an ordering of all nodes $(u_1, \dots, u_n)$ where $|\mathcal{T}_{u_1}| \leq \dots \leq |\mathcal{T}_{u_n}|$.

Power is fixed to nodes such that for all $u \in \{u_1, \dots, u_{|L|}\}$, $\mathcal{P}_u = l^*$; for all $v \in \{u_{|L|+1}, \dots, u_{|L|+|M|}\}$, $\mathcal{P}_v = m^*$; and for all $w \in \{u_{|L|+|M|+1}, \dots, u_n\}$, $\mathcal{P}_w = s^*$. This means that nodes in thinner tiers have greater power. Prices are then iteratively assigned over edges. Starting from r , moving backwards along edges, the price over edge $s_{ur} = 1$. For antecedent edges, $s_{vu} = \max_{\{w: (uw) \in E\}} (s_{uw}) + 1$, thus ensuring all sales are profitable with respect to purchases. Simulations in this model were run over these DAGs.

8.2.3 Decentralised Collaborative Finance Threshold Control

The financial threshold function is informed by centrality measures, utilising nexus supplier theory. Conditioning for other factors, this dictates that less embedded companies have less influence than others on a given company in the same supply chain. This embedding is from the perspective of the given company, and thus exists within its ego-network. In a supply chain, I define this to be the collection of companies the node is sharing any sort of information with, and circumscribe it by an information boundary, restricted by their visibility. The information boundary is defined according to geodesics such that a node u with visibility i shares information over $H_i(u)$.

Companies broadcast their financing policy f_u^t to each other when they share an ego-network. A node u combines these with its record of current expenses, P_u^0 , and weighs each value by some measure of centrality of the given node with respect to the ego-network of u . A node's centrality takes network information concerning that node and returns some quantification of importance to the network, compared to other nodes. To prevent over-sensitivity to current conditions, time series are smoothed over a moving average with time window, $\hat{w}$. For a given node, u , and some centrality measure, $C(u; H_i(u))$, this translates to

$$\hat{f}_u^\tau = C(u; H_i(u))P_u^0 + \sum_{v \in H_i(u)} C(v; H_i(u))f_v^\tau$$

$$f_u^t = \frac{\sum_{\tau=t-\hat{w}}^t \hat{f}_u^\tau}{\hat{w}}$$

When the sum of centralities equals 0, only the information contained in H_0 is used.

Four measures that any node u uses to determine f_u^t are examined. All four are measured so as to not bias results to any single benchmarking method. These are:

- Degree centrality, which is the total number of linked edges a node has, denoted C_d
- Betweenness centrality, which measures a ratio of paths through a node versus those that work around it, denoted C_b
- Eigencentrality, which measures the influence of a node, as an element in the eigenvector of the adjacency matrix, denoted C_e
- Closeness centrality, which measures a function of the distance of the node to all others, denoted C_c [125, 15].

8.2.4 Experimental and Analytical Procedure

The survival time of supply network simulations is measured using the simulation model with nexus financing thresholds, run over the topologies as described above. This is repeated over multiple simulations, where each is initialised in a given configuration, over a space of configurations. This includes the initial graph topology, visibility, centrality measure, λ , and e . Each supply network has a

diameter, ∇ , which is the length of the shortest geodesic in the graph topology, including the market and raw materials nodes. Visibility for each simulation is capped from above by the given ∇ , since if it is increased beyond ∇ for any graph, all nodes retain the same collaborative networks. Thus, I normalise according to ∇ for each graph to map each visibility to the relative visibility (RV) in the graph. This allows comparison of outcomes between graphs with different ∇ .

This returns histograms conditioned on RV, which is used to study the multimodality of system survival times, visible upon learning the probability distribution of survival times, as in Chapter 7 [257]. Using kernel density estimates (KDEs) [267, 243], as in Chapter 5, I non-parametrically estimate the probability distributions of these histograms. For the smoothing parameter, or bandwidth, h_D , of the KDE of a dataset D generated from a given simulation, a tuned version of the Silverman rule of thumb [285] is used, where,

$$h_D = 0.9 \min \left(\sigma_D, \frac{\text{IQR}_D}{1.34} \right) |D|^{-\frac{1}{5}} \tilde{h},$$

where for D , IQR_D is the interquartile range, σ_D is the standard deviation, and $|D|$ is the number of entries in the dataset. The tuning parameter is $\tilde{h}$, and is constant across each D for a given topology type.

To study the manner in which extended communication mitigates around failure cascades, breakpoint analysis is applied, specifically the binary segmentation algorithm [24] for a single breakpoint, B_D , onto the KDE of D . Parts of the KDE before B_D are denoted B_D^{pre} , categorised as short-lived, and those after B_D are denoted B_D^{post} , categorised as long-lived. I thus distinguish between outcomes of short-term and long-term surviving systems.

Meaningful clusters are analysed by conducting modality analysis on B_D^{pre} , similar to Chapter 7 [257], in this case done by counting prominent peaks. A peak, (x, y) , denoting coordinates, is a local maximum, and given an adjacent local minimum, (x', y') , its prominence is $\bar{x} = y - y'$. A prominent peak is where $\bar{x} > \Xi$, where the prominence threshold Ξ is a constant across each D . The number of prominent peaks of D is denoted Λ_D .

For given Ξ , $\tilde{h}$ is selected to maximise Λ_D , constrained such that $\min_D(\Lambda_D) = 1$, where $\tilde{h}$ is chosen to 5 decimal places of precision. This was chosen to fit the KDEs for each D in such a manner that maximises the size of the modality space, improving how precise the analysis of the richness of the survivability space is.

With this, the set of long-term survivors can be determined as all elements of the dataset with survival time greater than B_D , which was derived using the steps above, such that $D(B_D^{\text{post}}) = \{t \in D : t \geq B_D\}$. Simulations are capped at $t = 1000$ if they reach it without collapse, and record this as their survival time.

The variable of most interest is the RV, and is measured against the following pair of viability metrics:

- First is the LTVM, which is the proportion of survival times beyond the breakpoint. Given a space of simulations, X , it is defined as $\text{LTVM}^X = \frac{D(B_D^{\text{post}})}{D(0)}$.
- Second is the IVM, which is restricted to the proportion of survival times at timestep $t = 1000$. Given a space of simulations, X , it is defined as $\text{IVM}^X = \frac{D(1000)}{D(0)}$.

The LTVM is distinguished from the IVM because the survival times at $t = 1000$ include both legitimate failures at the timestep and all companies that would survive but were cut off due to us capping simulation times from above. For practical purposes, a long-term supply chain may be more desirable for temporary measures, while an indefinitely surviving supply chain may be more desirable for permanent institutional chains. The relationship of RV to Λ_D , is also investigated, seeing how the complexity of the survival space changes as this value is varied. Polynomial regression analysis is conducted on each of these sets of outputs, studying their gradient if linear, and conducting detailed curve analysis of inflections and notable points. The data itself is examined and compared with the regressions, studying P -values.

8.3 Results

The following contains results for the analysis, where RV_D of the supply chain is varied with respect to diameter as the input variable, and the output measures as the LTVMs, IVMs, and Λ_D .

Experiments were conducted over the space of model parameters in Table 8.2. Simulations were completed on an Armari Magnetar workstation, with a 3.5GHz Intel Xeon E5-1620 CPU, 64GB RAM, in Windows 10 Pro. Modelling and results were completed in Python 3 with the `os`, `pandas`, `tqdm`, `numpy`, `matplotlib`, `math`, `sklearn`, `itertools`, `pprint`, `art`, `simulation`, `networkx`, `random`, `sys`, `seaborn`, `statistics`, `time`, and `tabulate` packages.

8.3.1 Scale-Free Networks

First examined are the scale-free networks. These are parameterised such that $m = 1$ and $\delta \in \{4, 5, 6, 7\}$.

Fig. 8.3 shows the KDEs for each distribution of survival times conditioned on visibility for the scale-free graphs. Vertical lines denote B_D for each curve, colour-coded accordingly. It displays visibility curves rather than RV for visual interpretability. Table 8.3 contains the outputs of the various measures I investigate, conditioned against RV_D . The following subsections contain polynomial regression analysis of relevant pairs inside this table.

RV_D vs Λ_D^{SF} : For the scale-free diagrams, I study the relationship between relative visibility, RV_D and the number of peaks before the breakpoint, Λ_D . This has a 5th-degree polynomial, accurate with

Parameter	Values
Operation Fees o	1
Average Demand λ	5, 10, 15
Delivery Expenses e	0.25, 0.75
Loan Repayment Time	120
Firm Power Distribution	Heterogeneous
Financed	True
Paradigm	Proactive
Supply Chain Financing Annual Rates γ	20%
Bank Annual Interest Rates β	10%
Invoice Term τ	60
Window Size $\hat{w}$	60
Medium Power Market Share $f(m^*)$	3
Large Power Market Share $f(l^*)$	10
Topologies	ER Random, BA Scale-Free
Network Diameters ∇	4, 5, 6, 7
Network Size n	100

Table 8.2 Table of the parameters tested. Note, large power market shares that would be smaller than medium power market shares are not tested

$P = 0.027$ and standard error $\sigma_{\hat{D}} = 0.414$. The regression curve (Fig. 8.4(a)) is defined according to:

$$\begin{aligned} \widehat{\Lambda}_D^{\text{SF}}(\text{RV}_D) = & -780.0891\text{RV}_D^5 + 1956.8451\text{RV}_D^4 - 1723.1095\text{RV}_D^3 \\ & + 627.9353\text{RV}_D^2 - 90.5196\text{RV}_D + 11.3497. \end{aligned}$$

RV_D vs IVM_D^{SF} : For the scale-free graphs, measuring relative visibility, RV_D , to proportion of all failures at $t = 1000$, IVM_D^{SF} , gives a 1st degree polynomial regression, accurate with $P = 0.001$ but a standard error, $\sigma_{\hat{D}} = 0.255$. The regression curve (Fig. 8.4(b)) is defined according to

$$\widehat{\text{IVM}}_D^{\text{SF}}(\text{RV}_D) = 0.1032\text{RV}_D - 0.0026,$$

where $\widehat{\text{IVM}}_D^{\text{SF}}(1) = 0.1006$.

RV_D vs $\text{LTVM}_D^{\text{SF}}$: For the scale-free graph, measuring the proportion of values past the break point, $\text{LTVM}_D^{\text{SF}}$, gives significance, with $P = 0.041$ and $\sigma_{\hat{D}} = 0.451$, for a 4th order polynomial regression that. The curve (Fig. 8.4(c)) is defined as

$$\begin{aligned} \widehat{\text{LTVM}}_D^{\text{SF}}(\text{RV}_D) = & 0.0051 - 0.5797\text{RV}_D + 4.8083\text{RV}_D^2 \\ & - 8.1847\text{RV}_D^3 + 4.1274\text{RV}_D^4. \end{aligned}$$

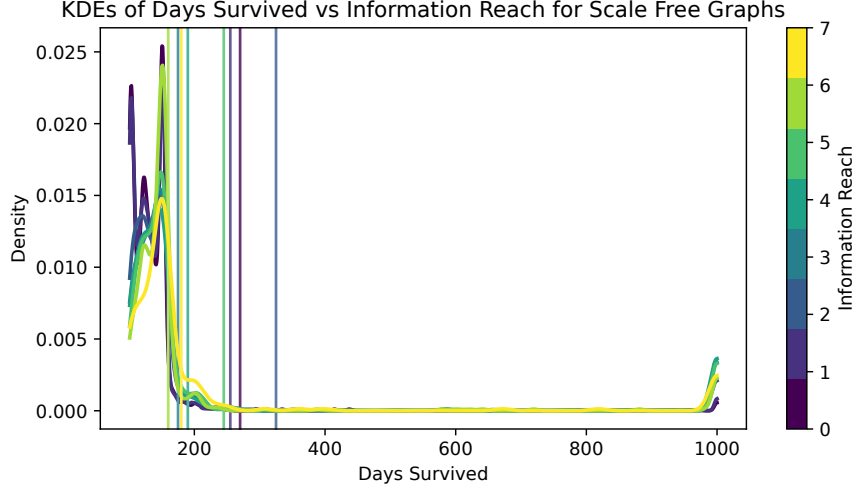


Fig. 8.3 KDEs of survival time per visibility for scale-free graphs, with vertical lines denoting breakpoints separating short-lived supply chains and long-lived supply chains

$(RV_D; RV_D \neq 0.4286)$ vs $LTVM_D^{SF}$: The regression (Fig. 8.4(d)) is defined according to

$$\widehat{LTVM_D^{SF}}(RV_D; RV_D \neq 0.4286) = -0.0172 + 0.1863RV_D,$$

and $P = 0$ and $\sigma_{\hat{D}} = 0.017$.

8.3.2 Random Networks

Next are the random networks, parameterised such that $(p, \delta) \in \{(0.1, 7), (0.2, 6), (0.2, 5), (0.3, 4)\}$.

Fig. 8.5 shows the KDEs for each distribution of survival times conditioned on visibility for the random graphs. To investigate the minimal quantity of remaining viability, I investigate the distributions of viable networks (Table 8.4). The following subsections contain polynomial regression analysis of relevant pairs inside this table.

RV_D vs Λ_D^R : The analysis is repeated for random graphs, comparing Λ_D^R to RV_D . The polynomial regression (Fig. 8.6(a)) delivers an 8th degree polynomial, accurate with $P = 0.031$, with $\sigma_{\hat{D}} = 0.425$. It is defined as

$$\begin{aligned} \widehat{\Lambda_D^R}(RV_D) = & -207895.4284RV_D^8 + 828388.3627RV_D^7 - 1358506.6894RV_D^6 \\ & + 1182795.9815RV_D^5 - 586913.5003RV_D^4 + 164780.6428RV_D^3 \\ & - 24049.5475RV_D^2 + 1395.0947RV_D + 10.0570. \end{aligned}$$

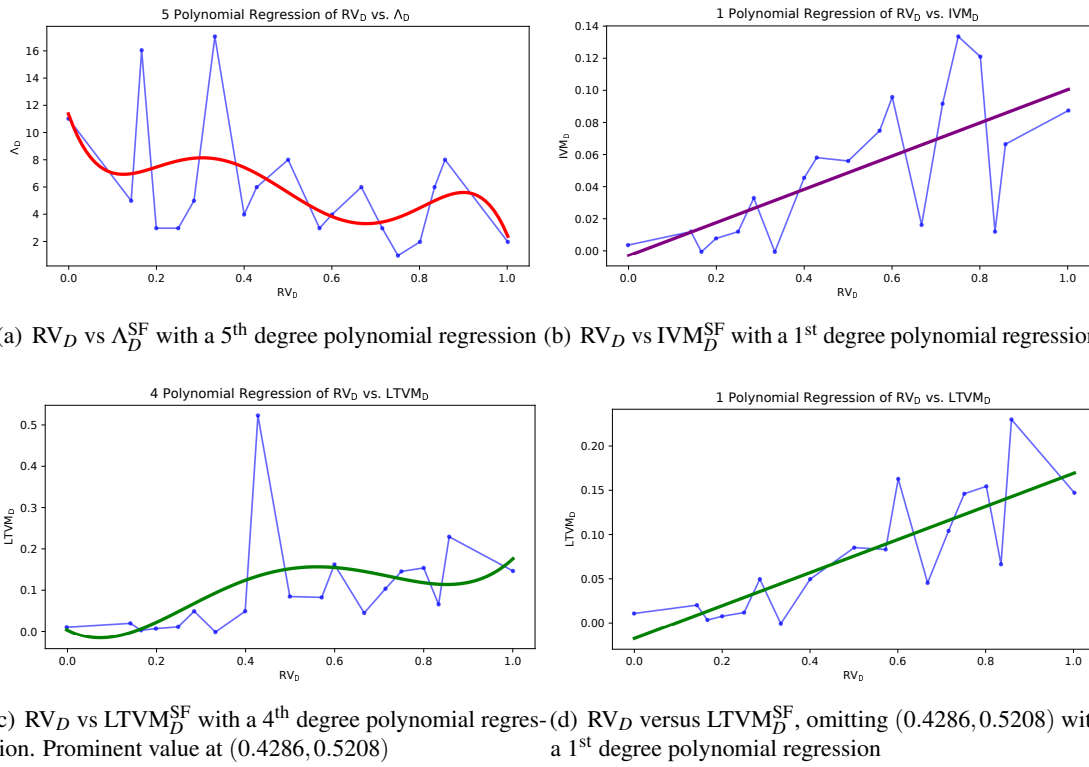


Fig. 8.4 Regressions of RV_D to Λ_D^{SF} , IVM_D^{SF} , and $LTVM_D^{SF}$ for scale-free networks

Table 8.3 All outputs investigated for the scale-free topologies

RV_D^{SF}	B_D^{SF}	Λ_D^{SF}	IVM_D^{SF}	$LTVM_D^{\text{SF}}$
0	270	11	0.0042	0.0115
0.1429	230	5	0.0125	0.0208
0.1667	280	16	0	0.0042
0.2	200	3	0.0083	0.0083
0.25	195	3	0.0125	0.0125
0.2857	315	5	0.0333	0.05
0.3333	375	17	0	0
0.4	340	4	0.0458	0.05
0.4286	150	6	0.0583	0.5208
0.5	225	8	0.0563	0.0854
0.5714	375	3	0.075	0.0833
0.6	190	4	0.0958	0.1625
0.6667	190	6	0.0167	0.0458
0.7143	260	3	0.0917	0.1042
0.75	335	1	0.1333	0.1458
0.8	205	2	0.1208	0.1542
0.8333	170	6	0.0125	0.0667
0.8571	160	8	0.0667	0.2292
1	180	2	0.0875	0.1469

Table 8.4 All outputs investigated for the random topologies

RV_D^{R}	B_D^{R}	Λ_D^{R}	IVM_D^{R}	$LTVM_D^{\text{R}}$
0	310	10	0.001042	0.003125
0.142857	570	18	0	0
0.166667	270	5	0.004167	0.016667
0.2	540	9	0	0
0.25	740	4	0	0
0.285714	395	21	0	0
0.333333	485	6	0	0
0.4	370	3	0.008333	0.008333
0.428571	775	16	0	0
0.5	355	1	0.0125	0.016667
0.571429	430	15	0	0
0.6	315	4	0.004167	0.0125
0.666667	650	5	0	0
0.714286	220	5	0.004167	0.016667
0.75	290	5	0.004167	0.0125
0.8	255	4	0.004167	0.0125
0.833333	565	6	0	0
0.857143	225	10	0	0.008333
1	365	5	0.004167	0.005208

RV_D vs IVM_D^{R} : I measure relative visibility, RV_D , to proportion of all failures at $t = 1000$, IVM_D^{R} , giving a 6th degree polynomial regression (Fig. 8.6(b)), accurate with $P = 0.05$ exactly, but a standard error, $\sigma_D = 0.473$. The curve is defined according to

$$\widehat{IVM_D^{\text{R}}}(RV_D) = -2.2411RV_D^6 + 6.9729RV_D^5 - 8.0382RV_D^4 + 4.1713RV_D^3 - 0.9332RV_D^2 + 0.0711RV_D + 0.0011.$$

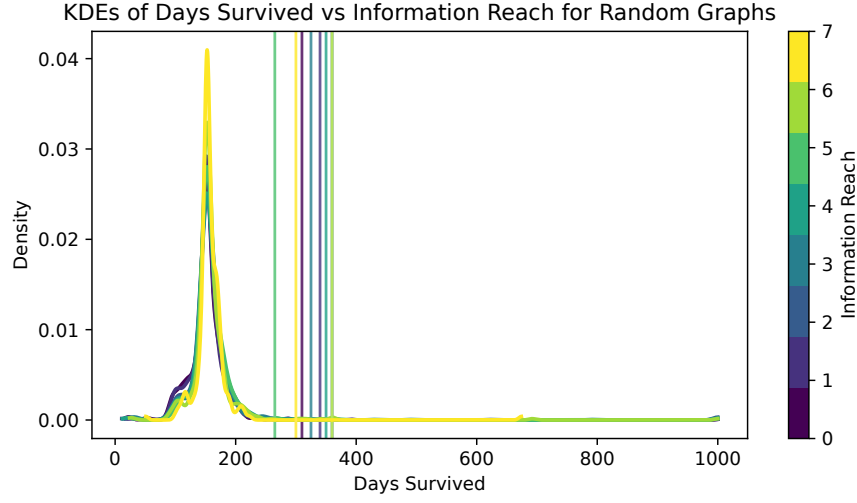


Fig. 8.5 KDEs of survival time per visibility for random graphs, with vertical lines denoting break-points separating short-lived supply chains and long-lived supply chains

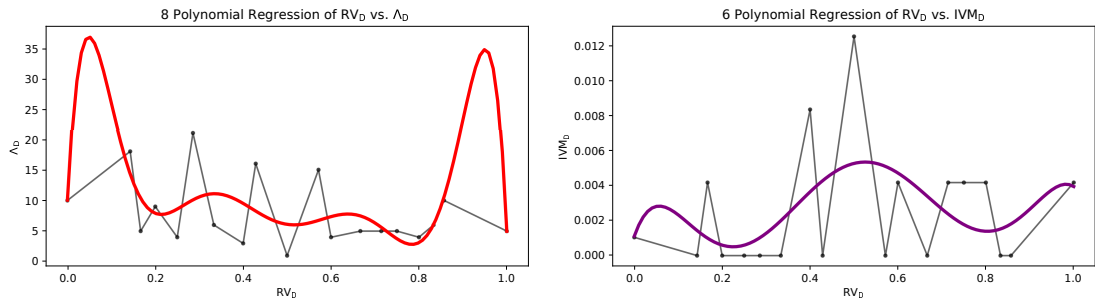
RV_D vs $LTVM_D^R$: I measure RV_D vs $LTVM_D^R$, giving a 7th degree polynomial (Fig. 8.6(c)) in this case, with $P = 0.040$, and $\sigma_{\bar{D}} = 0.449$, defined by:

$$\begin{aligned} \widehat{LTVM_D^R}(RV_D) = & 28.7872RV_D^7 - 102.8713RV_D^6 + 147.1856RV_D^5 \\ & - 107.4140RV_D^4 + 41.8254RV_D^3 - 8.1099RV_D^2 \\ & + 0.5993RV_D + 0.0029. \end{aligned}$$

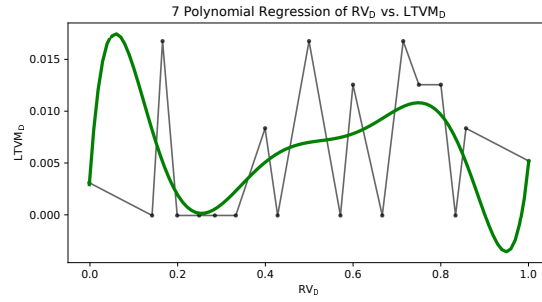
8.4 Discussion

In this chapter, I introduced a novel method of scalable decentralised control to maximise viability in a deep tier firm-to-firm upstream supply chain that uses limited visibility collaboration and nexus theory to autonomously generate financing thresholds to ensure optimal liquidity across the supply chain. These thresholds were used to inform supply chain financing, generating policies for adjusting cash position with financing based on company dynamics. It was found that this scalably generates viable supply chains, thus demonstrating the capacity for shared value generation in joint production and financial networks.

The test environment was complex deep-tier supply chain financing network topologies, generated using a novel method to construct theoretical complex acyclic firm-to-firm upstream supply networks. Then a testing methodology was introduced that uses dynamic complex network analysis and novel simulation-based metrics of viability to study the impact of this method. These metrics measure viability in proportion to simulations that survive past either a data-derived threshold (for finite viability) or up to a preselected simulation termination threshold (for indefinite viability). Analysis of the behaviour prior to the finite threshold was also conducted.



(a) RV_D vs Λ_D^R with an 8th degree polynomial regression (b) RV_D versus IVM_D^R with a 6th degree polynomial regression



(c) RV_D vs $LTVM_D^R$ with a 7th degree polynomial regression

Fig. 8.6 Regressions of RV_D to Λ_D^{SF} , IVM_D^{SF} , and $LTVM_D^{SF}$ for random networks

Regarding the relationships between output pairs, generalised curves were generated to be able to project particular behaviours to less formalised network topologies. This methodology preserved irreducible nonlinearities, examining the effects of varying visibility (and thus collaborative ranges) for this control method on long-term dynamics at both finite and indefinite scales. Random and scale-free supply networks were tested, basing off of known findings that supply networks are scale-free [55, 56, 318, 194].

The ER random complex network model was used to create random networks as a baseline (Fig. 8.1), and the BA scale-free complex network for the scale-free, structured networks (Fig. 8.2), each a classic generator of their respective topologies. A scale-free topology has equal degree distribution at all scales of magnification, and is structured. It is suggested that supply networks tend to have a scale-free topology [55–57, 319, 194, 229, 294].

Extending on Proselkov et al. [257], which had lattices and therefore no bottlenecks, I incorporated bottlenecks by using DAG topologies which are not trees, thus studying branches that are thinner than others. The model accounted for this by introducing the delivery expense parameter, e , into this simulation, affording branches greater security as their expenses matched their activity as controlled by their bottlenecks.

8.4.1 Scale-Free Networks

Following Fig. 8.3, for the scale-free topologies shows a rough region with high density before B_D , a long plateau with some noise, and then a prominent inflection towards and at $t = 1000$. This shows that imparting a local collaboration function to all members of the supply network, with variable visibility, also imparts viability to the whole supply network.

In the case of the scale-free topologies, in Fig. 8.4 per RV_D , and accounting for what may be outliers, long-term viable networks and indefinitely viable networks occur with approximately equal frequency. In addition, viability shows a positive correlation with visibility, observable by the angles of the regressions in Fig. 8.4(b), Fig. 8.4(c), Fig. 8.4(d). However, while regression is a useful tool to generalise findings, because data was collected over multiple network configurations and topologies, one cannot assume data variations are simply due to noise. Thus I see there is a nonlinear relationship between visibility and viability. Of note, $RV_D = 1$ does not maximise either viability measure. Centralised control is not only unscalable with network size, but distributed control, using limited range collaboration, yields superior viability of all types.

For the scale-free topologies, nonlinearities in the curves suggests target viabilities can be satisfied with smaller, possibly more attainable visibilities. IVM_D^S can reach nearly 14% (Fig. 8.4(b)) while $LTVM_D^S$ can reach over 52% (Fig. 8.4(c)), or over 22% omitting the prominent value (Fig. 8.4(d)). Solutions to such problems may depend on which sort of viability is more desirable. This is because variations for each viability are such that the globally maximal RV_D for the $LTVM_D^S$ does not equal that for the IVM_D^S , and local maxima or minima for each viability hold different coordinates.

The peak count in the scale-free topologies, Λ_D^{SF} , shows a negative incline with RV_D (Fig. 8.4(a)). Peaks of Λ_D^{SF} are local maximum likelihoods of survival time, appearing due to clustering. Such clustering denotes lower variability, and greater RV_D increases variability in survival times for short-lived supply networks. This shows reduction in peak counts for short-lived networks corresponds to reduced clustering in failure times of short-lived networks and a growth in viability. With further research to map this relationship, it may be thus possible to predict viability from the distribution of short-term survival, either from comparable networks or shorter-run simulations.

8.4.2 Random Networks

Fig. 8.5 shows curves that are much flatter and ordered than the corresponding region for the scale-free graphs, and an order of magnitude smaller, inducing a very small peak at the ends of the KDEs. Short-lived supply networks show strong clustering around a single point, suggesting a bottleneck. Therefore, random supply networks are most likely to break down at a single point, regardless of visibility, and have a very low, near-zero likelihood of being viable.

The max point reached by the random data is $(0.5, 0.0125)$, such that $\max IVM_D^R = 0.0125$. $\max IVM_D^{SF} / \max IVM_D^R = 10.664$, suggesting that scale-free structure is needed for the information contained through collaboration to have a strong impact (Table 8.4). Otherwise, the principles of relevancy do not hold, as in the random graph case. Other topologies with similar properties to the scale-free topology, such as hierarchy or scaling, may also enable viability.

Troughs in Λ_D^R tend to correspond to peaks, respectively, in either $\widehat{IVM}_D^R$ or $\widehat{LTVM}_D^R$ when at least one viability measure contains at that region a particularly prominent cluster, however the inverse is not necessarily true (Fig. 8.6(a)). This suggests that for random graph topologies, for their low viability, achieving maxima depends on minimal peaks. This follows the argument as in the scale-free graphs, which states greater variety in short-term configurations suggests more possible ways to achieve viability by passing the viability threshold situated at B_D . I propose this nonlinear relationship, thresholded by viability cluster prominence, is due to the choice of RV_D giving appropriate dependency to the structured elements of the random graphs, generated by the nodes m and r . This explains the irregularity and low viability associated with it.

Note that none of the regressions show any general incline, however the IVM_D^R performs best at $RV_D = 0.5$, no matter how small this value is (Fig. 8.6). The ER graph was constructed to show no scaling or hierarchical properties, however my operation to create a supply network DAG alters this, introducing such structure. Given the structuring imposed upon the network by the assignment of the market node m and the raw materials node r , it may be the case that any association in random graphs between visibility and indefinite viability is a function of the majority of nodes having both m and r visible. I propose 0.5 of the diameter is when this occurs. This requires further investigation.

8.5 Conclusions

Findings in this chapter show that companies can collaborate over financial policy with only nearby relevant partner to deliver better warning of changes both up and downstream. They achieve this by moderating their effective range of visibility, valuable because risk mitigation requires holistic views [42]. I therefore investigated the effect of varying visibility in this model. This, using information about real time partner dynamics, created a well-informed bidirectional mitigation function and tunable form of distributed control through supply chain financing. Thus, ripple effects were mitigated against, increasing resilience and survivability of the whole supply chain.

I found scale-free topology permits higher viability, and random does not. Generalising dynamics via polynomial regression and excluding possible outliers shows linear viability growth with visibility. However, studying outputs themselves suggests limited visibility and cooperation can outperform maximum visibility, such that distributed control outperforms centralised control.

I also found high sensitivity to visibility with exceptional performance of certain configurations. Due to the experimental procedure, these may not be outliers, suggesting opportunities to research precision supply network topology engineering for viability maximisation, facilitated by the research environment developed in this chapter.

This is a scalable method which increases the likelihood of long-lived supply chains, achieving supply network viability [167]. This is because agent dynamics have been coupled via financing without removing agency, since as information is shared, it is recontextualised and simplified as to how it pertains to the company sharing it [255].

The use of nexus information allowed nodes to dynamically adapt policies according to structural changes, thus being able to react agilely, be resilient to negative events and quickly recover, surviving over long-term global disruptions. This method was applied to create a viable supply chain, as defined in Ivanov [165].

Positive relationships between visibility and viability suggest the importance of collaboration, while the nonlinear phases suggest the value of doing so with only relevant partners. Further research into choice for collaborative groups would yield valuable insights into what these relevant partners are, which would enable greater viability and give information for business operations. This may have insights into scopes of impact for determining spatial motifs in the network — such motif analysis may be core to transforming agent-based dynamics into dynamical systems equations, though nonlinearities in the visibility-viability relationship would make explicitly doing so difficult, but relevant partner selection may inform this.

I also found that, in scale-free networks, there is a negative association between clustering in the unviable supply chain space and the size of the viable supply chain space. Less clustering suggests less dependency on key risks since network failure in the system results from ripple effects. Ripple effects are the result of shocks from particular network regions, propagating at a particular rate, so failure time is a function of the time of shock instigation and ripple propagation rate. The less this is clustered, the more variety in either of these factors, which must be due to variety in network

responses to shock. As this increases with visibility, this control function reduces determinism by imparting greater agency for firms, empowering players in the supply chain, which results in greater viability. More possible outcomes means a supply chain is more likely to survive past the viability threshold and thus in the long-term. I thus propose that supply chain managers, to maximise viability, should seek to maximise the variety of possible supply network configurations during the lifespan of the supply network by involving supply chain players as much as possible in determining dynamic supply chain control policy.

These methods used only current information, while Proselkov et al. [257] showed that using information from further into the past can have negative consequences on predictive capability (though it may have also been a consequence of the forecasting method). This suggests that nexus based metrics can outperform metrics that use temporal information. For policymakers, this means they need not focus on making historic information centrally available, rather on facilitating peer-to-peer communication or transaction platforms [312, 242]. In fact, given this, restrictions placed on visibility of actors to preserve privacy and competitiveness can be advantageous to operations.

8.6 Summary of Contributions

In this chapter, I proposed a scalable autonomous control system by introducing a collaboration layer, which provides a mean of extended communication reach for distributed controllers via local, intersecting decentralised control clusters. Through this was shown that local cooperation enables long-term survivability, and thus is a means to improve supply chain viability.

An intercommunication layer is added to the control layer of the model introduced in Chapter 7, which imparts nodes with decentralised local, scalable control functions according to their local topology. This creates a five-layer network simulation model which can be used to investigate the effects of introducing distributed decentralised financial control systems into a given empirical supply chain.

Two new measures of viability based on simulations were also introduced, focusing on long-term dynamics. These are defined in contrast to measures of short-term dynamics, which is explored using analysis. These provide means of defining autonomous control architecture, and ways to measure its effectiveness.

A collection of findings in the field of supply chain control were found:

- Distributed control with local decentralised cooperation to give extended communication reach introduces a very small amount of viability in random networks
- Random networks need more visibility for indefinite viability than for finite viability, but this can be limited
- Random networks are very sensitive to viability, with high variability

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- Distributed control with local decentralised cooperation to give extended communication reach introduces large amounts of viability in scale-free networks
 - Little visibility is needed for substantial viability in scale-free networks
 - Viability in scale-free networks reaches orders of magnitude above random networks.
 - Scale-free topology enables viability. 1 order of magnitude for the IVM and 2 for the LTVS versus random topologies.
 - In scale-free topology, peak count predicts for viability, which can be developed as an estimator of long-term dynamics.

Chapter 9

Conclusions

This chapter gives conclusions regarding the work presented in this thesis. In this thesis, a unified methodology was developed to describe, model, and investigate the dynamics of complex infrastructural networks which are autonomously controlled for criticality by distributed agents with information bounds. Cases were defined, and of these, simulations were run, from which was generated simulated data. This was used to investigate variations of the hypothesis that, by correctly restricting information to whatever is spatiotemporally relevant in a complex dynamic system, substantial accuracy and speed can be achieved in computing and delivering KPIs, to the point that methods of dynamic autonomous control become scalable and effective. The data was found to prove this hypothesis.

9.1 Research Outline

The research question driving this thesis was “How can information requirements for scalable autonomous control of infrastructural networks be reduced?”. It had several aspects that were completed to answer it. I defined a model of explicit failure propagation in telecoms; a method of network changepoint detection; and a model with both a telecom data packet routing layer and an intercommunication layer to model both functional and communication dynamics, with failure conditions. I then defined a model of supply chain dynamics with abstracted supply chain financing distributed control, controlling routing according to a heuristic of internode power and liquidity, and studying the space of the model’s adaptive network dynamics. This had a distributed control layer that mitigated against breakdown via dynamic cash injection. Building on this model, I modified it to function over general complex network topologies, and to regulate the distributed control with a decentralised local collaboration layer, combining work from the rest of the thesis. This culminated in a tractable model of distributed scalable autonomous infrastructural complex network control.

Telecoms and Analytics In the telecom models, risk is expressed as a function of temporal criticality. In the telecom dynamics model, the time boundary was based off of empirical measures obtained through time series analysis of whole simulation dynamics. I defined measures of decentralised

dynamic criticality, each requiring information that limits their use to certain cases. They measured impact in a dynamic system. The timescales of this impact were limited, and these time boundaries were investigated in different ways.

In the failure propagation model, the time boundary was based off of theoretical definitions governing failure propagation within the model.

This was further studied using changepoint detection methods over empirical datasets, using network information to learn time epochs over the whole system.

Using each model's respective time boundary, time series of ground truth criticality were defined for each model. These were used to compare decentralised information bounded criticality estimates (IB-CMs).

To achieve decentralised measures that use current dynamics, I used centrality based CMs for each node, and to permit scalability I bounded information for each node, creating the IB-CMs. In the failure propagation model, I tested the similarity of measures specialised to measure criticality, against accuracy. In the telecom model, I tested the relative value of limiting or not limiting information boundaries for accuracy and compute speed, using instead standard centrality measures for generalisability, creating value of information measures in decentralised analytics systems.

Supply Chain and Control Following on from information valuation in telecoms, I designed a model of supply chain and supply chain finance dynamics. This was analysed in detail to characterise the model with respect to survivability, studying the influence of different model agent types on its dynamics, and how failure was distributed in this case. Within this model, risk was controlled for by cash injection when firms go below a certain dynamic threshold, creating a distributed control function.

Given a concrete understanding of this model, it was further elaborated with a decentralised distributed control layer. This was studied in complex networks models, focusing on the relationship between information reach and viability, measurable because of the distributed decentralised control function. Here, risk is still measured as a function of criticality, measured with local IB-CMs. Novel measures of long-term viability were introduced and used to assess the performance of the model under different control paradigms.

9.2 Summary of Contributions

In this thesis, I researched the behaviours and structures of scalable methods for dynamic failure risk management in complex network infrastructures. I identified that such methods require each aspect of this problem to be broken down and solved in combination. My novel contribution is the combination of the methods and principles that each solve an intersection of the problems of failure risk management, scalability, complex network infrastructures, and dynamicity, all in a format that is

researchable, such that the underlying mechanics and dynamics can be understood. The contributions described here are matched to the themes described in Figure 1.7.

Researchability I studied failure risk management using simulation models to prevent undue harm to human welfare. To this end, four models of simulated infrastructural systems dynamics were proposed: scalability is measured against validation techniques to determine effectiveness of methods; complex network infrastructures are modelled using multi-agent systems; and dynamicity is managed with ad-hoc and post-hoc analytics methods designed to be representative of applied systems.

Failure risk management In scalable systems, decentralised and distributed control, and analytics techniques are used; to study it in complex network infrastructures, nexus theory for centrality measures was applied to understand the relationship between analytics and their physical system interpretations; and in a dynamic context, the method this describes is autonomous control.

Scalability Effective scalable methods of any sort in a dynamic context are determined by sufficiently encapsulating the scope of relevant information; and in complex network infrastructures, require information limitation.

Dynamicity The dynamic behaviours in complex network infrastructures are modelled using dynamic networks.

By interlacing each of these techniques and principles, I generated novel methods and insights. The theoretical background underpinning the combination of these principles was discussed in Chapter 2 (RO1).

My research question was answered in the particular case of dynamic control against network risk, that is, viability, of infrastructural complex networks. Information bounded network control and analytics solutions using distributed and decentralised architectures were proposed and tested on bespoke multilayer agent-based network simulation models designed to exhibit complex nonlinear dynamics that can lead to endogenously generated system failure, but with nonlinear relationships of dynamics between agents.

In Chapters 4, 5 and 6, analytical methodologies were studied using telecom models as a case study. This case was selected for the restriction it places on information and communication dynamics, such that timeliness and distance of provided information for analytic agents can be investigated in a simulation environment. In Chapters 7 and 8, control methodologies were studied using supply chain financing models. This case was selected for the monetary dynamics tied to material unit value, where cash can be used as a control lever. As financial control can be generalised, financial control measures have less bias, enabling reliable investigation into control topologies.

Throughout this thesis, novel dynamic criticality and viability measures, as well as analytical constructions, were proposed to measure the efficacy of these architectures, with respect to their

operation under limited information, expressed as a study into the value of relevant information given the context of the dynamics investigated.

Two models of telecoms infrastructure dynamics were defined: one an augmented SIS model (Chapter 4); and the other based off of a routing model, adding an intercommunication layer (Chapter 6). The former represents the propagation of data packets, which contribute linearly to congestion. On this, a collection of bespoke criticality measures were defined (**RO3**), their effectiveness at estimating network criticality measured according to a theoretical measure of expected criticality, defined according to theoretical propagation rates (**RO4**).

A method of interpreting and measuring criticality in an information limited, distributed manner was proposed (Chapter 4), compared against three decentralised measures using network topological metrics of criticality and centrality, studied within a telecoms infrastructure network (**RO3**). These are particularly useful for informing the prioritisation of maintenance, repair, or reinforcement of an asset, where sooner expected failures with greater network impact need to be addressed sooner.

In Chapter 4, a way to isolate criticality from a single node was defined using dynamic network analysis (**RO2**). The impact timeframe was defined as a novel function of the theoretical propagation rate, as according to a mean field approximation of the cascade dynamics of the network (**RO4**). This can be used to estimate impact thresholds in empirical models. From the dynamic information imparted by this, a method for determining the network's true criticality was developed, with this compared to three novel dynamic IB-CMs that estimate local criticality. These are each dynamic modifications of Wehmuth, Local, and Bridging centrality, of which Local was found to be most accurate (**RO3**).

A dynamically defined decentralised measure of system criticality was generated out of their dynamic outputs, showing which IB-CM was most accurate. This was done according to a novel means of measuring the accuracy of such a decentralised measure against true criticality (**RO4**). Thus was defined a 2-layer multiplex network model with an analytics layer, applicable to telecoms infrastructural systems for determining criticality in simulation based dynamic prediction (**RO2**).

In Chapter 5, investigation of changepoints of epochs of a telecom backbone infrastructural dataset showed that they are irregular over the course of a day (**RO2**). A pair of decentralised methods were defined, and characterised the dynamics of a day, the information of which can be applied in further simulation model research and control systems (**RO3**). Findings were shown to closely track the work and rest periods of human activity, suggesting a way to measure national productivity.

These methods and findings not only contribute to understanding irregular changepoints for epochs in telecoms and infrastructural networks, but also demonstrate the effectiveness of kernel density estimation for changepoint detection to remove human bias, which is valuable for computational agent driven control and analytics.

This non-constant behaviour of infrastructural systems motivated the construction of the second telecoms analytics model. In Chapter 6, a model was defined where congested data packets propagate according to nodal interaction from a telecom data packet routing model, and nodes communicate

their state over the infrastructural network, such that congestion propagates according to emergent dynamics, while node perceptions of network states are topology dependant (**RO2**). Several communication paradigms that describe communication dynamics of agents computing approximations of live decentralised criticality were explained further. This includes: instant communication, with no lag; constant communication, with lag but updates are every timestep; and periodic communication, with lag and updates only on centrality computing timesteps. A novel measure of the value of the information used to obtain an IB-CM was defined, based on the accuracy of the new decentralised dynamic measure of criticality IB-CM over each CP, combined with the timeliness of calculation (**RO4**).

In this second case, using an empirically derived dynamic criticality benchmark, findings show that if one intentionally designs a dynamic IB-CM to be specific for capturing criticality, it can exhibit remarkably high accuracy, and that simple to compute IB-CMs provide the best trade-off between accuracy and compute speed. Specifically, the weighted degree centrality showed the highest value of information, and did so for the periodic communication paradigm (**RO4**). However, they are not necessarily the most accurate, depending on the rate of communication within the system. This suggests that fitness for purpose can drastically affect the performance of a given IB-CM.

The functionality-oriented dynamics of the routing layer and the second layer of intercommunication that uses network structure gives an asymmetric precedence to more central nodes due to the greater currentness of their information, as a function of their topology. Thus their information is more valuable as it is up-to-date. If they were controllers, all else being equal, these agents would have more power to affect the network.

To investigate control, in Chapter 7, I defined a new model, where agent functionality depends on dynamics of flow units, such that they have some value. This value is represented as cash, which has its own dynamic in the system. In this new model, which represents supply chain financing dynamics, agents are now able to make decisions to capture this value, enabling the investigation of this thesis' research question.

Specifically, this is a deep-tier supply financing model, where agents have asymmetric power (**RO2**). This is a multilayer agent-based model of supply chain financing dynamics, specifically of cash-constrained demand-driven material flow, where cascade failure in the form of the ripple effect is mitigated against by autonomous distributed financing control (**RO3**). Failure cascades are generated by financial bullwhips, triggered by late payments due to power imbalances among firms. The dynamics of such cascades were illustrated for two deep-tier topologies: a lattice (constant tier width); and diamond (thin tiers at the ends of the supply chain). These operate and break down as complex adaptive systems.

The effects of varied financial conditions, market demand conditions, and power imbalances upon supply network cascade failure were found in the context of this simulation model. Furthermore, the distributions of survival time under different conditions of different network objects, including firms

and the supply chain itself were described. It was found that survival time is strongly clustered to be early and that longer survival time is conditioned for by the homogeneity of tiers and firm power.

This forms a four-layer network model, where material flow is its own layer, driven by demand in its layer, and controlled by cash in another layer. The fourth layer is made of the financial controllers situated at agents, though the network at this layer is disjoint (**RO2**). This contributes to the theory of multilayer systems control — by imparting proactive financing as the control function, the system lifespan tends to extend, however, without interactivity of the controllers that captures network structure, the network derived dynamics are not suitably captured by controllers (**RO3**). This motivated the fourth network model of this thesis.

Building on the work presented in this thesis, a scalable autonomous control system was proposed in Chapter 8. This is achieved by introducing a collaboration layer connected to the control layer, where edges exist where agents communicate their target functions, edges being attached following geodesic distance within the infrastructural network layers (**RO2**). This provides a means of extended communication reach for distributed controllers via local, intersecting decentralised control clusters, thus a dynamic of communication is introduced, but the complexity of communication remains fixed for each agent as the network changes size (**RO3**). This model showed that by introducing control functions that operate as their own network dynamic linked to infrastructural dynamics, autonomy becomes possible.

Methods of measuring long-term and indefinite survivability using dynamic network analysis of a given supply network topology were defined, which were used to assess the long-term survival capacity of the control paradigm introduced there (**RO4**). Through this, it was demonstrated that local cooperation enables long-term survivability, and thus is a means to improve supply chain viability — a system's capacity to survive into the future (**RO4**). Together, these provide means of defining autonomous control architecture and ways to measure its effectiveness.

This unifies the research themes of my thesis, by studying local criticality methods in information asymmetric environments over an autonomous supply chain financing system, while also decomposing epochs — at macro rather than micro timescales. A form of scalable distributed autonomous control was proposed, where the interplay of the boundaries of decentralisation and distribution in that context is explored, accounting for the challenges of communication with distant partners, utilising the principle of information relevancy such that the resultant network viability is actually improved through information limitation.

9.3 Limitations

Investigations in this thesis are defined by the limitations of studying critical failure of infrastructures. Computational capacity was one limiting factor, where investigations were conducted over single machines. To achieve richer results, higher compute capacity must be applied to be able to test larger and more complex models in tractable times. This includes empirical datasets, of which rich ones are

becoming more available [252]. This may involve cloud or other remote computing, which would require redesigning code to permit multi-agent, dynamic analytics in such environments.

Changes would be necessary for multi-machine studies. Motivation for this includes the fact that results on the value of information may be a consequence of the experimental procedure using simulation over a singular machine, and may require retesting over an emulated environment.

While Chapter 4 investigated node failure, the rest of the thesis investigates normal function. Thus, further work is needed to integrate the methodology of Chapter 4 and learn criticality to simulate critical node failure and its effects, in isolation, of the other models.

The heterogeneity of computation times among nodes was also not investigated, rather studying this KPI in aggregate, despite its presence. Methods will need to be developed to study this effect, which may affect monitoring interval selection, and give further findings on digital control scalability.

Another observation was that the value of information across all CEs and CPs in the telecoms models is very low. The overall value of information for distributed measures should be investigated, since it is likely, given they further reduce computational load of any single asset and shorten computation reach, they would effectively increase the value of information in a computational system. Methods to compute value of information for distributed measures, in ways that are not network-centric, must be created to confirm this.

The telecoms chapters studied cyclic graphs, while the supply chain studies did not, constructing DAG chains. Models must be developed that represent supply dynamics in cyclic systems to generate results more representative of reality.

Scalable autonomy in infrastructural systems is not a singular state, but occupies a spectrum of functions, as they are embedded into systems [319]. There is therefore no singular solution to implementing it, but a space of them. This means that the abstract and simulated environment studied within this thesis, while giving insights into how scalable autonomous systems would function, do not complete the picture. This brings me to the future works, and ways to extend beyond the studies within this thesis.

9.4 Broader Impact and future works

Researchers can use the methods and findings in this thesis to conduct further studies into network risk (particularly criticality), as metrics to measure performance in industrial applications of network risk management, and as fundamental architectures to help manage long-term risk.

Further work will apply the models and measures proposed in this thesis to a spectrum of network topologies, to develop a theory on the impact of network structure on dynamic measure accuracy for those models that have not incorporated this. The theoretical effects defined in this thesis may be used as benchmarks, and different performance of different topologies can be compared according to the outputs already found. These would need measures of network structure to account for the different contexts — for example, simulations can be run for other topologies, with outputs scaled

or conditioned according to some measure of systemic network risk, such as Ricci curvature [269], though this is but one example.

Three novel, distributed measures of criticality were introduced in this thesis, and can be used to predict the behaviour of dynamic processes occurring over a network. Their performance is compared and tested on a simulated diffusive data transfer network, demonstrating the distributed dynamic criticality measures are highly accurate at predicting the accumulation of data packet loads within a network. These measures can be useful in predictive maintenance and routing for telecommunications and autonomous vehicles, as well as informing businesses of partner criticality in supply networks.

These novel measures of criticality can be applied to advanced simulation models or empirical use cases, and new ones can be developed according to the same principles. For example, simulations built from the logistical simulation software, AnyLogic, may have their network dynamics measured according to the criticality estimates defined in this thesis to study their dynamic criticality.

The analytics frameworks for telecoms, developed in this thesis, describing methods of live information bounded data collection and criticality estimation within a simulation environment, can also be applied to other systems with dynamic flow and independent cognitive agents.

For example, in business networks, for some supervisory body, such as a peer-to-peer lending platform, the criticality of potential debtors can be detailed on the platform to potential lenders to help inform their decision. This may be especially valuable in a blockchain enabled system where information is decentrally stored, thus reducing computational and informational overhead.

Alternatively, it may be used for predictive software maintenance of a telecoms infrastructure to help deliver updates while maintaining service more effectively. The short delivery and implementation times of algorithmic updates, as well as the fast pace of global code development [111] make this a valuable use case. For this, criticality measures, which have been designed to minimise computational load, can quickly give dynamic measures that can be incorporated into master functions that automate code delivery. Higher criticality can inform a nonlinear choice function that contextualises risk versus demand and service requirements, and with this, a computational system may choose to automatically implement a given update.

Generally, these measures can be implemented into any system with an infrastructural service requirement, where assets in the network mutually rely on one another to provide this service, and low-latency can give major advantages to operation. In both defined cases and any others that satisfy these conditions (for example, water delivery networks in urban environments), the advantage of a simulated space — if expertly redeveloped to represent system dynamics — is that measures need not only be applied momentarily, but the value of information can be more easily assessed for designing new features of the analytical architecture. This would be achieved by simulating system dynamics, and measuring similarity to the benchmark measure within the simulation, and can thus help determine if limited scope measurement is an appropriate policy.

In the value of information study with the realistic telecoms data packet routing model, it was found that information is most valuable in the periodic paradigm, suggesting diminishing returns in

reducing information latency versus improving handling of limited information. This justifies further decentralisation of control functions of telecoms infrastructures. For such decentralised systems, such as edge computational networks, for example, autonomous vehicle networks [61], centrality estimates should be used as either a computational input variable or a computationally light and human-comprehensible variable for human-integrated autonomous systems. Decentralised computing of this sort is being completed in cases such as secure inter-domain routing based on blockchain [216] — metrics generated through centrality estimation, as defined in this thesis, may be introduced and calculated, informing smart contract systems while inducing low computational burden or lag.

Results on the value of information can be adapted and extended to be used for financial-cost-based data traffic rerouting algorithms by converting the value of information measure into a real financial value, with this mapping developed in collaboration with operations specialists. This would introduce an information-based valuation measure that extends my supply chain work in this thesis.

This study simulated normally operating networks. Further research will simulate critical node failure. It is expected that this will introduce variability in computation times, which may affect monitoring interval selection. Future work will also use advanced CEs [256] and produce case specific, data derived CEs for maximum relevancy. Using these for network control functions will allow one to learn the value of information function by removing measure dependency. This will be achieved through the validation methods in this thesis, measured against a performance metric. This can give useful findings in studies of network homophily and tools for policymakers when constructing or designing networks with communication.

These results combined suggest that increased heterogeneity in both topology and characteristic powers, while improving the relative survivability of larger companies, overall reduces systemic survivability. Since large companies fail after small ones, this may explain the large company risk skew — their advantages undermine the system, whose survival dominates theirs. Further investigations into decentralised collaboration for different levels of agent heterogeneity will shed more light on this effect. This could be achieved by extending the existing study to incorporate heterogeneous materials and agent functions. For results to be meaningful, they should be distributed realistically.

Results on supply chains suggest that focal company supply chain managers should both seek to uplift the cash positions and promote the growth of their suppliers to improve their own growth while minimising predatory late payment tactics. This closely follows existing literature [266], but that the results hold at a complex systems level further reinforces them as essential for the long-term resilience of supply chains. Organising supply chain financing according to reverse factoring is one way to satisfy both the liquidity requirements of focal firms and suppliers, though care must be taken that financing remains accessible to all members of the chain in equal measure [156].

In reactive systems, all functions depend on cash available within the system at the present time. Greater delays in repayment were found not to directly change total cash, but rather to change its distribution. I conclude that predictive functions depend on distribution of cash, and more specifically, work better when there is a stronger coupling between material and cash flows. Longer terms also

mean larger fees upon repayment, overall reducing total system cash, so managers should make sure to avoid increasing their liquidity as much as possible through late payments, while policymakers may wish to deliver incentives for earlier payments to allow for this.

Policymakers may take notice that higher rates for alternative financing may actually improve supply chain resilience in supply ecosystems where tiers are highly asymmetric, shown by the comparative performance of the diamond to the lattice supply networks. Supply chain managers may work together with policymakers to support them in increasing tier flexibility by adding new suppliers and buyers at tiers with few options to protect their supply chain and keep financing widely accessible.

Findings also show that for low growth supply ecosystems with high interchange ability, high bank loan interest rates reduce survivability, so policymakers should try to either induce the creation of more key suppliers and economic growth through the support of disruptive technology ([83]) or suppress bank rates, possibly with reduced regulation ([58]).

Managers must engage in some degree of predictive behaviours to optimise profits, but evidence shown here suggests that limited solutions can actually be detrimental. Managers must ensure that they are interpreting data correctly, which would be aided by greater data sharing and communication between partner firms, and greater investment in supply analytics.

Results on the relationship between predictive methods and information gathering in this context can also be used to inform when and how to construct supply chain digital twins or any other representative models for maximising resilience and viability. This can be achieved by conducting value of information analysis [210], and using outputs from simulations into functions within the analytical framework.

The supply chain financing model represents supply networks using network modelling, however modern supply chains are often better described as intertwined supply networks (ISNs) ([170]) that are complex with heterogeneous nodal functions. Material and financial flow (with demand flow) are in opposite directions, and nodes are buyer-suppliers, so by a measure my model is an ISN, however these are respectively single directional flows. Further work may include loops of similar flows in complex topologies. Further heterogeneity would be achieved through information imbalances and equipping actors with limited visibility, as in [256], used to inform proactive financing methods in line with the low-certainty-need supply networks concept ([169]). This visibility could include mutual information sharing within a local region, for example, using the principles illustrated by [281] of local value generation from collaboration, but applied to price setting rather than product creation. Proactive financing could be redefined to follow time series forecasting methods or optimise to certain functions, such as the cash-to-cash cycle [154] restricted by other functions, such as the liquidity coverage ratio ([30]).

I have proposed a means of making autonomous control feasible via limited range communication, which is of value for systems engineers and policymakers of systemic infrastructures. I have also provided a basis for further work to learn associations between short-term dynamics and long-term

ones such that the long-term viability measures and indefinite viability measure can be utilised empirically by introducing notions of the boundary of long-term and short-term network survival.

Future research using dynamic network analysis may be on the temporal clustering of nodal failures, and its relationship with information reach within a given topology. In the supply chain case, this may give insights into how visibility affects risk propagation by applying telecoms literature on “burstiness” [232]. This would impart insights into how visibility affects the dynamics of localised cascades.

The particular outputs for the scale-free networks suggest there are phases to how visibility affects viability, so particular choices of visibility can optimise for viability more efficiently. I propose this is due to synchronisation of financial shock propagation and information propagation for the control response. I am currently researching network theoretical representations of the multispectral nature of these two propagating effects [92]. This will permit true optimisation for viability, opening up new possibilities for business network management.

Furthermore, there is a current trend, post-COVID-19, towards regionalisation of supply chains [28, 136]. My findings, combined with evidence of the interconnectedness of risk in the global financial system [199, 70], suggest this is conditioning against viability, since it restricts capacity for full supply network coverage with limited cooperation while retaining the ability for ripple effects to propagate over the financial system. I thus recommend to policymakers against this trend, and to move towards reintegration with interlocking collaboration regions. To investigate this further, empirical research is needed to infer viability in real systems as well as richer modelling research. For modelling, I propose multilayer financial and supply chain models [293] with disconnected supply chain layers to study the effect of supply decoupling without financial risk decoupling.

Further research should investigate more advanced financial thresholding functions that appropriately incorporate both historic and current dynamics so as to maximise supply chain viability [261, 152, 153, 156], and to determine precise boundaries of each to generalise into other fields.

I studied forms of scale-free and random network topologies, however by the method of constructing the DAG chains, these may not have been truly random networks. This may justify the need for a way to meaningfully simulate cyclic graphs, which is the focus of ongoing further work.

A notable feature of my results is that they show trends, but have particular sensitivities to changes in visibility. This paves the way for further research, which if this simulation were applied to an empirical supply chain, tuned to follow its dynamics, then meaningful perturbation analysis could be conducted to determine various metrics about its quality. This could then form the basis for determining how to structure a limited visibility information sharing scheme for the purpose of supporting empirical supply chain viability from a financial perspective. This would pave the way to directly applicable scalable autonomous control solutions for real-life infrastructural systems.

References

- [1] Abbasi, M. and Varga, L. (2022). Steering supply chains from a complex systems perspective. *European Journal of Management Studies*, 27(1):5–38.
- [2] Acharya, V. V., Pedersen, L. H., Philippon, T., and Richardson, M. (2017). Measuring systemic risk. *Review of Financial Studies*, 30(1):2–47.
- [3] Agca, S., Babich, V., Birge, J. R., and Wu, L. (2021). Credit Shock Propagation Along Supply Chains: Evidence from the CDS Market. *Management Science*, 0(0).
- [4] Ahuja, B. K., Magnanti, T. L., and Orlin, J. B. (1993). *Theory, Algorithms, and Applications*. Prentice hall, Upper Saddle River, New Jersey.
- [5] Albert, R. and Barabasi, A.-L. (2002). Statistical mechanics of complex networks. *Reviews of Modern Physics*, 74(1):47–97. arXiv: cond-mat/0106096.
- [6] Albert, R., Jeong, H., and Barabasi, A.-L. (2000). Error and attack tolerance of complex networks. *Nature*, 406:378–382.
- [7] Ali, M. M., Babai, M. Z., Boylan, J. E., and Syntetos, A. A. (2017). Supply chain forecasting when information is not shared. *European Journal of Operational Research*, 260(3):984–994.
- [8] Aliev, H., Kim, H., and Choi, S. (2020). A Scalable and Secure Group Key Management Method for Secure V2V Communication. *Sensors*, 20(21):6137.
- [9] Alirezaie, M., Hoffman, W., Zabihi, P., Rahn timer, H., and Pentland, A. (2024). Decentralized Data and Artificial Intelligence Orchestration for Transparent and Efficient Small and Medium-Sized Enterprises Trade Financing. *Journal of Risk and Financial Management*, 17(1):38.
- [10] Allen, F. and Gale, D. (2000). Financial Contagion. *Journal of Political Economy*, 108(1):1–33.
- [11] Allen, P. M. and Varga, L. (2006). A co-Evolutionary Complex Systems Perspective on Information Systems. *Journal of Information Technology*, 21(4):229–238.
- [12] Altfi (2019). Peer-to-Peer Lending: State of the Market. Technical report, Altfi.
- [13] Aminikhanghahi, S. and Cook, D. J. (2017). A survey of methods for time series change point detection. *Knowledge and Information Systems*, 51(2):339–367.
- [14] Amiri, Z., Heidari, A., Navimipour, N. J., and Unal, M. (2023). Resilient and dependability management in distributed environments: a systematic and comprehensive literature review. *Cluster Computing*, 26(2):1565–1600.
- [15] Anderson, W. N. and Morley, T. D. (1985). Eigenvalues of the Laplacian of a graph. *Linear and Multilinear Algebra*, 18(2):141–145.

- [16] Anthony Jnr, B. and Abbas Petersen, S. (2021). Examining the digitalisation of virtual enterprises amidst the COVID-19 pandemic: a systematic and meta-analysis. *Enterprise Information Systems*, 15(5):617–650.
- [17] Antonelli, G. (2013). Interconnected dynamic systems: An overview on distributed control. *IEEE Control Systems*, 33(1):76–88.
- [18] Arenas, A., Diaz-Guilera, A., Kurths, J., Moreno, Y., and Zhou, C. (2008). Synchronization in complex networks. *Physics Reports*, 469(3):93–153. arXiv: 0805.2976.
- [19] Arlot, S., Celisse, A., and Harchaoui, Z. (2019). A Kernel Multiple Change-point Algorithm via Model Selection. *Journal of Machine Learning Research*, 20:1–56.
- [20] Arora, S. and Brintrup, A. (2021). How does the position of firms in the supply chain affect their performance? An empirical study. *Applied Network Science*, 6(1).
- [21] Ashraf, M. H. and Yalcin, M. G. (2024). Supply Chain “Flows” Management. In *The Palgrave Handbook of Supply Chain Management*, pages 341–362. Palgrave Macmillan.
- [22] Ayalon, O. and Toch, E. (2017). Not Even Past: Information Aging and Temporal Privacy in Online Social Networks. *Human-Computer Interaction*, 32(2):73–102.
- [23] Bai, G., Li, Y., Fang, Y., Zhang, Y.-A., and Tao, J. (2020). Network approach for resilience evaluation of a UAV swarm by considering communication limits. *Reliability Engineering & System Safety*, 193:106602.
- [24] Bai, J. (1997). Estimating Multiple Breaks One at a Time. *Econometric Theory*, 13(3):315–352.
- [25] Balchanos, M. G., Domercant, J. C., Tran, H. T., and Mavris, D. N. (2014). Metrics-based analysis and evaluation framework for engineering resilient systems. In *2014 7th International Symposium on Resilient Control Systems (ISRCS)*, pages 1–7. IEEE.
- [26] Barabási, A.-L. (2016). *Network Science*. Cambridge University Press.
- [27] Barabási, A.-L. and Albert, R. (1999). Emergence of Scaling in Random Networks. *Science*, 286(5439):509.
- [28] Barthe-Dejean, G. (2021). Shifting paradigms: Regionalisation and the post-COVID-19 risk matrix. *Journal of Risk Management in Financial Institutions*, 14:355–366.
- [29] Basel Committee on Banking Supervision (2000). Principles for the Management of Credit Risk. Technical report, Basel Committee on Banking Supervision.
- [30] Basel Committee on Banking Supervision (2013). *Basel III: the liquidity coverage ratio and liquidity risk monitoring tools*. Bank for Internat. Settlements.
- [31] Basel Committee on Banking Supervision (2017). High-level summary of Basel III reforms. Technical report, Bank for International Settlements.
- [32] Basel Committee on Banking Supervision. and Bank for International Settlements. (2006). *International convergence of capital measurement and capital standards : a revised framework, comprehensive version*. Bank for International Settlements.
- [33] Batagelj, V. and Zaversnik, M. (2003). An O(m) Algorithm for Cores Decomposition of Networks. *arXiv:cs/0310049*. arXiv: cs/0310049.

- [34] Battiston, S., Delli Gatti, D., Gallegati, M., Greenwald, B., and Stiglitz, J. E. (2007). Credit chains and bankruptcy propagation in production networks. *Journal of Economic Dynamics and Control*, 31(6):2061–2084.
- [35] Bavelas, A. (1948). A Mathematical Model for Group Structures. *Human Organization*, 7(3):16–30.
- [36] Bavelas, A. (1950). Communication Patterns in Task-Oriented Groups. *The Journal of the Acoustical Society of America*, 22(6):725–730.
- [37] Beaumier, G. and Cartwright, M. (2023). Cross-Network Weaponization in the Semiconductor Supply Chain. *International Studies Quarterly*, 68(1).
- [38] Beer, R. D. and Williams, P. L. (2015). Information Processing and Dynamics in Minimally Cognitive Agents. *Cognitive Science*, 39(1):1–38.
- [39] Behera, R. K., Rath, S. K., Misra, S., Damaševičius, R., and Maskeliunas, R. (2019). Distributed centrality analysis of social network data using MapReduce. *Algorithms*, 12(8).
- [40] Benveniste, L., Capozza, D. R., and Seguin, P. J. (2001). The value of liquidity. *Real Estate Economics*, 29(4):633–660.
- [41] Benzekki, K., El Fergougui, A., and Elbelrhiti Elalaoui, A. (2016). Software-defined networking (SDN): a survey. *Security and Communication Networks*, 9(18):5803–5833.
- [42] Bimpikis, K., Fearing, D., and Tahbaz-Salehi, A. (2018). Multisourcing and miscoordination in supply chain networks. *Operations Research*, 66(4):1023–1039.
- [43] Birge, J. R., Capponi, A., and Chen, P. C. (2023). Disruption and Rerouting in Supply Chain Networks. *Operations Research*, 71(2):750–767.
- [44] Birnbaum, Z. W. (1968). On the Importance of Different Components in a Multicomponent System. Technical report, WASHINGTON UNIV SEATTLE LAB OF STATISTICAL RESEARCH, SEATTLE.
- [45] Bisdikian, C., Branch, J., Leung, K. K., and Young, R. I. (2009). A Letter Soup for the Quality of Information in Sensor Networks. In *2009 IEEE International Conference on Pervasive Computing and Communications*.
- [46] Bloom, P. N. and Perry, V. G. (2001). Retailer power and supplier welfare. *Journal of Retailing*, 77(3):379–396.
- [47] Boccaletti, S., Latora, V., Moreno, Y., Chavez, M., and Hwang, D. (2006). Complex networks: Structure and dynamics. *Physics Reports*, 424(4-5):175–308.
- [48] Bodó, B., Brekke, J. K., and Hoepman, J.-H. (2021). Decentralisation: a multidisciplinary perspective. *Internet Policy Review*, 10(2).
- [49] Borgatti, S. P. and Everett, M. G. (2006). A Graph-theoretic perspective on centrality. *Social Networks*, 28(4):466–484.
- [50] Borgatti, S. P. and Li, X. (2009). On social network analysis in a supply chain context. *Journal of Supply Chain Management*, 45(2):5–22.
- [51] Borůvka, O. (1926). O jistém problému minimálním. *Práce Moravské přírodovědecké společnosti*, 3(3):37–58.

- [52] Braden, R. (1989a). Requirements for Internet Hosts – Application and Support.
- [53] Braden, R. (1989b). Requirements for Internet Hosts – Communication Layers.
- [54] Brauer, F., van den Driessche, P., Wu, J., Morel, J. M., Takens, F., and Teissier, B., editors (2008). *Mathematical Epidemiology*, volume 1945 of *Lecture Notes in Mathematics*. Springer Berlin Heidelberg, Berlin, Heidelberg.
- [55] Brintrup, A. and Ledwoch, A. (2018). Supply network science: Emergence of a new perspective on a classical field. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 28(3):033120.
- [56] Brintrup, A., Ledwoch, A., and Barros, J. (2016). Topological robustness of the global automotive industry. *Logistics Research*, 9(1):1.
- [57] Brintrup, A., Wang, Y., and Tiwari, A. (2017). Supply Networks as Complex Systems: A Network-Science-Based Characterization. *IEEE Systems Journal*, 11(4):2170–2181.
- [58] Buchak, G., Matvos, G., Piskorski, T., and Seru, A. (2018). Fintech, regulatory arbitrage, and the rise of shadow banks. *Journal of Financial Economics*, 130(3):453–483.
- [59] Callaway, D. S., Newman, M. E. J., Strogatz, S. H., and Watts, D. J. (2000). Network robustness and fragility: Percolation on random graphs. *Physical Review Letters*, 85(25):5468–5471.
- [60] Cao, B., Li, M., Liu, X., Zhao, J., Cao, W., and Lv, Z. (2021). Many-Objective Deployment Optimization for a Drone-Assisted Camera Network. *IEEE Transactions on Network Science and Engineering*, 8(4):2756–2764.
- [61] Cao, Z., Cheng, Y., Zhou, Z., Lu, A., Hu, Y., Liu, J., Zhang, M., and Li, Z. (2024). Patching in Order: Efficient On-Device Model Fine-Tuning for Multi-DNN Vision Applications. *IEEE Transactions on Mobile Computing*, 23(12):14484–14501.
- [62] Carmi, S., Havlin, S., Kirkpatrick, S., Shavitt, Y., and Shir, E. (2007). A model of Internet topology using k-shell decomposition. *Proceedings of the National Academy of Sciences*, 104(27):11150–11154.
- [63] Cassar, A. and Duffy, N. (2002). Contagion of financial crises under Local and global networks. In Luna, F. and Perrone, A., editors, *Agent-Based Methods in Economics and Finance*, volume 17, chapter 5, pages 111–131. Springer, New York, 1 edition.
- [64] Castellón, P. A. and Petit, C. (2008). Chain reaction or slow burn. *IBM Institute for Business Value*, pages 1–14.
- [65] Ceci, E. and Barbarossa, S. (2018). Small Perturbation Analysis of Network Topologies. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 4194–4198, Calgary, AB. IEEE.
- [66] Cetinkaya, E. K. and Sterbenz, J. P. (2013). A Taxonomy of Network Challenges. In *2013 9th International Conference on the Design of Reliable Communication Networks (DRCN)*, Budapest, Hungary. IEEE.
- [67] Chae, S. X., Choi, T. Y., and Hur, D. (2017). Buyer Power and Supplier Relationship Commitment: A Cognitive Evaluation Theory Perspective. *Journal of Supply Chain Management*, 53(2):39–60.
- [68] Chen, D., Lü, L., Shang, M.-S., Zhang, Y.-C., and Zhou, T. (2012). Identifying influential nodes in complex networks. *Physica A: Statistical Mechanics and its Applications*, 391(4):1777–1787.

- [69] Chen, F., Ryan, J. K., and Simchi-Levi, D. (2000). The Impact of Exponential Smoothing Forecasts on the Bullwhip Effect. *Naval Research Logistics*, 47(4):269–286.
- [70] Chen, N., Li, S., and Lu, S. (2023a). The extreme risk connectedness of the global financial system: G7 and BRICS evidence. *Journal of Multinational Financial Management*, 69.
- [71] Chen, N., Liu, X., and Yao, D. D. (2016). An optimization view of financial systemic risk modeling: Network effect and market liquidity effect. *Operations Research*, 64(5):1089–1108.
- [72] Chen, T. K., Liao, H. H., and Kuo, H. J. (2013). Internal liquidity risk, financial bullwhip effects, and corporate bond yield spreads: Supply chain perspectives. *Journal of Banking and Finance*, 37(7):2434–2456.
- [73] Chen, X., Wang, C., and Li, S. (2023b). The impact of supply chain finance on corporate social responsibility and creating shared value: a case from the emerging economy. *Supply Chain Management*, 28(2):324–346.
- [74] Chen, Y., Qi, D., Hui, H., Yang, S., Gu, Y., Yan, Y., Zheng, Y., and Zhang, J. (2023c). Self-triggered coordination of distributed renewable generators for frequency restoration in islanded microgrids: A low communication and computation strategy. *Advances in Applied Energy*, 10:100128.
- [75] Cheng, F., Niu, B., Xu, N., and Zhao, X. (2024). Resilient distributed secure consensus control for uncertain networked agent systems under hybrid DoS attacks. *Communications in Nonlinear Science and Numerical Simulation*, 129:107689.
- [76] Cheng, X., Hu, Y., and Varga, L. (2022). 5G network deployment and the associated energy consumption in the UK: A complex systems’ exploration. *Technological Forecasting and Social Change*, 180.
- [77] Cheng, Y., Olson, D. L., and Dolgui, A. (2021). Financing with preferential credit to coordinate the capital-constraint supply chain. *International Journal of Production Research*.
- [78] Cho, W., Ke, J. y. F., and Han, C. (2019). An empirical examination of the use of bargaining power and its impacts on supply chain financial performance. *Journal of Purchasing and Supply Management*, 25(4).
- [79] Choi, T. Y., Dooley, K. J., and Rungtusanatham, M. (2001). Supply networks and complex adaptive systems: Control versus emergence. *Journal of Operations Management*, 19(3):351–366.
- [80] Choi, T. Y. and Hong, Y. (2002). Unveiling the structure of supply networks: Case studies in Honda, Acura, and DaimlerChrysler. *Journal of Operations Management*, 20(5):469–493.
- [81] Choi, T. Y., Narayanan, S., Novak, D., Olhager, J., Sheu, J. B., and Wiengarten, F. (2021). Managing extended supply chains. *Journal of Business Logistics*, pages 200–206.
- [82] Chopra, S. and Sodhi, M. S. (2004). Managing Risk to Avoid Supply-Chain Breakdown. *MIT Sloan Management Review*.
- [83] Christensen, C., Craig, T., and Hart, S. (2001). The Great Disruption. *Foreign Affairs*, 80(2):80–95.
- [84] Cohen, R., Erez, K., ben Avraham, D., and Havlin, S. (2000). Resilience of the Internet to random breakdowns. *Physical Review Letters*, 85(21):4626–4628. arXiv: cond-mat/0007048.
- [85] Colias, M. (2014). GM gently squeezes its suppliers as collaboration in cost-cutting encouraged. *Automotive News*.

- [86] Cuñat, V. (2007). Trade credit: Suppliers as debt collectors and insurance providers. *Review of Financial Studies*, 20(2):491–527.
- [87] Cvetkovic, D., Doob, M., and Sachs, H. (1995). *Spectra of graphs: theory and application*. Academic Press, Michigan.
- [88] Cybenko, G. and Hallman, R. (2021). Resilient Distributed Adaptive Cyber-Defense Using Blockchain. In *Game Theory and Machine Learning for Cyber Security*, pages 485–498. Wiley.
- [89] D’Andrea, R. and Dullerud, G. E. (2003). Distributed Control Design for Spatially Interconnected Systems. *IEEE Transactions on Automatic Control*, 48(9):1478–1495.
- [90] de Jong, K., Panja, D., Karssenberg, D., and van Kreveld, M. (2022). Scalability and composability of flow accumulation algorithms based on asynchronous many-tasks. *Computers & Geosciences*, 162:105083.
- [91] Deering, S. E. (1989). Host Extensions for IP Multicasting. *RFC Editor*.
- [92] Dekker, M. M., Schram, R. D., Ou, J., and Panja, D. (2022a). Hidden dependence of spreading vulnerability on topological complexity. *Physical Review E*, 105(5).
- [93] Dekker, M. M., van Lieshout, R. N., Ball, R. C., Bouman, P. C., Dekker, S. C., Dijkstra, H. A., Goverde, R. M., Huisman, D., Panja, D., Schaafsma, A. A., and van den Akker, M. (2022b). A next step in disruption management: combining operations research and complexity science. *Public Transport*, 14(1):5–26.
- [94] Devadasan, P., Zhong, H., and Nof, S. Y. (2013). Collaborative intelligence in knowledge based service planning. *Expert Systems with Applications*, 40(17):6778–6787.
- [95] Dijkstra, E. W. (1959). A Note on Two Problems in Connexion with Graphs. *Numerische Mathematik*, 1:269–271.
- [96] Dinh, T. N., Xuan, Y., Thai, M. T., Park, E. K., and Znati, T. (2010). On Approximation of New Optimization Methods for Assessing Network Vulnerability. In *2010 Proceedings IEEE INFOCOM*, pages 1–9, San Diego, CA, USA. IEEE.
- [97] Dinitz, Y. (1970). Algorithm for solution of a problem of maximum flow in a network with power estimation. *Doklady Akademii Nauk SSSR*, 11(5):1277–1280.
- [98] Dodds, P. S., Watts, D. J., and Sabel, C. F. (2003). Information exchange and the robustness of organizational networks. *Proceedings of the National Academy of Sciences*, 100(21):12516–12521.
- [99] Dolgui, A. and Ivanov, D. (2021). Ripple effect and supply chain disruption management: new trends and research directions.
- [100] Dolgui, A., Ivanov, D., and Rozhkov, M. (2020a). Does the ripple effect influence the bullwhip effect? An integrated analysis of structural and operational dynamics in the supply chain. *International Journal of Production Research*, 58(5):1285–1301.
- [101] Dolgui, A., Ivanov, D., and Sokolov, B. (2018). Ripple effect in the supply chain: an analysis and recent literature. *International Journal of Production Research*, 56(1-2):414–430.
- [102] Dolgui, A., Ivanov, D., and Sokolov, B. (2020b). Reconfigurable supply chain: the X-network.
- [103] Dong, L., Qiu, Y., Xu, F., and Louis, S. (2022). Blockchain-Enabled Deep-Tier Supply Chain Finance. *Manufacturing & Service Operations Management*, pages 1–48.

- [104] Dorogovstev, S. and Mendes, J. F. F. (2003). *Evolution of Networks From Biological Nets to the Internet and WWW*. Oxford University Press, Oxford.
- [105] Dorogovtsev, S. N., Goltsev, A. V., and Mendes, J. F. F. (2006). k-core organization of complex networks. *Physical Review Letters*, 96(4):040601. arXiv: cond-mat/0509102.
- [106] Dorogovtsev, S. N., Goltsev, A. V., and Mendes, J. F. F. (2008). Critical phenomena in complex networks. *Reviews of Modern Physics*, 80(4):1275–1335.
- [107] Durairajan, R., Barford, P., Sommers, J., and Willinger, W. (2015). InterTubes: A Study of the US Long-haul Fiber-optic Infrastructure. In *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication - SIGCOMM '15*, pages 565–578, London, United Kingdom. ACM Press.
- [108] Durugbo, C., Hutabarat, W., Tiwari, A., and Alcock, J. R. (2011). Modelling collaboration using complex networks. *Information Sciences*, 181(15):3143–3161.
- [109] Edali, M. and Yasarcan, H. (2014). A mathematical model of the Beer game. *Jasss*, 17(4):1–7.
- [110] Endsley, M. R. (1995). Toward a Theory of Situation Awareness in Dynamic Systems. *Human Factors Journal*. *Human Factors Journal*, 37(1):32–64.
- [111] Eng, C. (2024). State of Software Security 2024: Addressing the Threat of Security Debt. Technical report, Veracode.
- [112] Ercsey-Ravasz, M. and Toroczkai, Z. (2010). Centrality scaling in large networks. *Physical Review Letters*, 105(3).
- [113] Erdős, P. and Rényi, A. (1959). On Random Graphs I. *Publicationes Mathematicae (Debrecen)*, 6:290–297.
- [114] Estrada, E. and Hatano, N. (2008). Communicability in complex networks. *Physical Review E*, 77(3):036111.
- [115] Estrada, E., Higham, D. J., and Hatano, N. (2009). Communicability betweenness in complex networks. *Physica A: Statistical Mechanics and its Applications*, 388(5):764–774.
- [116] ETSI (2018). 5G; Procedures for the 5G System.
- [117] Euler, L. (1741). Solutio problematis ad geometriam situs pertinentis. *Commentarii academiae scientiarum Petropolitanae*, 8:128–140.
- [118] Fang, Y. and Zio, E. (2013). Hierarchical Modeling by Recursive Unsupervised Spectral Clustering and Network Extended Importance Measures to Analyze the Reliability Characteristics of Complex Network Systems. *American Journal of Operations Research*, 03(01):101–112.
- [119] Farris, M. T. and Hutchison, P. D. (2002). Cash-to-cash: The new supply chain management metric. *International Journal of Physical Distribution and Logistics Management*, 32(4):288–298.
- [120] Fernández, E., Iglesias-Antelo, S., López-López, V., Rodríguez-Rey, M., and Fernandez-Jardon, C. M. (2019). Firm and industry effects on small, medium-sized and large firms’ performance. *BRQ Business Research Quarterly*, 22(1):25–35.
- [121] Ferraris, A., Santoro, G., and Dezi, L. (2017). How MNC’s subsidiaries may improve their innovative performance? The role of external sources and knowledge management capabilities. *Journal of Knowledge Management*, 21(3).

- [122] Fiedler, A. (2022). An agent-based negotiation protocol for supply chain finance. *Computers & Industrial Engineering*, 168:108136.
- [123] Ford, L. R. and Fulkerson, D. R. (1956). Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404.
- [124] Françoisse, K., Kivimäki, I., Mantrach, A., Rossi, F., and Saerens, M. (2017). A bag-of-paths framework for network data analysis. *Neural Networks*, 90:90–111.
- [125] Freeman, L. C. (1977). A Set of Measures of Centrality Based on Betweenness. *Sociometry*, 40(1):35.
- [126] Freeman, L. C. (1978). Centrality in social networks conceptual clarification. *Social Networks*, 1(3):215–239.
- [127] Freeman, L. C., Borgatti, S. P., and White, D. R. (1991). Centrality in valued graphs: A measure of betweenness based on network flow. *Social Networks*, 13(2):141–154.
- [128] Garreau, D. and Arlot, S. (2018). Consistent change-point detection with kernels. *Electronic Journal of Statistics*, 12(2):4440–4486.
- [129] Gatti, D. D., Di Guilmi, C., Gaffeo, E., Giulioni, G., Gallegati, M., and Palestrini, A. (2005). A new approach to business fluctuations: Heterogeneous interacting agents, scaling laws and financial fragility. *Journal of Economic Behavior and Organization*, 56(4 SPEC. ISS.):489–512.
- [130] Gelsomino, L. M., Mangiaracina, R., Perego, A., and Tumino, A. (2016). Supply chain finance: a literature review. *International Journal of Physical Distribution and Logistics Management*, 46(4):348–366.
- [131] Ghataoura, D. S., Mitchell, J. E., and Matich, G. E. (2011). Networking and application interface technology for wireless sensor network surveillance and monitoring. *IEEE Communications Magazine*, 49(10):90–97.
- [132] Giannoccaro, I., Nair, A., and Choi, T. (2018). The Impact of Control and Complexity on Supply Network Performance: An Empirically Informed Investigation Using NK Simulation Analysis. *Decision Sciences*, 49(4):625–659.
- [133] Gilbert, E. N. (1959). Random Graphs. *The Annals of Mathematical Statistics*, pages 1141–1144.
- [134] Giuggioli, L., Potts, J. R., and Harris, S. (2011). Animal Interactions and the Emergence of Territoriality. *PLoS Computational Biology*, 7(3):e1002008.
- [135] Gofman, M., Segal, G., Wu, Y., Giglio, S., Gutierrez, R., Herskovic, B., Julio, B., Kaniel, R., Kilic, M., Kogan, L., Konermann, P., Novy-Marx, R., Moreira, A., Savov, A., Shaliastovich, I., Tu, J., Tuzel, S., Yaron, A., and Zhang, H. (2020). Production Networks and Stock Returns: The Role of Vertical Creative Destruction. *The Review of Financial Studies*, 33(12):5856–5905.
- [136] Goldberg, P. and Reed, T. (2023). Is the Global Economy Deglobalizing? In *BPEA Conference Drafts*, pages 1–51.
- [137] Grafius, D. R., Varga, L., and Jude, S. (2020). Infrastructure Interdependencies: Opportunities from Complexity. *Journal of Infrastructure Systems*, 26(4).
- [138] Grigoris, F., Hu, Y., and Segal Gill (2023). Counterparty Risk: Implications for Network Linkages and Asset Prices. *The Review of Financial Studies*, 36(2):814–858.

- [139] Grimm, J. H., Hofstetter, J. S., and Sarkis, J. (2016). Exploring sub-suppliers' compliance with corporate sustainability standards. *Journal of Cleaner Production*, 112:1971–1984.
- [140] Gross, T. and Feudel, U. (2004). Analytical search for bifurcation surfaces in parameter space.
- [141] Gross, T. and Sayama, H. (2009). *Adaptive Networks*. Springer-Verlag Berlin Heidelberg, 1 edition.
- [142] Gupta, V., He, B., and Sethi, S. P. (2015). Contingent sourcing under supply disruption and competition. *International Journal of Production Research*, 53(10):3006–3027.
- [143] Han, J. and Shin, K. (2016). Evaluation mechanism for structural robustness of supply chain considering disruption propagation. *International Journal of Production Research*, 54(1):135–151.
- [144] Harjes, F., Scholz-Reiter, B., and Kaviani Mehr, A. (2011). Elman networks for the prediction of inventory levels and capacity utilization. *International Journal of Applied Mathematics and Informatics*, 5(4):283–290.
- [145] Harland, C., Brenchley, R., and Walker, H. (2003). Risk in supply networks. *Journal of Purchasing and Supply Management*, 9(2):51–62.
- [146] Harsono, I. and Suprapti, I. A. P. (2024). The Role of Fintech in Transforming Traditional Financial Services. *Accounting Studies and Tax Journal (COUNT)*, 1(1):81–91.
- [147] Havlin, S., Araujo, N. A. M., Buldyrev, S. V., Dias, C. S., Parshani, R., Paul, G., and Stanley, H. E. (2010). Catastrophic Cascade of Failures in Interdependent Networks. *arXiv:1012.0206 [cond-mat, physics:physics]*. arXiv: 1012.0206.
- [148] Herrera, M., Perez-Hernandez, M., Kumar Jain, A., and Kumar Parlikad, A. (2020). Critical link analysis of a national Internet backbone via dynamic perturbation. In *Advanced Maintenance Engineering, Services and Technologies*.
- [149] Herrera, M., Proselkov, Y., Perez-Hernandez, M., and Parlikad, A. K. (2021). Mining Graph-Fourier Transform Time Series for Anomaly Detection of Internet Traffic at Core and Metro Networks. *IEEE Access*, 9:8997–9011.
- [150] Ho Seung, Y., Choi, T. Y., and Kim, D. (2021). Integrating Sourcing and Financing Strategies in Multi-Tier Supply Chain Management. *International Journal of Production Economics*, 234(January):1–45.
- [151] Hobbs, J. E. (2020). Food supply chains during the COVID-19 pandemic. *Canadian Journal of Agricultural Economics*, 68(2):171–176.
- [152] Hofmann, E. (2007). The Flow of Financial Resources: an Inevitable Part of Supply Chain. *Strategic Supply Chain Design*, pages 173–201.
- [153] Hofmann, E., Maucher, D., Piesker, S., and Richter, P. (2011a). Strengthening Internal Financing Power Using Cash-to-Cash Cycle Optimization. In *Ways Out of the Working Capital Trap*, pages 13–20. Springer Berlin Heidelberg.
- [154] Hofmann, E., Maucher, D., Sabrina, P., and Richter, P. (2011b). *Ways Out of the Working Capital Trap - Empowering Self-Financing Growth Through Modern Supply Management*. Springer.
- [155] Hofmann, E. and Rutschmann, E. (2018). Big data analytics and demand forecasting in supply chains: a conceptual analysis. *International Journal of Logistics Management*, 29(2):739–766.

- [156] Hofmann, E., Templar, S., Rogers, D. S., Choi, T. Y., Leuschner, R., and Korde, R. Y. (2021). Supply Chain Financing and Pandemic: Managing Cash Flows to Keep Firms and Their Value Networks Healthy. *Rutgers Business Review*, 6(1):1–23.
- [157] Holland, P. W. and Leinhardt, S. (1977). A dynamic model for social networks. *The Journal of Mathematical Sociology*, 5(1):5–20.
- [158] Holme, P. and Saramäki, J. (2012). Temporal networks. *Physics Reports*, 519(3):97–125.
- [159] Hou, X., Ren, Z., Wang, J., Cheng, W., Ren, Y., Chen, K.-C., and Zhang, H. (2020). Reliable Computation Offloading for Edge-Computing-Enabled Software-Defined IoV. *IEEE Internet of Things Journal*, 7(8):7097–7111.
- [160] Hwang, W., Cho, Y.-r., Zhang, A., and Ramanathan, M. (2006). Bridging Centrality: Identifying Bridging Nodes In Scale-free Networks. Technical report, University of Buffalo, Buffalo.
- [161] Ilin, T. and Varga, L. (2015). The uncertainty of systemic risk. *Risk Management*, 17(4):240–275.
- [162] Inomata, S. and Hanaka, T. (2024). Measuring exposure to network concentration risk in global supply chains: Volume versus frequency. *Structural Change and Economic Dynamics*, 68:177–193.
- [163] Irshad, A., Chaudhry, S. A., Ghani, A., and Bilal, M. (2021). A secure blockchain-oriented data delivery and collection scheme for 5G-enabled IoD environment. *Computer Networks*, 195:108219.
- [164] Ivanov, D. (2020a). Predicting the impacts of epidemic outbreaks on global supply chains: A simulation-based analysis on the coronavirus outbreak (COVID-19/SARS-CoV-2) case. *Transportation Research Part E: Logistics and Transportation Review*, 136.
- [165] Ivanov, D. (2020b). Viable supply chain model: integrating agility, resilience and sustainability perspectives - lessons from and thinking beyond the COVID-19 pandemic. *Annals of Operations Research*.
- [166] Ivanov, D. (2022). Blackout and supply chains: Cross-structural ripple effect, performance, resilience and viability impact analysis. *Annals of Operations Research*.
- [167] Ivanov, D. (2023a). Intelligent digital twin (iDT) for supply chain stress-testing, resilience, and viability. *International Journal of Production Economics*, 263.
- [168] Ivanov, D. (2023b). Two views of supply chain resilience. *International Journal of Production Research*, pages 1–15.
- [169] Ivanov, D. and Dolgui, A. (2019). Low-Certainty-Need (LCN) supply chains: a new perspective in managing disruption risks and resilience.
- [170] Ivanov, D. and Dolgui, A. (2020). Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. *International Journal of Production Research*, 58(10):2904–2915.
- [171] Ivanov, D., Sokolov, B., and Dolgui, A. (2014). The Ripple effect in supply chains: Trade-off ‘efficiency-flexibility- resilience’ in disruption management. *International Journal of Production Research*, 52(7):2154–2172.
- [172] Jackson, B., Scargle, J., Barnes, D., Arabhi, S., Alt, A., Gioumousis, P., Gwin, E., San, P., Tan, L., and Tun Tao Tsai (2005). An algorithm for optimal partitioning of data on an interval. *IEEE Signal Processing Letters*, 12(2):105–108.

- [173] Jackson, I., Jesus Saenz, M., and Ivanov, D. (2024). From natural language to simulations: applying AI to automate simulation modelling of logistics systems. *International Journal of Production Research*, 62(4):1434–1457.
- [174] Jogalekar, P., Woodside, M., and Member, S. (2000). Evaluating the Scalability of Distributed Systems. *IEEE Transactions on Parallel and Distributed Systems*, 11(6):589–603.
- [175] Kalaiarasan, R., Olhager, J., Agrawal, T. K., and Wiktorsson, M. (2022). The ABCDE of supply chain visibility: A systematic literature review and framework. *International Journal of Production Economics*, 248.
- [176] Katehakis, M. N., Yang, J., and Zhou, T. (2020). Dynamic inventory and price controls involving unknown demand on discrete nonperishable items. *Operations Research*, 68(5):1333–1355.
- [177] Katz, L. (1953). A new status index derived from sociometric analysis. *Psychometrika*, 18(1):39–43.
- [178] Kazil, J., Masad, D., and Crooks, A. (2020). Utilizing Python for Agent-Based Modeling: The Mesa Framework. In *Lecture Notes in Computer Science*, pages 308–317. Springer International Publishing.
- [179] Kembro, J., Näslund, D., and Olhager, J. (2017). Information sharing across multiple supply chain tiers: A Delphi study on antecedents. *International Journal of Production Economics*, 193:77–86.
- [180] Kermack, W. O. and McKendrick, A. G. (1927). A Contribution to the Mathematical Theory of Epidemics. *The Royal Society*, 115(772):22.
- [181] Kermarrec, A.-M., Le Merrer, E., Sericola, B., and Trédan, G. (2011). Second order centrality: Distributed assessment of nodes criticality in complex networks. *Computer Communications*, 34(5):619–628.
- [182] Killick, R., Fearnhead, P., and Eckley, I. A. (2012). Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500):1590–1598. arXiv: 1101.1438.
- [183] Kim, Y., Choi, T. Y., Yan, T., and Dooley, K. (2011). Structural investigation of supply networks: A social network analysis approach. *Journal of Operations Management*, 29(3):194–211.
- [184] Kivela, M., Arenas, A., Barthelemy, M., Gleeson, J. P., Moreno, Y., and Porter, M. A. (2014). Multilayer networks. *Journal of Complex Networks*, 2(3):203–271.
- [185] Kong, L., Zheng, G., and Brintrup, A. (2024). A federated machine learning approach for order-level risk prediction in Supply Chain Financing. *International Journal of Production Economics*, 268:109095.
- [186] Kosasih, E. E., Papadakis, E., Baryannis, G., and Brintrup, A. (2024). A review of explainable artificial intelligence in supply chain management using neurosymbolic approaches. *International Journal of Production Research*, 62(4):1510–1540.
- [187] Kücher, A., Mayr, S., Mitter, C., Duller, C., and Feldbauer-Durstmüller, B. (2020). Firm age dynamics and causes of corporate bankruptcy: age dependent explanations for business failure. *Review of Managerial Science*, 14(3):633–661.
- [188] Kullback, S. and Leibler, R. A. (1951). On Information and Sufficiency. <https://doi.org/10.1214/aoms/1177729694>, 22(1):79–86.

- [189] Kumari, A., Gupta, R., Tanwar, S., and Kumar, N. (2020). A taxonomy of blockchain-enabled softwarization for secure UAV network. *Computer Communications*, 161:304–323.
- [190] Landau, E. (1895). Zur relativen Wertbemessung der Turnierresultate. *Deutsches Wochensach*, 11:366–369.
- [191] Leavitt, H. J. (1951). Some effects of certain communication patterns on group performance. *The Journal of Abnormal and Social Psychology*, 46(1):38–50.
- [192] Lebichot, B. and Saerens, M. (2018). A bag-of-paths node criticality measure. *Neurocomputing*, 275:224–236.
- [193] Ledwoch, A., Brintrup, A., Mehnen, J., and Tiwari, A. (2018a). Systemic Risk Assessment in Complex Supply Networks. *IEEE Systems Journal*, 12(2):1826–1837.
- [194] Ledwoch, A., Yasarcan, H., and Brintrup, A. (2018b). The moderating impact of supply network topology on the effectiveness of risk management. *International Journal of Production Economics*, 197:13–26.
- [195] Lee, H. L., Padmanabhan, V., and Whang, S. (1997). The bullwhip effect in supply chains. *Sloan Management Review*, pages 93–102.
- [196] Lee, H. L., Padmanabhan, V., and Whang, S. (2004). Information distortion in a supply chain: The bullwhip effect.
- [197] Lee, J. M.-y. and Wong, E. Y.-c. (2021). Suez Canal blockage: an analysis of legal impact, risks and liabilities to the global supply chain. *MATEC Web of Conferences*, 339:01019.
- [198] Lehmann, K. A. and Kaufmann, M. (2003). Decentralized algorithms for evaluating centrality in complex networks. Technical report, Eberhard Karls University of Tübingen.
- [199] Leibrock, M. (2022). Interconnectedness Revisited. Technical report, DTCC.
- [200] Leuschner, R., Choi, T. Y., Rogers, D. S., Hofmann, E., and Templar, S. (2023). ‘To fund’ as a new purpose of supply chain management: Making a case for supply chain financing. *Journal of Purchasing and Supply Management*, 29(5).
- [201] Lewis, T. G. (2009). *Network science: theory and practice*. John Wiley & Sons, Hoboken, N.J. OCLC: ocn265094260.
- [202] Li, M., Alam, Z., Bernardes, E., Giannoccaro, I., Skilton, P. F., Rahman, M. S., and Bari, P. (2021a). Out of Sight, out of Mind? Modeling the Impacts of Financial Squeeze on Extended Supply Chain Networks. *Journal of Business Logistics*, 42(2):233–263.
- [203] Li, Y., Chen, K., Collignon, S., and Ivanov, D. (2021b). Ripple effect in the supply chain network: Forward and backward disruption propagation, network health and firm vulnerability. *European Journal of Operational Research*, 291(3):1117–1131.
- [204] Li, Y. and Zobel, C. W. (2020). Exploring supply chain network resilience in the presence of the ripple effect. *International Journal of Production Economics*, 228.
- [205] Likic, V. and Shafi, K. (2018). Battlespace Mobile/Ad Hoc Communication Networks: Performance, Vulnerability and Resilience. In Sarker, R., Abbass, H., Dunstall, S., Kilby, P., Davis, R., and Young, L., editors, *Data and Decision Sciences in Action*, pages 303–314. Springer, Cham.
- [206] Liu, X., Zhao, J., Li, J., Cao, B., and Lv, Z. (2022). Federated Neural Architecture Search for Medical Data Security. *IEEE Transactions on Industrial Informatics*, 18(8):5628–5636.

- [207] Lohmer, J., Bugert, N., and Lasch, R. (2020). Analysis of resilience strategies and ripple effect in blockchain-coordinated supply chains: An agent-based simulation study. *International Journal of Production Economics*, 228.
- [208] Lü, L., Chen, D., Ren, X.-L., Zhang, Q.-M., Zhang, Y.-C., and Zhou, T. (2016). Vital nodes identification in complex networks. *Physics Reports*, 650:1–63.
- [209] Lv, Z., Chen, D., Feng, H., Zhu, H., and Lv, H. (2022). Digital Twins in Unmanned Aerial Vehicles for Rapid Medical Resource Delivery in Epidemics. *IEEE Transactions on Intelligent Transportation Systems*, 23(12):25106–25114.
- [210] Macias-Aguayo, J., McFarlane, D., Schönfuß, B., and Salter, L. (2022). A catalogue of digital solution areas for logistics SMEs. *IFAC-PapersOnLine*, 55(10):1828–1833.
- [211] Malak-Rawlikowska, A., Milczarek-Andrzejewska, D., and Falkowski, J. (2019). Farmers’ bargaining power and input prices: What can we learn from self-reported assessments? *Social Sciences*, 8(2).
- [212] Maloni, M. and Benton, W. C. (2000). Power Influences in The Supply Chain. *Journal of Business Logistics*, 21:16–36.
- [213] Manjula, K. A. and Karthikeyan P (2010). Distributed Computing Approaches for Scalability and High Performance. *International Journal of Engineering Science and Technology*, 2(6):2328–2336.
- [214] Marsden, P. V. (2002). Egocentric and sociocentric measures of network centrality. *Social Networks*, 24(4):407–422.
- [215] Martynyuk, O., Drozd, O., Van Thuong, B., Stepova, H., Sugak, L., and Martynyuk, D. (2023). Some Method of Behavioral Energy Testing of Distributed Computer Systems. In *2023 13th International Conference on Dependable Systems, Services and Technologies (DESSERT)*, pages 1–7. IEEE.
- [216] Mastilak, L., Helebrandt, P., Galinski, M., and Kotuliak, I. (2022). Secure Inter-Domain Routing Based on Blockchain: A Comprehensive Survey. *Sensors*, 22(4):1437.
- [217] Masuda, N., Porter, M. A., and Lambiotte, R. (2017). Random walks and diffusion on networks. *Physics Reports*, 716-717:1–58.
- [218] Mauthe, A., Hutchison, D., Cetinkaya, E. K., Ganchev, I., Rak, J., Sterbenz, J. P., Gunkelk, M., Smith, P., and Gomes, T. (2016). Disaster-resilient communication networks: Principles and best practices. In *2016 8th International Workshop on Resilient Networks Design and Modeling (RNDM)*, pages 1–10, Halmstad, Sweden. IEEE.
- [219] Mayer, A. H., Rodrigues, V. F., Costa, C. A. d., Righi, R. d. R., Roehrs, A., and Antunes, R. S. (2021). FogChain: A Fog Computing Architecture Integrating Blockchain and Internet of Things for Personal Health Records. *IEEE Access*, 9:122723–122737.
- [220] Mezei, J., Byanjankar, A., and Heikkilä, M. (2018). Credit Risk Evaluation in Peer-to-peer Lending With Linguistic Data Transformation and Supervised Learning. In *51st Hawaii International Conference on System Sciences*.
- [221] Milgram, S. (1967). The Small World Problem. *Psychology Today*, 1(1):61–67.
- [222] Miller, T. and Wongsaroj, S. (2017). The Domino Effect: the impact of late payments. Technical report, Sage.

- [223] Mohar, B., Alavi, E. Y., Chartrand, G., Oellermann, O. R., and Schwenk, A. J. (1991). The Laplacian spectrum of graphs. *Wiley*, pages 871–898.
- [224] Moriahra, Y. (2020). Vertical dis-integration and vertical re-integration: Limits to the modern production system. *Japanese Political Economy*, 46(2-3):176–199.
- [225] Moura, J. and Hutchison, D. (2019). Cyber-Physical Systems Resilience: State of the Art, Research Issues and Future Trends. *arXiv*, page 42.
- [226] Murfin, J. (2014). Big Box Retailers Squeeze Smaller Suppliers by Borrowing from Them.
- [227] Naghshbandi, S. N., Varga, L., and Dolan, T. (2020a). Review of emergent behaviours of systems comparable to infrastructure systems and analysis approaches that could be applied to infrastructure systems. Technical report, National Infrastructure Commission.
- [228] Naghshbandi, S. N., Varga, L., Purvis, A., McWilliam, R., Minisci, E., Vasile, M., Troffaes, M., Sedighi, T., Guo, W., Manley, E., and Jones, D. H. (2020b). A Review of Methods to Study Resilience of Complex Engineering and Engineered Systems. *IEEE Access*, 8:87775–87799.
- [229] Nair, A. and Vidal, J. M. (2011). Supply network topology and robustness against disruptions - An investigation using multi-agent model. *International Journal of Production Research*, 49(5):1391–1404.
- [230] Nanda, S. and Kotz, D. (2008a). Localized Bridging Centrality for Distributed Network Analysis. In *2008 Proceedings of 17th International Conference on Computer Communications and Networks*, pages 1–6, St. Thomas, US Virgin Islands. IEEE.
- [231] Nanda, S. and Kotz, D. (2008b). Localized Bridging Centrality for Distributed Network Analysis. In *17th International Conference on Computer Communications and Networks 2008*, pages 1–6. IEEE.
- [232] Neuts, M. F. (1993). The burstiness of point processes. *Communications in Statistics. Stochastic Models*, 9(3):445–466.
- [233] Newman, M., Barabasi, A.-L., and Watts, D. J. (2006a). *The Structure and Dynamics of Networks*. Princeton University Press.
- [234] Newman, M. E. J. (2003). The Structure and Function of Complex Networks. *SIAM Review*, 45(2):167–256.
- [235] Newman, M. E. J. (2010). *Networks: an introduction*. Oxford University Press, Oxford ; New York. OCLC: ocn456837194.
- [236] Newman, M. E. J., Barabasi, A.-L., and Watts, D. J. (2006b). *The Structure and Dynamics of Networks*. Princeton University Press, 1st edition.
- [237] Newman, M. J. (2005). A measure of betweenness centrality based on random walks. *Social Networks*, 27(1):39–54.
- [238] Ogawa, K., Tashiro, O., Kito, T., Yin, L., Chong, J., and Yamauchi, K. (2020). Evolution of Supply Chain Finance in Japan. Technical report, Fintech Association of Japan.
- [239] Orlin, J., Punnen, A., and Schulz, A. (2004). Approximate Local Search in Combinatorial Optimization. *SIAM J. Comput.*, 33:587–596.
- [240] Özelkan, E. C. and Çakanyildirim, M. (2009). Reverse bullwhip effect in pricing. *European Journal of Operational Research*, 192(1):302–312.

- [241] Page, L., Brin, S., Motwani, R., and Winograd, T. (1998). The PageRank Citation Ranking: Bringing Order to the Web. Technical Report, Stanford InfoLab.
- [242] Pan, X., Guo, S., and Chu, J. (2021). P2P supply chain financing, R&D investment and companies' innovation efficiency.
- [243] Parzen, E. (1962). On the Estimation of Probability Density Functions and Mode. *Ann. Math. Statist.*, 33:1065–1076.
- [244] Pathak, S. D., Day, J. M., Nair, A., Sawaya, W. J., and Murat Kristal, M. (2007a). Complexity and Adaptivity in Supply Networks: Building Supply Network Theory Using a Complex Adaptive Systems Perspective *. *Decision Sciences*, 38(4):547–580.
- [245] Pathak, S. D., Dilts, D. M., and Biswas, G. (2007b). On the evolutionary dynamics of supply network topologies. *IEEE Transactions on Engineering Management*, 54(4):662–672.
- [246] Patil, C. and Prabhu, V. (2024). Supply chain cash-flow bullwhip effect: An empirical investigation. *International Journal of Production Economics*, 267:109065.
- [247] Perko, I. (2017). Behaviour-based short-term invoice probability of default evaluation. *European Journal of Operational Research*, 257(3):1045–1054.
- [248] Petersen, M. A. and Rajan, R. G. (1997). Trade Credit: Theories and Evidence. *The Review of Financial Studies*, 10(3):661–691.
- [249] Peterson, I. (1995). *Fatal Defect: Chasing Killer Computer Bugs*, volume 20. Times Books.
- [250] Pettit, T. J., Fiksel, J., and Croxton, K. L. (2010). Ensuring Supply Chain Resilience: Development of a Conceptual Framework. *Journal of Business Logistics*, 31(1):1–21.
- [251] Pfohl, H.-C. (2022). *Logistics Systems: Business Fundamentals*. Springer, Berlin, 1 edition.
- [252] Pichler, A., Diem, C., Brintrup, A., Lafond, F., Magerman, G., Buiten, G., Choi, T. Y., Carvalho, V. M., Farmer, J. D., and Thurner, S. (2023). Building an alliance to map global supply networks. *Science*, 382(6668):270–272.
- [253] Porter, M. A. and Gleeson, J. P. (2015). Dynamical Systems on Networks: A Tutorial. *arXiv:1403.7663 [cond-mat, physics:nlin, physics:physics]*. arXiv: 1403.7663.
- [254] Porter, M. E. and Kramer, M. R. (2011). Creating Shared Value. *Harvard Business Review*.
- [255] Proselkov, Y., Herrera, M., Hernandez, M. P., Parlikad, A. K., and Brintrup, A. (2022). The value of information for dynamic decentralised criticality computation. In *14th IFAC Workshop on Intelligent Manufacturing Systems IMS*, volume 55, pages 408–413, Tel-Aviv, Israel.
- [256] Proselkov, Y., Herrera, M., Parlikad, A. K., and Brintrup, A. (2020). Distributed Dynamic Measures of Criticality for Telecommunication Networks. In *Service Oriented, Holonic and Multi-agent Manufacturing Systems for Industry of the Future*, pages 1–12, Paris, France. Springer.
- [257] Proselkov, Y., Zhang, J., Xu, L., Hofmann, E., Choi, T. Y., Rogers, D., and Brintrup, A. (2023). Financial ripple effect in complex adaptive supply networks: an agent-based model. *International Journal of Production Research*, pages 1–23.
- [258] Proselkov, Y., Zhang, J., Xu, L., Hofmann, E., Choi, T. Y., Rogers, D., and Brintrup, A. (2024). Financial ripple effect in complex adaptive supply networks: an agent-based model. *International Journal of Production Research*, 62(3):823–845.

- [259] Punzo, G., Tewari, A., Butans, E., Vasile, M., Purvis, A., Mayfield, M., and Varga, L. (2020). Engineering Resilient Complex Systems: the Necessary Shift toward Complexity Science. *IEEE Systems Journal*, 14(3):3865–3874.
- [260] Rafaeli, S. and Hutchison, D. (2003). A survey of key management for secure group communication. *ACM Computing Surveys*, 35(3):309–329.
- [261] Randall, W. S. and Farris, M. T. (2009). Supply chain financing: Using cash-to-cash variables to strengthen the supply chain. *International Journal of Physical Distribution and Logistics Management*, 39(8):669–689.
- [262] Reynolds, D. A. (2009). Gaussian Mixture Models. In *Encyclopedia of Biometrics*.
- [263] Rezapour, S., Farahani, R. Z., and Pourakbar, M. (2017). Resilient supply chain network design under competition: A case study. *European Journal of Operational Research*, 259(3):1017–1035.
- [264] Rivera-Gonzalez, C., Holguin-Veras, J., and Calderon, O. (2023). Supply-chain-focused measures of centrality and spread in metropolitan areas. *Journal of Transport Geography*, 107:103553.
- [265] Rogers, D., Leuschner, R., and Choi, T. Y. (2016). The Rise of FinTech in Supply Chains. *Harvard Business Review*.
- [266] Rogers, D., Leuschner, R., and Choi, T. Y. (2020). *Supply Chain Financing: Funding the Organization and the Supply Chain*. World Scientific Publishing Co., London, United Kingdom.
- [267] Rosenblatt, M. (1956). Remarks on Some Nonparametric Estimates of a Density Function. *The Annals of Mathematical Statistics*, 27(3):832–837.
- [268] Salazar, J. C., Nejjari, F., Sarrate, R., Weber, P., and Theilliol, D. (2016). Reliability Importance Measures for Availability Enhancement in Drinking Water Networks. In *2016 3rd Conference on Control and Fault-Tolerant Systems (SysTol)*, Barcelona, Spain.
- [269] Sandhu, R. S., Georgiou, T. T., and Tannenbaum, A. R. (2016). Ricci curvature: An economic indicator for market fragility and systemic risk. *Science Advances*, 2(5).
- [270] Satoshi Nakamoto (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. Technical report, Bitcoin.
- [271] Savitzky, A. and Golay, J. E. M. (1964). Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry*, 36(8):1627–1639.
- [272] Saxena, A. and Iyengar, S. (2020). Centrality Measures in Complex Networks: A Survey. *arXiv*.
- [273] Scheidt, D. H. (2002). Intelligent Agent-Based Control. *Johns Hopkins APL Technical Digest*, 23(4).
- [274] Schmitt, A. J., Sun, S. A., Snyder, L. V., and Shen, Z. J. M. (2015). Centralization versus decentralization: Risk pooling, risk diversification, and supply chain disruptions. *Omega (United Kingdom)*, 52:201–212.
- [275] Sedgwick, D., Kuryulko, D., LaReau, J., and Kranz, R. (2008). As dollar dives, automakers juggle global strategies.
- [276] Seidman, S. B. (1983). Network structure and minimum degree. *Social Networks*, 5(3):269–287.

- [277] Shao, B. B., Shi, Z. M., Choi, T. Y., and Chae, S. (2018). A data-analytics approach to identifying hidden critical suppliers in supply networks: Development of nexus supplier index. *Decision Support Systems*, 114:37–48.
- [278] Sharp, K. and Matschinsky, F. (2015). Translation of Ludwig Boltzmann’s Paper “On the Relationship between the Second Fundamental Theorem of the Mechanical Theory of Heat and Probability Calculations Regarding the Conditions for Thermal Equilibrium” *Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften. Mathematisch-Naturwissen Classe. Abt. II, LXXVI* 1877, pp 373-435 (Wien. Ber. 1877, 76:373-435). Reprinted in *Wiss. Abhandlungen*, Vol. II, reprint 42, p. 164-223, Barth, Leipzig, 1909. *Entropy*, 17(4):1971–2009.
- [279] Shen, B., Cao, Y., and Xu, X. (2020a). Product line design and quality differentiation for green and non-green products in a supply chain. *International Journal of Production Research*, 58(1):148–164.
- [280] Shen, B., Wang, X., Cao, Y., and Li, Q. (2020b). Financing decisions in supply chains with a capital-constrained manufacturer: competition and risk. *International Transactions in Operational Research*, 27(5):2422–2448.
- [281] Shen, B., Xu, X., Chan, H. L., and Choi, T. M. (2021). Collaborative innovation in supply chain systems: Value creation and leadership structure. *International Journal of Production Economics*, 235.
- [282] Sheng, H., Cong, R., Yang, D., Chen, R., Wang, S., and Cui, Z. (2022). UrbanLF: A Comprehensive Light Field Dataset for Semantic Segmentation of Urban Scenes. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(11):7880–7893.
- [283] Shibuya, Y. and Babich, V. (2021). Multi-tier Supply Chain Financing with Blockchain. *Georgetown McDonough School of Business Research Paper No. 3787044*, pages 1–43.
- [284] Shulgin, B. (1998). Pulse vaccination strategy in the SIR epidemic model. *Bulletin of Mathematical Biology*, 60(6):1123–1148.
- [285] Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Routledge, London.
- [286] Sokolov, B., Ivanov, D., Dolgui, A., and Pavlov, A. (2016). Structural quantification of the ripple effect in the supply chain. *International Journal of Production Research*, 54(1):152–169.
- [287] Sornette, D. (2006). Endogenous versus Exogenous Origins Crises. In Alberverio, S., Jentsch, V., and Kantz, H., editors, *Extreme Events in Nature and Society*, pages 95–119. Springer, Berlin.
- [288] Spielman, D. A. (2011). Algorithms, Graph Theory, and Linear Equations in Laplacian Matrices. In *Proceedings of the International Congress of Mathematicians 2010 (ICM 2010)*, pages 2698–2722, Hyderabad, India. Published by Hindustan Book Agency (HBA), India. WSPC Distribute for All Markets Except in India.
- [289] Stanley, R. P. (1973). Acyclic orientations of graphs. *Discrete Mathematics*, 5(2):171–178.
- [290] Stevens, S. S. (1946). On the Theory of Scales of Measurement. *Science, New Series*, 103(2684):677–680.
- [291] Strom, S. (2015). Big Companies Pay Later, Squeezing Their Suppliers.
- [292] Surana, A., Kumara, S., Greaves, M., and Raghavan, U. N. (2005). Supply-chain networks: A complex adaptive systems perspective. *International Journal of Production Research*, 43(20):4235–4265.

- [293] Tabachová, Z., Diem, C., Borsos, A., Burger, C., and Thurner, S. (2023). Estimating the impact of supply chain network contagion on financial stability. *arXiv preprint*.
- [294] Thadakamalla, H. P., Raghavan, U. N., Kumara, S., and Albert, R. (2004). Survivability of Multiagent-Based Supply Networks: A Topological Perspective. *IEEE Intelligent Systems*, 19(5).
- [295] Tran, H. T., Domerçant, J. C., and Mavris, D. N. (2016). A Network-based Cost Comparison of Resilient and Robust System-of-Systems. *Procedia Computer Science*, 95:126–133.
- [296] Truong, C., Oudre, L., and Vayatis, N. (2020). Selective review of offline change point detection methods. *Signal Processing*, 167:107299.
- [297] Tseliou, G., Samdanis, K., Adelantado, F., Perez, X. C., and Verikoukis, C. (2016). A capacity broker architecture and framework for multi-tenant support in LTE-A networks. In *2016 IEEE International Conference on Communications (ICC)*, pages 1–6, Kuala Lumpur, Malaysia. IEEE.
- [298] Ullah, A., Wang, B., Sheng, J., Long, J., Khan, N., and Ejaz, M. (2022). A novel relevance-based information interaction model for community detection in complex networks. *Expert Systems with Applications*, 196:116607.
- [299] Uppala, M. and Desmarais, B. A. (2023). Contagion, Confounding, and Causality: Confronting the Three C’s of Observational Political Networks Research. *Political Analysis*, 31(3):472–479.
- [300] van Mieghem, P. (2014). *Performance Analysis of Complex Networks and Systems*. Cambridge University Press, Cambridge.
- [301] Veres, A. and Boda, M. (2005). *Complex Dynamics in Communication Networks*. Springer: Complexity.
- [302] Vilalta, R., Lopez, V., Giorgetti, A., Peng, S., Orsini, V., Velasco, L., Serral-Gracia, R., Morris, D., De Fina, S., Cugini, F., Castoldi, P., Mayoral, A., Casellas, R., Martinez, R., Verikoukis, C., and Munoz, R. (2017). TelcoFog: A Unified Flexible Fog and Cloud Computing Architecture for 5G Networks. *IEEE Communications Magazine*, 55(8):36–43.
- [303] Virakwan Hai Kelian, Mohd Nazri Mohd Warip, R. Badlishah Ahmad, Phaklen Ehkan, Fazrul Faiz Zakaria, and Mohd Zaizu Ilyas (2023). Toward Adaptive and Scalable Topology in Distributed SDN Controller. *Advanced Research in Applied Sciences and Engineering Technology*, 30(1):115–131.
- [304] von Mises, R. and Pollaczek-Geiringer, H. (1929). Praktische Verfahren der Gleichungsauflösung. *Zamm-zeitschrift Fur Angewandte Mathematik Und Mechanik*, 9:152–164.
- [305] Waller, M., Johnson, M., and Davis, T. (1999). Vendor-Managed Inventory in the Retail Supply Chain. *Journal of Business Logistics*, 20:183–203.
- [306] Waller, M. A. and Fawcett, S. E. (2013). Data Science, Predictive Analytics, and Big Data: A Revolution That Will Transform Supply Chain Design and Management. *Journal of Business Logistics*, 34(2).
- [307] Wang, J., Liu, Y. H., Jiao, Y., and Hu, H. Y. (2009). Cascading dynamics in congested complex networks. *The European Physical Journal B*, 67:95–100.
- [308] Watts, D. J. and Dodds, P. S. (2007). Influentials, Networks, and Public Opinion Formation. *Journal of Consumer Research*, 34(4):441–458.
- [309] Watts, D. J. and Strogatz, S. H. (1998). Collective dynamics of ‘small-world’ networks. *Nature*.

- [310] Wehmuth, K. and Ziviani, A. (2011a). Distributed location of the critical nodes to network robustness based on spectral analysis. In *2011 7th Latin American Network Operations and Management Symposium*, pages 1–8, Quito, Ecuador. IEEE.
- [311] Wehmuth, K. and Ziviani, A. (2011b). Distributed location of the critical nodes to network robustness based on spectral analysis. In *2011 7th Latin American Network Operations and Management Symposium - (LANOMS 2011)*, pages 1–8. IEEE.
- [312] Wei, Q. and Zhang, Q. (2016). P2P Lending Risk Contagion Analysis Based on a Complex Network Model. *Discrete Dynamics in Nature and Society*, 2016:8.
- [313] Wen, G., Lv, Y., Zheng, W. X., Zhou, J., and Fu, J. (2023). Joint Robustness of Time-Varying Networks and Its Applications to Resilient Consensus. *IEEE Transactions on Automatic Control*, 68(11):6466–6480.
- [314] Wen, G., Yu, W., Lv, Y., and Wang, P. (2021). *Cooperative Control of Complex Network Systems with Dynamic Topologies*. CRC Press, Boca Raton, 1 edition.
- [315] West, D. (2000). Neural network credit scoring models. *Computers & Operations Research*, 27(11):1131–1152.
- [316] Xiao, S., Xiao, G., and Cheng, T. H. (2008). Tolerance of Intentional Attacks in Complex Communication Networks. *IEEE Communications Magazine*, page 7.
- [317] Xu, L., Almahri, S., Mak, S., and Brintrup, A. (2024a). Multi-Agent Systems and Foundation Models Enable Autonomous Supply Chains: Opportunities and Challenges. In *18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024*, Vienna, Austria.
- [318] Xu, L., Mak, S., Proselkov, Y., and Brintrup, A. (2023a). Autonomous supply chains: Definition, characteristics, conceptual framework and autonomy levels. *Submitted for publication*.
- [319] Xu, L., Mak, S., Proselkov, Y., and Brintrup, A. (2024b). Towards Autonomous Supply Chains: Definition, Characteristics, Conceptual Frameworks, and Autonomy Levels. *Journal of Industrial Information Integration*, 42.
- [320] Xu, L., Proselkov, Y., Schoepf, S., Minarsch, D., Minaricova, M., and Brintrup, A. (2023b). Implementation of Autonomous Supply Chains for Digital Twinning: a Multi-Agent Approach. In *22nd IFAC World Congress*, volume 56, pages 11076–11081, Yokohama, Japan. Elsevier B.V.
- [321] Yan, G., Eidenbenz, S., Thulasidasan, S., Datta, P., and Ramaswamy, V. (2010). Criticality analysis of Internet infrastructure. *Computer Networks*, 54(7):1169–1182.
- [322] Yan, R., Yuan, W., Su, X., and Zhang, Z. (2023). FLPA: A fast label propagation algorithm for detecting overlapping community structure. *Expert Systems with Applications*, 234:120971.
- [323] Yan, T., Choi, T. Y., Kim, Y., and Yang, Y. (2015). A Theory of the Nexus Supplier: A Critical Supplier From A Network Perspective. *Journal of Supply Chain Management*, 51(1):52–66.
- [324] Yang, Q., Xing, J., and Wang, P. (2012). An agent-based framework for collaboration among critical mechanical devices working in a cluster. *Proceedings of the Institution of Mechanical Engineers. Part I: Journal of Systems and Control Engineering*, 226(9):1249–1261.
- [325] Yang, Y., Que, X., Qian, Q., and Fang, H. (2023). An agent-based model of supply chain financing: agents’ strategies, performance, and risk contagions. *Journal of Simulation*, pages 1–18.

- [326] Yao, L., Li, X., Zheng, R., and Zhang, Y. (2022). The Impact of Air Pollution Perception on Urban Settlement Intentions of Young Talent in China. *International Journal of Environmental Research and Public Health*, 19(3):1080.
- [327] Yao, Y.-C. (1984). Estimation of a Noisy Discrete-Time Step Function: Bayes and Empirical Bayes Approaches. *Annals of Statistics*, 12(4):1434–1447.
- [328] Yazıcı, V., Kozat, U., and Sunay, M. (2014). A new control plane for 5G network architecture with a case study on unified handoff, mobility, and routing management. *IEEE Communications Magazine*, 52(11):76–85.
- [329] Yi, S., Li, C., and Li, Q. (2015). A survey of fog computing: Concepts, applications and issues. In *Mobidata 2015 - Proceedings of the 2015 Workshop on Mobile Big Data, co-located with MobiHoc 2015*, volume 2015-June, pages 37–42.
- [330] You, K., Tempo, R., and Qiu, L. (2017). Distributed Algorithms for Computation of Centrality Measures in Complex Networks. *IEEE Transactions on Automatic Control*, 62(5):2080–2094. arXiv: 1507.01694.
- [331] Yu, W., Wen, G., Chen, G., and Cao, J. (2016). *Distributed Cooperative Control of Multi-agent Systems*. Wiley.
- [332] Zhang, J. and Thomas, L. C. (2016). The effect of introducing economic variables into credit scorecards: an example from invoice discounting. *The Journal of Risk Model Validation*, 9(1):57–78.
- [333] Zhao, K., Zuo, Z., and Blackhurst, J. V. (2019). Modelling supply chain adaptation for disruptions: An empirically grounded complex adaptive systems approach. *Journal of Operations Management*, 65(2):190–212.
- [334] Zheng, G., Kong, L., and Brintrup, A. (2023). Federated machine learning for privacy preserving, collective supply chain risk prediction. *International Journal of Production Research*.